

# Birthday Cutoffs and Hereditary Sets

The set-theoretic content of bounded surreal arithmetic:  
bi-interpretations, elementary inclusions, axiom spectra,  
outer models, and surcomplex orientation

September 22, 2026

## Abstract

Expand the ordered field of surreal numbers of birthday below an ordinal  $\lambda$  by its birthday function. The global connection of this kind of structure with set theory is the subject of a bi-interpretation theorem announced by Chen, Hamkins and Yang. This report, merged from five manuscripts that prove overlapping versions of one local theorem, asks *which set universe is recovered from a particular birthday cutoff, and what that recovery implies*.

For an epsilon number  $\lambda$  let  $\kappa(\lambda)$  be the least cardinal not below  $\lambda$ . One parameter-free interpretation recovers exactly  $H_{\kappa(\lambda)}$ . At cardinal cutoffs, singular ones included, this is a uniform bi-interpretation with  $H_\lambda$ ; at noncardinal cutoffs it is a bi-interpretation with  $(H_{\kappa(\lambda)}, \in, \lambda)$ , and the cutoff is recovered by a definable collector. Thus the field of birthdays below  $\varepsilon_0$  already recovers all hereditarily countable sets. Small arithmetic certificates justify the reverse interpretation without Replacement in  $H_\kappa$ , and a second route codes graphs modulo bisimulation. The forward interpretation extends to the ring cutoffs, including  $\omega$ .

Consequences within one universe: for epsilon numbers  $\lambda < \eta$  the inclusion of birthday structures is elementary exactly when both are cardinals and  $H_\lambda \prec H_\eta$ ; elementary embeddings correspond to those of the hereditary universes; one sentence detects noncardinal cutoffs and another the strong-limit cardinals; birthday eternity is Replacement at cardinal cutoffs, and with fiber packing it selects the worldly cardinals; in ZF the graph coding reaches exactly the sets with well-orderable transitive closure. Across universes: old arithmetic is absolute, a bounded structure is unchanged exactly when no bounded subset is added, the first new birthday is the least ordinal with a new subset and is a ground-model cardinal, explicit formulas detect new subsets, and, assuming a measurable cardinal, Prikry forcing leaves the enriched field unchanged while singularizing its cutoff. Under GBC the full structure has no nontrivial elementary self-embedding. For surcomplex numbers with conjugation and coordinate birthday, each real elementary embedding has exactly two lifts, and one nonreal parameter, but no set of real parameters, defines an orientation.

Two later parts work below ZFC. Over Zermelo set theory without Replacement, bounding the birthdays of definable surreal families is ordinal bounding (with a hierarchy axiom, Replacement), and kernel models with atoms have exact Collection spectra; over ZF, universality of the surreals for free fields on all sets is exactly the first Kinna–Wagner principle; named atom symmetries preserving Replacement are classified. A last part gives numbers-only theories bi-interpretable with ZFC, also in the omnific integers, and the boundary at NBG.

Built from five manuscripts written independently in relation to the `VladimirReshetnikov/Surreal` project, all pinned to commit `4f26455`, and merged after placement commit `d4e71b7`; Parts VI–VII (batch 89) add two more, pinned to `2b7b388`, placed by `23adb85`; Part VIII (batch 90) one, pinned to `8dc2592`, placed by `12076b2`; Part IX (batch 95) two, pinned to `7223374` and `bd1de45`, placed by `350b9a9`.

An AI-assisted research draft with written proofs and finite regression checks; not refereed; not a claim of any Lean formalization. The global bi-interpretation is prior announced work of Chen–Hamkins–Yang.

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## Part I

# The cutoff theorem

## 1 The question, the results, and their provenance

The field structure of the surreal numbers and their construction history carry very different information. Real-closed-field quantifier elimination makes all inclusions between real closed subfields elementary in the ordered field language. A birthday function remembers much more: together with the order it reads individual sign coordinates, and thereby encodes subsets, relations and well-founded membership graphs. The issue addressed here is not merely whether this coding is possible globally, but how its strength changes when only birthdays below one ordinal are allowed, and what the resulting dictionary says about elementary inclusions, set-existence axioms, outer models of set theory, and the surcomplex numbers.

### 1.1 What is already known, and credited

**The global theorem is prior work.** Chen, Hamkins, and Yang announced that the surreal ordered field with birthday order is bi-interpretable with the set-theoretic universe, and described a first-order theory of surreal arithmetic intended to be bi-interpretable with ZFC. Hamkins’s September and November 2025 slides give the global coding by pointed well-founded extensional relations, together with birthday-growth principles; the March 2026 workshop announcement and slides repeat the global theorem and label the axiomatic work as still being checked [6, 7, 8]. This report does *not* claim that theorem, its pointed-graph coding idea, or the association of birthday growth with set-existence axioms as new. The collection’s **foundations-and-computation/foundations** report already records the March 2026 announcement, at slide level and with its status boundary, in its subsection **found:sub:announcement**, and lists it among its non-claims as work in progress that is not a proved dependency of that report. Of the five manuscripts merged here, sources 02, 06 and 09 credit the announcement prominently. Sources 05 and 07 do not cite it; source 07 presented the full-class bi-interpretation as a proposed new contribution. That presentation is corrected here: [theorem 18.1](#) states the announced theorem with its credit, and the argument printed there, taken from source 07, is a written proof of the announced theorem, not a new result.

**Other prior inputs.** If  $\lambda$  is an epsilon number,  $\mathbf{No}_{<\lambda}$  is a real closed subfield. The original van den Dries–Ehrlich paper must be used together with its erratum [3, 4]; Bournez–Guilmant state the corrected ring, field and real-closedness results in their Theorems 1.1–1.2 [5]. The collection’s foundations report states the same classical fact in its subsection **found:sub:cutoffs**, including that  $\varepsilon_0$  is already an epsilon number, so that no large cardinal is needed. The sign construction and recursive arithmetic are classical [1, 2]. Mostowski collapse, well-founded recursion, hereditary-size universes and forcing are standard [13]; quantifier elimination for real closed and algebraically closed fields is standard [14, 15]. The general obstruction to parameter-free bi-interpretation caused by complex conjugation, and the role of a named imaginary unit, were explained by Jeřábek in 2021 [9]; every use of that obstruction below ([section 19](#)) is credited to him. Bi-interpretation methodology follows Freire–Hamkins [10]. The Prikry forcing facts are imported from Poveda–Rinot–Sinapova [11], and the class form of Kunen’s inconsistency from Hamkins–Kirmayer–Perlmutter [12]. Rangel–Mariano’s algebraic set theory of surreal numbers and Bertram’s realization of numbers as sets of

ordinals are separate programs with different languages and aims; nothing here settles or refutes them [16, 17].

## 1.2 The principal statements

We use  $\text{TC}(\{x\})$  for the transitive closure containing  $x$  and all its iterated members, and write

$$H_\kappa = \{x : |\text{TC}(\{x\})| < \kappa\}.$$

For an epsilon number  $\lambda$  (an ordinal with  $\omega^\lambda = \lambda$  in ordinary ordinal exponentiation) put

$$\mathcal{B}_\lambda = (\mathbf{No}_{<\lambda}; 0, 1, +, -, \cdot, <, \mathbf{b}),$$

where  $\mathbf{b}(x)$  is the birthday of  $x$ , regarded as the all-plus surreal ordinal of that length. “Uniform” means that fixed finite formulas work at every cutoff in the stated class, not that a separate language is chosen for each cardinal. The source numbers in brackets are explained in [section 1.3](#).

**Theorem 1.1** (Cardinal rounding and cutoff recovery; [theorem 6.3](#)). *Let  $\lambda$  be an epsilon number and let*

$$\kappa(\lambda) = \min\{\kappa : \kappa \text{ is a cardinal and } \lambda \leq \kappa\}.$$

*There is one parameter-free interpretation  $\mathcal{I}$  such that*

$$\mathcal{I}(\mathcal{B}_\lambda) \cong (H_{\kappa(\lambda)}, \in).$$

*If  $\lambda$  is a cardinal,  $\mathcal{B}_\lambda$  and  $(H_\lambda, \in)$  are uniformly bi-interpretable. If  $\lambda$  is not a cardinal,  $\mathcal{B}_\lambda$  and  $(H_{\kappa(\lambda)}, \in, \lambda)$  are uniformly bi-interpretable, and the named ordinal  $\lambda$  is definable, as an interpreted set, in  $\mathcal{B}_\lambda$  itself.*

This is the most general form in the cluster, and it is source 09’s. The other four sources prove cases of it with their own codes:

- source 02: the forward interpretation onto  $H_{\kappa(\lambda)}$  at every epsilon number, one way only (it explicitly claims no reverse interpretation);
- source 05: the bi-interpretation at uncountable *regular* cardinals, with graphs modulo bisimulation ([section 7](#));
- sources 06 and 07: the bi-interpretation at every uncountable cardinal, singular ones included; source 07 also proves the forward interpretation onto  $H_{\kappa(\lambda)}$  at every epsilon number, which it calls a *cardinality plateau* and states forward only.

This is a cardinal-rounding statement, not a rank-cutoff statement: for example  $\kappa(\varepsilon_0) = \omega_1$ , whereas  $V_{\varepsilon_0}$  is not the recovered universe. At a singular cardinal  $\kappa$  the recovered structure is still  $H_\kappa$ ; no regularity assumption is hidden in the theorem.

**Theorem 1.2** (Elementary-inclusion spectrum; [theorem 9.3](#)). *For epsilon numbers  $\lambda < \eta$ ,*

$$\mathcal{B}_\lambda \prec \mathcal{B}_\eta \iff \lambda, \eta \text{ are cardinals and } H_\lambda \prec H_\eta.$$

*The ordered-field reducts, in contrast, always form an elementary inclusion.*

Source 09 proves the equivalence at all epsilon numbers. Source 02 proves the forward implication and left the converse as its first open question; that question is answered by source 09 ([remark 9.4](#)). Sources 05 (regular cardinals), 06 and 07 (uncountable cardinals) prove the equivalence between cardinal cutoffs, and 05 and 06 add the correspondence of elementary embeddings ([theorem 9.10](#)).

**Theorem 1.3** (Growth spectrum; [theorem 11.2](#)). *For an epsilon-number cutoff, birthday eternity Et fails at every noncardinal cutoff. At an uncountable cardinal  $\kappa$ ,*

$$\mathcal{B}_\kappa \models \text{Et} \iff H_\kappa \models \text{Replacement}.$$

*Fiber packing Pack, and also the translated Power Set sentence  $\text{Pow}^\mathcal{I}$ , hold exactly at the uncountable strong-limit cardinal cutoffs. Consequently*

$$\mathcal{B}_\lambda \models \text{Et} + \text{Pack} \iff \lambda \text{ is a cardinal and } V_\lambda \models \text{ZFC}.$$

The eternity, packing and worldly statements are source 09's. The translated Power Set sentence is source 02's, at every epsilon number, and source 06's at uncountable cardinals ([theorem 10.7](#)).

**Theorem 1.4** (Extension rigidity at a strong-limit cutoff; [theorem 16.4](#)). *Let  $M \subseteq N$  be transitive models of ZFC with the same ordinals, and let  $\kappa$  be an uncountable cardinal in both, strong limit in  $M$ . Then*

$$\mathcal{B}_\kappa^M \prec \mathcal{B}_\kappa^N \iff \mathcal{B}_\kappa^M = \mathcal{B}_\kappa^N \iff \mathcal{P}^M(\alpha) = \mathcal{P}^N(\alpha) \quad (\alpha < \kappa).$$

*Without the strong-limit hypothesis, a new subset of  $\alpha < \kappa$  still prevents elementarity whenever  $(2^{|\alpha|})^M < \kappa$ .*

The detector is sharp for the old-power-set *parameter* ([proposition 16.2](#)), not a claim that every other possible detector needs that much space. The report also proves:

- the forward interpretation at ring cutoffs, including  $\mathbf{No}_{<\omega}$  onto the hereditarily finite sets ([proposition 8.1](#), source 02);
- one sentence true exactly at noncardinal epsilon cutoffs ([theorem 6.2](#), sources 02 and 09);
- a separation of  $\mathcal{B}_{\omega^-}$  from  $\mathcal{B}_{\omega^+}$  by one Replacement instance ([theorem 11.5](#), source 06);
- the exact range of the graph coding over ZF without Choice ([theorem 13.2](#), source 06);
- the transfer of elementary equivalence, the interpretation of full second-order arithmetic, the independence property of the bit formula, an omitted parameter-free type, and nondefinability of birthday in the field reduct ([section 12](#), sources 05, 06, 07);
- the exact preservation criterion across outer models, the first new birthday, three explicit new-subset detectors, and the full-class statement  $\mathcal{B}^M \prec \mathcal{B}^N \iff M = N$  ([Part III](#), sources 02, 05, 06, 07, 09);
- conditional on a measurable cardinal, an unchanged enriched field whose cutoff changes cofinality, the preserved  $H_\kappa = V_\kappa^M$ , and the exact onset of new elements above the cutoff ([section 17](#), sources 02 and 06);
- under GBC, no nontrivial class elementary self-embedding of the full birthday structure, by transfer of Kunen's inconsistency ([theorem 18.3](#), source 07);
- for surcomplex numbers with conjugation and coordinate birthday: exactly two elementary lifts of each real elementary embedding (source 06), the exact parameter threshold for a definable orientation (source 06), the mutual-interpretation versus bi-interpretation distinction (sources 05, 06, 09, after Jeřábek), and, under GBC, the elementary self-embeddings of the full structure (source 07) ([section 19](#)).
- in Part VI (source 10, added in batch 89), over Zermelo set theory with Foundation and without Replacement or Choice: simplest separators of set cuts without Replacement, the equivalence of six definable-family schemes with ordinal bounding and, over an explicit hierarchy axiom, with full Replacement and Collection (the weak-base form of [theorem 10.4](#)); the exact Collection spectra of kernel urelement models; and a language-sensitive separation that pure surreal structures cannot see ([theorems 24.1 to 24.3](#));

- in Part VII (source 11, added in batch 89), over ZF: a set injects into **No** iff it injects into the power set of an ordinal; universality of **No** and **Oz** for free algebras on all sets is exactly the first Kinna–Wagner principle; optimal set-sized real-closed targets; sharp transcendence bounds in two permutation models and, in ZFA, an ordered field with no surreal embedding (theorems 43.5, 44.3 and 49.1);
- in Part VIII (source 12, added in batch 90), in the finite-kernel urelement model with a named action of a pure group on the atoms: Replacement holds exactly when all orbits are finite and only finitely many atoms lie in finitely repeated finite orbit types (for finitely many named permutations, already in the research-report collection); Replacement for outputs with at most  $k$  atoms is decided by the centralizer orbits of size at most  $k$ ; two involutions realize every finite level; and pure-valued Collection, hence the whole pure surreal universe, survives every such naming (theorems 58.1, 59.1, 61.3 and 63.1).

### 1.3 Five manuscripts, one report

This report is built from five manuscripts, written independently on September 22, 2026, all pinned to repository commit 4f2645599121fa872c7104995e47f86f0382351f and placed in the collection by commit d4e71b7. Their numbers are those of the placement batch and are local to this directory.

| Source | Manuscript                                                                                                                                                    | Contribution to this report                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                     |
|--------|---------------------------------------------------------------------------------------------------------------------------------------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 09     | <i>The Set-Theoretic Content of Bounded Surreal Arithmetic: hereditary-size universes, coherent bi-interpretations, axiom spectra, and extension rigidity</i> | The base. Its text is the spine of Part I (sections 2 to 6), the elementary-inclusion classification (section 9), the eternity, packing and worldly calibration (section 10), the cross-model coherence, detector and extension rigidity (section 16), the orientation obstruction (theorem 19.4), and most of section 20. Files prefixed 09-bounded-arithmetic-.                                                                                                                                                                               |
| 02     | <i>What a Birthday Cutoff Remembers: hereditary set universes, elementary obstructions, and forcing invisibility</i>                                          | The one-way forward theorem at all epsilon numbers (a special case of the base), ring cutoffs (section 8), the collector sentence (printed once with 09's), the translated Power Set sentence at every epsilon number, the preservation criterion at epsilon cutoffs and its countable corollary, the Prikry invisibility theorem with its preserved $H_\kappa$ and the onset above the cutoff (section 17), the concrete example in Appendix C. Files prefixed 02-birthday-cutoffs-, and its source audit 02-birthday-cutoffs-SOURCE_AUDIT.md. |

| Source | Manuscript                                                                                                                          | Contribution to this report                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                 |
|--------|-------------------------------------------------------------------------------------------------------------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 05     | <i>Birthdays Recover Sets: bi-interpretability, elementary embeddings, and forcing in surreal and surcomplex fields</i>             | The second route by codes modulo bisimulation (section 7), the explicit embedding formula and critical-point corollary, the separation of $\mathcal{B}_{\omega_1}$ and $\mathcal{B}_{\omega_2}$ , nondefinability of birthday, the explicit missing-row detector New, the full-class statement $\mathcal{B}^M \prec \mathcal{B}^N \iff M = N$ , the first new birthday for arbitrary same-ordinal outer models, the countably closed example, and the surcomplex forcing contrast. Files prefixed <code>05-birthdays-recover-sets-</code> . |
| 06     | <i>What Birthday-Enriched Surreal Fields Remember: hereditary-size interpretations, choice, forcing, and surcomplex orientation</i> | The bi-interpretation at singular cardinals (printed once with the base), the second route by prefix-pair arithmetic tables (section 3.4), definitional equivalence with $(\mathbf{No}_{<\kappa}; <, \mathbf{b}, G_\varpi)$ , transfer of elementary equivalence, the $\sqsubset_\omega$ Replacement instance, the ZF range theorem, the four-way preservation criterion, the Prikry corollaries, the two complex lifts, and the orientation threshold. Files prefixed <code>06-hereditary-size-orientation-</code> .                       |
| 07     | <i>The Birthday Expansion of the Surreal Field: recovering set theory, detecting forcing, and rigidity of surcomplex embeddings</i> | The written proof of the announced full-class theorem (theorem 18.1), the radix pairing, the $\varepsilon_0$ localization (printed once as a case of the base), recovery of models of ZFC, stage agreement at every ordinal, the two-parameter outer-model witness, the independence property and omitted type, the o-minimal nondefinability, Kunen transfer (theorem 18.3) and the surcomplex elementary self-embeddings. Files prefixed <code>07-birthday-expansion-</code> .                                                            |

**Why one report.** All five prove the same spine: the ordered field of surreals below a cutoff, expanded by birthday, interprets the hereditary sets  $H_\kappa$  by sign bits read from order and birthday, a bounded ordinal pairing polynomial, and pointed well-founded relation codes, and in most versions is bi-interpretable with them. They differ in the cutoff class and in whether the result is one-way or two-way. Source 09 has the widest cutoff class (all epsilon numbers, which include all uncountable cardinals by lemma 2.2) and the strongest conclusion (two-way, with the correct pointed target), so it is the base.

**Printed once.** The prefix and bit formulas (all five), the bounded pairing (all five, in four variants compared in remark 4.4), the rooted collapse and exact range (02, 06, 07, 09), the arithmetic certificates (05, 06, 07, 09), the bi-interpretation at cardinal cutoffs (05 regular; 06, 07, 09 all), the  $\varepsilon_0$  case (02, 07, 09), the collector sentence (02, 09), the elementary-inclusion classification (02 one direction; 05, 06, 07, 09), the translated Power Set threshold (02, 06, with 09's version inside  $H_\kappa$ ), absoluteness of old arithmetic (02, 05, 07, 09), the preservation criterion (02, 06, 07, 09), the first

new birthday (02, 05, 07), the outer-model non-elementarity (05, 07, 09), the Prikry invisibility (02, 06), rigidity of the bounded structures (05, 06, 07, 09), the surcomplex real-axis recovery (all five) and the two-element automorphism group (05, 06, 07, 09). Each is printed once and credited to every source that proves it.

**Kept twice, as different proofs.** The reverse arithmetic has two routes: source 09’s option-graph certificates built from finite signed words ([theorem 3.3](#)), and the prefix-pair tables of sources 05, 06 and 07 ([section 3.4](#)), whose single- $\mu$  size bound is source 06’s argument at singular cutoffs. The coding of sets has two routes: source 09’s extensional rooted codes with isomorphism as equality ([section 5](#)), and source 05’s codes modulo bisimulation ([section 7](#)). The first new birthday has two existence arguments: the new-subset lemma of 05 and 07 for any outer model, and the generic filter of 02 for set forcing ([remark 15.5](#)). The new-subset detectors are three different formulas ([section 16](#)).

**Where the merge had to choose.** (i) Hypotheses: whenever a result is proved by several sources under different hypotheses, the weakest proved hypothesis is printed, and the proof printed is one that uses only it; where this attaches a source’s argument to a weaker hypothesis than the source stated, the fact is said at the point of use ([remarks 3.7](#) and [16.7](#)). (ii) The first new birthday is stated for arbitrary same-ordinal transitive outer models (05, 07), with 02’s set-forcing theorem as a special case. (iii) The full-class bi-interpretation is printed as the announced theorem of Chen–Hamkins–Yang, not as a result of source 07. (iv) Symbols were renamed as listed in [section 1.4](#).

**Two later parts.** [Added 3 October 2026, batch 89.] Parts VI and VII print two further manuscripts, written independently of the five above and of each other, both pinned to commit `2b7b388ba`, delivered in commit `7c0f2d9f9` and placed by commit `23adb85f9`. Source 10 (batch-89 manuscript 03, *Surreal Cuts, Replacement, and Atom-Support Spectra*; files `10-cuts-replacement-`, labels `hset:zr:`) is [Part VI](#); source 11 (batch-89 manuscript 04, *Surreal Universality Without Choice*; files `11-choicelless-universality-`, labels `hset:kw:`) is [Part VII](#). The numbers 10 and 11 continue this directory’s local sequence after 09. Each part keeps its own setting, hypotheses and notation convention and ends with its non-claims and provenance ([sections 39](#) and [53](#)). Nothing in Parts I–V was renumbered; dated notes in Parts II and V record where the new parts bear on them.

**A further part.** [Added 4 October 2026, batch 90.] Part VIII prints source 12 (batch-90 manuscript 05, *Named Symmetries and Replacement*; files `12-named-symmetries-`, labels `hset:ns:`), pinned to commit `8dc2592e9`, delivered in commit `a162e4386` and placed by commit `12076b2e8`: [Part VIII](#). It continues source 10 at the finite cutoff, answering its [question 38.6](#) for named group actions, and it keeps its own setting, notation convention ([convention 54.1](#)) and closing provenance ([section 66](#)). Nothing earlier was renumbered; dated notes in Part VI point to it.

**A ninth part.** [Added 4 October 2026, batch 95.] Part IX merges two manuscripts written independently on one assignment: source 13 (batch-95 manuscript 01, *Surreal and Omnific Foundations*, pinned to commit `722337445`; the base; its prefix `13-omnific-foundations-` names no shipped file) and source 14 (batch-95 manuscript 02, *Surreal-Only Foundations and the Set-Class Boundary*, pinned to `bd1de458b`; files `14-surreal-only-nbg-`), both delivered in commit `ec91f8c7c` and placed by commit `350b9a954`; it is [Part IX](#), with labels `hset:sf:`. Both re-derive [theorem 18.1](#)

and give it a theory-level form, in **No** and natively in the omnific integers, and both prove the boundary at class theory. The part keeps its own notation convention ([convention 67.2](#)) and closing provenance ([section 80](#)). Nothing earlier was renumbered; dated notes after [theorem 18.1](#), after the “Minimal signature” question and after [question 21.7](#) point to it.

## 1.4 Conventions and renamed symbols

**Convention 1.5** (One meaning per symbol). The following conventions hold throughout, including in material taken from sources that used other letters.

| Notion                                     | Here                                                                                                          | In the sources                                                                                                                                                                          |
|--------------------------------------------|---------------------------------------------------------------------------------------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Birthday of $x$ , as a surreal ordinal     | $\mathbf{b}(x)$                                                                                               | $\mathbf{b}$ (02); $b$ (05, 07, 09); $b$ for the ordinal and $\mathbf{b}$ for its surreal (06)                                                                                          |
| Cutoff carrier                             | $\mathbf{No}_{<\lambda}$                                                                                      | $F_\lambda$ (02), $S_\kappa$ (05, 06), $\mathbf{No}_{<\kappa}$ (07), $K_\lambda$ (09)                                                                                                   |
| Birthday structure                         | $\mathcal{B}_\lambda$ ; full class $\mathcal{B}$                                                              | $\mathfrak{B}_\lambda$ (02), $\mathcal{S}_\kappa$ (05), $\mathcal{A}_\kappa$ (06), $\mathcal{B}_\kappa$ (07, 09)                                                                        |
| Cardinal ceiling                           | $\kappa(\lambda)$                                                                                             | $\theta(\lambda)$ , the least <i>uncountable</i> cardinal $\geq \lambda$ (07); equal for epsilon $\lambda$                                                                              |
| Pairing length                             | $\Lambda(\theta) = \theta^2 + \theta$                                                                         | $B_\theta$ (02); $L(a, d)$ (05); $\tau(\delta)$ (06); $q^2$ (07)                                                                                                                        |
| Code                                       | $C = (\delta, r, t)$ : domain, relation word, root                                                            | $(\theta, d, e, r)$ (02); $(\lambda, c, r)$ with $\lambda$ a node count (05); $(\delta, e, r)$ (06); $(q, d, e, r)$ (07)                                                                |
| Interpreted $=, \in$                       | $\sim, \in_{\mathcal{I}}$                                                                                     | $\text{Eq}, \text{Mem}$ (02); $\equiv, \in^*$ (06); $\simeq, \in_*$ (07)                                                                                                                |
| Ordinal code                               | $\text{OC}(a)$                                                                                                | $\text{OC}$ (02), $\text{Sync}$ (05), $\text{OrdCode}$ (06), $\text{OCode}$ (07), $O(a)$ (09)                                                                                           |
| Collector                                  | $\text{Coll}(C)$ ; sentence $\exists C \text{ Coll}(C)$                                                       | $\text{CollectOrd}$ (02), $\text{Coll}$ (09)                                                                                                                                            |
| Replacement instance at $\sqsupset_\omega$ | $\text{Rep}_{\sqsupset}$                                                                                      | $\text{Collect}_{\sqsupset}$ (06)                                                                                                                                                       |
| Translated Power Set                       | $\text{Pow}^{\mathcal{I}}$                                                                                    | $\text{Power}$ (02), $\text{Pow}^\#$ (06)                                                                                                                                               |
| Eternity, fiber packing                    | $\text{Et}, \text{Pack}$                                                                                      | $\text{E}, \text{P}$ (09)                                                                                                                                                               |
| Conjugation                                | $c$                                                                                                           | $c$ (02, 06); overline (05, 07); $\sigma$ (09)                                                                                                                                          |
| Coordinate birthday                        | $\text{cb}(x + iy) = \max(\mathbf{b}x, \mathbf{b}y)$                                                          | $\beta$ (02, 05, 09); $B$ (06, 07)                                                                                                                                                      |
| Surcomplex structures                      | $\mathcal{C}_\lambda$ (no $i$ named), $\mathcal{C}_\lambda^i$ ( $i$ named); full $\mathcal{C}, \mathcal{C}^i$ | $\mathfrak{K}_\lambda$ with $i$ (02); $\mathcal{C}_\kappa, \mathcal{C}_\kappa^-$ (05); $\mathcal{K}_\kappa$ (06); $\mathcal{C}_b, \mathcal{C}_b^i$ (07); $\mathcal{C}_\lambda^\pm$ (09) |
| First new birthday                         | $\beta = \beta(M, N)$                                                                                         | $\delta$ (02, 05, 07)                                                                                                                                                                   |
| New-subset formula of 05                   | $\text{New}(m, \theta, \nu)$                                                                                  | $\text{New}(c, \theta, \lambda)$ , $\lambda$ an enumeration length (05)                                                                                                                 |
| Eternity bound                             | $\varrho$                                                                                                     | $\beta$ (09)                                                                                                                                                                            |

Three collisions with the collection’s shared notation are resolved as follows ([docs/NOTATION.md](#)). The collection reserves  $F_\Gamma$  and  $K_\Gamma$  for the real and complex Hahn workspaces and  $K$  for a set-sized *complex* field; sources 02 and 09 used  $F_\lambda$  and  $K_\lambda$  for birthday cutoffs (09’s  $K_\lambda$  a *real* field), so this report writes  $\mathbf{No}_{<\lambda}$  and  $\mathbf{No}_{<\lambda}[i]$  instead. The collection’s  $H_\alpha$  is a subgroup of a value group (bounded multiples of  $\alpha$ ); here  $H_\kappa$  is always the universe of sets of hereditary size below  $\kappa$ , and

no value group occurs. The collection’s  $\mathcal{O}$  is a valuation ring, hence the ordinal code is written OC. The letter  $\beta$  is reserved for the first new birthday of an outer model, as in the sibling report `foundations-and-computation/surreal-fields-across-universes`; that report uses  $\delta$  for the first new *ordinal-sequence length*, a notion not named here, while here  $\delta$  is only the domain ordinal of a code. Sources 02, 05 and 07 wrote  $\delta$  for the first new birthday.

Two further words have two meanings in the sources. “Collect” named a sentence collecting the literal ordinals in source 02 and a Replacement instance in source 06; the two are  $\exists C \text{ Coll}(C)$  and  $\text{Rep}_{\sqsupset}$  here. A sans-serif P was fiber packing in source 09 and Power Set in sources 02 and 06; they are **Pack** and  $\text{Pow}^{\mathcal{I}}$  here, and **Pack** is *not* the Power Set axiom. Birthday precedence and prefix simplicity are different relations (`found:rem:simplicity` of the foundations report); prefix is defined below from order and birthday, and is never identified with precedence.

[Added 3 October 2026, batch 89: Parts VI and VII keep these conventions and add their own (`conventions 24.4` and `40.1`), where their sources’  $H$ ,  $SG$ ,  $M_{\kappa}$ ,  $S_{\kappa}$ ,  $K_{\kappa}$ ,  $C_{\alpha}$ ,  $\beta$  and  $\delta$  are renamed.]

## 1.5 Relation to the collection and corrected repository statements

**Foundations.** The foundations report records the Chen–Hamkins–Yang announcement (`found:sub:announcement`), the epsilon-number cutoffs (`found:sub:cutoffs`) and the distinction between simplicity and birthday precedence (`found:rem:simplicity`). Its inventory notes that the field language has no birthday symbol; [proposition 12.5](#) supplies the standard nondefinability proof. None of the interpretation theorems of this report was in the collection before it.

**Surcomplex automorphisms.** The report `surcomplex/surcomplex-field-automorphisms` proves  $\text{Aut}(K/F) = \{\text{id}, c\}$  for  $K = F(i)$  with  $F$  real closed (`saut:thm:axis`) and, in its section `saut:sec:exponential`, subsection “Simplicity restores the canonical surreal numbers”, that a bijection preserving numerical order and prefix simplicity is the identity, so that a real-axis-preserving automorphism whose real restriction preserves simplicity is the identity or conjugation. Since order and birthday define prefix ([lemma 4.1](#)), the bounded automorphism statements of sources 05, 06, 07 and 09 are consequences of those results; they are printed once, as [lemma 19.3](#) and [theorem 19.4](#), and credited there. What is new relative to that report is only source 07’s classification of the class elementary self-embeddings of the full structure, which are not assumed surjective ([theorem 19.7](#)), and source 06’s two-lift classification of elementary embeddings between different cutoffs ([theorem 19.5](#)). No question named in that report is answered or refuted; in particular its question whether  $\{\text{id}, c\}$  is a proper subgroup of its group  $\text{Aut}_L$  of field automorphisms commuting with  $L(z) = \log|z|$  (equivalently, whether a nontrivial exponential automorphism of **No** exists) is untouched.

**Product birthdays.** No argument here uses a bound  $b(xy) \leq b(x) \otimes b(y)$ . That bound is Gonshor’s product-birthday conjecture; the collection’s `surreal/gonshor-product-birthdays` report proves it on a specified ring of ordinal-support series and states that the general conjecture is not settled. Sources 05, 06, 07 and 09 all say that they avoid it, and the size estimates below use only field closure of the epsilon-number cutoffs.

**Sibling report.** The report `foundations-and-computation/surreal-fields-across-universes`, placed in the same batch, studies the external saturation of the *pure* ordered field  $\mathbf{No}^M$  inside an outer universe, without a birthday symbol. Its setting meets this report’s outer-model

part ([Part III](#)): the absoluteness of old arithmetic, the first new birthday, and the forcing examples. The two languages give opposite phenomena: the pure-field inclusion stays elementary but can lose saturation, while the birthday expansion loses elementarity ([remark 15.8](#)).

**Other reports.** The `foundations-and-computation/computable-surreals` report concerns countable effective subfields and is a different program; the undecidability remarks of [section 12](#) are standard. No report of the collection is contradicted by this one.

**Repository statements of the sources, corrected.** All five sources compared themselves with the collection at the pin `4f2645599121fa872c7104995e47f86f0382351f`, which is 32 commits before the placement commit `d4e71b7`.

- Sources 05 and 07 report that the catalogue at the pin described forty catalogued reports and four newly placed drafts, and that the formalization ledger counted forty-four main texts. This was true at the pin. Before this batch was placed the catalogue listed 44 research reports in five families.
- Sources 05 and 07 report that no matching reconstruction theorem was located in the material they inspected, and source 07 that the uniform bi-interpretation had not been located in earlier published work or in the repository. Source 05 read only the first 230 lines of the foundations report and source 07 its introduction; the Chen–Hamkins–Yang announcement is recorded in that report at the pin and now. The full-class statement is therefore credited to the announcement ([theorem 18.1](#)), and the local refinements of 05 and 07 carry the same priority caveat as those of 02, 06 and 09.
- Sources 05 and 06 cite line ranges and section structure of the surcomplex automorphisms report at the pin. That report has since been rewritten; its subsection “Simplicity restores the canonical surreal numbers” still exists, now in its section `saut:sec:exponential`, and the relative result is its `saut:thm:axis`.
- Source 06 reports that a GitHub search for “Hamkins” returned no indexed matches. That was a search-index observation; a direct search of the repository finds the announcement in the foundations report.
- Sources 02 and 07 report that the incoming-report directory contained only its README at the pin; that was true at the pin.

Source 02’s own source audit, kept byte-identical as `02-birthday-cutoffs-SOURCE_AUDIT.md`, contains no statement about the repository that is false; its closing description of the elementary-inclusion criterion as “necessary, not claimed sufficient” describes source 02 and is superseded inside this report by [theorem 9.3](#).

## 2 Foundations and the two ordinal languages

### 2.1 Ambient theory and class conventions

The proofs take place in ZFC, except in [section 13](#) (ZF) and in [section 18.2](#) (GBC, stated there). All cutoff structures are sets. The statements about the full surreal class are definable-class statements, understood one first-order formula at a time. Neither a set of all surreals nor an internally available truth predicate for the entire universe is assumed. Quotient interpretations use definable equivalence relations; the Mostowski collapse supplies set representatives when we compare them with  $H_\kappa$ . Global Choice is not used outside [section 18.2](#).

When a transitive model  $M$  is explicitly mentioned, its surreals are the sign sequences belonging to  $M$ , and cardinalities with superscript  $M$  are computed there. A statement proved internally in an arbitrary model of ZFC does not assert that the model's well-founded relations are externally well founded. The outer-model comparison theorems explicitly use transitive models.

## 2.2 The carrier and the birthday function

A surreal is a function  $s : \alpha \rightarrow \{-, +\}$  for an ordinal  $\alpha$ . Comparison is lexicographic, with termination between  $-$  and  $+$ . The all-plus sequence of length  $\alpha$  is the embedded ordinal  $\alpha$ . Throughout,  $b(s)$  is the *surreal ordinal* corresponding to the length of  $s$ . Thus it is a function on the surreal carrier, not an extra ordinal sort. Set-theoretic lengths and their surreal embeddings are distinguished by context.

For an ordinal  $\lambda$  put  $\mathbf{No}_{<\lambda} = \{x \in \mathbf{No} : b(x) < \lambda\}$ , a set. For an epsilon number  $\lambda$  define

$$\mathcal{B}_\lambda = (\mathbf{No}_{<\lambda}; 0, 1, +, -, \cdot, <, b), \quad \mathcal{B} = (\mathbf{No}; 0, 1, +, -, \cdot, <, b).$$

The language includes no exponentiation, support predicate, membership predicate, or second-order quantifier.

**Lemma 2.1** (Birthday order and birthday function). *On each  $\mathcal{B}_\lambda$ , and on  $\mathcal{B}$ , the birthday function and the birthday-precedence relation  $x \prec_b y \iff b(x) < b(y)$  are definitionally interconvertible. So is the equal-birthday relation  $E_b(x, y) \iff b(x) = b(y)$ . Moreover, the embedded ordinals are exactly the fixed points of  $b$ .*

*Proof.* Given the function, precedence and equal birthday are immediate. Conversely, equality of birthdays is expressed by neither  $x \prec_b y$  nor  $y \prec_b x$ . In the class of sign sequences of length  $\alpha$ , the largest in numerical order is the all-plus sequence: any other sequence of that length has a first minus sign and is smaller. Therefore  $b(x)$  is the unique largest member of  $x$ 's equal-birthday class,

$$y = b(x) \iff E_b(x, y) \wedge \forall z (E_b(x, z) \rightarrow z \leq y),$$

and this uses only  $E_b$  and the order. A sign sequence is fixed by this function exactly when it is all plus. Each cutoff contains the all-plus representative of each of its birthdays, so the definitions work in every  $\mathcal{B}_\lambda$ .  $\square$

The interconvertibility with precedence is proved in sources 02, 06 and 09; the sufficiency of the coarser equal-birthday relation is source 07's observation. In particular the results below concern a definitional variant of the birthday-precedence language of the Chen–Hamkins–Yang announcement, not an essentially stronger coding apparatus. We abbreviate  $\text{Ord}(a)$  by  $b(a) = a$ . Quantification over embedded ordinals in  $\mathbf{No}_{<\lambda}$  is precisely quantification over  $\alpha < \lambda$ .

## 2.3 Natural arithmetic, not ordinary ordinal arithmetic

For ordinal arguments, surreal addition and multiplication are the Hessenberg operations

$$\alpha + \gamma = \alpha \oplus \gamma, \quad \alpha \gamma = \alpha \otimes \gamma.$$

They are strictly increasing in each positive argument where appropriate, and  $\alpha + 1$  agrees with ordinary ordinal successor. We use ordinary ordinal exponentiation only in metamathematical bounds and in explicitly set-theoretic constructions. It is not silently identified with a field operation.

**Lemma 2.2** (Closure and cardinal estimates). *Every uncountable cardinal is an epsilon number. For infinite ordinals  $\alpha, \gamma$ , natural sums and products have cardinality at most  $\max(\aleph_0, |\alpha|, |\gamma|)$ . In particular, if  $\kappa$  is an uncountable cardinal and  $\alpha, \gamma < \kappa$ , their natural sum and product are below  $\kappa$ .*

*Proof.* In Cantor normal form, a natural sum joins finitely many exponent terms; a natural product uses finitely many natural sums of their exponents. The cardinality estimate follows from the corresponding estimate for  $\omega^\xi$ : for infinite  $\xi$ , its cardinality is at most  $\max(\aleph_0, |\xi|)$ . One can see the latter by representing ordinals below  $\omega^\xi$  by finite Cantor-normal-form lists with exponents below  $\xi$  and natural-number coefficients. These observations also cover finite arguments after adjoining  $\aleph_0$  to the bound.

If  $\kappa$  is uncountable and initial,  $\omega^\alpha < \kappa$  for every  $\alpha < \kappa$ . Continuity gives  $\omega^\kappa \leq \kappa$ . The opposite inequality follows from  $\omega^\alpha \geq \alpha$  and taking suprema. Hence  $\omega^\kappa = \kappa$ . Neither step uses regularity of  $\kappa$ .  $\square$

All five sources prove the first assertion; van den Dries and Ehrlich already note that every uncountable cardinal is an epsilon number [3], as source 06 records. For a general epsilon number, closure of natural arithmetic below  $\lambda$  also follows from the classical field-cutoff theorem. This simple closure is all the ordinal arithmetic needed by the relation codes.

## 2.4 Hereditary size rather than rank

**Lemma 2.3** (Basic hereditary-size facts). *Let  $\kappa$  be an uncountable cardinal. A sign sequence of length  $\alpha$  belongs to  $H_\kappa$  exactly when  $\alpha < \kappa$ . Every subset of an ordinal  $\alpha < \kappa$  belongs to  $H_\kappa$ . Also  $H_\kappa \subseteq V_\kappa$ , and  $\text{Ord} \cap H_\kappa = \kappa$ .*

*Proof.* The transitive closure of the usual set-coded graph of a sign sequence has cardinality at most  $\max(\aleph_0, |\alpha|)$ , and it contains enough of the domain ordinal to have cardinality at least  $|\alpha|$  when  $\alpha$  is infinite. The same estimate applies to a subset of  $\alpha$ , and  $\text{TC}(\{\alpha\}) = \alpha + 1$ .

If  $|\text{TC}(\{x\})| \leq \mu$  for an infinite  $\mu < \kappa$ , the membership relation on this transitive closure is a well-founded relation of size at most  $\mu$ . Its rank function takes values below  $\mu^+$ . This follows by well-founded induction, using regularity of the successor cardinal  $\mu^+$ : the supremum of at most  $\mu$  ordinals below  $\mu^+$  remains below  $\mu^+$ . Thus  $\text{rank}(x) < \mu^+ \leq \kappa$ , and  $x \in V_\kappa$ .  $\square$

The statement that every subset of  $\alpha$  belongs to  $H_\kappa$  does *not* say that the collection  $\mathcal{P}(\alpha)$  is an element of  $H_\kappa$ . This distinction drives the growth results. Three notions must not be interchanged: a family whose individual members are small, a family that is a small set, and a family that belongs to  $H_\kappa$ . For example, all subsets of  $\omega$  belong to  $H_{\omega_1}$ , but their collection does not (source 06).

## 3 Small certificates for surreal arithmetic

A reverse interpretation cannot simply say “perform the usual set-theoretic recursion inside  $H_\kappa$ .” At a singular cutoff, Replacement may fail in that structure. We instead prove that each individual arithmetic computation has a certificate of the same hereditary cardinal scale as its inputs. Two routes are given: source 09’s option-graph certificates (theorem 3.3) and the prefix-pair tables of sources 05, 06 and 07 (section 3.4).

### 3.1 Legal cuts can be checked using prefixes

For sets of sign sequences  $L < R$ , let  $\{L \mid R\}$  denote their simplest separator. The classical simplicity theorem supplies this separator and the bound

$$b(\{L \mid R\}) \leq \sup\{b(a) + 1 : a \in L \cup R\}. \quad (3.1)$$

The supremum of the empty set is zero. These statements can equivalently be proved by the sign-by-sign cut construction [2].

**Lemma 3.1** (Prefix test for a cut value). *A sign sequence  $z$  equals  $\{L \mid R\}$  if and only if it separates  $L$  and  $R$  and no proper sign prefix of  $z$  separates them.*

*Proof.* The forward implication is immediate. For the converse, suppose  $y$  is a separator shorter than  $z$ . If  $y$  is a prefix of  $z$ , the stated test fails. If it is not, their longest common prefix lies numerically between them and is a proper prefix of  $z$ . The set of separators is convex, so this common prefix is another separator. Again the test fails. Thus  $z$  has minimum birthday among separators, and uniqueness of the simplest separator gives the result.  $\square$

The check in this lemma quantifies only over the displayed option sets and the prefixes of  $z$ . In set coding it is a bounded check: one does not quantify over all sign sequences of earlier birthdays.

### 3.2 Well-founded option graphs of small size

A labeled option graph has a set  $D$  of vertices, left and right successor relations, and a rank label that strictly decreases along every option edge. A valuation of such a graph assigns a sign sequence to each vertex so that its value passes lemma 3.1 for the values at its options. When the graph is numeric, this valuation exists and is unique.

**Lemma 3.2** (Small numeric graphs have small certificates). *Suppose  $\mu$  is infinite and a numeric, well-founded option graph has at most  $\mu$  vertices, with its structural data of hereditary size at most  $\mu$ . Then it has a rank-labeled valuation certificate of hereditary size at most  $\mu$ .*

*Proof.* The graph rank of every vertex is below  $\mu^+$ , by the successor-cardinal argument in lemma 2.3. Applying (3.1) by induction shows that a vertex value has birthday at most that vertex's graph rank. Thus each rank label and each sign value has hereditary size at most  $\mu$ . There are at most  $\mu$  vertices and at most  $\mu^2 = \mu$  edges, so the union of all these data still has hereditary size at most  $\mu$ . The existence construction here takes place in the ambient ZFC universe, not in a possibly deficient  $H_\kappa$ .  $\square$

### 3.3 The computation graph for addition and multiplication

Take all sign prefixes of the inputs, including the inputs themselves. With

$$\mu = \max(\aleph_0, |b(x)|, |b(y)|),$$

this prefix family and its data have hereditary size at most  $\mu$ . For addition, use finite signed words of prefix atoms  $a(a)$ ; an option moves one atom to one of its sign-prefix options, reversing left and right for a negative atom. The root word is  $a(x) + a(y)$ . This is the usual disjunctive-sum construction of the numeric game for  $x + y$ .

For multiplication, use finite signed words of pair atoms  $\mathbf{p}(a, a')$ , with  $a, a'$  prefixes of the respective inputs. An option of a positive pair atom is a signed three-term word

$$\mathbf{p}(a^\circ, a') + \mathbf{p}(a, a'^\circ) - \mathbf{p}(a^\circ, a'^\circ). \quad (3.2)$$

It is a left option when  $(a^\circ, a'^\circ)$  is left–left or right–right, and a right option for the two mixed choices. Negative atoms exchange the sides and negate the replacement word. An option of a word replaces one atom by one of these words. The root is  $\mathbf{p}(x, y)$ . This spells out Conway’s multiplication recursion without first constructing a large universe of games.

There are at most  $\mu$  possible finite words. These graphs are well founded. For addition, use the natural sum, over atoms of the word, of  $\omega^{b(a)}$ . For multiplication put

$$\rho(a, a') = b(a) \oplus b(a')$$

and use the natural sum of  $\omega^{\rho(a, a')}$  over the pair atoms. Every pair atom in (3.2) has strictly smaller  $\rho$  than the replaced atom. A finite sum of lower powers of  $\omega$  is smaller than the replaced power, so each move strictly decreases the displayed measure. The standard numeric-game arithmetic theorem identifies the values of these graphs with  $x + y$  and  $xy$ ; their options are exactly the classical recursive options, including the signs in the negative summands [1].

**Theorem 3.3** (Hereditary-size arithmetic certificates). *There are fixed first-order set-theoretic formulas defining surreal addition, negation, and multiplication on sign sequences such that, for inputs of hereditary size at most an infinite  $\mu$ , a correct computation has a certificate of hereditary size at most  $\mu$ . Verification of a supplied certificate is absolute between transitive sets containing its data. Consequently, for every uncountable cardinal  $\kappa$ , these formulas define in  $H_\kappa$  exactly the external surreal operations on its sign sequences. No assumption  $H_\kappa \models \text{Replacement}$  is required.*

*Proof.* For  $+$  and multiplication, existentially quantify a set of word vertices containing the root and closed under the explicit option rule, its exact option relations, an ordinal rank labeling decreasing on edges, and a valuation by sign sequences passing the prefix cut test at every vertex. The node labels specify the input prefixes and the finite signed words; all checks of option generation quantify over those input prefix families and over entries of the finite words. Exactness requires both that each generated option is present and that every recorded edge is generated by the rule. The predicates for sign sequence, prefix, numerical comparison, finite word, and decreasing ordinal rank have bounded verifications on the supplied sets. In particular, a supplied prefix family is not accepted merely because its members are prefixes: for each ordinal  $\xi \leq \text{dom}(s)$  it must contain the unique restriction  $s \upharpoonright \xi$ , and every recorded prefix must be one of these restrictions. The same completeness check is imposed on the prefix families of vertex values used by the cut test. These are bounded checks against  $\text{dom}(s) + 1$  and the supplied families. A word length is a finite ordinal; this is checked by requiring every ordinal at most that length to be zero or a successor. Thus a missing prefix or a nonstandard-length word cannot evade an option requirement. The verifier is consequently a fixed bounded set-theoretic predicate, after including these domains and prefix families as certificate data. Their cardinality and hereditary-size bounds remain  $\mu$ .

Any valid certificate is sound: external well-founded induction on its rank labels identifies each vertex value with the value of its generated numeric game. This also proves uniqueness of the root value. A certificate exists by the preceding graph construction and lemma 3.2, with hereditary size at most  $\mu$ . Negation simply reverses all signs and has the same size estimate.

For inputs in  $H_\kappa$ , choose the infinite bound  $\mu < \kappa$  given above. The external certificate belongs to  $H_\kappa$ ; conversely any certificate there is externally sound by bounded absoluteness. Hence the

existential formulas have the same intended values. The uses of cardinal regularity were at  $\mu^+$ , not at  $\kappa$ .  $\square$

*Remark 3.4* (Scope of the certificate theorem). This is not a proof that every class-recursive surreal operation is absolute in every hereditary-size structure. It proves the specific finite-signature claim needed here. Inversion is definable from multiplication by  $xy = 1$ ; its existence in the epsilon-number cutoffs, and real closedness, are supplied by the classical cutoff theorem. No normal-form birthday estimates or unproved sharp product-birthday conjecture enter the argument (section 1.5).

### 3.4 Second route: prefix-pair tables

Sources 05, 06 and 07 certify arithmetic by a single table indexed by pairs of input prefixes. Source 05 states it at regular cardinals; source 06 isolates the single-cardinal size bound that makes it work at every uncountable cardinal, singular ones included, and source 07 uses the same bound. We print this route with 06's bound.

For a sign sequence  $x$  write  $\text{Pf}_x = \{x \upharpoonright \alpha : \alpha \leq \text{b}(x)\}$  for its prefix set, including  $x$ , and let  $L_x$  and  $R_x$  be its proper prefixes below and above  $x$ ; then  $x = \{L_x \mid R_x\}$ . Say that  $w$  is the *simplest separator* of two families when every left member is below  $w$ , every right member is above  $w$ , and no sign sequence of smaller length has the same property. For any candidate  $w \in H_\kappa$ , all sign sequences shorter than  $w$  also lie in  $H_\kappa$  (lemma 2.3); so this minimality test is absolute between  $H_\kappa$  and  $V$  whenever the families are in  $H_\kappa$ . Uniqueness of a least-birthday separator follows from signs: two distinct separators of the same least birthday would have a common prefix at their first difference strictly between them, still separating and of smaller birthday.

**Lemma 3.5** (Hereditary size of arithmetic tables). *Let  $\kappa$  be an uncountable cardinal, let  $x, y \in \mathbf{No}_{<\kappa}$  and set*

$$\mu = \max\{\aleph_0, |\text{b}(x)|, |\text{b}(y)|\} < \kappa.$$

*The functions on  $\text{Pf}_x \times \text{Pf}_y$  given by  $T_{x,y}(u, v) = u + v$  and  $U_{x,y}(u, v) = uv$  have graphs of hereditary size at most  $\mu$ ; in particular they belong to  $H_\kappa$ .*

*Proof.* There are at most  $\mu$  pairs of prefixes. Every prefix belongs to  $\mathbf{No}_{<\mu^+}$ , which is a field by lemma 2.2 and the classical cutoff theorem. Therefore every value in either table has birthday below  $\mu^+$  and hereditary size at most  $\mu$ . A set of at most  $\mu$  objects, each of hereditary size at most  $\mu$ , has hereditary size at most  $\mu$ ; the finite layers needed for ordered pairs and triples do not change this.

This argument is carried out in the ambient ZFC universe to establish the existence of a particular witness in  $H_\kappa$ . It does not invoke Replacement inside  $H_\kappa$ , and it uses a *single* infinite cardinal  $\mu$ , not a union of sizes approaching a singular  $\kappa$ . It also works when  $\mu^+ = \kappa$ .  $\square$

**Theorem 3.6** (Table definitions of the operations; sources 05, 06, 07). *Let  $\text{Add}(x, y, z)$  say that there is a function  $T$  on  $\text{Pf}_x \times \text{Pf}_y$  with  $T(x, y) = z$  such that every entry obeys*

$$T(u, v) = \{T(u^L, v), T(u, v^L) \mid T(u^R, v), T(u, v^R)\} \quad (u^L \in L_u, v^L \in L_v, u^R \in R_u, v^R \in R_v),$$

*the braces meaning the simplest-separator predicate. Let  $\text{Mult}(x, y, z)$  say that there is a function  $U$  on  $\text{Pf}_x \times \text{Pf}_y$  with  $U(x, y) = z$  such that  $U(u, v)$  is the simplest separator of the left options  $U(u', v) + U(u, v') - U(u', v')$  with  $(u', v') \in L_u \times L_v \cup R_u \times R_v$  and the right options of the same form with  $(u', v') \in L_u \times R_v \cup R_u \times L_v$ , where each sum and difference is certified by  $\text{Add}$  and sign reversal. Then for every uncountable cardinal  $\kappa$  and all  $x, y, z \in \mathbf{No}_{<\kappa}$ ,*

$$H_\kappa \models \text{Add}(x, y, z) \iff z = x + y, \quad H_\kappa \models \text{Mult}(x, y, z) \iff z = xy.$$

*Proof.* The actual tables are in  $H_\kappa$  by lemma 3.5 and satisfy the canonical Conway recursions, proving existence. Conversely, any accepted table is identified with the actual one by external well-founded induction on the pair of prefix lengths, ordered so that decreasing either coordinate decreases the pair: every option entry is earlier, the simplest-separator test is absolute, and the separator is unique. For multiplication, addition has already been identified. Each invocation of Add has its own hereditarily small witness; first-order satisfaction does not require that a single set collect the witnesses to all invocations.  $\square$

The two routes certify the same operations; the table route uses fewer auxiliary objects, the option-graph route makes the verifier an explicitly bounded predicate. Either suffices for the reverse interpretation of section 6.

*Remark 3.7* (Regularity in source 05). Source 05 states its size bookkeeping and its table lemma for uncountable regular  $\kappa$  and declines to infer a singular-cardinal theorem. Its table argument uses only the field closure of  $\mathbf{No}_{<\mu^+}$  and the single bound  $\mu$ , exactly as in lemma 3.5; regularity of  $\kappa$  is not used there. The singular case is sources 06 and 07's theorem, and the base's.

## 4 Definable sign access and a bounded pairing map

### 4.1 Recovering the sign tree from the two orders

For  $x, y, z$  let  $\text{Between}(z; x, y)$  say that  $z$  lies in the closed numerical interval with endpoints  $x$  and  $y$ . Define

$$\text{Pre}(x, y) \iff \forall z ((\text{Between}(z; x, y) \wedge z \neq x) \longrightarrow b(x) < b(z)). \quad (4.1)$$

**Lemma 4.1** (Definable prefix relation). *In every  $\mathcal{B}_\lambda$ , and in  $\mathcal{B}$ , (4.1) holds exactly when  $x$  is a sign prefix of  $y$ , allowing equality.*

*Proof.* If  $x$  is a prefix of  $y$ , every other point between them must continue the sign sequence of  $x$ ; a point diverging earlier would lie outside that interval. Its birthday is therefore larger.

If  $x$  is not a prefix of  $y$  and  $y$  is a proper prefix of  $x$ , the endpoint  $z = y$  violates the formula. In the remaining case the longest common prefix of  $x, y$  lies strictly between them and has birthday smaller than that of  $x$ . This prefix belongs to the same cutoff. Finally, for  $x = y$  the formula is vacuous, as required.  $\square$

*Remark 4.2* (Four prefix formulas). Each source defines prefix from  $<$  and  $b$  by its own formula; all are correct. Source 02 requires  $b(x) \leq b(y)$  and that every  $z$  with  $b(z) < b(x)$  compares with  $x$  as with  $y$ ; source 06 adds the symmetric comparison. Sources 05 and 07 require  $b(x) \leq b(y)$  and that no  $z$  strictly between  $x$  and  $y$  has  $b(z) \leq b(x)$  (05) or  $b(z) < b(x)$  (07). In the betweenness form the length conjunct cannot be dropped: for  $x = 1$  and  $y = 0$  no surreal of birthday at most 1 lies strictly between them, but 1 is not a prefix of 0 (source 05, which checks this case in its finite suite).

Define the bit predicate by

$$\begin{aligned} \text{Bit}(s, \alpha) \iff & \text{Ord}(\alpha) \wedge \alpha < b(s) \wedge \\ & \exists t (\text{Pre}(t, s) \wedge b(t) = \alpha \wedge t < s). \end{aligned} \quad (4.2)$$

The unique prefix of length  $\alpha$  lies below  $s$  precisely when the sign of  $s$  at position  $\alpha$  is plus. Thus

$$\{\alpha < \delta : \text{Bit}(s, \alpha)\}, \quad b(s) = \delta, \quad (4.3)$$

runs through *all* subsets of  $\delta$ , once each, for every  $\delta < \lambda$ . This uses a first-order quantifier over numbers of one birthday, not a second-order quantifier in the language. The exact length matters: trailing minus signs are not discarded, so the code has an explicit domain.

## 4.2 Pairing without reconstructing ordinary ordinal multiplication

For a positive embedded ordinal  $\theta$ , put

$$p_\theta(\xi, \zeta) = \theta\xi + \zeta, \quad \Lambda(\theta) = \theta^2 + \theta. \quad (4.4)$$

All operations in (4.4) are field operations, hence natural ordinal operations on these arguments.

**Lemma 4.3** (Bounded pairing). *The map  $p_\theta$  injects  $\theta \times \theta$  into  $\Lambda(\theta)$ . For every epsilon number  $\lambda$  and every  $0 < \theta < \lambda$ ,  $\Lambda(\theta) < \lambda$ . For infinite  $\theta$ ,  $|\Lambda(\theta)| = |\theta|$ .*

*Proof.* Suppose  $\xi < \xi' < \theta$  and  $\zeta, \zeta' < \theta$ . Since  $\xi + 1 \leq \xi'$  in ordinal order, positivity and distributivity give

$$\theta\xi + \zeta < \theta\xi + \theta = \theta(\xi + 1) \leq \theta\xi' \leq \theta\xi' + \zeta'.$$

For fixed  $\xi$ , cancellation gives injectivity in  $\zeta$ . Also  $\theta\xi + \zeta < \theta^2 + \theta$ . Closure below  $\lambda$  follows from its being a field cutoff. The cardinal estimate is lemma 2.2, together with  $\theta \leq \Lambda(\theta)$ .  $\square$

Source 02 uses the same pairing and bound (its  $B_\theta$ ). Consequently a sign sequence of birthday  $\Lambda(\theta)$  codes a binary relation on  $\theta$ . To make this code canonical for a given relation, require every plus position to be in the image of  $p_\theta$ ; unused positions are minus. This last requirement is the first-order formula

$$\forall \rho < \Lambda(\theta) (\text{Bit}(r, \rho) \longrightarrow \exists \xi, \zeta < \theta [\rho = p_\theta(\xi, \zeta)]).$$

Here and below bounded ordinal quantifiers include the predicate Ord.

For a graph between two different ordinals  $\delta, \epsilon < \lambda$ , use  $\theta = \delta + \epsilon + 1$ , which is above both. Coding the graph in the  $\delta \times \epsilon$  rectangle of this square again stays below  $\lambda$ . The construction requires no definable global pairing on the entire class of ordinals, and no reconstruction of ordinary ordinal arithmetic.

*Remark 4.4* (Four pairings in the sources). The other sources pack relations differently; each variant works with the codes below after the obvious changes. Source 05 uses the rectangle map  $p_a(i, j) = aj + i$  for  $i < a$ ,  $j < d$ , with length  $a(d + 1)$ , which is the same idea with the roles of the coordinates exchanged. Source 06 uses the polynomial of lemma 4.5, and source 07 the radix pairing of lemma 4.6.

**Lemma 4.5** (A one-polynomial pairing; source 06). *On ordinal surreals put  $\varpi(\alpha, \gamma) = (\alpha + \gamma)^2 + \alpha$ . Then  $\varpi$  is injective on pairs of ordinals, increasing in each coordinate, and maps  $\kappa \times \kappa$  into  $\kappa$  for every uncountable cardinal  $\kappa$ .*

*Proof.* Put  $\sigma = \alpha + \gamma$ . Since  $0 \leq \alpha \leq \sigma$ ,  $\sigma^2 \leq \varpi(\alpha, \gamma) \leq \sigma^2 + \sigma$ . If  $\sigma < \sigma'$  are ordinals, then  $\sigma + 1 \leq \sigma'$  and  $\sigma^2 + \sigma < (\sigma + 1)^2 \leq \sigma'^2$ . The intervals belonging to different sums are therefore disjoint. Equal values force equal sums, then equal first coordinates by subtraction, and finally equal second coordinates. Monotonicity follows from positivity and monotonicity of addition and squaring on nonnegative elements, and the bound is lemma 2.2.  $\square$

**Lemma 4.6** (Admissible radices; source 07). *Let  $\text{Rad}(q)$  say that  $q$  is an ordinal,  $q > 1$ , and  $a + a' < q$  for all ordinals  $a, a' < q$ . The ordinals satisfying  $\text{Rad}$  are exactly  $q = \omega^\rho$  with  $\rho > 0$ ; for each of them  $p_q(a, a') = qa + a'$  injects  $q \times q$  into  $q^2$ . Both statements are absolute between  $\mathcal{B}$  and every initial cutoff containing the ordinals in question. Below an epsilon number  $\lambda$  the admissible radices are cofinal and  $q^2 < \lambda$  for each of them.*

*Proof.* For  $q = \omega^\rho$  every ordinal below  $q$  has Cantor normal form with exponents below  $\rho$ , and natural addition introduces no new exponent. If  $q > 1$  is not a single monomial with coefficient one, split its normal form into two nonzero proper parts (splitting a coefficient if needed) to get  $a, a' < q$  with  $a \oplus a' = q$ ;  $q = 1$  is excluded. For  $a < q$  the exponents of  $q \otimes a$  are all at least  $\rho$ , those of  $a' < q$  are below  $\rho$ , and cancellation of natural addition recovers  $a$  and  $a'$ ; moreover  $qa + a' < q(a + 1) \leq q^2$ . For  $\xi < \lambda$  the radix  $\omega^{\xi+1}$  exceeds  $\xi$  and lies below  $\lambda$ , and  $(\omega^\rho)^2 = \omega^{\rho \oplus \rho} < \lambda$  since  $\lambda = \omega^\lambda$  is closed under natural addition below itself. All ordinals below an argument of an initial cutoff are present there, so the quantified test is unchanged.  $\square$

## 5 Pointed relation codes and the exact interpreted universe

### 5.1 The definable domain of codes

A prospective code is a triple

$$C = (\delta, r, t), \quad 0 < \delta < \lambda, \quad t < \delta, \quad b(r) = \Lambda(\delta),$$

with  $\delta, t$  embedded ordinals and with unused bits of  $r$  equal to minus. Write

$$E_C(i, j) \iff i, j < \delta \wedge \text{Bit}(r, p_\delta(i, j)).$$

The orientation is *child belongs to parent*:  $i E_C j$  will mean that the object at node  $i$  is a member of the object at node  $j$ .

Require extensionality,

$$\forall i, j < \delta [\forall k < \delta (E_C(k, i) \leftrightarrow E_C(k, j)) \longrightarrow i = j], \quad (5.1)$$

and well-foundedness in the following form:

$$\begin{aligned} \forall u [b(u) = \delta \longrightarrow ((\exists j < \delta \text{Bit}(u, j)) \longrightarrow \\ \exists j < \delta [\text{Bit}(u, j) \wedge \forall i < \delta (\text{Bit}(u, i) \longrightarrow \neg E_C(i, j))]]). \end{aligned} \quad (5.2)$$

By (4.3), this is *actual* well-foundedness of the set relation, not just well-foundedness against a restricted collection of subsets.

Finally require generation by the root:

$$\begin{aligned} \forall u [b(u) = \delta \longrightarrow (\text{Bit}(u, t) \wedge \forall i, j < \delta [\text{Bit}(u, j) \wedge E_C(i, j) \longrightarrow \text{Bit}(u, i)] \\ \longrightarrow \forall i < \delta \text{Bit}(u, i))]. \end{aligned} \quad (5.3)$$

Thus no proper downward-closed subset of the graph contains the root. Let  $\text{Code}(C)$  be the conjunction of these conditions. The triple is an ordinary finite-dimensional interpretation domain in the single-sorted structure  $\mathcal{B}_\lambda$ . Source 02 uses four-tuples with an extra domain word that selects a subset of  $\theta$  as nodes, and source 07 four-tuples with a radix; in both, a “clean” or “no junk” condition plays the role of the unused-bit convention.

**Lemma 5.1** (Rooted collapse). *Every valid code has a unique Mostowski collapse  $\pi_C : \delta \rightarrow T_C$  onto a transitive set. If  $x_C = \pi_C(t)$ , then*

$$T_C = \text{TC}(\{x_C\}), \quad |\delta| = |\text{TC}(\{x_C\})|.$$

*Proof.* Define by well-founded recursion

$$\pi_C(j) = \{\pi_C(i) : i E_C j\}.$$

Extensionality and well-founded induction show that  $\pi_C$  is injective: if a collision existed, choose one minimizing the maximum of the two node ranks; comparison of predecessor values gives equal predecessor sets, and extensionality identifies the nodes. Its image is transitive, and the displayed recursion gives an isomorphism. Uniqueness follows by well-founded induction as well; this is the usual Mostowski-collapse argument.

The nodes reachable from  $t$  by finitely many downward edges form a downward-closed subset containing  $t$ . By (5.3) this is all of  $\delta$ . Under the collapse, these are exactly  $x_C$  and its iterated members. Thus the image is  $\text{TC}(\{x_C\})$ , giving the cardinal equality.  $\square$

Root generation is not needed merely to encode arbitrary sets. It is useful here because it makes the hereditary-size cost exact rather than just an upper bound polluted by unrelated graph components.

## 5.2 Equality and membership are definable

For codes  $C = (\delta, r, t)$  and  $D = (\epsilon, s, u)$ , define  $C \sim D$  when a coded bijection  $f : \delta \rightarrow \epsilon$  preserves  $E$ , sends  $t$  to  $u$ , and reflects  $E$ . Such a graph is expressed by a sign sequence in a square of width  $\delta + \epsilon + 1$ , as in lemma 4.3. The familiar function, bijection, and edge-preservation clauses use only bounded ordinal quantifiers and bits. Therefore  $\sim$  is a single first-order relation.

For membership, let  $C \in_I D$  mean that there is a coded injection  $f : \delta \rightarrow \epsilon$  such that

$$E_D(f(t), u), \tag{5.4}$$

$$E_C(i, j) \longleftrightarrow E_D(f(i), f(j)) \quad (i, j < \delta), \tag{5.5}$$

$$E_D(v, f(j)) \longrightarrow \exists i < \delta [E_C(i, j) \wedge f(i) = v] \quad (j < \delta, 0 \leq v < \epsilon). \tag{5.6}$$

In (5.6), the quantifier on  $v$  simply ranges over the embedded ordinals below  $\epsilon$ . The clause says that the image contains *all* predecessors of its nodes, not merely some of them. Sources 06 and 07 use the same downward-closed image clause; source 02 instead defines membership through an isomorphism of the root with the cone of a predecessor, using a definable reachability relation.

**Lemma 5.2** (Correctness of equality and membership). *For valid codes,*

$$C \sim D \iff x_C = x_D, \quad C \in_I D \iff x_C \in x_D.$$

*Proof.* A root-preserving isomorphism gives equal collapsed roots by uniqueness of collapse. Conversely, if the roots agree, the two images are the same transitive closure by lemma 5.1, and  $\pi_D^{-1}\pi_C$  gives the required isomorphism. Its graph has size at most  $\max(|\delta|, |\epsilon|)$  and is codable within the same cutoff.

For membership, (5.5) and (5.6), followed by induction along  $E_C$ , imply  $\pi_D(f(j)) = \pi_C(j)$  for all  $j$ . The root condition then gives  $x_C \in x_D$ . Conversely, if  $x_C \in x_D$ , the inclusion  $\text{TC}(\{x_C\}) \subseteq \text{TC}(\{x_D\})$

transports through the collapse maps to an injection satisfying all three clauses. Its image is predecessor-closed, so (5.6) really does hold. The graph is again small enough to be coded below the cutoff.  $\square$

The predecessor-full clause cannot be omitted; Appendix B gives the counterexample, which the finite suites of sources 06, 07 and 09 check.

Let  $\mathcal{I}(\mathcal{B}_\lambda)$  be the quotient of valid codes by  $\sim$ , with the membership relation just defined. All quantifiers in its interpreted formulas are replaced by quantifiers over triples satisfying Code. No quotient class is assumed to be an element of the original structure.

### 5.3 Cardinal rounding

**Theorem 5.3** (Exact collapse range). *For every epsilon number  $\lambda$ , the map  $[C] \mapsto x_C$  is an isomorphism*

$$\mathcal{I}(\mathcal{B}_\lambda) \cong (H_{\kappa(\lambda)}, \in).$$

*More precisely, the ordinals  $\delta$  supporting rooted codes for a given set  $x$  are exactly those with*

$$0 < \delta < \lambda, \quad |\delta| = |\text{TC}(\{x\})|. \quad (5.7)$$

*Proof.* Every code has  $\delta < \lambda \leq \kappa(\lambda)$ . Hence its collapse has hereditary size  $|\delta| < \kappa(\lambda)$ , by lemma 5.1. This proves one inclusion of the range.

Conversely, suppose  $x \in H_{\kappa(\lambda)}$ . Its nonempty transitive closure has some cardinal  $\nu < \kappa(\lambda)$ . By the definition of cardinal ceiling,  $\nu < \lambda$ : if  $\lambda$  is a cardinal this is immediate; otherwise  $\nu \leq |\lambda| < \lambda$ . Choose a bijection from  $\nu$  onto  $\text{TC}(\{x\})$ , transfer membership to  $\nu$ , and designate the node for  $x$ . The resulting graph is extensional, well founded, and generated by that node. It has a sign code of length  $\Lambda(\nu) < \lambda$ . Thus  $x$  occurs in the range.

For any other  $\delta < \lambda$  equipotent to this transitive closure, the same construction works with a bijection from  $\delta$ . Necessity in (5.7) was already proved by rooted collapse. Correctness of both relations is lemma 5.2.  $\square$

Choice is used here only to well-order one transitive closure at a time; the interpretation keeps all valid codes and never selects one code per set. Source 02 proves this theorem one way, at every epsilon number, as a four-dimensional interpretation; source 07 proves the same range at every epsilon number (its cardinality plateaux, with  $\theta(\lambda)$  in place of  $\kappa(\lambda)$ ), also stating it forward only; sources 05 and 06 prove it at regular, respectively all uncountable, cardinals.

*Remark 5.4* (Why the bound is sharp; source 02). No set of hereditary size  $\kappa(\lambda)$  can be represented by these codes: its collapse would require at least  $\kappa(\lambda)$  distinct nodes, while each domain is an ordinal below  $\lambda \leq \kappa(\lambda)$ . This obstruction is intrinsic to the stated interpretation. It is not a claim that no different interpretation in  $\mathcal{B}_\lambda$  could represent a larger structure.

**Example 5.5** (The empty set and a first nonempty code). The empty set uses one node, no edges, and root zero. With  $\delta = 1$ , the canonical relation string has birthday  $\Lambda(1) = 2$  and both signs minus. Thus a code for the empty set is  $(1, --, 0)$ , not the surreal zero itself. For  $\{\emptyset\}$  take  $\delta = 2$ , root 1, and the single edge  $0E1$ . The relation string has length 6 and its only plus is at  $p_2(0, 1) = 1$ . Numeric values and the sets they code must not be identified. A three-node example with two labelings is in Appendix C.

**Corollary 5.6** (Countable cutoffs recover all countable sets; sources 02, 07, 09). *For every countable epsilon number  $\lambda$ , in particular  $\varepsilon_0$ ,  $\mathcal{I}(\mathcal{B}_\lambda) \cong (H_{\omega_1}, \in)$ .*

*Proof.* A countable epsilon number is strictly larger than  $\omega$  and strictly below  $\omega_1$ , so its cardinal ceiling is  $\omega_1$ . Concretely, every hereditarily countable set has a rooted code on a finite ordinal or on  $\omega$ , and  $\Lambda(\omega) = \omega^2 + \omega < \varepsilon_0$ .  $\square$

The recovered sets include countable ordinals far larger than  $\varepsilon_0$ . Those larger ordinals are recovered as *collapsed relations*, not as all-plus scalar sign sequences in the cutoff: every countable ordinal  $\alpha$  has a membership graph labeled by natural numbers, whose relation word has birthday below  $\omega^2 + \omega$ . Labels need not preserve ordinal order; the collapse recovers that order from the edges. One should therefore not infer a low set-theoretic rank of the interpreted objects from a low birthday of the numerical codes (source 07).

## 6 Bi-interpretation and recovery of the cutoff

### 6.1 The ordinal and sign-graph bridges

For an actual embedded ordinal  $a < \lambda$ , there is a distinguished code  $\text{OC}(a)$  with domain  $a + 1$ , root  $a$ , and edge relation  $iEj$  iff  $i < j$ . Its sign relation code is uniquely determined by the unused-bit convention. The formula defining  $\text{OC}(a)$  is uniform, and its collapse is the von Neumann ordinal  $a$ . Because  $\lambda$  is a limit and a ring cutoff, both  $a + 1$  and  $\Lambda(a + 1)$  are below  $\lambda$ .

**Definition 6.1** (Visible interpreted ordinals). For a set code  $C$  define

$$\text{VisOrd}(C) \iff \exists a [\text{Ord}(a) \wedge C \sim \text{OC}(a)].$$

Under the collapse isomorphism, these are exactly the ordinals below  $\lambda$ , not necessarily all ordinals of  $H_{\kappa(\lambda)}$ .

There is also a definable bridge  $\text{SignGraph}(s, C)$  saying that  $C$  represents the set-coded sign function of the scalar  $s$ : its domain is  $\text{OC}(\text{b}(s))$ , its values are the interpreted ordinals 0 and 1, and at each actual  $\alpha < \text{b}(s)$  it has value 1 exactly when  $\text{Bit}(s, \alpha)$  holds. Application of a set-coded function and the Kuratowski ordered-pair construction are ordinary fixed formulas in the interpreted membership language. The ordinal bridge supplies their arguments. Thus  $\text{SignGraph}$  is a fixed first-order formula, not an external assignment operation.

For each  $s$  there is exactly one equivalence class of codes  $C$  with  $\text{SignGraph}(s, C)$ ; conversely, every sign function whose length is below  $\lambda$  arises this way. Existence follows from the hereditary-size estimate for sign graphs and [theorem 5.3](#).

### 6.2 A first-order test for a cardinal cutoff

Define the *visible-ordinal collector* formula

$$\text{Coll}(C) \iff \text{Code}(C) \wedge \forall D [\text{Code}(D) \longrightarrow (D \in_{\mathcal{I}} C \leftrightarrow \text{VisOrd}(D))]. \quad (6.1)$$

It says that one interpreted set collects all the scalar ordinals. This is not the assertion that the scalar carrier itself is a set inside itself.

**Theorem 6.2** (The cutoff is a set exactly in the noncardinal case; sources 02, 09). *For an epsilon number  $\lambda$ ,*

$$\mathcal{B}_\lambda \models \exists C \text{ Coll}(C) \iff \lambda \text{ is not a cardinal.}$$

*When it exists, the collector is unique modulo  $\sim$  and its collapse is exactly the ordinal  $\lambda$ .*

*Proof.* The elements specified in (6.1) are exactly all ordinals  $a < \lambda$ . In the ambient set universe the unique set with precisely those elements is  $\lambda$ . Hence a collector exists precisely when  $\lambda \in H_{\kappa(\lambda)}$ , equivalently  $|\lambda| < \kappa(\lambda)$ , since  $\text{TC}(\{\lambda\}) = \lambda + 1$ . For a cardinal  $\lambda$  the right side fails; otherwise  $\kappa(\lambda) = (|\lambda|)^+$  and it holds. Uniqueness follows from extensionality of the interpreted universe.  $\square$

Source 02's sentence `CollectOrd` is this sentence written with its own codes. The theorem recovers the *ordinal value* of a noncardinal cutoff as an interpreted object. It is not a construction of a surreal number with birthday  $\lambda$  inside  $\mathbf{No}_{<\lambda}$ . This collector is the additional information needed to recover a noncardinal cutoff after interpreting its hereditary-size universe. Merely saying that  $\mathcal{B}_\lambda$  interprets  $H_{\kappa(\lambda)}$  would lose it.

### 6.3 The reverse interpretation

If  $\lambda = \kappa$  is a cardinal, interpret surreals in  $H_\kappa$  as all set-coded functions from an ordinal into  $\{0, 1\}$ . By lemma 2.3, these are exactly the sign sequences of length below  $\kappa$ . Define the order by first disagreement, birthday by the all-one function on the domain ordinal, and arithmetic by the certificate formulas of theorem 3.3 (or the tables of theorem 3.6). Call this interpretation  $\mathcal{J}$ .

If  $\lambda$  is not a cardinal, work instead in the pointed structure  $(H_{\kappa(\lambda)}, \in, \lambda)$ . Use the same definitions, with the extra domain restriction that the sign length is below the named ordinal  $\lambda$ . The certificate verifier may use auxiliary sign sequences anywhere in  $H_{\kappa(\lambda)}$ ; it is not required to carry out a large recursion inside the smaller interpreted sign domain. The root outputs are below  $\lambda$  by classical cutoff closure. This gives a uniform interpretation  $\mathcal{J}_{\text{cut}}$  in the language with one named cutoff. Without the named cutoff, interpreting sign sequences in  $H_{\omega_1}$  gives  $\mathbf{No}_{<\omega_1}$ , not  $\mathbf{No}_{<\varepsilon_0}$ ; this is why source 07 stated its plateau theorem forward only.

**Theorem 6.3** (Uniform local bi-interpretation). *The interpretations just constructed prove all assertions of theorem 1.1. The maps witnessing the two bi-interpretations are definable, not merely external bijections.*

*Proof.* Start on the set side. A code in the interpreted surreal structure gives an extensional well-founded relation on an ordinal. Its Mostowski-collapse root defines the map back to the original set. The graph of the collapse belongs to the same  $H_\kappa$ : its domain, range, and all values have hereditary size bounded by a fixed infinite cardinal below  $\kappa$ . Consequently this correspondence is defined by a first-order formula in  $H_\kappa$ , even when Replacement is not an axiom of that structure. The collapse equation is a bounded verification of the supplied function. Every set has a code, and equality and membership are respected by lemma 5.2. In the pointed case, the collector collapses to the named cutoff, so the correspondence preserves that constant as well.

Start on the surreal side. The formula  $\text{SignGraph}(s, C)$  identifies  $s$  with the sign function representing it in the copy of the set universe. In the cardinal case all sign functions in the interpreted  $H_\lambda$  occur. In the noncardinal case, the collector defines  $\lambda$ , and the restricted reverse interpretation uses exactly lengths below that ordinal. Thus  $\text{SignGraph}$  is a definable bijection to the twice-interpreted surreal structure. It preserves numerical order and birthday directly from the

sign rules, and it preserves arithmetic by the certificate theorem. This proves the required definable isomorphisms in both directions.  $\square$

At cardinal cutoffs sources 06 and 07 give the same two round trips with their own codes (06 by a formula `Eval` for the collapse and a sign-function bridge; 07 by an ordinal-code comparison and the interpreted sign name  $\langle b(x), A_x \rangle$ ), and source 05 gives them with graphs modulo bisimulation (section 7).

**Corollary 6.4** (The first epsilon cutoff and hereditary countability). *The structure  $\mathcal{B}_{\varepsilon_0}$  is parameter-free bi-interpretable with  $(H_{\omega_1}, \in)$ .*

*Proof.* The theorem first gives the pointed structure  $(H_{\omega_1}, \in, \varepsilon_0)$ . The ordinal  $\varepsilon_0$  is parameter-free definable in  $H_{\omega_1}$  as the least nonzero fixed point of ordinary  $\alpha \mapsto \omega^\alpha$ . For any countable arguments, the functions witnessing ordinal arithmetic and the relevant well-orders are countable sets, hence belong to  $H_{\omega_1}$ . Their well-founded recursive verification is absolute. The least-fixed-point specification therefore picks out the actual  $\varepsilon_0$  there. Naming it is a definitional expansion, which may be removed.  $\square$

The same argument applies to any noncardinal epsilon number that is parameter-free definable, with its correct ordinal value, in the recovered hereditary-size universe. It is not an assertion that every ordinal can be named parameter-free.

*Remark 6.5* (Questions of the sources answered here). Source 02 claimed only a one-way interpretation and warned that a two-way theorem “should specify whether the target carries a predicate for that segment or a constant for  $\lambda$ ”. Theorem 6.3, source 09’s, is exactly such a theorem, with a constant; at  $\varepsilon_0$  it is even unpointed. Source 06’s question on noncardinal cutoffs, source 07’s question on recovering a noncardinal cutoff, and the noncardinal half of source 05’s question on optimal cutoffs are answered for epsilon-number cutoffs by the same theorem; the singular-cardinal half of 05’s question is answered by sources 06 and 07. The questions stay open for cutoffs that are not epsilon numbers (section 21).

## 6.4 Coherence of the interpretation

**Proposition 6.6** (Coherence under cutoff inclusions). *If  $\lambda \leq \eta$  are epsilon numbers, the inclusion of scalar cutoffs induces, under  $\mathcal{I}$ , the actual inclusion*

$$H_{\kappa(\lambda)} \longrightarrow H_{\kappa(\eta)}.$$

*A code keeps the same collapsed value, and all sign-graph bridges agree. For cardinal cutoffs the reverse interpretations also commute with the actual inclusions.*

*Proof.* For a code already below  $\lambda$ , the larger structure has no additional sign sequences of birthday  $\delta$  for any  $\delta < \lambda$ : both contain *all* sequences of each such length in the same ambient universe. Thus the subset tests in well-foundedness and root generation are unchanged. Equality and membership agree by their collapse characterizations; any needed witness graph has a code below  $\lambda$  as proved above. The collapse is defined from the same relation, so its value is unchanged. The sign-graph bridge refers to the same prefixes and bits. Finally the reverse arithmetic formulas compute the same external operations.  $\square$

Sources 02 (at epsilon numbers, forward), 06 and 07 (at cardinal cutoffs, both directions) prove the same compatibility. If two noncardinal cutoffs have the same cardinal ceiling, the induced inclusion on recovered *unpointed* set universes is the identity. Their recovered distinguished cutoffs nevertheless differ. This is precisely why elementary inclusion will usually fail.

## 7 Second route: graphs modulo bisimulation

Source 05 codes sets by well-founded graphs that need not be extensional or generated by their root, and takes *bisimilarity* as equality. This avoids any extensionality or canonical-enumeration requirement: two different terminal nodes both represent  $\emptyset$ , and a disjoint component below no root changes nothing. We write its codes with the pairing of lemma 4.3; source 05 uses its rectangle map (remark 4.4), which changes nothing below. Source 05 proved the resulting bi-interpretation at uncountable regular cardinals; the argument below uses regularity nowhere (the certificates of section 3 are the only arithmetic input), and we state it at the cardinal cutoffs of theorem 1.1, where the base already proves the conclusion by the first route. The value of the second route is its different equality test.

A *graph code* is a triple  $G = (\nu, e, t)$  with  $\nu > 0$  and  $t < \nu$  embedded ordinals and  $b(e) = \Lambda(\nu)$ ; its relation is  $E_G(i, j) \iff i, j < \nu \wedge \text{Bit}(e, p_\nu(i, j))$ , again oriented from a member to a set. Let  $\text{WF}(G)$  be the well-foundedness clause (5.2). No extensionality or root generation is required. The *decoration* of  $G$  is the unique function with

$$v_G(j) = \{v_G(i) : i E_G j\}, \quad \text{dec}(G) = v_G(t);$$

ordinary well-founded recursion suffices, and the Mostowski collapse is its extensional special case.

**Proposition 7.1** (Exactly the hereditary sets; source 05). *For an uncountable cardinal  $\kappa$ , the decorations of graph codes in  $\mathcal{B}_\kappa$  are exactly the members of  $H_\kappa$ .*

*Proof.* Every member of  $\text{TC}(\{v_G(j)\})$  is a value at a node reachable by a finite predecessor chain, so it lies in the range of  $v_G$ , of cardinality at most  $|\nu| < \kappa$ . Conversely, pull membership on a transitive set containing  $a \in H_\kappa$  back to an ordinal  $\nu < \kappa$  along a bijection and root it at the label of  $a$ ; induction gives decoration  $a$ .  $\square$

For graph codes  $G = (\nu, e, t)$  and  $G' = (\nu', e', t')$ , a relation  $Q \subseteq \nu \times \nu'$  is a *bisimulation* when every  $(i, j) \in Q$  satisfies

$$\begin{aligned} i' E_G i &\implies \exists j' (j' E_{G'} j \wedge (i', j') \in Q), \\ j' E_{G'} j &\implies \exists i' (i' E_G i \wedge (i', j') \in Q). \end{aligned}$$

Code  $Q$  by one sign sequence  $q$  in a square of width  $\nu + \nu' + 1$ , and let  $\text{Bis}(G, G', q)$  be the resulting first-order formula. Define

$$G \approx G' \iff \exists q [\text{Bis}(G, G', q) \wedge (t, t') \in Q_q], \quad G \in_\approx G' \iff \exists j < \nu' [E_{G'}(j, t') \wedge G \approx G'[j]],$$

where  $G'[j] = (\nu', e', j)$  is the same graph rerooted at  $j$ .

**Theorem 7.2** (Equality is bisimilarity; source 05). *For well-founded graph codes,  $G \approx G' \iff \text{dec}(G) = \text{dec}(G')$  and  $G \in_\approx G' \iff \text{dec}(G) \in \text{dec}(G')$ . Consequently the graph codes modulo  $\approx$ , with  $\in_\approx$ , form a parameter-free interpretation of  $(H_\kappa, \in)$  in  $\mathcal{B}_\kappa$ .*

*Proof.* Suppose  $Q$  is a bisimulation relating the roots. By well-founded induction on  $i$  along  $E_G$ , every  $j$  with  $(i, j) \in Q$  satisfies  $v_G(i) = v_{G'}(j)$ : a member  $v_G(i')$  of  $v_G(i)$  is matched by the forth clause with a predecessor  $j'$  of  $j$  and equals  $v_{G'}(j')$  by induction, and the back clause gives the other inclusion. Conversely, if the root values agree, the set relation  $Q = \{(i, j) : v_G(i) = v_{G'}(j)\}$  is a bisimulation relating the roots, because equal decorated sets supply the forth and back witnesses without selecting unique matches; it has a code in the same cutoff. Membership follows by applying this to the rerooted graph. Hence  $\approx$  is an equivalence relation, a conclusion established semantically rather than presumed from the shape of the definition, and [proposition 7.1](#) gives surjectivity.  $\square$

**Example 7.3** (Duplicates and irrelevant components). One terminal node represents  $\emptyset$ . A graph with two terminal nodes and a root above both represents  $\{\emptyset\}$ , as does a graph with only one terminal child. Their rooted graphs are not isomorphic, but their roots are bisimilar. Graph isomorphism is therefore *not* the right equality for these codes; source 05’s finite suite exercises exactly this case.

The round trips of source 05 are the analogues of those in [theorem 6.3](#). The set-side map is the decoration relation “there is a function  $v$  on  $\nu$  satisfying the recursion with  $v(t) = a$ ”, whose witness has hereditary size at most  $\max(\aleph_0, |\nu|)$ . On the surreal side, for an ordinal  $\alpha$  let  $\text{OC}(\alpha)$  be the graph on  $\alpha + 1$  with  $iEj \iff i < j$  and root  $\alpha$ , so that  $G \approx \text{OC}(\alpha)$  iff  $\text{dec}(G) = \alpha$ ; and for a surreal  $x$  let  $A_x$  be the interpreted set whose members are the classes  $\text{OC}(\gamma)$  for the ordinals  $\gamma < \text{b}(x)$  with  $\text{Bit}(x, \gamma)$ . The number  $x$  is then matched with the interpreted sign name  $\langle \text{b}(x), A_x \rangle$ , which externally decodes to  $\langle \text{b}(x), \{\gamma : x(\gamma) = +\} \rangle$ . Source 07 uses the same sign-name round trip with extensional codes.

*Remark 7.4* (Internal versions and models of ZFC; sources 05, 07). All constructions above are theorems of ZFC and can be carried out inside any model  $M \models \text{ZFC}$ , with  $M$ ’s own ordinals, subsets and well-founded relations; for the full class  $\mathbf{No}^M$  they recover  $M$  itself. For a nonstandard  $M$  this is an internal statement: graph well-foundedness is  $M$ ’s notion, not external well-foundedness. For set models  $M, N$  of ZFC, writing  $\mathcal{B}^M$  and  $\mathcal{B}^N$  for the internal birthday structures regarded externally, source 07 derives

$$M \equiv N \iff \mathcal{B}^M \equiv \mathcal{B}^N, \quad M \cong N \iff \mathcal{B}^M \cong \mathcal{B}^N,$$

by transporting sentences and isomorphisms through the uniform interpretations and their definable composites. These full-class and internal statements are instances of the announced Chen–Hamkins–Yang theorem ([theorem 18.1](#)), not results of sources 05 or 07.

## 8 Ring cutoffs and the finite boundary

The classical ring statement is weaker than the field statement: the infinite ring cutoffs  $\mathbf{No}_{<\lambda}$  are exactly those with  $\lambda = \omega^{\omega^\alpha}$  [[5](#), Theorems 1.1–1.2]. Source 02 observes that the forward interpretation needs only ring operations.

**Proposition 8.1** (Ring version; source 02). *The interpretation  $\mathcal{I}$  and its exact image remain valid for every infinite ordinal  $\lambda$  for which  $\mathbf{No}_{<\lambda}$  is a subring of  $\mathbf{No}$ , when  $\mathcal{B}_\lambda$  is regarded as a birthday-enriched ordered ring. Its image is  $H_{\kappa(\lambda)}$ , where now  $\kappa(\lambda)$  is the least infinite cardinal  $\geq \lambda$ . In particular*

$$\mathcal{I}(\mathbf{No}_{<\omega}; +, -, \cdot, <, \text{b}) \cong (H_\omega, \in),$$

*the hereditarily finite sets, and  $\mathcal{I}(\mathcal{B}_{\omega^\omega}) \cong (H_{\omega_1}, \in)$ .*

*Proof.* Every formula of the interpretation uses only the ordered-ring operations and birthday. The bounds  $\Lambda(\theta)$  and  $\theta + \epsilon + 1$  remain in the cutoff by ring closure, and an infinite ring cutoff is a limit ordinal, so the ordinal codes also fit. All later coding and collapse arguments use only these facts and the cardinal-ceiling calculation. For  $\lambda = \omega$  every domain is finite, all codes for its subsets and finite isomorphisms have finite length, and the represented transitive closures are exactly the finite ones.  $\square$

Real closedness matters for the comparison with decidable field theories and for the surcomplex transfer, but it is not what enables the set coding. The decisive ingredients are complete sign layers, definable reading of those signs, and enough ordinal arithmetic to fit relation codes below the cutoff. Only the forward interpretation is claimed at ring cutoffs that are not epsilon numbers. At  $\omega$  the cutoff is the dyadic ring, not a field, and it would be incorrect to reuse any real-closed-field argument there. Arbitrary ordinal cutoffs that are not ring cutoffs are not classified.

## Part II

# Consequences inside one universe

## 9 The exact elementary-inclusion spectrum

The elementary-inclusion result is stronger than the observation that a birthday expansion is not governed by real-closed-field quantifier elimination. It identifies exactly which initial inclusions are possible.

**Lemma 9.1** (Noncardinal lower cutoffs never include elementarily; sources 02, 09). *If  $\lambda < \eta$  are epsilon numbers and  $\lambda$  is not a cardinal, then  $\mathcal{B}_\lambda \not\prec \mathcal{B}_\eta$ .*

*Proof.* Choose a code  $C$  in  $\mathcal{B}_\lambda$  for its visible-ordinal collector. Its collapse is  $\lambda$ , and it satisfies  $\text{Coll}(C)$  there. By coherence it still collapses to  $\lambda$  in  $\mathcal{B}_\eta$ . But  $\lambda$  itself is now a scalar ordinal, since  $\lambda < \eta$ . Its interpreted ordinal code is visible, and is not a member of the set coded by  $C$ , because  $\lambda \notin \lambda$ . Thus  $\text{Coll}(C)$  is false in the larger structure. This single formula, with the three entries of  $C$  as parameters, witnesses failure of elementarity.  $\square$

Source 02's witness is the weaker formula “the coded set contains every literal ordinal”, which is also true in  $\mathcal{B}_\lambda$  and false in  $\mathcal{B}_\eta$  for the same code. As source 02 observes,  $\mathcal{B}_{\epsilon_0}$  and  $\mathcal{B}_{\epsilon_1}$  interpret the same  $H_{\omega_1}$ , yet their inclusion is not elementary: interpretation does not discard the additional first-order information of the original structure.

**Lemma 9.2** (A cardinal lower cutoff cannot include elementarily into a noncardinal cutoff). *If  $\lambda$  is a cardinal and  $\eta$  is not, then  $\mathcal{B}_\lambda \not\prec \mathcal{B}_\eta$ .*

*Proof.* The parameter-free sentence  $\exists C \text{ Coll}(C)$  is false in the first structure and true in the second, by [theorem 6.2](#).  $\square$

**Theorem 9.3** (Classification of elementary initial inclusions). *For epsilon numbers  $\lambda < \eta$ ,*

$$\mathcal{B}_\lambda \prec \mathcal{B}_\eta \iff \lambda, \eta \text{ are cardinals and } H_\lambda \prec H_\eta.$$

*Proof.* The preceding two lemmas force both cutoffs to be cardinals; they are uncountable, since no countable cardinal is an epsilon number. For cardinal cutoffs, an elementary inclusion of the surreal structures induces an elementary inclusion of their interpreted set structures. By [proposition 6.6](#) this is the actual inclusion  $H_\lambda \subseteq H_\eta$ . Conversely, an elementary inclusion of those set structures induces an elementary inclusion of the uniformly interpreted sign structures. The reverse interpretation and its definable sign-graph isomorphism identify this with the original inclusion  $\mathcal{B}_\lambda \subseteq \mathcal{B}_\eta$ .  $\square$

*Remark 9.4* (Source 02’s open question, answered). Source 02 proved the forward implication (its necessary conditions) and asked whether they could be sharpened to a sufficient criterion, noting that any proof would have to control the reverse interpretation locally. Source 09’s [theorem 9.3](#) is that criterion: the reverse interpretation of [theorem 6.3](#) and the coherence of [proposition 6.6](#) give the converse. Between cardinal cutoffs the equivalence is also proved by source 05 (regular cardinals) and sources 06 and 07 (all uncountable cardinals).

**Corollary 9.5** (Two kinds of elementary extension). *The ordered field inclusion  $\mathbf{No}_{<\lambda} \subseteq \mathbf{No}_{<\eta}$  is elementary for all epsilon numbers  $\lambda < \eta$ . Adding birthday changes its elementary status according to [theorem 9.3](#). In particular,*

$$\mathbf{No}_{<\varepsilon_0} \prec \mathbf{No}_{<\omega_1}, \quad \mathcal{B}_{\varepsilon_0} \not\prec \mathcal{B}_{\omega_1}.$$

*Proof.* Both fields are real closed, so ordered-field quantifier elimination gives the first assertion. The birthday assertions are instances of the theorem.  $\square$

Despite the last failure, both structures in that example are separately bi-interpretable with  $H_{\omega_1}$ . There is no contradiction: a bi-interpretation does not make *every particular inclusion* elementary. The smaller structure defines an interpreted cutoff set; that formula changes its meaning under the displayed inclusion.

**Example 9.6** (Two field-equivalent cutoffs separated by one sentence; sources 05, 07). Let  $\chi$  say that every set admits an injection into  $\omega$  (defined as the least infinite ordinal). Then  $H_{\omega_1} \models \chi$  and  $H_{\omega_2} \models \neg\chi$ , since  $\omega_1 \in H_{\omega_2}$ . Hence the parameter-free sentence  $\chi^{\mathcal{I}}$  gives  $\mathcal{B}_{\omega_1} \not\equiv \mathcal{B}_{\omega_2}$ , although their ordered-field reducts are elementarily equivalent and the field inclusion is elementary.

**Corollary 9.7** (Transfer of elementary equivalence; source 06). *For uncountable cardinals  $\kappa, \lambda$ ,  $\mathcal{B}_\kappa \equiv \mathcal{B}_\lambda \iff H_\kappa \equiv H_\lambda$ . The complete theories  $\text{Th}(H_\kappa, \in)$  and  $\text{Th}(\mathcal{B}_\kappa)$  are computably many-one reducible to each other by translations independent of  $\kappa$  (source 05, at regular cardinals).*

*Proof.* Translate sentences through the uniform interpretations in both directions; the translations are effective syntactic operations.  $\square$

*Remark 9.8* (Bi-interpretation versus elementary equivalence; source 05). Distinct membership structures can be bi-interpretable without being isomorphic or elementarily equivalent in their own language; Freire and Hamkins give relative-consistency instances involving  $H_{\omega_1}$  and  $H_{\omega_2}$  [10]. [Example 9.6](#) does not contradict this: the theorem here supplies one uniform interpretation between each cutoff and its own hereditary-size structure, and does not classify all interpretations between two different hereditary-size structures.

**Corollary 9.9** (Comparison with the full class). *For an epsilon number  $\lambda$ , interpreted formula by formula,*

$$\mathcal{B}_\lambda \prec \mathcal{B} \iff \lambda \text{ is a cardinal and } H_\lambda \prec V.$$

*Proof.* The same collector obstruction excludes a noncardinal lower cutoff. For a cardinal lower cutoff, the uniform interpretation and reverse sign-graph interpretation extend to definable classes (theorem 18.1). The induced set map is  $H_\lambda \subseteq V$ , so the elementary equivalence follows for each formula. This uses no internal satisfaction class for  $V$ .  $\square$

The condition  $H_\lambda \prec H_\eta$  in theorem 9.3 is not replaced here by a conjectural large-cardinal characterization. It is an exact reduction of one elementary-inclusion problem to another familiar one, together with the complete exclusion of every noncardinal initial cutoff. If both cutoffs are strongly inaccessible,  $H_\rho = V_\rho$  and the condition reads  $V_\lambda \prec V_\eta$  (source 05).

## 9.1 Elementary embeddings

Sources 05 (regular cardinals) and 06 (all uncountable cardinals) prove that the uniform bi-interpretation also transfers elementary embeddings that are not inclusions. We state the theorem at uncountable cardinals and give source 05's explicit formula. Write  $\text{name}(x) = \langle b(x), A_x \rangle$  with  $A_x = \{\alpha < b(x) : x(\alpha) = +\}$  for the set-theoretic sign name of  $x$ .

**Theorem 9.10** (Elementary-embedding correspondence; sources 05, 06). *For uncountable cardinals  $\kappa, \lambda$  there is a canonical bijection*

$$\text{Emb}_{\text{el}}((H_\kappa, \in), (H_\lambda, \in)) \longleftrightarrow \text{Emb}_{\text{el}}(\mathcal{B}_\kappa, \mathcal{B}_\lambda),$$

*respecting identities and composition. For  $j : H_\kappa \rightarrow H_\lambda$  the corresponding map  $e_j$  is given by*

$$\text{name}(e_j(x)) = j(\text{name}(x)) = \langle j(b(x)), j(A_x) \rangle,$$

*and for  $e : \mathcal{B}_\kappa \rightarrow \mathcal{B}_\lambda$  the corresponding  $j_e$  is  $j_e(x_C) = x_{e(C)}$ , with  $e$  applied coordinatewise to codes. In particular  $e_j(\alpha) = j(\alpha)$  on ordinals and  $e_j(x)(j(\alpha)) = x(\alpha)$  for  $\alpha < b(x)$ .*

*Proof.* Sign names, their order, birthday and arithmetic are uniformly defined in the membership language (section 6), so an elementary  $j$  maps sign names to sign names and preserves every translated formula; this proves elementarity of  $e_j$ . Since the code domain and  $\sim$  are parameter-free, an elementary  $e$  maps codes to codes and equivalent codes to equivalent codes, so  $j_e$  is well defined, and applying  $\mathcal{I}$  to each membership formula proves it elementary. The definable round-trip isomorphisms commute with elementary maps, because an elementary map preserves their defining formulas and their unique outputs modulo the equivalence relations; hence  $e_{j_e} = e$  and  $j_{e_j} = j$ , and composition and identities are respected. The name of an ordinal  $\alpha$  is  $\langle \alpha, \alpha \rangle$ , which gives  $e_j(\alpha) = j(\alpha)$ , and elementarity of Bit gives the last identity.  $\square$

**Remark 9.11** (Do not replace  $j(A)$  by the pointwise image). For a nontrivial elementary set embedding,  $j(A)$  need not equal  $j^{\text{ptw}}A$ . The formula uses the actual image *set*  $j(A)$ : the sign sequence  $e_j(x)$  has domain  $j(b(x))$  and may have coordinates outside  $j^{\text{ptw}}b(x)$ , which a pointwise transport of signs would leave undetermined.

**Corollary 9.12** (Critical ordinals and the identity case; source 05). *Corresponding embeddings move exactly the same ordinals; when a least moved ordinal exists it is the same for both. An elementary embedding  $\mathcal{B}_\kappa \rightarrow \mathcal{B}_\lambda$  fixing every ordinal below  $\kappa$  is the canonical inclusion.*

*Proof.* If  $e$  fixes all ordinals, then  $b(e(x)) = e(b(x)) = b(x)$ , and elementarity of the bit formula gives the same signs below that birthday.  $\square$

The correspondence is a classification by reduction, not an existence theorem for such embeddings: it proves no new large-cardinal consistency statement and does not refute rank-into-rank hypotheses. A proper canonical inclusion moves no ordinal, so “proper embedding” must not be confused with “embedding having a critical point”. Every automorphism of  $\mathcal{B}_\kappa$  is the identity (lemma 19.3); that statement overlaps the collection’s existing simplicity rigidity and is not counted as new.

## 10 Birthday growth and set-existence axioms

### 10.1 Birthday eternity

**Definition 10.1** (The eternity scheme Et; source 09). For every formula  $\varphi(a, x, \vec{p})$  in the birthday-field language, include the assertion that if  $\gamma$  is an embedded ordinal and

$$\forall a < \gamma \exists! x \varphi(a, x, \vec{p}),$$

then some embedded ordinal  $\varrho$  satisfies

$$\forall a < \gamma \forall x (\varphi(a, x, \vec{p}) \longrightarrow b(x) < \varrho).$$

The tuple  $\vec{p}$  consists of arbitrary parameters from the structure.

This is a precisely specified version of birthday boundedness for definable ordinal-indexed functions. The analogy with the eternity principle in the Chen–Hamkins–Yang announcements is intentional; the cutoff classification below is proved directly, rather than imported from an unexpanded axiom scheme in the slides.

**Theorem 10.2** (Eternity forces cardinality of an epsilon cutoff). *If  $\lambda$  is a noncardinal epsilon number, then  $\mathcal{B}_\lambda \not\models \text{Et}$ .*

*Proof.* Put  $\mu = |\lambda|$ , so  $\mu < \lambda$  and  $\kappa(\lambda) = \mu^+$ . Choose a bijection  $f : \mu \rightarrow \lambda$ . Its graph has hereditary size at most  $\mu$ , so it belongs to  $H_{\mu^+}$  and has a code in  $\mathcal{B}_\lambda$ . Fix one such code as a parameter. Using the interpreted graph of  $f$  and the ordinal bridge, a single formula now defines a scalar map on the actual ordinal  $\mu$ : at  $a < \mu$  its value is the embedded ordinal  $f(a) < \lambda$ . Its birthdays exhaust  $\lambda$ , hence have no bound below  $\lambda$ . This violates the corresponding instance of Et.  $\square$

At a singular cardinal  $\kappa$ , an external cofinal map into  $\kappa$  generally has hereditary size  $\kappa$  and need not belong to  $H_\kappa$ . Thus the same argument cannot simply be repeated with  $\mu = \text{cf}(\kappa)$ . Whether an internally definable cofinal map exists is the point of the next theorem.

### 10.2 The axioms that all hereditary-size universes already satisfy

**Lemma 10.3** (The fixed background axioms). *For every uncountable cardinal  $\kappa$ ,  $H_\kappa$  satisfies Extensionality, Empty Set, Pairing, Union, Infinity, Foundation, every Separation instance, and Choice. Moreover the actual transitive closure of each of its elements belongs to  $H_\kappa$ , and its cardinality is correctly computed there.*

*Proof.* Transitivity gives Extensionality and Foundation. Empty Set, Pairing, Union, and Infinity preserve hereditary size below  $\kappa$ . For any fixed formula and parameters, the externally defined subset of a set  $a \in H_\kappa$  satisfying that formula in the set structure  $H_\kappa$  has hereditary size bounded

by that of  $a$  (up to finitely many additional objects). It therefore belongs to  $H_\kappa$ , proving Separation, including formulas with unbounded quantifiers in that structure.

An ambient choice function for a family  $a \in H_\kappa$  has its domain and chosen values inside the transitive closure of  $a$ , and its graph has the same infinite hereditary-cardinality bound. It belongs to  $H_\kappa$ . This proves Choice for set families, not a class-choice principle.

Writing  $T = \text{TC}(\{x\})$ , the transitive closure of  $T$  adds at most the set  $T$  to its already transitive data, so  $T \in H_\kappa$ . Its description as the least transitive set containing  $x$  as an element is correct internally: the actual least such set is available there. A bijection between  $T$  and its least equipotent ordinal has hereditary size below  $\kappa$ , and any bijection asserted internally is externally a bijection. Minimality among ordinals is therefore also computed correctly.  $\square$

### 10.3 Replacement is exactly eternity at cardinal cutoffs

**Theorem 10.4** (Eternity–Replacement equivalence). *For every uncountable cardinal  $\kappa$ ,*

$$\mathcal{B}_\kappa \models \text{Et} \iff H_\kappa \models \text{every instance of Replacement.}$$

*Proof.* Assume first Replacement in  $H_\kappa$ . A definable scalar function on an ordinal  $\gamma < \kappa$  corresponds under the bi-interpretation to a set-theoretically definable function assigning to  $a < \gamma$  its unique sign graph. Replacement collects those sign graphs in one set of  $H_\kappa$ . The union of their domains, with one added successor for each length, is bounded below  $\kappa$ : otherwise the transitive closure of the collected set would contain at least  $\kappa$  ordinals. More explicitly,

$$\varrho = \sup\{\text{dom}(s) + 1 : s \text{ is an image sign graph}\} < \kappa$$

bounds their birthdays strictly. This proves the corresponding eternity instance; the empty-domain case is vacuous.

Conversely, assume eternity and consider a formula defining, in  $H_\kappa$ , a function  $F : a \rightarrow H_\kappa$ , where  $a \in H_\kappa$  and the formula has fixed parameters. Choose a bijection  $e : \delta \rightarrow a$  with  $\delta < \kappa$ . Its graph belongs to  $H_\kappa$ . For each  $i < \delta$ , put externally

$$\nu_i = |\text{TC}(\{F(e(i))\})|,$$

regarded as an initial ordinal. By lemma 10.3, this is a uniquely defined ordinal of  $H_\kappa$ , with the correct external value. The definition of  $F$ , the parameter  $e$ , and the hereditary-cardinality formula translate into a definable scalar map  $i \mapsto \nu_i$  in  $\mathcal{B}_\kappa$ . Here the ordinal bridge is available for every ordinal below  $\kappa$ . No definable choice of a code for  $F(e(i))$  is required: its *hereditary cardinal* is unique.

Eternity gives a  $\varrho < \kappa$  bounding all these ordinals. Ambient Replacement forms the actual image set  $A = F[a]$ , and its transitive closure has cardinality at most

$$\max(\aleph_0, |\delta|, |\varrho|) < \kappa.$$

Indeed it is contained, up to the image set itself, in the union of  $|\delta|$  transitive closures, each of cardinality at most  $\max(\aleph_0, |\varrho|)$ . Thus  $A \in H_\kappa$ , proving the desired Replacement instance in that structure. This proof uses a fixed cardinal bound below  $\kappa$ ; it never assumes that an arbitrary union of fewer than  $\kappa$  small sets is small.  $\square$

**Corollary 10.5.** *If  $\kappa$  is uncountable and regular, then  $\mathcal{B}_\kappa \models \text{Et}$ .*

*Proof.* Any externally given function on an ordinal below  $\kappa$  has fewer than  $\kappa$  birthdays in its image. Regularity bounds them below  $\kappa$ . This argument applies in particular to definable functions.  $\square$

This corollary genuinely uses regularity; the equivalence of the theorem does not.

[Added 3 October 2026, batch 89: the same principle on a weak base. Source 10 (Part VI) proves over Zermelo set theory with Foundation, without Replacement or Choice, that bounding the birthdays of definable set-indexed families of surreals, even strictly inside  $(-1, 0)$ , is equivalent to ordinal bounding, and, over an explicit hierarchy axiom, to full Replacement and Collection (theorems 28.2 and 29.1). That is not this theorem: it is an internal scheme equivalence over a weaker base theory, uses only the order of sign sequences, and does not apply to  $H_\kappa$  unless  $H_\kappa$  satisfies that base theory. Remark 24.5 compares the two and records where they combine.]

## 10.4 Power Set and packing a birthday fiber

**Lemma 10.6** (Power Set in a hereditary-size universe; sources 02, 06, 09). *For every infinite cardinal  $\kappa$ ,*

$$H_\kappa \models \text{Power Set} \iff \kappa \text{ is strong limit } (2^\nu < \kappa \text{ for all cardinals } \nu < \kappa).$$

*For an individual  $\alpha < \kappa$ , the actual power set  $\mathcal{P}(\alpha)$  is an element of  $H_\kappa$  exactly when  $2^{|\alpha|} < \kappa$ . Regularity is not required.*

*Proof.* Every subset of a set  $a \in H_\kappa$  is already in  $H_\kappa$ , and the internal subset relation is the actual one by transitivity. So an internal power set of  $a$ , if it exists, is the full actual power set  $\mathcal{P}(a)$ . Its hereditary size is  $2^{|\alpha|}$  up to harmless finite or countable bounds. This proves necessity (apply it to the cardinals  $\nu < \kappa$ ) and the individual assertion. Conversely, for  $x \in H_\kappa$  with  $\kappa$  uncountable, the transitive closure of  $\mathcal{P}(x)$  has size at most  $\max(\aleph_0, |\text{TC}(\{x\})|, 2^{|x|})$ , which is below a strong-limit  $\kappa$ . If  $\kappa = \omega$ , power sets of hereditarily finite sets are hereditarily finite.  $\square$

**Theorem 10.7** (The translated Power Set sentence; sources 02, 06). *Let  $\text{Pow}^\mathcal{I}$  be the  $\mathcal{I}$ -translation of the Power Set axiom  $\forall a \exists p \forall x (x \in p \leftrightarrow \forall y (y \in x \rightarrow y \in a))$ ; it is one fixed parameter-free sentence of the birthday-field language, not a scheme. For every epsilon number  $\lambda$ ,*

$$\mathcal{B}_\lambda \models \text{Pow}^\mathcal{I} \iff \lambda \text{ is a strong-limit cardinal.}$$

*Among uncountable regular cutoffs it holds exactly at the strongly inaccessible ones.*

*Proof.* By theorem 5.3 and lemma 10.6,  $\text{Pow}^\mathcal{I}$  holds exactly when  $\kappa(\lambda)$  is strong limit. If  $\lambda$  is not a cardinal, write  $\nu = |\lambda|$ ; then  $\kappa(\lambda) = \nu^+$  and Cantor's theorem gives  $2^\nu \geq \nu^+$ , so the condition fails. If  $\lambda$  is a cardinal, its ceiling is itself. Adding regularity yields strong inaccessibility.  $\square$

Source 02 proves this at every epsilon number, source 06 at uncountable cardinals (its  $\text{Pow}^\#$ ). It would be incorrect to remove the words “among uncountable regular cutoffs”: a singular strong-limit cardinal such as  $\beth_\omega$  also satisfies  $\text{Pow}^\mathcal{I}$ , and section 17 shows that the same enriched field can have a regular cutoff in one universe and a singular one in another. The sentence detects the strong-limit property, not regularity, Replacement, or full ZFC.

**Definition 10.8** (The fiber-packing principle Pack; source 09). *For every embedded ordinal  $\alpha$ , there exist a positive embedded ordinal  $\theta$  and a scalar  $r$ , with*

$$\chi = \theta + \alpha + 1, \quad \text{b}(r) = \Lambda(\chi),$$

such that

$$\forall u [b(u) = \alpha \longrightarrow \exists \xi < \theta \forall \zeta < \alpha (\text{Bit}(u, \zeta) \leftrightarrow \text{Bit}(r, p_\chi(\xi, \zeta)))] \quad (10.1)$$

This is one first-order sentence, using the previously defined bit and pairing formulas; it is *not* the Power Set axiom. Its rows list every sign string of birthday  $\alpha$ ; repetitions are allowed. Unused coordinates can be ignored. It does not use ordinary ordinal concatenation or a primitive power-set operation.

**Theorem 10.9** (Exact packing threshold). *In  $\mathcal{B}_\lambda$ , the instance of (10.1) for a particular  $\alpha < \lambda$  holds exactly when*

$$2^{|\alpha|} < \kappa(\lambda), \quad \text{equivalently, } 2^{|\alpha|} < \lambda,$$

where the cardinal on the left is compared as an ordinal in the second inequality. Consequently Pack holds exactly when  $\lambda$  is an uncountable strong-limit cardinal.

*Proof.* A packing table with  $\theta < \lambda$  rows gives a surjection from  $\theta$  onto  $\mathcal{P}(\alpha)$ , so  $2^{|\alpha|} \leq |\theta| < \kappa(\lambda)$ . Conversely, if that power-set cardinal is  $\nu < \kappa(\lambda)$ , then  $\nu < \lambda$  by cardinal rounding. Choose an enumeration of the fiber by  $\theta = \nu$  and encode its table in the square of width  $\chi = \theta + \alpha + 1$ . The square code stays below  $\lambda$  by closure, so the required  $r$  exists. The equivalence of the two inequalities follows because no cardinal lies in  $[\lambda, \kappa(\lambda))$ .

For a cardinal  $\lambda$ , validity for all  $\alpha < \lambda$  is exactly the strong-limit property. If  $\lambda$  is not a cardinal, let  $\mu = |\lambda| < \lambda$  and use  $\alpha = \mu$ . Cantor's theorem gives  $2^\mu \geq \mu^+ = \kappa(\lambda)$ , so this instance fails.  $\square$

Notice that for a noncardinal cutoff  $\lambda$ , the recovered  $H_{\kappa(\lambda)}$  and the scalar domain do not have the same ordinal height. Assertions about *all* interpreted ordinals and assertions about only scalar ordinals must be kept separate. The failure at  $\alpha = |\lambda|$  is enough to settle the full packing principle.

| Cutoff $\lambda$                        | Recovered universe    | $\exists C \text{ Coll}(C)$ | $\text{Pow}^\mathcal{T}$ , Pack |
|-----------------------------------------|-----------------------|-----------------------------|---------------------------------|
| Countable epsilon number                | $H_{\omega_1}$        | true                        | false                           |
| Noncardinal, $ \lambda  = \nu$          | $H_{\nu^+}$           | true                        | false                           |
| Uncountable successor cardinal $\kappa$ | $H_\kappa$            | false                       | false                           |
| Singular strong-limit cardinal $\kappa$ | $H_\kappa$            | false                       | true                            |
| Strongly inaccessible $\kappa$          | $H_\kappa = V_\kappa$ | false                       | true                            |

The table is source 02's, with the packing column added from source 09.

## 11 Worldly cutoffs and separation examples

### 11.1 The exact conjunction

Call an uncountable cardinal  $\kappa$  *worldly* here when  $V_\kappa \models \text{ZFC}$ . This definition does not impose regularity. We need the following familiar set-theoretic fact, for which a proof is included to fix the precise meaning.

**Lemma 11.1** (When hereditary size becomes cumulative rank). *For an uncountable cardinal  $\kappa$ ,*

$$H_\kappa \models \text{ZFC} \iff H_\kappa = V_\kappa \text{ and } V_\kappa \models \text{ZFC}.$$

Moreover,  $V_\kappa \models \text{ZFC}$  by itself implies  $H_\kappa = V_\kappa$ .

*Proof.* Suppose  $H_\kappa \models \text{ZFC}$ . Its internal cumulative hierarchy exists to every ordinal  $\alpha < \kappa$ . Every subset of each of its sets belongs to  $H_\kappa$ , so its internal power sets are actual power sets. Induction on  $\alpha$  shows that its internal  $V_\alpha$  is the actual  $V_\alpha$ ; the limit step is ordinary union over the actual smaller ordinals. Thus every  $V_\alpha$  for  $\alpha < \kappa$  belongs to  $H_\kappa$ , and transitivity gives  $V_\kappa \subseteq H_\kappa$ . The reverse inclusion is [lemma 2.3](#).

Conversely, assume  $V_\kappa \models \text{ZFC}$ . For every  $\alpha < \kappa$ , that model can well-order  $V_\alpha$  and supply a bijection with some ordinal  $\delta < \kappa$ . These are actual sets and actual bijections, because the model is transitive. Hence  $|V_\alpha| < \kappa$ . The transitive closure of any  $x \in V_\kappa$  is contained in some such  $V_\alpha$ , and therefore has size below  $\kappa$ . Again  $H_\kappa = V_\kappa$  follows. The remaining implication is immediate from equality.  $\square$

**Theorem 11.2** (Worldly-cutoff characterization). *For an epsilon number  $\lambda$ ,*

$$\begin{aligned} \mathcal{B}_\lambda \models \text{Et} + \text{Pack} &\iff \lambda \text{ is a cardinal and } H_\lambda \models \text{ZFC} \\ &\iff \lambda \text{ is a worldly cardinal.} \end{aligned}$$

*For regular uncountable  $\lambda$ , this holds exactly when  $\lambda$  is strongly inaccessible.*

*Proof.* Either growth principle excludes noncardinal epsilon cutoffs. At cardinal cutoffs, [theorem 10.4](#) and [lemmas 10.3](#) and [10.6](#) identify the two missing ZFC requirements as Replacement and Power Set. The equivalence with the rank formulation is [lemma 11.1](#). For regular uncountable cardinals, eternity is automatic and packing is exactly strong limit, giving the usual inaccessible condition.  $\square$

This proves [theorem 1.3](#). It does not identify the two formulas with every formulation of a theory called SA. It is an exact theorem about the explicit scheme and sentence defined here.

## 11.2 The two growth principles do different work

**Example 11.3** (Eternity without fiber packing). At  $\kappa = \omega_1$ , regularity gives eternity. The birthday- $\omega$  fiber has  $2^{\aleph_0} \geq \aleph_1$  members, so no ordinal below  $\omega_1$  can index a packing of it. Hence

$$\mathcal{B}_{\omega_1} \models \text{Et}, \quad \mathcal{B}_{\omega_1} \not\models \text{Pack}.$$

In the decoded structure, every individual real is present, but the set of all reals is not an element of  $H_{\omega_1}$ .

**Example 11.4** (Fiber packing without eternity). Let  $\kappa = \beth_\omega = \sup_{n < \omega} \beth_n$ , with  $\beth_0 = \aleph_0$  and  $\beth_{n+1} = 2^{\beth_n}$ . This is a strong-limit cardinal: for every  $\mu < \kappa$  some  $\beth_n$  is at least  $\mu$ , and then  $2^\mu \leq \beth_{n+1} < \kappa$ . Thus packing holds.

In  $H_\kappa$ , each finite initial segment of the beth recursion is present and correctly computed. The necessary finite sequence of power sets, cardinals, and bijections has hereditary size below  $\kappa$ , and Power Set is correct there by [lemma 10.6](#). A single formula therefore defines  $n \mapsto \beth_n$  on the actual  $\omega$ . Transporting this formula through the ordinal bridge gives a definable scalar function whose birthdays are cofinal in  $\kappa$ . Consequently

$$\mathcal{B}_{\beth_\omega} \models \text{Pack}, \quad \mathcal{B}_{\beth_\omega} \not\models \text{Et}.$$

The infinite sequence of all these ordinals is precisely what cannot be collected as an element of  $H_\kappa$ .

Source 06 isolates the same phenomenon as one explicit sentence, and uses it to separate  $\beth_\omega$  from its successor. Let  $\vartheta(n, \gamma)$  be the membership formula saying that a finite sequence of length  $n + 1$  begins with  $\omega$ , ends with  $\gamma$ , and at each step takes the least ordinal equinumerous with the power set of the previous entry; and let

$$\text{Rep}_\beth : [\forall n \in \omega \exists! \gamma \vartheta(n, \gamma)] \longrightarrow \exists y \forall \gamma [\gamma \in y \leftrightarrow \exists n \in \omega \vartheta(n, \gamma)],$$

a single Replacement instance.

**Theorem 11.5** (A singular Replacement boundary; source 06). *With  $\kappa = \beth_\omega$ ,  $H_\kappa \not\models \text{Rep}_\beth$  and  $H_{\kappa^+} \models \text{Rep}_\beth$ , the antecedent being true in both. Consequently*

$$\mathcal{B}_{\beth_\omega} \not\prec \mathcal{B}_{\beth_\omega^+},$$

although  $\mathcal{B}_{\beth_\omega} \models \text{Pow}^I$  and the ordered-field inclusion is elementary.

*Proof.* Every finite beth computation, with the power sets and bijections at its finitely many stages, has hereditary size below  $\kappa$  and belongs to both structures, where it is correctly computed; so  $\vartheta(n, \gamma)$  picks out the actual  $\beth_n$  in both. A set collecting all  $\beth_n$  has every ordinal below  $\kappa$  in its transitive closure, so it is not in  $H_\kappa$ ; the actual set  $\{\beth_n : n < \omega\}$  has hereditary size  $\kappa$  and lies in  $H_{\kappa^+}$ . Transfer by [theorem 9.3](#) (or the translation  $\text{Rep}_\beth^I$ ) gives the failure of elementarity, and [theorem 10.7](#) the Power Set assertion.  $\square$

The theorem gives one singular example with failed Replacement. It does *not* show that every singular  $\kappa$  fails Replacement in  $H_\kappa$ : an externally cofinal sequence can fail to be definable over  $H_\kappa$  or fail to be an element of it. Nor does internal ZFC imply external inaccessibility; [corollary 17.2](#) gives a sharp example.

**Example 11.6** (A noncardinal field cutoff). Both principles fail at  $\varepsilon_0$ . Packing fails at the ordinal  $\omega$ . Eternity fails by [theorem 10.2](#); one may also use the definable finite power-tower sequence cofinal in  $\varepsilon_0$ , with ordinary ordinal exponentiation computed in the decoded  $H_{\omega_1}$ . Nevertheless the field is real closed and its birthday expansion is bi-interpretable with hereditary countability.

These are actual structures exhibiting independence of the two displayed growth requirements within this family. They are not an assertion of independence from an unspecified list of all the other axioms of SA.

### 11.3 Singular cutoffs satisfying both principles

**Proposition 11.7** (A conditional source of singular worldly cutoffs). *If  $\Omega$  is strongly inaccessible, there is a cardinal  $\kappa < \Omega$  of cofinality  $\omega$  such that*

$$V_\kappa \prec V_\Omega.$$

*In particular,  $\kappa$  is worldly and  $\mathcal{B}_\kappa$  satisfies both growth principles.*

*Proof.* Enumerate all existential formulas in the language of set theory. Construct strictly increasing infinite cardinals  $\alpha_n < \Omega$ . Given  $\alpha_n$ , consider the first  $n$  formulas and all their finite parameter tuples from  $V_{\alpha_n}$ . Whenever  $V_\Omega$  has a witness, choose one. There are fewer than  $\Omega$  such requirements, since  $|V_{\alpha_n}| < \Omega$ . Regularity bounds their witness ranks below  $\Omega$ . Choose a cardinal  $\alpha_{n+1} < \Omega$  above that bound and above  $\alpha_n$ .

Set  $\kappa = \sup_n \alpha_n$ . Regularity of  $\Omega$  gives  $\kappa < \Omega$ ; it is a cardinal of cofinality  $\omega$ . Every finite tuple in  $V_\kappa$  appears in one stage, and every existential formula is treated at all sufficiently late stages. Thus every existential statement true in  $V_\Omega$  with parameters in  $V_\kappa$  has a witness in  $V_\kappa$ . The Tarski–Vaught criterion gives elementarity. Since  $V_\Omega$  satisfies ZFC, so does  $V_\kappa$ . Apply [theorem 11.2](#).  $\square$

The reflection construction is standard set-theoretic reasoning, included as an illustration rather than as a novelty claim, and it needs the stated inaccessible. It demonstrates why replacing “worldly” by “inaccessible” in [theorem 11.2](#) would be wrong. No inaccessible cardinal is assumed in the unconditional cutoff theorems.

## 12 Model-theoretic consequences

### 12.1 Full second-order arithmetic and undecidability

**Proposition 12.1** (The standard natural numbers and all their subsets; sources 06, 07, 09). *In  $\mathcal{B}$  and in every  $\mathcal{B}_\lambda$  ( $\lambda$  an epsilon number) there is a parameter-free interpretation of the two-sorted structure  $(\mathbb{N}, \mathcal{P}(\mathbb{N}); 0, 1, +, \cdot, <, \in)$  with its full actual subset sort. The number sort consists of the ordinals below the least nonzero limit ordinal  $w$  (externally  $w = \omega$ ), the subset sort of the  $s$  with  $b(s) = w$ , and membership is  $\text{Bit}(s, n)$ .*

*Proof.* “ $q$  is a nonzero limit ordinal” is  $\text{Ord}(q) \wedge q > 0 \wedge \forall a < q \exists d (a < d < q)$  with ordinal guards, and  $w$  is its least solution. The ordinals below  $w$  are the standard finite ordinals, and natural arithmetic on them is ordinary arithmetic. The fiber  $b(s) = \omega$  is in bijection with all subsets of  $\omega$  by (4.3); no restriction to definable subsets is imposed.  $\square$

When working inside a model  $M$ , the subset sort is  $\mathcal{P}(\omega)^M$ , as it must be. Sources 06 and 07 state the proposition, and the next two results, at uncountable cardinals and source 09 at  $\varepsilon_0$ ; the proofs use only  $\omega < \lambda$ , which holds at every epsilon number. This is an ordinary first-order interpretation of a two-sorted structure, not a replacement of first-order by second-order syntax. It does not make set theory definable in the pure ordered-field reduct, where the birthday structure is absent.

**Corollary 12.2** (Undecidability; sources 05, 06, 07, 09). *True first-order arithmetic reduces effectively to the complete theory of every  $\mathcal{B}_\lambda$ , and formula by formula to  $\mathcal{B}$ . Consequently these complete theories are neither decidable nor computably enumerable.*

*Proof.* Relativize arithmetic quantifiers to the number sort. If the target theory were computably enumerable, the inverse image would enumerate true arithmetic; a complete computably enumerable theory is decidable, and true arithmetic is not.  $\square$

These are logical-complexity consequences, not new algorithms. A code can contain an arbitrary noncomputable subset of an ordinal, and first-order definability in the birthday expansion must not be confused with decidability from a finite symbolic representation.

### 12.2 Independence and an omitted type

**Proposition 12.3** (Independence property and a countable omitted type; source 07). *The formula  $\text{Bit}(s, n)$  has the independence property in every  $\mathcal{B}_\lambda$ . The parameter-free partial type  $\{x < w \wedge$*

$\text{Ord}(x)\} \cup \{x > n : n \in \mathbb{N}\}$ , with each standard  $n$  named by its definition, is finitely satisfiable but omitted.

*Proof.* For distinct finite ordinals  $n_1, \dots, n_k$  and any  $S \subseteq \{1, \dots, k\}$ , a length- $\omega$  sign code positive exactly at the  $n_j$  with  $j \in S$  realizes the pattern. Finite parts of the type are realized by large finite ordinals; a realization of the whole type would be a standard natural number above all standard natural numbers.  $\square$

*Remark 12.4* (Saturation of the pure field, and the sibling report). For regular uncountable  $\kappa$ , the pure real closed field  $\mathbf{No}_{<\kappa}$  realizes every one-variable type over fewer than  $\kappa$  parameters: by quantifier elimination a nonalgebraic type over the real closed subfield generated by the parameters is a cut, regularity bounds the birthdays of its fewer than  $\kappa$  endpoints, and (3.1) supplies a separator. Source 07 records this as the usual saturation property of the pure field, not a new result. Naming birthday nevertheless produces the omitted type of [proposition 12.3](#). External saturation of the pure field  $\mathbf{No}^M$  inside an outer universe is the subject of the sibling report [foundations-and-computation/surreal-fields-across-universes](#); this report does not study it.

[Added 2 October 2026, batch 80: the order-theoretic case of this remark, which uses neither the field structure nor uncountability, is proved and used in the report [foundations-and-computation/surreal-well-orders](#) (batch-80 manuscripts 08, 09, 11 and 12). Its Lemma 7.2 (`swo:sw:stageeta`, its source 11's) shows, by the birthday bound (3.1), that for every regular infinite  $\kappa$ ,  $\kappa = \omega$  included,  $\mathbf{No}_{<\kappa}$  fills every cut  $A < B$  with  $|A \cup B| < \kappa$ , even avoiding any prescribed set of fewer than  $\kappa$  points; it calls this property  $\eta_\kappa$ . Its Note 7.4 (`swo:sk:prop:cutoffeta`, its source 08's Proposition 6.1) states it for regular  $\kappa$  with  $2^{<\kappa} = \kappa$  and cites this remark. Its Proposition 12.1 (`swo:sw:permutationeta`) shows that the lexicographic order of the well-orderings of length  $\kappa$  of an  $\eta_\kappa$ -order of cardinality  $\kappa$  is again  $\eta_\kappa$ ; with Lemma 7.2 this applies to  $\mathbf{No}_{<\kappa}$  when  $2^{<\kappa} = \kappa$ . No result here uses that report.]

### 12.3 Birthday is not visible in the field reduct

**Proposition 12.5** (Nondefinability of birthday; sources 05, 07). *On  $\mathbf{No}_{<\lambda}$ , for  $\lambda$  an epsilon number, neither birthday nor prefix simplicity is definable, even with parameters, in the ordered-field reduct, nor in any o-minimal expansion of it. The same holds formula by formula on  $\mathbf{No}$ .*

*Proof.* If birthday were definable, its fixed-point set, the embedded ordinals, would be definable. It is infinite and contains no interval: between ordinals  $\alpha < \gamma$  lies  $\alpha + \frac{1}{2}$ , which is not an ordinal. An o-minimal structure (in particular a real closed field, by quantifier elimination) has only finite unions of points and intervals as definable subsets of the line. If prefix simplicity were definable, the ordinals would be defined by “every proper prefix of  $x$  lies below  $x$ ”, with the same contradiction.  $\square$

Source 05 states the proposition at regular cardinals and source 07 at uncountable cardinals; the proof uses only real closedness (o-minimality). The collection's foundations report notes that the field language has no birthday symbol; this proposition is the corresponding standard nondefinability proof, new to the collection but not a novelty claim.

## 12.4 How little arithmetic is needed initially

Let  $G_\varpi(\alpha, \gamma, \rho)$  be the ternary relation on  $\mathbf{No}_{<\kappa}$  saying that  $\alpha, \gamma, \rho$  are ordinals and  $\rho = \varpi(\alpha, \gamma)$ , with  $\varpi$  the polynomial of lemma 4.5.

**Corollary 12.6** (Reconstruction from ordinal pairing; source 06). *For every uncountable cardinal  $\kappa$ , all operations and constants of  $\mathcal{B}_\kappa$  are uniformly definable in  $(\mathbf{No}_{<\kappa}; <, b, G_\varpi)$ . Hence the two structures are definitionally equivalent.*

*Proof.* Run the coding of section 5 with relation words of length  $\varpi(\delta, \delta) + 1$  whose plus positions are the  $\varpi(u, v)$ , as source 06 does; ordinal successor is definable from the ordinal order. The code formulas then use field arithmetic only to evaluate  $\varpi$  on ordinals, which  $G_\varpi$  supplies, so this reduct still interprets  $H_\kappa$ . Translate the arithmetic definitions of section 3 through that interpretation and use the surreal-side round trip to put their graphs on the original carrier. The converse is the polynomial formula for  $G_\varpi$ .  $\square$

No minimal-language claim is made: the corollary does not remove birthday data or show that arithmetic is definable from order and prefix simplicity alone. It is partial progress on the minimal-signature questions of sources 02, 05 and 07, which stay open (section 21).

## 13 The exact Choice boundary of ordinal graph coding

This section, source 06's, works in  $ZF$ . It considers the full class of sign sequences and the graph interpretation of section 5 without a cutoff. All its defining formulas remain first-order formulas about individual sets or surreals, and class assertions are read formula by formula.

**Definition 13.1.** Let  $\text{HWO} = \{x : \text{TC}(\{x\}) \text{ admits a well-ordering}\}$ . One well-ordering of the whole transitive closure is required, with no definability condition such as the one in HOD.

**Theorem 13.2** (Choice-sensitive range; source 06). *Over  $ZF$ , the pointed ordinal-graph interpretation has quotient structure isomorphic to  $(\text{HWO}, \in)$ . Its external collapse map is surjective onto  $V$  if and only if the Axiom of Choice holds.*

*Proof.* The Mostowski collapse of a valid code  $(\delta, r, t)$  is injective with domain the ordinal  $\delta$ , and its range is  $\text{TC}(\{x_C\})$  by root generation; transporting the ordinal order well-orders that closure. Conversely, a well-order of  $\text{TC}(\{x\})$  gives a bijection from an ordinal, along which membership pulls back to a valid code for  $x$ ; no simultaneous choice for a family is used. Equality and membership remain correct in  $ZF$ : all subsets of a fixed ordinal exist and have characteristic words, so the well-foundedness test still quantifies over every subset, and Mostowski collapse is a theorem of  $ZF$ . If Choice holds every transitive closure is well-orderable. Conversely, surjectivity well-orders every  $\text{TC}(\{x\})$  and hence every  $x \subseteq \text{TC}(\{x\})$ , and the well-ordering principle implies Choice.  $\square$

**Proposition 13.3** (Internal Choice is a different assertion; source 06). *In  $ZF$ , HWO is transitive, contains every ordinal and every subset of an ordinal, and internally satisfies the statement that every set can be well-ordered. If  $\mathcal{P}(\omega)$  is not well-orderable in the ambient universe,  $(\text{HWO}, \in)$  fails the Power Set axiom at  $\omega$ .*

*Proof.* If  $y \in x \in \text{HWO}$  then  $\text{TC}(\{y\}) \subseteq \text{TC}(\{x\})$  is well-orderable. For  $a \subseteq \alpha$ ,  $\text{TC}(\{a\}) \subseteq \{a\} \cup \alpha$ . For  $x \in \text{HWO}$  the restriction to  $x$  of a well-order of  $\text{TC}(\{x\})$ , and every subset of  $x$ , lie in HWO,

since finite products and finite subsets of a well-orderable set are well-orderable in ZF; so the internal well-order has a witness. Every subset of  $\omega$  is in HWO, so an internal power set of  $\omega$  would be the actual  $\mathcal{P}(\omega)$ , which is then well-orderable, contrary to the hypothesis.  $\square$

The surjectivity statement of [theorem 13.2](#) is an external comparison of ranges, not the value of the internal Choice sentence. The theorem is about the specified ordinal-graph construction; no claim is made that every possible interpretation of sets in surreals has this boundary. Global Choice is unnecessary here and for the local ZFC bi-interpretations: existence of a code is proved object by object and the quotient identifies all codes.

*Remark 13.4* (Source 09’s fourth target, answered). Source 09 listed “the choice-free analogue” among its targets and observed that ordinal-indexed relations naturally recover the sets with well-orderable transitive closures. [Theorem 13.2](#), source 06’s, is that analogue for the full class; source 07 records the same range in a remark without a separate proof. Whether a richer coding signature can escape the boundary, and a choice-free version of the *cutoff* theorems, remain open.

[Added 2 October 2026, batch 80: a class-level counterpart. The report `foundations-and-computation/surreal-well-orders` proves in its Theorem 13.1 (`swo:cf:choice`, its source 11’s) that over GB without set choice, Global Choice holds if and only if **No** has a class well-order, if and only if it has a set-like one, if and only if there is a class bijection  $\text{Ord} \rightarrow \mathbf{No}$ . From a class well-order of the sign sequences, the proof well-orders every  $V_\alpha$  by set-valued recursion, coding a subset of  $V_\beta$  by a sign sequence along the well-order of  $V_\beta$  already built. Its Remark 13.4 (`swo:rem:answered-choice`) compares this with the ordinal-graph coding of [theorem 13.2](#) and notes that the theorem here concerns a different statement and is unchanged. The two open problems just named are not affected.]

[Added 3 October 2026, batch 89: an injection counterpart over ZF. Source 11 ([Part VII](#)) proves that a set  $X$  injects into **No** iff it injects into  $\mathcal{P}(\alpha)$  for some ordinal  $\alpha$  ([theorem 42.1](#)), so that universality of **No** for free fields on all sets is exactly the first Kinna–Wagner principle ([theorem 43.5](#)). The sets injecting into **No** form a wider class than HWO:  $\mathcal{P}(\omega)$  always injects into **No** ([remark 40.3](#)). An injection names elements; it does not recover a set as the coding here does. [Theorem 13.2](#) and the two open problems of [remark 13.4](#) are unchanged.]

## Part III

# Outer models

Throughout this part,  $M \subseteq N$  are transitive models of ZFC with the same ordinals. They may be set models viewed from an ambient metatheory, or class models with formulas interpreted one at a time. Superscripts indicate the model in which a construction is performed, and all references to  $M$ -cardinality are internal to  $M$ . The intended examples are set-forcing extensions, but no result below assumes forcing unless it says so.

## 14 Old arithmetic and cross-model coherence

**Lemma 14.1** (An old simplest separator stays simplest; sources 02, 05, 07). *If  $L, R \in M$  are separated sets of old sign sequences and  $w \in M$  is their simplest separator computed in  $M$ , then  $w$  is their simplest separator in  $N$ .*

*Proof.* Comparisons of old sign sequences are absolute, so  $w$  still separates. Suppose  $z \in N$  separates and is shorter. If  $z$  is a prefix of  $w$  it is a restriction of  $w$  to an ordinal, hence old, contradicting minimality in  $M$ . Otherwise  $z$  and  $w$  first disagree before  $z$  ends; their common prefix  $p$  is a restriction of  $w$ , hence old, lies strictly between  $z$  and  $w$ , and is a shorter separator by convexity of the set of separators, with all its comparisons with  $L, R$  holding in  $M$ . This again contradicts minimality.  $\square$

Source 02 gives a constructive variant: build the separator sign by sign, appending  $+$  if some  $\ell \in L$  is at least the current word and  $-$  otherwise, and taking unions at limits; every test involves only the given sets and the previous prefix, so the stopping place and every sign agree in  $M$  and  $N$ .

**Proposition 14.2** (Absolute old field structure; sources 02, 05, 07). *The inclusion of old sign sequences  $\mathbf{No}^M \subseteq \mathbf{No}^N$  preserves and reflects order and preserves birthday, constants, negation, addition, multiplication and inverses. Its ordered-field reduct is an elementary inclusion, and so is the pure-field inclusion  $\mathbf{No}^M[i] \subseteq \mathbf{No}^N[i]$ . The same holds for  $\mathbf{No}_{<\lambda}^M \subseteq \mathbf{No}_{<\lambda}^N$  at every common epsilon number  $\lambda$ .*

*Proof.* Order, length, constants and sign reversal are absolute. For addition induct on pairs of old prefixes: all prefixes of an old number are old, the options are identified by induction, and lemma 14.1 identifies their simplest separator. Multiplication follows by the same induction using addition, and inverses by uniqueness. Both real fields are real closed and both complex fields algebraically closed of characteristic zero, so quantifier elimination makes the inclusions elementary [14, 15]. For proper classes the equivalences are applied one formula at a time.  $\square$

This is the usual absoluteness of the canonical construction, not an assertion about arbitrary chosen presentations of isomorphic fields. It does not say that the outer universe cannot distinguish the inner one once extra structure is named: the rest of this part shows that birthday does.

**Proposition 14.3** (Cross-model coherence; source 09). *Let  $\kappa$  be an uncountable cardinal of both models. There are actual inclusions*

$$\mathcal{B}_\kappa^M \subseteq \mathcal{B}_\kappa^N, \quad H_\kappa^M \subseteq H_\kappa^N,$$

*compatible with the interpretations and their definable isomorphisms. Consequently*

$$\mathcal{B}_\kappa^M \prec \mathcal{B}_\kappa^N \iff H_\kappa^M \prec H_\kappa^N.$$

*Proof.* A sign sequence of  $M$  remains the same sign sequence in  $N$ ; order and birthday are absolute. A small arithmetic certificate existing in  $M$  remains valid in  $N$ , so the field operations agree. An old set of hereditary size below  $\kappa$  comes with an old bijection witnessing that bound; since  $\kappa$  is still a cardinal, it belongs to  $H_\kappa^N$ .

For code validity, one must not claim that the two models have the same subsets of the coding ordinal. Instead, a relation well founded in  $M$  has an ordinal rank function in  $M$ , which remains decreasing in  $N$ . Its well-foundedness therefore persists despite the extra subsets in  $N$ . Root generation says that every node is connected to the root by a finite downward path, as established in lemma 5.1; these paths are absolute. Extensionality is bounded and absolute. Failure of any of the code conditions in  $M$  also persists, using its old witness. Thus validity on old codes is unchanged.

The old Mostowski collapse still satisfies its bounded defining equations in  $N$  and hence is the same collapse. Equality and membership of decoded old sets are unchanged. Witnesses to

their equality and membership can be constructed in  $M$  from those collapses, so their required graph codes already exist there. The reverse sign-graph interpretation likewise sends old objects to their actual images. Applying each interpretation to an elementary inclusion now proves the equivalence.  $\square$

Unlike two cutoffs in a single universe,  $\mathcal{B}_\kappa^M$  and  $\mathcal{B}_\kappa^N$  may have different birthday fibers even at the same small ordinal. Their ordered field reducts nevertheless always include elementarily (proposition 14.2).

## 15 The preservation criterion and the first new birthday

**Theorem 15.1** (Exact preservation criterion; sources 02, 06, 07, 09). *For every common ordinal  $\gamma$ ,*

$$(\mathbf{No}_{<\gamma})^M = (\mathbf{No}_{<\gamma})^N \iff \forall \alpha < \gamma \ \mathcal{P}(\alpha)^M = \mathcal{P}(\alpha)^N. \quad (15.1)$$

*If  $\gamma = \lambda$  is an epsilon number, these conditions are equivalent to the literal equality  $\mathcal{B}_\lambda^M = \mathcal{B}_\lambda^N$  of the canonically presented structures, including all operations and the birthday function. If  $\gamma = \kappa$  is an uncountable cardinal in both models, they are also equivalent to  $H_\kappa^M = H_\kappa^N$ .*

*Proof.* Taking plus positions is an absolute bijection between sign sequences of length  $\alpha$  and subsets of  $\alpha$ ; this gives (15.1). Once the carriers agree, proposition 14.2 gives equality of all operations; conversely, equal structures have equal carriers.

For  $H_\kappa$ : a subset of  $\alpha < \kappa$  has hereditary size below  $\kappa$  in either model, so  $H_\kappa^M = H_\kappa^N$  implies that no bounded subset is new. Conversely assume no new bounded subsets. Old sets of hereditary size below  $\kappa$  stay so in  $N$  (proposition 14.3). For  $x \in H_\kappa^N$  take in  $N$  a valid code  $(\delta, r, t)$  with  $\delta < \kappa$ . Its relation word has length  $\Lambda(\delta) < \kappa$ , so it is old, as are  $\delta$  and  $t$ . The coded relation is well founded in  $N$ , hence in  $M$  (an old nonempty subset without a minimal element would also be one in  $N$ ); extensionality is absolute; the collapse computed in  $M$  satisfies its recursion in  $N$ , so it is the collapse there. Hence  $x \in M$ , with hereditary size below  $\kappa$  witnessed in  $M$ .  $\square$

Source 07 proves (15.1) for every common ordinal, source 02 the epsilon-number clause, source 06 the cardinal clause with  $H_\kappa$ , and source 09 the structure equality inside theorem 16.4. The right side of (15.1) must *not* be replaced by “no new short sequences of ordinals”: a short sequence with unbounded range can be new while every bounded subset below a large cutoff stays old, which is exactly what the Prikry example of section 17 exploits. Nor is “adds no real” identified with “adds no surreal” (example 15.7).

**Corollary 15.2** (Countable cutoffs detect exactly new reals; source 02). *If  $\lambda$  is a countable epsilon number, then  $\mathcal{B}_\lambda^M = \mathcal{B}_\lambda^N \iff \mathcal{P}(\omega)^M = \mathcal{P}(\omega)^N$ .*

*Proof.* For infinite  $\alpha < \lambda$  a bijection between  $\omega$  and  $\alpha$  lies in  $M$  and remains one in  $N$ , so a new subset of  $\alpha$  gives a new subset of  $\omega$ ; finite ordinals have no new subsets; and  $\omega < \lambda$ .  $\square$

**Lemma 15.3** (New sets imply a new ordinal subset; sources 05, 07). *If  $M \subsetneq N$ , then  $N$  has a subset of some ordinal that is not in  $M$ .*

*Proof.* Suppose not, and let  $a \in N$ . In  $N$  enumerate  $\text{TC}(\{a\})$  by an ordinal and code the pulled-back membership relation by a subset of an ordinal through an old injective pairing (for example the natural-arithmetic rectangle map). By hypothesis the code, hence the relation, is in  $M$ . Well-founded

in  $N$ , it is well founded in  $M$ ; its decoration computed in  $M$  is the decoration in  $N$  by uniqueness of recursion, and its root value is  $a$ . So  $a \in M$ , and  $M = N$ .  $\square$

**Theorem 15.4** (First new birthday; sources 02, 05, 07). *Let  $M \subsetneq N$  and*

$$\beta = \beta(M, N) = \min\{\alpha : \mathcal{P}(\alpha)^N \neq \mathcal{P}(\alpha)^M\}.$$

*Then  $\beta$  exists and:*

- (i) *every surreal of  $N$  of birthday below  $\beta$  belongs to  $M$ ;*
- (ii)  *$N$  has a surreal of birthday exactly  $\beta$  not in  $M$ , all of whose proper prefixes are in  $M$ ; thus  $\beta$  is the least birthday of an element of  $\mathbf{No}^N \setminus \mathbf{No}^M$ ;*
- (iii)  *$\beta$  is an infinite cardinal of  $M$ .*

*Proof.*  $\beta$  exists by lemma 15.3. A sign sequence of birthday  $\alpha$  is determined by  $\alpha$  and its plus set, which is old when  $\alpha < \beta$ ; this proves (i), and a new subset of  $\beta$  gives the new sign of (ii), whose proper prefixes are old by (i). No finite ordinal has a new subset. If  $\beta$  were not an  $M$ -cardinal,  $M$  would have a bijection  $f : \nu \rightarrow \beta$  with  $\nu < \beta$ ; for a new  $A \subseteq \beta$  the pullback  $f^{-1}[A]$  would be old by minimality, and then  $A = f[f^{-1}[A]]$  would be old.  $\square$

**Remark 15.5** (Source 02's version). Source 02 proved the theorem for a set-forcing extension  $N = M[G] \neq M$ , obtaining existence of  $\beta$  from the generic filter: a ground-model well-order of the forcing set turns  $G$  into a new subset of an ordinal. Sources 05 and 07 remove the forcing hypothesis by lemma 15.3. No claim that all cardinals are preserved is needed. The theorem does not assert that  $\beta$  is regular, and it does not identify the signs of birthday at most  $\omega$  with real numbers.

**Example 15.6** (Adding a real). If  $N = M[g]$  adds a Cohen subset  $g$  of  $\omega$ , no finite subset is new and  $g$  is, so  $\beta = \omega$ . The sign with plus set  $g$  is a new surreal of birthday  $\omega$ , and each of its finite prefixes is old. Such an extension changes  $\mathbf{No}_{<\kappa}$  for every common uncountable cardinal  $\kappa$ .

**Example 15.7** (No new reals, but new surreals; sources 05, 07). Force over  $M$  with countable partial functions from  $\omega_1^M$  to 2, ordered by extension. The union of a descending  $\omega$ -chain is a condition, so the forcing is countably closed; the successive-decision argument shows that it adds no new countable sequence of ordinals, hence no subset of any old countable ordinal, and preserves  $\omega_1$ . The generic union differs from every old subset of  $\omega_1^M$  by a density argument [13]. Hence  $\beta = \omega_1^M$ : all countable-birthday surreals are unchanged, there are new signs of birthday  $\omega_1^M$ , and the full birthday inclusion still fails elementarity (theorem 16.5). Agreement on the reals, or even on every countable sign sequence, does not exhaust what the birthday expansion remembers.

**Remark 15.8** (The sibling report). The sibling report `foundations-and-computation/surreal-fields-across-universes` works in the same setting of transitive  $M \subseteq N$  with the same ordinals and uses the same letter  $\beta$  for the first new birthday; it writes  $\delta$  for the first new ordinal-sequence length, which is not named here. It studies the external saturation of the *pure* ordered field  $\mathbf{No}^M$  inside  $N$ , which is elementary in  $\mathbf{No}^N$  by proposition 14.2; the present report studies the birthday expansion, whose inclusion is non-elementary as soon as  $M \neq N$  (theorem 16.5). The first-new-birthday theorem and the absoluteness of old arithmetic are shared ground; the countably closed and Priky examples appear in both.

**Remark 15.9** (Gap pairs of the cutoff fields; batch 37). Added in batch 37 (28 September 2026). Part VIII of the sibling report computes the external gaps of this report's cutoff fields as *pure* ordered fields (no birthday symbol). Let  $M \subseteq N$  have the same ordinals and the same cofinalities, and let  $\theta$  be a common uncountable regular cardinal. By its Lemma 26.1 (`univ:gs:lem:boundary`),

the gap pairs of  $(\mathbf{No}_{<\theta})^M$  in  $N$  are exactly the symmetric pairs  $(\mu, \mu)$ , for  $\mu$  the  $N$ -cofinality of the length of a fresh sign (a new sign sequence all of whose proper initial segments are old) of length below  $\theta$ , together with the boundary pairs  $(\mu, \theta)$  and  $(\theta, \mu)$  for every infinite regular  $\mu \leq \theta$ , which come from old signs of length  $\theta$ . Pairs of the first kind, all with  $\mu < \theta$ , are absent exactly when  $(\mathbf{No}_{<\theta})^M = (\mathbf{No}_{<\theta})^N$ , that is, by [theorem 15.1](#), when no subset of an ordinal below  $\theta$  is new. With  $M = N$  this shows that the internal field  $\mathbf{No}_{<\theta}$  already has asymmetric gaps such as  $(\omega, \theta)$ . This is consistent with [remark 12.4](#), since each such gap has a side of cofinality  $\theta$ . The sibling's Theorem 26.2 and Corollary 26.3 (`univ:gs:thm:cutoff`, `univ:gs:cor:setfields`) use these fields, at  $\theta = \aleph_{\omega+2}$ , for  $2^{\aleph_0}$  pairwise nonisomorphic real closed fields of one cardinality with one saturation spectrum.

[Added 3 October 2026, batch 81: the case  $M = N$  holds for every infinite cardinal. Source 17 of the report `foundations-and-computation/surreal-well-orders` computes the gap pairs of the ordered set  $\mathbf{No}_{<\theta}$  for every infinite cardinal  $\theta$ , including singular  $\theta$  and  $\theta = \omega$ : with  $r = \text{cf } \theta$  they are  $(r, r)$  and  $(\rho, r)$ ,  $(r, \rho)$  for every infinite regular  $\rho < \theta$ , and deleting fewer than  $2^{<\theta}$  points does not change them (its Theorem 29.4, `sw:gs:thm:alphabet-gaps`, with the same witnesses,  $\rho$  pluses followed by  $\theta$  minuses). At uncountable regular  $\theta$  this is the case  $M = N$  of `univ:gs:lem:boundary` above, which that report records with a pointer.]

## 16 Detecting new subsets

### 16.1 The old-power-set detector at a cutoff

**Theorem 16.1** (Uniform detector for an added subset; source 09). *Let  $\kappa$  be an uncountable cardinal of both models. Suppose  $\alpha < \kappa$ ,  $N$  has a subset of  $\alpha$  not in  $M$ , and*

$$\nu = (2^{|\alpha|})^M < \kappa.$$

*Then  $\mathcal{B}_\kappa^M \not\prec \mathcal{B}_\kappa^N$ . One fixed interpreted formula witnesses the failure using parameters coding  $\alpha$  and  $A = \mathcal{P}^M(\alpha)$ .*

*Proof.* The cardinal bound gives  $A \in H_\kappa^M$ . In the membership language consider

$$\exists y [\forall v \in y (v \in \alpha) \wedge y \notin A]. \quad (16.1)$$

This is false in  $H_\kappa^M$ , because all its subsets of  $\alpha$  lie in  $A$ . It is true in  $H_\kappa^N$ : any newly added subset of  $\alpha$  has hereditary size below the still-cardinal  $\kappa$  and is not an element of the old set  $A$ . Translate (16.1) through  $\mathcal{I}$  and choose old codes for the two parameters. Cross-model coherence gives the claimed failure in the birthday structures.  $\square$

The membership-language formula (16.1) is  $\Sigma_1$ : its matrix has only bounded quantifiers. Its fully expanded birthday-field translation need *not* be existential or  $\Sigma_1$  in that different language, because code validity includes the bit-quantified well-foundedness test. No quantifier-complexity claim is made across that translation.

**Proposition 16.2** (Sharpness of the detector's parameter cost; source 09). *The old parameter  $\mathcal{P}^M(\alpha)$  belongs to  $H_\kappa^M$  exactly when  $(2^{|\alpha|})^M < \kappa$ . In the rooted coding scheme, every code for that parameter has node-domain cardinality  $|\text{TC}(\{\mathcal{P}^M(\alpha)\})|^M$ , which equals the power-set cardinal for infinite  $\alpha$ . In that infinite case, a code of birthday cardinality no larger than that cardinal can be chosen.*

*Proof.* The first statement is lemma 10.6 inside  $M$ . The exact node-domain cardinality is (5.7). For infinite  $\alpha$ , the transitive closure consists of the power set, its subsets, and the ordinal data beneath  $\alpha$ , giving cardinality  $2^{|\alpha|}$ . Choosing its initial ordinal as the node domain gives relation birthday  $\Lambda(\nu)$ , of cardinality  $\nu$  by lemma 4.3.  $\square$

This is not a theorem that elementarity must hold when the bound fails. Another formula can detect a change at a smaller cutoff. It says exactly when this particular, universally applicable old-power-set witness is available.

**Example 16.3** (A conditional continuum-sized illustration). Suppose  $M$  satisfies CH,  $N$  preserves  $\omega_2^M$ , and  $N$  adds a real. Then the old set of reals has hereditary size  $\omega_1^M$ , so the detector works at the cutoff  $\omega_2^M$ . That old set is not an element of  $H_{\omega_1^M}^M$ , so the same parameter cannot be used at the cutoff  $\omega_1^M$ . No claim of elementarity at the smaller cutoff follows.

## 16.2 Rigidity under extensions at a strong limit

**Theorem 16.4** (Strong-limit extension rigidity; source 09). *Let  $\kappa$  be an uncountable cardinal of both models, strong limit in  $M$ . Then*

$$\begin{aligned} \mathcal{B}_\kappa^M \prec \mathcal{B}_\kappa^N &\iff \mathcal{B}_\kappa^M = \mathcal{B}_\kappa^N \\ &\iff \mathcal{P}^M(\alpha) = \mathcal{P}^N(\alpha) \text{ for every } \alpha < \kappa. \end{aligned}$$

*Proof.* If some subset of an  $\alpha < \kappa$  is new, the strong-limit property makes its old power-set parameter small enough, and theorem 16.1 excludes elementarity. Conversely, if all these power sets agree, the structures are equal by theorem 15.1, hence the inclusion is elementary. Equality of the scalar domains forces equality of the corresponding plus-sets, proving the remaining direction.  $\square$

This proves theorem 1.4. It applies, for example, to any forcing extension satisfying the stated cardinal-preservation hypotheses, but uses no particular forcing construction. It is a theorem about *standard cutoff structures at a fixed common cardinal*, not a prohibition on abstract elementary extensions supplied by model theory. Source 02 explicitly declined to claim that a proper enlargement at a fixed cutoff is non-elementary merely because it is proper; the theorem above and theorem 16.1 give sufficient conditions, and nothing more is claimed. Their ordered field reducts still include elementarily even when the birthday expansion fails the detector; a set-theoretic change can be invisible to ordered-field formulas yet visible to one birthday-field formula with old parameters.

## 16.3 The full class, and an explicit row-search formula

For the full classes write  $\mathcal{B}^M = (\mathbf{No}^M; 0, 1, +, -, \cdot, <, b)$  and likewise  $\mathcal{B}^N$ ; by proposition 14.2 the old signs form a substructure.

**Theorem 16.5** (An outer universe is detectable; sources 05, 07).  $\mathcal{B}^M \prec \mathcal{B}^N \iff M = N$ . *When  $M \neq N$  the failure is witnessed with old parameters: by the  $\mathcal{I}$ -translation of (16.1) with codes for  $\alpha$  and  $\mathcal{P}(\alpha)^M$ , for any  $\alpha$  with a new subset (source 07), and by the formula New of theorem 16.6 (source 05).*

*Proof.* If  $M \neq N$ , lemma 15.3 gives an ordinal  $\alpha$  with a new subset. Old valid codes remain valid in  $N$  and decode to the same sets (as in proposition 14.3, without a cardinal bound), and (16.1) is false in  $M$  and true in  $N$  with those parameters.  $\square$

[Added 4 October 2026, batch 93: Part III of [definable-surreals-and-omnific-integers](#) (its source 06; AI-assisted, unrefereed, not formalized) proves a parameter-free quotient interpretation of  $(V, \in)$  in  $(\mathbf{No}; +, \cdot, <, \mathbf{Oz}, \Omega)$ , a signature without birthday order (its Theorem 37.1, [dsn:op:thm:V-interpretation](#)), and the transfer  $M \equiv N \iff \mathfrak{S}_\Omega^M \equiv \mathfrak{S}_\Omega^N$  together with, for nested  $M \subseteq W$ ,  $M \preceq W \iff \mathfrak{S}_\Omega^M \preceq \mathfrak{S}_\Omega^W$  (its Theorem 40.4, [dsn:op:thm:theory-faithful](#)); a dated note there uses the argument of this theorem to show that for models of the same height both sides hold only when  $M = W$ . No bi-interpretation is claimed for that signature (its Question 42.1, [dsn:op:q:biinterpretation](#)). Added 4 October 2026, batch 95: Part IX reduces the specific form of that question to the definability of birthday precedence in that signature ([remark 67.5](#) and [proposition 74.2](#)).]

Source 05's detector reads the birthday language directly. Fix a nonzero ordinal  $\theta$  and in  $M$  an enumeration  $(A_j)_{j < \nu}$ ,  $\nu > 0$ , of  $\mathcal{P}(\theta)^M$  (a surjection suffices), and code it by one old sign  $m$  of birthday  $\Lambda(\theta + \nu + 1)$  whose bit at  $p_{\theta+\nu+1}(j, \xi)$  is plus exactly when  $\xi \in A_j$ . Let

$$\text{New}(m, \theta, \nu) : \exists x \left[ b(x) = \theta \wedge \forall j < \nu \exists \xi < \theta \neg (\text{Bit}(x, \xi) \leftrightarrow \text{Bit}(m, p_{\theta+\nu+1}(j, \xi))) \right],$$

together with the ordinal and length conditions on  $\theta, \nu, m$ ; it asks for a row of length  $\theta$  different from every old row.

**Theorem 16.6** (An explicit new-subset detector; source 05). *With such parameters,  $\mathcal{B}^M \models \neg \text{New}(m, \theta, \nu)$ , and  $\mathcal{B}^N \models \text{New}(m, \theta, \nu)$  if and only if  $\mathcal{P}(\theta)^N \neq \mathcal{P}(\theta)^M$ .*

*Proof.* Every sign of birthday  $\theta$  in  $M$  has an old plus set, which is some row  $A_j$ . The old sign  $m$  has the same bits in  $N$ , so it still has exactly the old rows. A new subset of  $\theta$  gives a witness in  $N$ ; conversely a witness has a plus set differing from every old subset, hence new.  $\square$

Source 05 writes the matrix with its rectangle map and length  $\theta(\nu + 1)$ ; the formula is the same up to that pairing. The enumeration is a set in  $M$  and the matrix one sign of set length; no class of ground-model signs is named. The formula is not claimed to be existential after the abbreviations are expanded, since the prefix and bit predicates have quantifiers of their own. The detector branches off after the pairing lemma and does not need the arithmetic certificates.

*Remark 16.7* (The bounded version). At a cutoff the matrix must fit. Source 05 states: if a common ordinal  $\kappa$  is an uncountable regular cardinal in both models, the old matrix has  $\theta, \nu < \kappa$  and length below  $\kappa$ , and  $N$  has a new subset of  $\theta$ , then  $\mathcal{B}_\kappa^M \not\preceq \mathcal{B}_\kappa^N$ , witnessed by New. Its proof uses regularity only through closure of natural arithmetic below  $\kappa$ , which holds at every uncountable cardinal ([lemma 2.2](#)); so the conclusion holds whenever  $\kappa$  is an uncountable cardinal of both models and  $(2^{|\theta|})^M < \kappa$ , the hypothesis of [theorem 16.1](#). Without a bound on the old enumeration the corollary cannot be inferred from  $\theta < \kappa$  alone, because the set of old subsets of a small ordinal may have cardinality at least  $\kappa$ .

*Remark 16.8* (The conclusions use parameters). These theorems say that the *canonical inclusion* is not elementary. They do not by themselves show that the two structures have different parameter-free complete theories, nor do they exclude unrelated embeddings or isomorphisms. If the extension changes a parameter-free set-theoretic sentence, its translation does distinguish the theories (sources 05, 07).

## 17 Prikry forcing: an unchanged field with a singularized cutoff

### 17.1 The imported forcing theorem

We use the classical ordinary Prikry forcing associated with a normal measure  $U$  on a measurable cardinal  $\kappa$ . Conditions consist of a finite increasing stem  $s \subset \kappa$  and a measure-one set  $A \in U$  above that stem; an extension may lengthen the stem using points of  $A$  and shrink  $A$ , a direct extension only shrinks  $A$ . Its standard properties are: the generic stem union is cofinal of order type  $\omega$  in  $\kappa$ ; the forcing adds no bounded subsets of  $\kappa$ ; and it preserves cardinals. These are imported from the exposition of Poveda–Rinot–Sinapova, especially Lemma 2.10 and Section 3.1 [11], and are not reproved in full. For orientation, source 02 sketches the bounded-subset consequence of the Prikry property: decide a name for a subset of  $\alpha < \kappa$  by successive direct extensions, using that intersections of fewer than  $\kappa$  measure-one sets have measure one; bounded subset preservation keeps the cardinals below  $\kappa$  and  $\kappa$  itself, and the  $\kappa^+$ -chain condition the cardinals above.

**Theorem 17.1** (Forcing invisibility, conditional; sources 02, 06). *Suppose a transitive model  $M$  of ZFC has a measurable cardinal  $\kappa$ , and let  $N = M[G]$  be an ordinary Prikry extension at  $\kappa$ . Then*

$$\mathcal{B}_\kappa^M = \mathcal{B}_\kappa^N, \quad H_\kappa^M = H_\kappa^N = V_\kappa^M,$$

*with the same carrier, functions and relations, although*

$$M \models \text{cf}(\kappa) = \kappa, \quad N \models \text{cf}(\kappa) = \omega.$$

*The common membership structure satisfies ZFC in both universes.*

*Proof.* In  $M$  the measurable  $\kappa$  is strongly inaccessible, hence an epsilon number by lemma 2.2; it remains a cardinal, hence an epsilon number, in  $N$ , although singular there. Prikry forcing adds no subset of any  $\alpha < \kappa$ , so theorem 15.1 gives both equalities of structures. Since  $\kappa$  is inaccessible in  $M$ ,  $H_\kappa^M = V_\kappa^M$  (sets in  $V_\kappa$  have small transitive closures by strong limit and regularity, and a well-founded relation on fewer than  $\kappa$  nodes has rank below  $\kappa$ ), and  $V_\kappa^M \models \text{ZFC}$ . The generic sequence witnesses  $\text{cf}^N(\kappa) = \omega$ . Truth of each fixed first-order formula in one unchanged set structure is absolute between transitive models, by induction on formulas.  $\square$

**Corollary 17.2** (No uniform regularity detector; sources 02, 06). *Relative to the existence of such a pair of models, no first-order sentence in the language of  $\mathcal{B}_\kappa$  holds, in every universe and at every uncountable cardinal cutoff, exactly when the external cutoff is regular, or exactly when it is strongly inaccessible. More strongly, no invariant determined by the isomorphism type of the enriched structure alone determines the external cofinality of its cutoff in a universe-independent way.*

*Proof.* Such a sentence or invariant would take different values on the two identical structures of theorem 17.1.  $\square$

This is a relative-consistency separation using a measurable cardinal: no measurable cardinal is constructed or proved to exist, and ZFC is not claimed to prove the existence of such a pair of universes. The statement concerns fixed first-order sentences; unrestricted second-order satisfaction is *not* asserted to be unchanged, since the family of external subsets of the carrier changes. The equality established in  $N$  is with the *old*  $V_\kappa^M$ ; it is not claimed that  $H_\kappa^N = V_\kappa^N$ , and the rank hierarchy of the extension can contain new objects. The corollary also shows why diagnosing

singularity by failure of some unspecified Replacement instance is unreliable: the small universe keeps every axiom of ZFC while the ambient universe acquires an unbounded sequence invisible to it.

**Proposition 17.3** (Where the missing sequence lives; sources 02, 06). *In the situation of [theorem 17.1](#), the first new birthday is exactly  $\beta(M, N) = \kappa$ . The Prikry sequence  $h : \omega \rightarrow \kappa$ , and its range, have hereditary size  $\kappa$ , so they lie in  $H_{\kappa^+}^N \setminus H_\kappa^N$ ; any faithful pointed code for them needs at least  $\kappa$  nodes. For every epsilon number  $\lambda$  with  $\kappa < \lambda < \kappa^+$ , of which there are some, the carrier  $\mathbf{No}_{<\lambda}$  changes and its interpreted universe  $H_{\kappa^+}$  contains the new sequence.*

*Proof.* No subset below  $\kappa$  is new, while the range of  $h$  is a new subset of  $\kappa$  (already by regularity of  $\kappa$  in  $M$ ); [theorem 15.4](#) gives  $\beta = \kappa$ , and every  $\lambda > \kappa$  contains that sign layer. The values of  $h$  are cofinal ordinals below  $\kappa$ , so the transitive closure of  $h$  contains all of  $\kappa$  and has size  $\kappa$ . For  $\kappa < \lambda < \kappa^+$  the cardinal ceiling is  $\kappa^+$  in both models. Iterating  $\alpha \mapsto \omega^\alpha$   $\omega$  times from above  $\kappa$  gives a fixed point of cardinality  $\kappa$ , an epsilon number in the interval.  $\square$

The proposition distinguishes the *appearance of new elements* from a *change of the complete first-order theory*; it proves the former and does not assert the latter for every enlarged cutoff. By contrast, an extension adding a real changes  $\mathbf{No}_{<\kappa}$  at every common uncountable cardinal ([example 15.6](#)); the preservation criterion distinguishes the two kinds of extension without referring to the names of their forcing notions. The surcomplex enrichment is likewise unchanged in the Prikry situation ([theorem 19.8](#)).

## Part IV

# The full class and the surcomplex numbers

## 18 The full class: the announced theorem and Kunen rigidity

### 18.1 The announced global bi-interpretation

**Theorem 18.1** (Chen–Hamkins–Yang, announced [6, 7, 8]). *In ZFC there are parameter-free first-order interpretations  $\mathcal{I}$  and  $\mathcal{J}$  such that, formula by formula,  $\mathcal{I}(\mathcal{B})$  is definably identified with  $(V, \in)$ ,  $\mathcal{J}(V, \in)$  with  $\mathcal{B}$ , and both composites have definable isomorphisms to the original structures.*

**Status and credit.** This is the global theorem announced by Chen, Hamkins and Yang, stated there for the surreal field with birthday order; by [lemma 2.1](#) that language and the birthday-function language are definitionally equivalent. Their slides describe the coding of sets by pointed well-founded extensional relations and then by sign sequences, and mark the work as still being checked; the collection’s foundations report records the announcement in `found:sub:announcement`. The written argument below is source 07’s; source 05 gives the same statement with its bisimulation codes, and internally in every model of ZFC ([remark 7.4](#)). It is printed as *a proof of the announced theorem*, not as a new result, and it is not an audit of an unpublished manuscript of Chen–Hamkins–Yang or of their proposed axiomatization.

[Added 4 October 2026, batch 95: Part IX prints two further written proofs of this theorem, by sources 13 and 14 ([theorems 71.1](#), [71.3](#) and [72.2](#) and [proposition 72.4](#)), and the same structure-level bi-interpretation natively for the omnific integers with birthday precedence ([corollary 72.3](#)). It also

gives explicit recursively axiomatized theories whose only individuals are numbers, bi-interpretable with ZFC (theorems 73.4 and 73.6), built from the interpretation by a general round-trip recipe (theorem 73.1). They are not the Chen–Hamkins–Yang theory, whose intrinsic axioms no report of the collection verifies (question 79.2); so the statement above, that this proof is not an audit of their proposed axiomatization, stands.]

*Proof (source 07).* Use the codes of section 5 without a cutoff. Every set has a valid code: choose an injection of  $\text{TC}(\{a\})$  into an ordinal (one application of Choice) and code the transported membership relation. The correctness of  $\sim$  and  $\in_{\mathcal{I}}$  (lemma 5.2) and of the collapse is proved for set-sized graphs and applies to each code. The reverse interpretation takes sign names  $\langle \alpha, A \rangle$  with  $A \subseteq \alpha$  and defines the arithmetic by the certificates of section 3, which now exist without a size restriction. The two composites are the collapse relation and the sign-name relation  $x \mapsto \langle b(x), A_x \rangle$  of sections 6 and 7; both are single formulas, and their correctness is proved one formula at a time, with no truth predicate for  $V$  and no class of chosen representatives.  $\square$

The local refinements proved in Part I were stated for  $V$  by the announcement; the cutoff theorems, the noncardinal pointed version, and the local certificates at singular cutoffs are what sources 02, 05, 06, 07 and 09 propose to add, with the priority caveats of section 20. Source 09 already noted that its refinements may overlap unpublished details of the Chen–Hamkins–Yang project.

## 18.2 Transfer of Kunen’s inconsistency

An automorphism and a non-surjective elementary self-embedding must not be conflated. For automorphisms of the full class the argument of lemma 19.3 applies (the ordinals form a definable well-ordered class). It fails for a non-surjective elementary self-embedding  $j$ , whose restriction to the ordinals need not be an automorphism, and which satisfies  $b(j(x)) = j(b(x))$  rather than the externally birthday-fixed equation  $b(j(x)) = b(x)$ . Literature on birthday-fixed surreal morphisms, such as the structures of Rangel–Mariano [16], must be compared with this distinction in mind (source 07).

**Lemma 18.2** (Transferring an elementary embedding; source 07). *Let  $j : \mathcal{B} \rightarrow \mathcal{B}$  be a class elementary embedding. For each set  $a$  choose any code  $C$  with  $x_C = a$  and put  $\hat{j}(a) = x_{j(C)}$ , with  $j$  applied coordinatewise. This is a well-defined class elementary embedding  $(V, \in) \rightarrow (V, \in)$ , and  $\hat{j}$  is the identity only if  $j$  is.*

*Proof.* Elementarity preserves validity and  $\sim$ , so the value does not depend on the code, and  $\hat{j}$  is a class defined from  $j$  without choosing representatives. For each set-theoretic formula and codes for its parameters, theorem 18.1 and elementarity of  $j$  give  $\varphi(\vec{a}) \iff \varphi(\hat{j}(\vec{a}))$ . The definable sign-name composite is natural under elementary maps; if  $\hat{j}$  is the identity, it identifies  $j(x)$  and  $x$  with the same interpreted sign name, so  $j(x) = x$ .  $\square$

**The class-theoretic hypothesis.** The classical Kunen inconsistency rules out a nonidentity elementary embedding  $V \rightarrow V$  under Choice, in the form that permits the embedding as a class parameter in Separation and Replacement (Hamkins–Kirmayer–Perlmutter [12]). We work, as source 07 does, in GBC for convenience; global choice is stronger than needed. The class  $\hat{j}$  is definable from  $j$  and inherits availability as a class parameter. Elementarity may be read formula

by formula; for a single class formulation one may use  $\Delta_0$ -elementarity with membership-cofinality, which holds because  $\hat{j}(\alpha) \geq \alpha$  and  $\hat{j}(V_\alpha) = V_{\hat{j}(\alpha)}$ .

**Theorem 18.3** (No nontrivial class elementary self-embedding; source 07). *In GBC, or under the class-parameter Separation and Replacement hypotheses just described, every class elementary self-embedding of  $\mathcal{B}$  is the identity.*

*Proof.* A nonidentity  $j$  would give a nonidentity elementary  $\hat{j} : V \rightarrow V$  by lemma 18.2, contradicting the cited Kunen theorem.  $\square$

The set-theoretic input is the classical Kunen inconsistency; this is not a new proof of it. The theorem concerns the full birthday-expanded class. It does not assert that the pure surreal ordered field has no nontrivial embeddings or automorphisms; it does not classify the elementary self-embeddings of the bounded  $\mathcal{B}_\kappa$  (which theorem 9.10 reduces to those of  $H_\kappa$ ); and it does not prohibit external self-maps of a countable nonstandard model's internal surreal structure, which need not be classes of that model with the required Separation and Replacement properties.

[Added 4 October 2026, batch 89: how much the class-parameter hypothesis carries in a setting with atoms. In the urelement kernel models ( $M_\kappa^{\text{ker}}$  of Part VI, written  $M_\kappa(A)$  there) of the research-report collection's SetTheory/Cardinals/docs/reports/ordinals-and-order-types/naming-elementary-embeddings (batch-89 manuscript 05), every injection of the atoms lifts to an elementary self-embedding that fixes every pure set and is amenable (its restriction to each set is a set), and these lifts are exactly the elementary self-embeddings with transitive range (its Theorems 3.2 and 3.4, `nee:thm:lifts`, `nee:thm:initial`). Naming finitely many such lifts keeps Replacement in the expanded language exactly under a condition on the components of the named atom injections (its Theorem 5.1, `nee:thm:profile`). This does not touch theorem 18.3: those embeddings fix every pure set, so every ordinal and every surreal number, while the theorem concerns embeddings of the pure class  $\mathcal{B}$ , which lemma 18.2 turns into embeddings of  $V$ . That report is AI-assisted, unrefereed and not formalized.]

## 19 Surcomplex numbers and the orientation obstruction

### 19.1 The enriched surcomplex structures

For an epsilon number  $\lambda$ ,  $\mathbf{No}_{<\lambda}[i]$  is algebraically closed because  $\mathbf{No}_{<\lambda}$  is real closed. Let  $c(x + iy) = x - iy$  be conjugation and

$$\text{cb}(x + iy) = \max(\text{b}(x), \text{b}(y)),$$

the *coordinate birthday*, whose value is an embedded real ordinal. Put

$$\mathcal{C}_\lambda = (\mathbf{No}_{<\lambda}[i]; 0, 1, +, -, \cdot, c, \text{cb}), \quad \mathcal{C}_\lambda^i = (\mathcal{C}_\lambda, i),$$

and  $\mathcal{C}$ ,  $\mathcal{C}^i$  for the full classes on  $\mathbf{No}[i]$ . The coordinate birthday is a stated convention, not the birthday of a combinatorial-game representative of a complex object, and not asserted to be a pre-existing universal definition of a “surcomplex birthday”. Conjugation,  $\text{cb}$  and  $i$  are explicit extra structure; none is asserted to be definable in the pure algebraically closed field, whose inclusions are elementary in the pure language [14].

**Lemma 19.1** (Independence of orientation; source 06).  *$\text{cb}$  is unchanged if  $i$  is replaced by  $-i$ , and its restriction to  $\mathbf{No}_{<\lambda}$  is  $\text{b}$ .*

*Proof.* The replacement changes coordinates  $(x, y)$  to  $(x, -y)$ , and negation reverses every sign without changing length, so  $b(-y) = b(y)$ ; also  $b(0) = 0$ .  $\square$

**Proposition 19.2** (Recovery of the real birthday structure; all five sources). *Both  $\mathcal{C}_\lambda$  and  $\mathcal{C}_\lambda^i$  interpret  $\mathcal{B}_\lambda$  without parameters on the fixed field of  $c$ , with order*

$$x < y \iff \exists t [c(t) = t \wedge t \neq 0 \wedge t^2 = y - x]$$

for real  $x, y$ . The structure  $\mathcal{C}_\lambda^i$  is uniformly bi-interpretable with  $\mathcal{B}_\lambda$ , hence with  $(H_\lambda, \in)$  at cardinal cutoffs and with  $(H_{\kappa(\lambda)}, \in, \lambda)$  otherwise; the same holds for  $\mathcal{C}^i$ ,  $\mathcal{B}$  and  $(V, \in)$  formula by formula.

*Proof.* The fixed field is  $\mathbf{No}_{<\lambda}$ , real closed, where positivity is being a nonzero square; field operations restrict, and  $\text{cb}$  restricts to  $b$ . Conversely interpret complex numbers as pairs of reals with  $(x, y)(u, v) = (xu - yv, xv + yu)$ ,  $c(x, y) = (x, -y)$ ,  $i = (0, 1)$  and  $\text{cb}(x, y) = (\max(bx, by), 0)$ . With  $i$  named,  $(x, y) \mapsto x + iy$  is definable with inverse

$$z \mapsto \left( \frac{z + c(z)}{2}, \frac{z - c(z)}{2i} \right),$$

and the real composite is  $x \mapsto (x, 0)$ . These are the definable isomorphisms of a bi-interpretation; compose with [theorem 6.3](#) and [theorem 18.1](#).  $\square$

Thus cardinal rounding, recovery of a noncardinal cutoff, and all the set-theoretic distinctions above transfer to the oriented surcomplex structures, and adjoining  $i$  creates no new set-theoretic size requirement. The real-axis recovery is standard quadratic-extension algebra; its role here is to identify which enrichment carries the foundational information. In the Prikry situation of [theorem 17.1](#),  $\mathcal{C}_\kappa$  and  $\mathcal{C}_\kappa^i$  are also unchanged, because both coordinates and all displayed operations are; so adjoining  $i$  and retaining conjugation does not repair the inability to recover the ambient cofinality (sources 02, 06).

## 19.2 Rigidity and the two-element automorphism group

**Lemma 19.3** (Rigidity of a standard birthday cutoff; sources 05, 06, 07, 09). *Every automorphism of  $\mathcal{B}_\lambda$  is the identity, and every automorphism of a transitive structure  $(H_\kappa, \in)$ , or  $(H_\kappa, \in, \lambda)$ , is the identity.*

*Proof.* The embedded ordinals are definable and numerically well ordered, so any order automorphism of this segment fixes every ordinal by transfinite induction; the automorphism therefore fixes all birthdays, preserves the definable bit predicate at each fixed coordinate, and fixes every sign sequence. For  $H_\kappa$ , external well-founded induction gives  $h(x) = \{h(y) : y \in x\} = x$ ; naming an ordinal does not change this.  $\square$

The first assertion is a case of the collection’s existing simplicity rigidity: the report `surcomplex/surcomplex-field-automorphisms` shows, in its section `saut:sec:exponential` (subsection “Simplicity restores the canonical surreal numbers”), that a bijection preserving numerical order and prefix simplicity is the identity, and prefix is definable from order and birthday ([lemma 4.1](#)). It is printed here once, not as a new result. Rigidity is not pointwise definability: a rigid structure can have elements that are not individually definable without parameters.

**Theorem 19.4** (The orientation obstruction; sources 05, 06, 09, after Jeřábek). *For every epsilon number  $\lambda$ ,  $\text{Aut}(\mathcal{C}_\lambda) = \{\text{id}, c\}$ . Consequently  $\mathcal{C}_\lambda$  is parameter-free mutually interpretable with, but not parameter-free bi-interpretable with,  $(H_\lambda, \in)$  at cardinal cutoffs, or  $(H_{\kappa(\lambda)}, \in, \lambda)$  otherwise; after naming  $i$  the bi-interpretation holds (proposition 19.2).*

*Proof.* An automorphism preserves the fixed field of  $c$ , its definable order and birthday, so its restriction is the identity by lemma 19.3; it sends  $i$  to a root of  $X^2 + 1$ , so it is the identity or conjugation, as in `saut:thm:axis` ( $\text{Aut}(F(i)/F) = \{\text{id}, c\}$  for real closed  $F$ ). Both preserve  $c$  and `cb` (lemma 19.1).

A parameter-free bi-interpretation induces mutually inverse homomorphisms between the automorphism groups: an automorphism preserves the definable interpretation domain and equivalence relation, induces an automorphism of the interpreted structure, and the definable round-trip isomorphism recovers it. The set structure is rigid (lemma 19.3), whereas conjugation is a nonidentity automorphism of  $\mathcal{C}_\lambda$ . This excludes *every* parameter-free bi-interpretation, not only the coordinate one. Mutual interpretation holds: recover the real birthday structure inside  $\mathcal{C}_\lambda$  and then its set universe; in the other direction use the reverse surreal interpretation and the pair field, omitting the constant  $i$ .  $\square$

The general real/complex obstruction, that real and complex fields with conjugation are mutually interpretable but need a named imaginary unit for a parameter-free bi-interpretation, was explained by Jeřábek in 2021 [9]; only source 06 credited it, and the credit applies to the versions in sources 05 and 09 as well. The group computation is a consequence of classical rigidity and of `saut:thm:axis`, not a new classification of the unrestricted field automorphisms of  $\mathbf{No}[i]$  studied in the automorphisms report. The point here is the distinction between mutual interpretability and bi-interpretability, and why an unnamed imaginary unit cannot be treated as harmless notation in the latter claim. The two roots  $i$  and  $-i$  have the same parameter-free type in  $\mathcal{C}_\lambda$ ; interpreting a rigid universe does not force the interpreting structure to be rigid, and it is the definability of both round trips that transmits rigidity (source 06).

### 19.3 Two lifts of each real embedding

**Theorem 19.5** (Two lifts, and only two; source 06). *Let  $\kappa, \lambda$  be uncountable cardinals. Every elementary embedding  $f : \mathcal{B}_\kappa \rightarrow \mathcal{B}_\lambda$  has exactly two elementary extensions  $\mathcal{C}_\kappa \rightarrow \mathcal{C}_\lambda$ ,*

$$\widehat{f}_\epsilon(x + iy) = f(x) + \epsilon i f(y), \quad \epsilon \in \{1, -1\},$$

*relative to reference roots of  $-1$ , and every elementary embedding  $\mathcal{C}_\kappa \rightarrow \mathcal{C}_\lambda$  is one of these. After naming the reference imaginary units only the positive lift remains. Hence*

$$\text{Emb}_{\text{el}}(\mathcal{C}_\kappa, \mathcal{C}_\lambda) \simeq \text{Emb}_{\text{el}}(H_\kappa, H_\lambda) \times \{1, -1\}, \quad \text{Emb}_{\text{el}}(\mathcal{C}_\kappa^i, \mathcal{C}_\lambda^i) \simeq \text{Emb}_{\text{el}}(H_\kappa, H_\lambda),$$

*the signs multiplying under composition, and  $\mathcal{C}_\kappa^i \prec \mathcal{C}_\lambda^i \iff H_\kappa \prec H_\lambda$  for  $\kappa < \lambda$  (source 05 proved the last equivalence, and the positive-lift correspondence, at regular cardinals).*

*Proof.* The displayed maps are field embeddings commuting with  $c$  and preserving `cb`, because  $f$  preserves order and birthdays, and they differ on  $i$ . For elementarity, name  $i$  and translate each complex formula into  $\mathcal{B}_\kappa$  by coordinates, as in proposition 19.2; elementarity of  $f$  gives elementarity of  $\widehat{f}_1$ , and composing with  $c$  gives the other lift in the language without  $i$ . Conversely an elementary embedding of the complex structures restricts to an elementary embedding of the

definable real structures and sends  $i$  to  $\pm i$ , and the field laws force the displayed form. Combine with [theorem 9.10](#).  $\square$

At  $\kappa = \lambda$  with  $f = \text{id}$  this gives  $\text{Aut}(\mathcal{C}_\kappa) = \{\text{id}, c\}$  again. The statement concerns the specified birthday-and-conjugation enrichment; it is emphatically not a classification of automorphisms of the pure algebraically closed field, which has far more symmetry (see the automorphisms report). It is a specialization combining the local theorem with coordinate complexification, not a new general Galois principle.

## 19.4 The exact parameter threshold for orientation

**Theorem 19.6** (Orientation threshold; source 06). *Let  $\kappa$  be an uncountable cardinal and  $A \subseteq \mathbf{No}_{<\kappa}[i]$  any set of parameters. The following are equivalent:*

- (i) *some root of  $X^2 + 1$  is definable in  $\mathcal{C}_\kappa$  over  $A$ ;*
- (ii) *a linear order of the whole carrier is definable in  $\mathcal{C}_\kappa$  over  $A$ ;*
- (iii)  *$A$  contains a nonreal element.*

*In particular neither an imaginary unit nor a linear order is definable over any collection of real parameters, and one nonreal parameter suffices for both.*

*Proof.* If all parameters are real, conjugation fixes them and exchanges the two roots, so neither root is definable; and a definable linear order  $\triangleleft$  would be preserved by  $c$ , while for nonreal  $z$  exactly one of  $z \triangleleft c(z)$  and  $c(z) \triangleleft z$  holds and  $c$  reverses it. If  $z \in A$  is nonreal,  $v = z - c(z)$  is nonzero and purely imaginary,  $-v^2$  is real and positive, and

$$j = \frac{z - c(z)}{\sqrt{-(z - c(z))^2}},$$

with the positive square root of the definable real field, is definable from  $z$  and satisfies  $j^2 = -1$ . The definable coordinates  $(\frac{w+c(w)}{2}, \frac{w-c(w)}{2j})$ , ordered lexicographically, give a definable linear order of the carrier; it is not a field order.  $\square$

Each defining formula uses finitely many parameters; allowing arbitrarily many real parameters does not help because conjugation fixes them all. The proof uses only that  $\mathbf{No}_{<\kappa}$  is real closed. A nonreal parameter removes the twofold obstruction to the coordinate round trip, but if it is kept as a named constant, its two real coordinates must be kept in the real interpretation and the corresponding sets in the  $H_\kappa$  interpretation. It is *not* inferred that an arbitrary expansion  $(\mathcal{C}_\kappa, z)$  is bi-interpretable with the *unexpanded*  $H_\kappa$ ; naming a root of  $-1$  avoids the issue because its coordinates  $(0, \pm 1)$  are already definable.

## 19.5 The full class, under GBC

**Theorem 19.7** (Surcomplex elementary self-embeddings; source 07). *In the class setting of [theorem 18.3](#), the class elementary self-embeddings of  $\mathcal{C}$  are exactly the identity and conjugation, and those of  $\mathcal{C}^i$  only the identity. In particular every such self-embedding is surjective.*

*Proof.* An elementary self-embedding of  $\mathcal{C}$  restricts to one of the definable real structure  $\mathcal{B}$ , which is the identity by [theorem 18.3](#); it sends  $i$  to  $\pm i$ , and the coordinates are fixed, so it is the identity or conjugation. Both are automorphisms. Naming  $i$  excludes conjugation.  $\square$

This is the one surcomplex result of this report that goes beyond the automorphism statements of the automorphisms report: it concerns elementary self-embeddings that are not assumed surjective, of the full structure, under GBC. With [lemma 19.3](#) in place of Kunen's theorem the same argument gives only the two-element *automorphism* group of the bounded structures. Nothing is asserted about unrestricted field embeddings, and the specified conjugation and coordinate birthday are not claimed to be recoverable from the pure surcomplex field.

## 19.6 Outer models on the surcomplex side

**Theorem 19.8** (The surcomplex outer-model contrast; sources 05, 07). *For transitive  $M \subseteq N$  with the same ordinals, the inclusion of the full surcomplex fields is elementary in the pure field language, and also after naming  $i$  and conjugation. After additionally naming  $cb$ , with or without  $i$ , it is elementary exactly when  $M = N$ . If  $M \neq N$ , the least coordinate birthday of a new surcomplex number is  $\beta(M, N)$ .*

*Proof.* The pure-field statement is [proposition 14.2](#). With  $i$  and  $c$  named, translate every formula into real coordinates; it becomes an ordered-field formula, so the inclusion is elementary. With  $cb$  named the real axis with its birthday is definable, so an elementary inclusion would restrict to  $\mathcal{B}^M \prec \mathcal{B}^N$ , which forces  $M = N$  by [theorem 16.5](#). If  $cb(x + iy) < \beta$ , both coordinates are old; a new real sign of birthday  $\beta$  is a new surcomplex number of coordinate birthday  $\beta$ .  $\square$

The result concerns field and coordinate structure, not complex analysis: it involves no contour integral, topology or total surcomplex exponential, and is not a consequence of an analytic continuation or a valuation argument.

## Part V

# Scope, questions and provenance

## 20 What the theorems do and do not imply

### 20.1 Logical strength versus field strength

Already  $\mathcal{B}_{\varepsilon_0}$  recovers  $H_{\omega_1}$ . Within that membership structure, the natural numbers and all subsets of the natural numbers form the usual full two-sorted structure of second-order arithmetic ([proposition 12.1](#)). The birthday-field interpretation thus includes quantification over all actual subsets of  $\omega$ , not merely over computable subsets, because the numbers coding those subsets are first-order elements. This does not make set theory definable in the pure ordered-field reduct, where birthday is not definable ([proposition 12.5](#)), and it does not give a decision procedure ([corollary 12.2](#)).

### 20.2 Four statements that must not be conflated

For a cutoff  $\lambda$  the following can have very different answers:

| Question                                         | Exact issue in this report                                                                   |
|--------------------------------------------------|----------------------------------------------------------------------------------------------|
| Is $\mathbf{No}_{<\lambda}$ a real closed field? | Classical epsilon-number cutoff theorem.                                                     |
| Which sets can be decoded?                       | Exactly $H_{\kappa(\lambda)}$ , by rooted relation codes.                                    |
| Does the decoded universe satisfy ZFC?           | At cardinal cutoffs, exactly the worldly condition.                                          |
| Is an initial inclusion elementary?              | Both cutoffs must be cardinals, followed by elementarity of their hereditary-size universes. |

An algebraic closure theorem alone does not answer any of the last three questions. Conversely, a transitive set model satisfying ZFC need not arise from a regular inaccessible cutoff ([proposition 11.7](#)), and the same enriched field can sit below a regular and a singular cutoff in two universes ([theorem 17.1](#)). Hereditary size, not rank, is what the birthday structure measures: the recovered universe is  $H_\kappa$ , which in general differs from  $V_\kappa$ ; the two agree, for example, at every worldly  $\kappa$  ([lemma 11.1](#)).

### 20.3 Choice and quotients

Ordinary Choice is used to enumerate an individual transitive closure and, in some proofs, to enumerate a particular fiber or choose a particular bijection. The interpretation does not choose one representative simultaneously for every set. It keeps all valid codes and quotients by a definable equivalence relation. This is why Global Choice is unnecessary outside [section 18.2](#).

The exact cardinal-rounding conclusion is stated in ZFC. Without Choice, not every transitive closure is well-orderable, and a code on an ordinal cannot represent every set; [theorem 13.2](#) identifies the exact range of the full-class coding in ZF. No choice-free version of the cutoff theorems is claimed, and nothing in the ZFC proofs justifies silently deleting Choice.

### 20.4 Set models, nonstandard models, and class truth

At a fixed cutoff, the structures in the main theorems are actual set structures with ordinary first-order satisfaction; their interpretations are not hypothetical syntactic models satisfying only a fragment of surreal arithmetic. A nonstandard model of ZFC can internally perform the same constructions, but its coded well-foundedness and its internal cardinal ceiling must be read in that model; an externally ill-founded model is not covered by an external Mostowski-collapse claim merely because it internally believes its code is well founded. Arbitrary nonstandard models of a proposed surreal-arithmetic axiom system need a separate analysis.

For the full class  $\mathbf{No}$ , one applies the translation to a fixed formula and fixed set parameters. Statements about “all formulas” are metatheoretic schemes. We do not form a set of all class functions, a proper-class membership quotient, or a truth predicate for  $V$  inside ZFC. The cutoff results do not require a class theory, a Grothendieck universe, or a large-cardinal existence axiom; GBC is used only in [section 18.2](#) and [theorem 19.7](#), a measurable cardinal only in [section 17](#), and an inaccessible only in [proposition 11.7](#).

### 20.5 Why uncountability and the epsilon hypothesis matter

For the hereditary-size bi-interpretation at cardinal cutoffs, uncountability ensures that countably sized computation certificates and the ordinary infinite set  $\omega$  remain below the cutoff. At  $\omega$  the

finite-birthday surreals form the dyadic ring, not a field, and only the forward interpretation onto the hereditarily finite sets is claimed ([proposition 8.1](#)). For noncardinal cutoffs, the epsilon hypothesis supplies closure of the bounded pairing terms and field operations. Arbitrary ordinal sign cutoffs that fail those closure conditions are not classified; the full birthday-field statement would then need a different language or partial operations. The reconstruction also applies to the canonical birthday function only, and uses the presence of every sign of each length below the cutoff: a small subfield with an inherited birthday may omit most subset codes, and no such generalization is asserted (source 05). For outer models all ordinals are assumed common; models of different heights are a different problem.

## 20.6 Non-claims of the sources

Every limitation, disclaimer and priority caveat stated by a source is kept. Most appear at their point of use above; this subsection lists them by source.

**Across the report.** The proofs are written mathematical arguments. They have not been independently refereed and have not been formalized in Lean; no Lean source is supplied. The finite programs check finite analogues only. No named longstanding conjecture is resolved. Priority is not certified by any of the five targeted literature and repository searches; “not found” is a bounded search result, not proof of absence.

[Added 3 October 2026, batch 89: this subsection lists the non-claims of sources 02, 05, 06, 07 and 09. Those of sources 10 and 11 are collected in [sections 39.4](#) and [53.3](#). Added 4 October 2026, batch 90: those of source 12 are in [section 66.4](#). Added 4 October 2026, batch 95: those of sources 13 and 14 are in [section 80.6](#).]

**Source 02.** The full-class interpretation is Chen–Hamkins–Yang’s prior announced work; its contributions are the bounded refinement and its consequences, with priority not certified. Interpreting  $H_{\kappa(\lambda)}$  is not interpreting the ambient universe and does not assert  $H_{\kappa(\lambda)} \models \text{ZFC}$ . The translated Power Set sentence is one axiom; it does not characterize Replacement, regularity or full ZFC. Its own elementary-inclusion conditions were necessary only (the converse is now [theorem 9.3](#)). The results concern concrete externally well-founded structures. Sharpness is intrinsic to the interpretation. The collector recovers an ordinal value, not a surreal of birthday  $\lambda$ . Prikry forcing is imported; no measurable cardinal is constructed; the statement is relative consistency and concerns fixed first-order sentences, not second-order satisfaction. The preservation criterion must not be replaced by “no new short sequences”, and a proper enlargement at a fixed cutoff is not claimed non-elementary merely because it is proper. The onset proposition concerns new elements, not a change of complete theory. In the surcomplex part  $i$ ,  $c$  and the coordinate birthday are explicit structure, not definable in the pure algebraically closed field, and no new automorphism classification is claimed. Only one-way interpretation was claimed at ring cutoffs and by source 02 generally. The Rangel–Mariano program is not settled. The finite tests do not prove the transfinite results.

**Source 05.** Its package was called candidate-original: not independently refereed or formally verified; no named problem settled; the audit targeted; priority unverified. It omitted the Chen–Hamkins–Yang credit, which is restored here. Classical inputs (sign arithmetic, cardinal closure, well-founded recursion, quantifier elimination, and the simplicity rigidity of the automorphisms

report) are credited. Regularity was its uniform hypothesis and it inferred no singular theorem;  $H_\kappa \neq V_\kappa$  in general and  $\mathcal{P}(\alpha)$  need not be a set in  $H_\kappa$ . The coordinate birthday is a convention, not a game birthday. No sharp product-birthday bound is used. The outer-model theorem needs transitive same-ordinal models; nonstandard models are handled internally. Its non-elementarity is witnessed with parameters, not by distinct parameter-free theories. The first new birthday is not asserted regular, and signs of birthday at most  $\omega$  are not identified with reals. The bounded detector needs an enumeration bound. No small-subfield generalization and no different ordinal heights are covered. The embedding correspondence is a reduction, not an existence theorem, and claims no large-cardinal consistency result; exponential automorphisms are untouched, and the collection's exponential faithfulness result is outside its proof dependencies. The detector formula is not claimed to be existential after expansion. Its Python checks are diagnostic only.

**Source 06.** Chen–Hamkins–Yang is prior work and the central coding idea is theirs; Jeřábek (2021) is prior for the conjugation obstruction; field closure and Prikry are imported. Novelty is provisional; not peer-reviewed or Lean-certified; no named open problem settled. The coordinate birthday is a convention, and the automorphism conclusions do not apply to the pure surcomplex field or classify its automorphisms. The  $\sqsupset_\omega$  theorem does not show that every singular  $\kappa$  fails Replacement, and internal ZFC does not imply external inaccessibility. The ZF boundary concerns this construction only, and internal Choice is not the range statement. Prikry gives relative consistency, not a ZFC theorem, and not  $H_\kappa^N = V_\kappa^N$ . A nonreal parameter does not give bi-interpretability with an unexpanded  $H_\kappa$ . Rigidity is not pointwise definability. Its reduct corollary is not a minimal-language claim. It left open noncardinal cutoffs (answered here for epsilon numbers), ill-founded models of a surreal-arithmetic theory, quantifier elimination for an order-and-simplicity language, and a universal embedding theorem without Global Choice. The finite tests do not verify the theorems. [Added 3 October 2026, batch 89: the last of these is answered in part by source 11; see the note after [question 21.7](#).]

**Source 07.** It omitted the Chen–Hamkins–Yang credit and presented the full-class bi-interpretation as a proposed contribution; that is corrected in [theorem 18.1](#), and it had no Jeřábek credit, now added. Its results were proposed contributions without priority certification, refereeing or Lean. It claims no new proof of Kunen's inconsistency, no rigidity of the pure field, no classification of bounded elementary self-embeddings, and no distinct parameter-free theories from non-elementarity; nothing about unrestricted field embeddings. Its plateau theorem is forward only, and its Choice remark says the construction is not a reconstruction of  $V$  without Choice. The Kunen transfer needs GBC or class-parameter hypotheses and does not forbid external self-maps of countable nonstandard models. Nonstandard models are handled internally. The reconstruction is a logical representation, not literal membership between chosen representatives; there is no canonical code, no decidability result, and no simple closed sign rules for arithmetic. Rangel–Mariano and Bertram are distinct programs whose categorical obstructions are not refuted. The finite checks are limited.

**Source 09.** The global result of Chen–Hamkins–Yang, the van den Dries–Ehrlich cutoff theorem with its erratum, and ordinary well-founded coding are prior work; the refinements may overlap unpublished details of the Chen–Hamkins–Yang project; no named conjecture is resolved; no Lean, no peer review. It did not credit Jeřábek; that credit is added. Cardinal rounding is not a rank cutoff. The principles **Et** and **Pack** are explicit schemes, not identified with any formulation of SA, and the independence examples are within this family only. The detector bound is sharp only for the old-power-set parameter; no quantifier-complexity claim is made across the translation; failure

of the bound does not imply elementarity. Extension rigidity concerns standard cutoff structures and does not forbid abstract elementary extensions. The orientation result is a consequence of classical rigidity, not a new automorphism classification. Choice is used object by object and no ZF version of its theorems was claimed ([theorem 13.2](#) is source 06's). Nonstandard models are internal; no class theory, Grothendieck universe or large cardinal is needed for the cutoff theorems. At  $\omega$  the cutoff is a ring; non-epsilon cutoffs are not classified. The singular worldly construction is a standard illustration needing an inaccessible.

## 21 Open questions, re-scoped

### Answered inside this report.

- Source 02's first question, a sufficient criterion for elementary birthday inclusions: answered by source 09 ([theorem 9.3](#), [remark 9.4](#)).
- Source 02's caution that a two-way theorem must name the cutoff: answered by source 09's pointed bi-interpretation ([remark 6.5](#)).
- Source 05's optimal cutoffs at singular cardinals: answered by sources 06 and 07; at noncardinal epsilon numbers: answered by source 09. Source 06's and source 07's noncardinal-cutoff questions: answered by source 09 for epsilon numbers.
- Source 09's fourth target, the choice-free analogue: answered for the full-class graph coding by source 06 ([remark 13.4](#)).
- Source 06's universal embedding theorem without Global Choice: answered in part, over ZF and with a ZFA counterexample, by source 11 ([Part VII](#)); see the note after [question 21.7](#). [Added 3 October 2026, batch 89.]

### Still open.

**Question 21.1** (Cutoffs that are not epsilon numbers). For which ordinals  $\eta$  that are not epsilon numbers, beyond the ring cutoffs of [proposition 8.1](#) (forward only), is there a uniform reconstruction from  $\mathbf{No}_{<\eta}$  with its inherited operations, and what exactly is recovered? (Re-scoped from sources 05, 06 and 07.)

**Question 21.2** (Minimal signature). How much of  $\{0, 1, +, -, \cdot, <, b\}$  can be deleted while retaining a uniform interpretation of hereditary sets? [Corollary 12.6](#) shows that  $<$ ,  $b$  and the graph of one ordinal pairing polynomial suffice; whether  $<$  and  $b$  alone, or addition with birthday, suffice is open, as is whether multiplication is necessary. (Sources 02, 05, 07; partial progress by 06.)

[Added 4 October 2026, batch 95: Part IX bears on the bi-interpretation form of this question. By its [proposition 74.2](#), a reduct of the full birthday structure, or any structure on  $\mathbf{No}$  or  $\mathbf{Oz}$  with set-definable primitives, has a bi-interpretation with  $V$  whose comparison sends each number to the codes of its own sign sequence exactly when it defines birthday precedence and the ring operations; source 13 asks which reducts do ([question 79.3](#)), and the definable-surreals report's Part III asks a related question for  $(\mathbf{No}; +, \cdot, <, \mathbf{Oz}, \Omega)$  ([remark 67.5](#)). The pure ordered ring of  $\mathbf{Oz}$  defines neither the ordinals nor birthday precedence, even with parameters, and has no bi-interpretation with  $V$  on the canonical numbers ([corollary 74.13](#) and [proposition 74.14](#)). On the omnific integers unary birthday already defines the quotient birthday  $b(a/b)$  ([proposition 74.5](#)). The one-way question asked here, whether  $<$  and  $b$  alone, or addition with birthday, interpret the hereditary sets, and whether multiplication is necessary, is not answered.]

**Question 21.3** (Elementary embeddings and explicit criteria). Replace the condition  $H_\kappa \prec H_\tau$  in [theorem 9.3](#) by useful explicit hypotheses in interesting families of cardinals (source 09). Classify the elementary self-embeddings of particular bounded  $\mathcal{B}_\kappa$ ; by [theorems 9.10](#) and [19.5](#) this is the corresponding problem for  $H_\kappa$  (source 07).

**Question 21.4** (Forcing and the bounded theory). Which changes of the complete bounded theory can occur under forcing that adds new bounded subsets, and how much of the cutoff predicate is first-order recoverable from an appropriately expanded  $H_\kappa$  (source 02)? The detectors of [section 16](#) show failure of elementarity with parameters; they do not decide changes of parameter-free theories.

**Question 21.5** (Quantifier complexity). What is the least quantifier complexity of a uniform formula detecting all proper same-ordinal outer-model inclusions in the birthday language (source 05)?

**Question 21.6** (Surcomplex reducts). What set-theoretic information is recoverable from  $(\mathbf{No}_{<\kappa}[i]; 0, 1, +, -, \cdot, \text{cb})$  when neither conjugation nor  $i$  is named (source 05)? [Theorem 19.6](#) concerns the language with conjugation.

**Question 21.7** (Axiomatics and formalization). Establish definitional equivalence, or its failure, between Et and Pack and particular published axiom formulations of surreal arithmetic, once finalized (source 09); analyse ill-founded models of such a theory, quantifier elimination for an order-and-simplicity language, and a universal embedding theorem without Global Choice (source 06); and formalize the size-certified interpretation, beginning with the prefix, bit and pairing lemmas, then the certificate verifier and its hereditary-size bound (sources 05, 06, 07, 09).

[Added 3 October 2026, batch 89: the third item, “a universal embedding theorem without Global Choice”, is answered in part by source 11 ([Part VII](#)). Over ZF, a set  $X$  carries an  $X$ -indexed algebraically independent family of positive infinite omnific integers, equivalently  $\mathbb{R}(T_X)$  embeds in  $\mathbf{No}$  over  $\mathbb{R}$ , iff  $X$  injects into the power set of an ordinal; so universality for the free fields on all sets is exactly the first Kinna–Wagner principle ([theorem 43.5](#)), with optimal set-sized real-closed targets  $U_\kappa$  ([theorem 44.3](#)). Every linear order on a well-orderable set order-embeds in  $\mathbf{No}$  ([proposition 45.3](#)). In the ordered Mostowski permutation model, a model of ZFA, an ordered field whose finitely generated coordinate subfields all embed in order has no field embedding into the internal  $\mathbf{No}$  ([theorem 49.1](#)). Still open: universality for prescribed orders and for general ordered fields without choice, and any transfer of the ZFA example to ZF ([questions 51.1](#), [51.2](#) and [51.5](#)). The other items of this question are not affected.] The collection’s Lean library already contains relevant sign-sequence infrastructure (prefix truncation, the failure of birthday precedence to be prefix simplicity, Hessenberg agreement of ordinal arithmetic), but none of the interpretation theorems of this report is formalized.

[Added 4 October 2026, batch 95: Part IX (sources 13 and 14) gives explicit recursively axiomatized theories in the birthday language, with only numerical individuals, each bi-interpretable with ZFC:  $\mathbf{T}_{\mathbf{No}}$ ,  $\mathbf{T}_{\mathbf{Oz}}$ ,  $\mathbf{SA}_{\text{rt}}$  and  $\mathbf{OA}_{\text{qb}}$  ([theorems 73.4](#), [73.6](#) and [74.7](#)), and a two-sorted class lift bi-interpretable with GBC, with GBc and with KM ([theorem 76.5](#) and [corollary 76.7](#)). They are built from the interpretation; they are not the finalized published formulations the first item asks about, so its comparison with Et and Pack stays open ([remark 67.4](#)). The items on ill-founded models and on formalization meet [questions 79.17](#) and [79.19](#), which are open.]

## 22 Provenance and verification

### 22.1 The sources' repository and literature audits

All five sources inspected the repository read-only at the pinned commit

4f2645599121fa872c7104995e47f86f0382351f

and modified nothing. Source 02 inspected the tree, the root and documentation indexes, the catalogue source and the foundations report, and records its scope in `02-birthday-cutoffs-SOURCE_AUDIT.md`, kept byte-identical. Source 05 records its requested line ranges, some truncated, in `data/05-birthdays-recover-sets-repository_audit.json`. Source 06 inspected listings, the first 200 lines of the documentation catalogue, the first 150 lines of the foundations README, and the README and first 240 lines of the automorphisms article. Source 07 inspected the tree, READMEs, catalogue, formalization ledger, the foundations introduction and the automorphisms README. Source 09 inspected the catalogue, the relevant parts of the foundations report and the scope of the automorphisms report. The corrections these audits need are in [section 1.5](#).

The sources' literature checks were: the Chen–Hamkins–Yang announcements and slides (sources 02, 06, 09; source 06 cites slide pages 103, 107 and 122, and source 02 pages 96–122, which are compatible); van den Dries–Ehrlich and its erratum, Bournez–Guilmant, Poveda–Rinot–Sinapova, and Rangel–Mariano (sources 02, 06, 09); Freire–Hamkins, Ehrlich's paper on the simplicity hierarchy [18] (record and abstract only) and Ehrlich–Kaplan [19] (background, not an input) (source 05, whose session could not load the publisher PDF of the erratum); Bertram and Hamkins–Kirmayer–Perlmutter (source 07); Jeřábek (source 06). These external checks were not repeated for this merge.

### 22.2 Dependency and novelty ledger

| Result                                                                   | Main inputs                                                    | Status here                                                                             |
|--------------------------------------------------------------------------|----------------------------------------------------------------|-----------------------------------------------------------------------------------------|
| Global birthday-field bi-interpretation ( <a href="#">theorem 18.1</a> ) | Pointed well-founded relation coding                           | Prior Chen–Hamkins–Yang announcement; written proof from source 07; not claimed new.    |
| Real-closed epsilon cutoffs                                              | Classical surreal field theory                                 | Prior van den Dries–Ehrlich, with erratum.                                              |
| Small arithmetic certificates, two routes                                | Option graphs (09); prefix-pair tables (05, 06, 07)            | Explicit local proof ingredients; no priority claim for the general locality principle. |
| Cardinal rounding and cutoff recovery                                    | Bounded pairing, rooted collapse, collector, sign-graph bridge | Proposed local refinement (09), with cases in 02, 05, 06, 07.                           |
| Elementary-inclusion spectrum and embeddings                             | Collector and coherent bi-interpretations                      | Proposed classification (09); embeddings (05, 06).                                      |
| Power Set, eternity, packing, worldly, $\beth_\omega$                    | Hereditary cardinal bounds                                     | Proposed calibrations (02, 06, 09); the underlying $H_\kappa$ facts are standard.       |

| Result                                                       | Main inputs                           | Status here                                                                       |
|--------------------------------------------------------------|---------------------------------------|-----------------------------------------------------------------------------------|
| ZF range                                                     | Collapse in ZF                        | Direct refinement of the coding (06); no claim about other interpretations.       |
| Preservation, first new birthday                             | Sign/subset bijection                 | Direct consequences (02, 05, 06, 07), tools rather than isolated discoveries.     |
| Detectors, extension rigidity, $M = N$                       | Old-power-set parameter; row search   | Proposed explicit refinements (05, 07, 09); outer-model reasoning is standard.    |
| Prikry invisibility                                          | Imported Prikry forcing               | Proposed application under a measurable cardinal (02, 06).                        |
| Kunen transfer                                               | Imported Kunen inconsistency (GBC)    | Proposed application (07).                                                        |
| Surcomplex orientation, automorphisms                        | Real-axis recovery, rigidity, Jeřábek | Consequence of classical rigidity and the automorphisms report.                   |
| Two lifts, orientation threshold, full-class self-embeddings | Coordinates, rigidity, Kunen          | Proved specializations (06, 07); the only surcomplex items new to the collection. |

The places where the argument could not be replaced by a general slogan are the arithmetic certificates (saying that the global construction is definable does not prove that the same formula computes arithmetic in an  $H_\kappa$  without Replacement), and the collector (a noncardinal cutoff is not recovered merely by interpreting the unpointed hereditary-size universe).

## 22.3 Finite checks

Each source supplied a standard-library Python program checking finite analogues of its coding conventions. All five were rerun for this merge with Python 3.14.4 on copies of the files; each passed, and each recorded output was reproduced exactly, apart from line endings. The recorded runs are:

| Source | Recorded finite checks                                                                                                                                                                                                                                                                                             |
|--------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 02     | 65,025 prefix comparisons and 1,538 bit positions through length 7; 173,880 natural-ordinal matrix addresses on 81 Cantor-normal-form samples; 687 relabellings of all 16 sets of $V_4$ ; 1,024 code pairs for equality and membership; 4 invalid codes rejected.                                                  |
| 05     | 65,025 prefix comparisons through birthday 7, including the counterexample of <a href="#">remark 4.2</a> ; 64 acyclic graphs on four nodes, 4,096 graph pairs, 65,536 root-pair checks each for bisimulation equality and membership; 64 matrix round trips; 49 rectangle packings; 30 missing-row detector tests. |
| 06     | 16,129 prefix and 642 bit instances on 127 words of length at most 6; pairing on 10,000 pairs; all 1,570 rooted relations on at most three vertices, 15 valid codes, 225 pairs each for equality and membership.                                                                                                   |
| 07     | 16,129 prefix comparisons, 642 sign checks, 819 radix addresses (3 radices), 42,849 graph pairs each for isomorphism and membership over 207 labeled graphs decoding 12 sets; the missing-range counterexample confirmed.                                                                                          |
| 09     | 16,129 prefix pairs, 642 bit positions, 11,440 pairing checks; 530 relations and 1,570 rooted relations on at most three nodes, 15 valid codes, 225 pairs each for equality and membership; the counterexample of <a href="#">Appendix B</a> confirmed.                                                            |

These tests can expose a reversed edge, a missing root or range condition, a bad pairing index, or an incorrect finite sign formula. They do not prove transfinite well-foundedness, cardinal bounds, bi-interpretability, Replacement equivalences, forcing statements, or any large-cardinal assertion; those rest on the written proofs and the explicitly imported classical theorems.

[Added 3 October 2026, batch 89: the finite programs of sources 10 and 11 are described in [sections 37](#) and [52](#). Both were rerun with Python 3.14.4 on copies; both passed and reproduced their recorded files exactly, apart from line endings: source 10 checked 8,256 endpoint intervals over the 127 sign words of length at most 6, 2,187 depth-two assignments (576 separated), 1,050 negative-code comparisons and 4,032 atom transpositions; source 11 passed 20,595 assertions with seed 20261003. Their provenance is in [sections 39.5](#) and [53.4](#).]

[Added 4 October 2026, batch 90: source 12’s finite program ([section 64](#)) was rerun with Python 3.14.4 on a copy, with an explicit output path; it passed 203 checks, 35 of them brute-force comparisons over all permutations of at most seven points, and reproduced its recorded file exactly, apart from line endings. Its provenance is in [section 66.5](#).]

[Added 4 October 2026, batch 95: source 14’s finite program ([section 78.3](#)) was rerun with Python 3.14.4 on a copy; it passed and reproduced its recorded file exactly, apart from line endings: 16,129 prefix pairs, 642 sign bits, 6,400 finite and 4,096 Cantor-normal-form ordinal pairs, 207 valid rooted graphs with 42,849 comparisons each for equality and membership, 961 packing round trips, 511 dyadic sign-birthday and 1,225 fraction-invariance checks, and 66,066 diagonal matrices. Source 13 ships no program. Their provenance is in [section 80.7](#).]

## 23 Conclusion

Birthday truncation has a precise set-theoretic effect. At an epsilon-number cutoff  $\lambda$ , rooted sign codes recover the hereditary-size universe at the cardinal ceiling of  $\lambda$ ; when  $\lambda$  is not itself a cardinal, its value survives as a definable interpreted set, which explains why noncardinal cutoff inclusions never become elementary even though all their ordered field reducts are elementary subfields. Packing a birthday fiber requires its full power-set cardinal to fit below the cutoff; bounding definable ordinal-indexed birthday growth is Replacement at cardinal cutoffs; together they detect worldly, not merely inaccessible, cardinals. Across universes, the birthday structure below a cardinal is unchanged exactly when no bounded subset is added, while any new set is detected by an explicit formula; yet a measurable cardinal can be singularized without changing the structure at all. The same information is available in surcomplex arithmetic once conjugation, coordinate birthday and an imaginary unit are specified; omitting the unit preserves mutual interpretation but leaves a two-element obstruction to bi-interpretation. The decisive structure is not the field alone: it is the field with a carefully specified memory of how its elements are born.

[Added 3 October 2026, batch 89: Parts VI and VII, which follow, add two further manuscripts: the weak-base form of the eternity–Replacement equivalence with the atom-support spectra of kernel models, and choiceless universality of the surreal numbers. Added 4 October 2026, batch 90: Part VIII, after them, adds a third, on named atom symmetries and Replacement in the finite-kernel model. Added 4 October 2026, batch 95: Part IX, last, merges two manuscripts on surreal-only foundations: theory-level bi-interpretations of numerical theories with ZFC, natively in the omnific integers, and the boundary at NBG.]

## Part VI

# Replacement-free surreal cuts, definable families and atom-support spectra

## 24 Source 10: question, results, notation and relation to Part II

This part was added in batch 89. It prints one further manuscript, written independently of the five merged above: *Surreal Cuts, Replacement, and Atom-Support Spectra: Exact principles for definable families, weak set theories, and changes of language*, “Prepared for Vladimir Reshetnikov with ChatGPT”, 3 October 2026. It is batch-89 manuscript 03 and *source 10* in this report’s local numbering; its shipped files carry the prefix `10-cuts-replacement-`. It was pinned to repository commit `2b7b388ba` (“Catalogue batches 81 to 86”), arrived in commit `7c0f2d9f9` and was placed in commit `23adb85f9`. Its provenance, the mechanical changes made in printing it, and its finite checks are in [section 39](#). Inside this part, “this manuscript”, “this article”, “here” and “we” refer to source 10, and its own Theorems A–C are [theorems 24.1](#) to [24.3](#).

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**Status and attribution (source 10).** This is an unrefereed research manuscript, not a claim of an independently certified breakthrough. Classical simplicity and small-support arguments are credited explicitly. The exact formulations, comparisons, and refinements are proved here, but their historical priority has not been established. No Lean or Rocq verification of the new manuscript is claimed.

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**Source 10’s summary.** We investigate the intersection of Elliot Glazer’s work on weak set theories and reflection with the sign-sequence and foundational developments in the ProveIt repository. The central distinction is between an actual set of surreal bounds and a formula assigning a unique surreal to each member of a set. Over Zermelo set theory with Foundation, actual set cuts have unique simplest interpolants, without Replacement. In contrast, interpolation for definable set-indexed families is equivalent to ordinal bounding, to Replacement for surreal-valued functions, and even to strict upper bounding inside the single interval  $(-1, 0)$ . On a separately specified cumulative-hierarchy base, these schemes are equivalent to full Replacement and Collection.

A complementary atom-support analysis explains why the conclusion cannot simply be transferred to urelement theories. We give a uniform proof and an exact domain-size analysis for the classical small-kernel models  $M_\kappa^{\text{ker}}$ . All have the same pure sets and the same pure surreal structures. For a limit cardinal  $\kappa$ , Collection first fails at domain size  $\text{cf } \kappa$ ; for a successor cardinal it holds without a domain-size restriction. Naming an external enumeration of  $\kappa$  atoms changes the picture: both Replacement and Collection first fail at  $\text{cf } \kappa$ , although Collection for pure-valued relations, including all surreal-valued relations, still holds. Complete mathematical proofs, a finite regression checker, an explicit  $V_{\omega+\omega}$  countermodel, and a formalization plan are supplied.

### 24.1 Why this intersection is substantive

Glazer’s paper on global choice over Zermelo set theory demonstrates that the precise language and axiom scheme matter when Replacement is absent [20]. His joint 2026 paper with Bokai Yao separates Collection from several reflection principles in urelement set theory formulated with

Replacement [21]. These provide two concrete prompts: which closure assertions actually require Replacement, and which failures remain invisible when one looks only at pure mathematical objects?

ProveIt has two relevant components: a surreal development organized around sign sequences, and an independent first-order set-theoretic development, including a Closure axiomatization equivalent to ZF [26, 28]. The repository also contains a substantial report on old surreal fields in larger universes [29]. The present question is different: rather than adding new pure sequences by passing between ZFC universes, we first weaken set-existence axioms and then vary atom supports while keeping *all pure sets fixed*.

The research question answered here is therefore:

What is the exact strength of surreal cut interpolation for definable set-indexed families, and how far can such a pure-structure principle detect failures of Replacement, Collection, or reflection in the ambient universe?

This is a question formulated and resolved within this investigation. We do not identify it with an already named, established open conjecture.

## 24.2 Three main results

The notation  $Z_f$  means Zermelo set theory with Foundation but without Replacement or Choice. The additional hierarchy axiom Hier (source 10's H, renamed in [convention 24.4](#)) will be stated explicitly in [section 25](#); it must not be suppressed.

**Theorem 24.1** (A local surreal test for ordinal bounding; source 10's Theorem A; [theorems 28.2](#) and [29.1](#)). *Over  $Z_f$ , the following schemes are equivalent: ordinal bounding for definable set-indexed functions; existence of set graphs for such surreal-valued functions; surreal upper bounding; Collection for surreal-valued relations; interpolation for definable set-indexed surreal cuts; and strict upper bounding of definable families inside  $(-1, 0)$ . Over  $Z_f + \text{Hier}$ , each is also equivalent to full Replacement and full Collection.*

The construction detecting unbounded ordinals is elementary: a minus sign followed by plus signs, with birthday  $\alpha + 2$ , encodes the ordinal  $\alpha$  inside  $(-1, 0)$ . The nontrivial foundational issue is showing that ordinary set-cut interpolation does *not* already use the principle being characterized.

**Theorem 24.2** (Exact Collection spectra of cardinal-support models; source 10's Theorem B; [theorem 32.2](#) and [corollary 33.4](#)). *Work in a full urelement universe with Choice and a set  $A$  of atoms, where  $|A| \geq \kappa$  and  $\kappa$  is infinite. Put*

$$M_\kappa^{\text{ker}} = \{x : |\ker(x)| < \kappa\}.$$

*Then  $M_\kappa^{\text{ker}}$  satisfies Replacement and Choice and contains all pure sets. If  $\kappa$  is a limit cardinal, including  $\omega$ , Collection holds for every domain of size below  $\text{cf } \kappa$  and fails for a domain of size exactly  $\text{cf } \kappa$ . If  $\kappa$  is a successor cardinal, full Collection holds.*

The finite-support and countable-support mechanisms are classical; the exact uniform spectrum is the refinement developed in this presentation. The connection to established work is recorded in [section 24.3](#). The same spectrum, for the same models, is proved independently in the research-report collection ([remark 24.6](#)).

**Theorem 24.3** (A language-sensitive separation invisible to surreals; source 10’s Theorem C; [theorem 34.2](#) and [corollary 35.3](#)). *Let  $E$  name, as an external predicate, a bijection from  $\kappa$  onto a set of  $\kappa$  atoms. In  $(M_\kappa^{\text{ker}}, E)$ , Replacement and Collection both hold for all domains of size below  $\text{cf } \kappa$  and both fail at size  $\text{cf } \kappa$ . Nevertheless, full Collection for pure-valued relations, and consequently every surreal principle in [theorem 24.1](#), holds even for formulas using  $E$  and impure parameters.*

Thus the restriction to pure objects is not merely a technicality. The same pure surreal universe can coexist with very different ambient set-existence principles.

### 24.3 What is established, and what is not

Conway’s cut construction and Gonshor’s sign-sequence presentation are established foundations of surreal theory [1, 2]; Bertram provides a recent related pure-set treatment [17]. The elementary simplicity theorem itself is not new. Here its construction is reorganized to exhibit a proof over a base without Replacement.

Small-kernel methods are also established. Hamkins and Yao discuss the finite-support model, crediting the Replacement argument to Lévy, and the countable-support model satisfying Collection [22, §1.5]. Glazer–Yao’s Proposition 1 identifies the common-atom-base obstruction to Collection; their small-kernel permutation models provide much stronger reflection separations than are attempted here [21]. Yao’s earlier work treats the broader axiomatization and forcing setting [23].

The present contribution is a self-contained theorem package: an explicit weak-base equivalence, a fixed-interval ordinal detector, the sharp support-capacity classification with domain-size thresholds, and its language-sensitive comparison with pure surreal principles. These are proposed as useful research results and formalization targets. The sources consulted do not certify their priority; in particular, a generalization of a classical support argument must not automatically be advertised as a breakthrough.

| Component                                   | Evidential status                                                   |
|---------------------------------------------|---------------------------------------------------------------------|
| Ordinary surreal simplicity                 | Classical result; a bounded-container proof is supplied.            |
| Weak-base equivalence and rank countermodel | Fully argued in this manuscript; not machine-checked.               |
| Support-model mechanism                     | Classical method, explicitly credited; generalized proofs supplied. |
| Exact thresholds and predicate expansion    | Proved in this manuscript; priority not independently established.  |
| Finite verification                         | Executed regression tests, not a proof of transfinite statements.   |
| Reflection separations of Glazer–Yao        | External results used for context, not claimed as our discoveries.  |

### 24.4 Notation of this part

**Convention 24.4** (Symbols of source 10). Source 10 was written without knowledge of this report, and several of its symbols already mean something else here. In this part:

| Notion in source 10                                                                                          | Here                                                           | Do not confuse with                                                                                                                                                                                      |
|--------------------------------------------------------------------------------------------------------------|----------------------------------------------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Zermelo set theory with Foundation, no Replacement, no Choice                                                | $Z_f$                                                          | —                                                                                                                                                                                                        |
| Hierarchy axiom (existence and coverage of the stages $V_\alpha$ )                                           | Hier (source: H)                                               | $H_\kappa$ , the hereditary-size universe of Parts I–V                                                                                                                                                   |
| Set-graph scheme                                                                                             | Gr (source: SG)                                                | SignGraph( $\cdot$ ), the sign-graph operator of <a href="#">section 6</a> ; Gr asserts that a definable family has a <i>set</i> graph                                                                   |
| Collection scheme, and its restriction to domains of size $< \tau$                                           | Collection, $\text{Collection}_{<\tau}$                        | the collector $\text{Coll}(C)$ of <a href="#">theorem 6.2</a>                                                                                                                                            |
| Ordinal bounding, upper bounding, surreal Collection, definable-cut interpolation, bounding inside $(-1, 0)$ | OB, UB, SC, DCI, NIB                                           | —                                                                                                                                                                                                        |
| Kernel (small-support) model $\{x :  \ker(x)  < \kappa\}$                                                    | $M_\kappa^{\ker}$ (source: $M_\kappa$ )                        | the transitive models $M, N$ of <a href="#">Part III</a> and $\mathcal{B}_\kappa^M$ ; the full-class rank models of the report <a href="#">foundations-and-computation/surreal-well-orders</a> , Part XV |
| Rank-initial countermodel $V_{\omega+\omega}$                                                                | $V_\theta, \theta = \omega + \omega$ (source: $M = V_\theta$ ) | the ground model $M$ of <a href="#">Part III</a>                                                                                                                                                         |
| Urelement set theory with Replacement                                                                        | ZFU <sub>R</sub>                                               | theories named ZFU that include Collection                                                                                                                                                               |
| Ambient urelement universe; kernel; atom base                                                                | $\mathcal{U}; \ker(x); V_\alpha(B)$                            | —                                                                                                                                                                                                        |
| Generic ordinal variables                                                                                    | $\zeta$ (source: $\beta$ ), $\sigma$ (source: $\delta$ )       | $\beta$ , reserved for the first new birthday, and $\delta$ , reserved for a code domain ( <a href="#">convention 1.5</a> )                                                                              |
| Ordinal detector in $(-1, 0)$                                                                                | $\nu_\alpha$                                                   | the hereditary cardinals $\nu_i$ in the proof of <a href="#">theorem 10.4</a>                                                                                                                            |
| Birthday bound $\sup\{b(x) + 1\}$ of a cut                                                                   | $\lambda$ , any ordinal                                        | the epsilon-number cutoffs $\lambda$ of Parts I–V                                                                                                                                                        |
| A separator of a cut                                                                                         | $c$                                                            | the conjugation $c$ of <a href="#">section 19</a>                                                                                                                                                        |
| Sign container $\{s : b(s) \leq \sigma\}$                                                                    | $T_\sigma$                                                     | TC                                                                                                                                                                                                       |
| Birthday                                                                                                     | $b(s)$ , the length of $s$                                     | the same function; here it is an ordinal, not a surreal ordinal                                                                                                                                          |

Only the order of sign sequences is used in this part: no field operation and no birthday *symbol* of a structure  $\mathcal{B}_\lambda$  occurs. Typography only: source 10's sans-serif theory names ZF, ZFC and ZFU<sub>R</sub> are set as in the rest of this report.

## 24.5 Relation to Part II and to the rest of the collection

*Remark 24.5* (The weak-base form of [Theorem 10.4](#)). Source 10 did not know this report. Its spine is the same principle as [theorem 10.4](#) (source 09's): *Replacement is exactly the bounding of birthdays of definable families of surreals*. The two statements live on different bases, and neither contains the other.

- [Theorem 10.4](#) is a theorem of ZFC about two set structures at an uncountable cardinal  $\kappa$ : the eternity scheme Et ([section 10](#)), a scheme of the birthday-field language evaluated in  $\mathcal{B}_\kappa$  for families indexed by embedded ordinals, holds iff the hereditary-size universe  $H_\kappa$  satisfies Replacement. Its proof uses the bi-interpretation of [Part I](#) and, in the converse direction, ambient Replacement and Choice.
- Source 10’s [theorems 28.2](#) and [29.1](#) are internal scheme equivalences over the weak theory  $Z_f$  in the ordinary membership language, with sign sequences as sets and only their order used. Over  $Z_f$  alone birthday bounding is ordinal bounding OB; it becomes full Replacement (and Collection) only over  $Z_f + \text{Hier}$ , and no Choice is used. [Corollary 28.3](#) is the per-family form, the counterpart of a single instance of Et. New beside Et is NIB: a strict bound can always be sought inside the one interval  $(-1, 0)$ .
- The bases differ:  $H_\kappa$  need not satisfy Power Set ([lemma 10.3](#) does not list it), which  $Z_f$  contains, and Hier can fail in  $H_\kappa$ . Where  $H_\kappa$  does satisfy  $Z_f + \text{Hier}$ , for example at an uncountable  $\kappa$  with  $\kappa = \beth_\kappa$ , where  $H_\kappa = V_\kappa$ , [theorem 29.1](#) applies inside  $H_\kappa$ , and together with [theorem 10.4](#) it gives:  $\mathcal{B}_\kappa \models \text{Et}$  iff  $H_\kappa$  satisfies any one of the six schemes of [theorem 28.2](#), read in  $H_\kappa$ . This combination is an observation of this merge, not a statement of either source.

Source 10’s  $V_{\omega+\omega}$  countermodel ([proposition 30.1](#)) has the same shape as the failures recorded in [section 10](#): every member of a definable family exists, but the family cannot be collected; compare the  $\beth_\omega$  instance  $\text{Rep}_\beth$  ([theorem 11.5](#)) and the cofinal power-tower sequence at  $\varepsilon_0$  that defeats Et. Source 10’s Replacement-free set-cut theorem ([theorem 27.1](#)) is a different statement from the Replacement-free arithmetic certificates of [section 3](#): the latter compute field operations inside  $H_\kappa$ , the former constructs the simplest separator of a set cut from the order alone.

*Remark 24.6* (The same Collection spectrum in two collections). Batch-89 manuscript 05, placed as the research-report collection’s new report `SetTheory/Cardinals/docs/reports/ordinal s-and-order-types/naming-elementary-embeddings` (*Naming Elementary Embeddings: Atom Components, Replacement, and Reflection*, pinned to the same commit; AI-assisted, unrefereed, not formalized), works with the same kernel models, written  $M_\kappa(A)$  there after Yao’s ideal construction. Its lemma “No new atoms in a unique base-language output” (Lemma 2.3, `nee:lem:none`) and its “Known kernel-ideal theorem, specialized” (Theorem 2.4, `nee:thm:base`) are [lemma 32.1](#) and [theorem 32.2](#). Its lemmas “Automatic small-domain Collection” (Lemma 7.1, `nee:lem:smallcoll`) and “The limit-cutoff obstruction is already bounded” (Lemma 7.2, `nee:lem:limitspectrum`), with its corollary “Exact local spectrum at limit cutoffs” (Corollary 7.3, `nee:cor:limitspectrum`), prove [lemma 33.1](#) and [theorem 33.2](#) for the same models and for their expansions by named atom injections, with the added information that the failing instance at a limit cardinal has a  $\Delta_0$  matrix. Its “Full Collection criterion” (Theorem 7.5, `nee:thm:collection`) proves full Collection at successor cardinals, as in [theorem 33.3](#), by a reservoir argument for named tuples. The two manuscripts were written independently on the same pin; neither is claimed first, and both credit the mechanism to Hamkins–Yao, Lévy, Glazer–Yao and Yao. This part prints source 10’s proofs. That report’s dated note after its Corollary 7.6 compares the two: its failing instance at a limit cardinal collects injection graphs with a  $\Delta_0$  matrix, where source 10 collects sets of atoms of prescribed cardinalities, and its successor-case reservoir consists of whole components, so that the permutations commute with the named maps. Source 10’s enumeration predicate  $E$  ([section 34](#)) and that report’s named lifts of atom injections are different expansions of the same models, and their theorems are different.

*Remark 24.7* (Full-class rank models). Batch-89 manuscript 01, placed as Part XV of the report `foundations-and-computation/surreal-well-orders` (its source 31), computes for the two-sorted full-class rank models  $(V_\lambda, \mathcal{P}(V_\lambda))$  at every limit  $\lambda > \omega$  the domains on which class Replacement and Collection hold: exactly those of size below  $\text{cf } \lambda$ . At  $\lambda = \omega + \omega$  this contains the

failure of [proposition 30.1](#) for the one *definable* family  $n \mapsto \omega + n$ . The failure of NIB there, and the weak-base equivalences, are source 10's own.

## 25 The base theories and the meaning of a family

### 25.1 Zermelo set theory without hidden Replacement

**Definition 25.1.**  $Z_f$  consists of Extensionality, Empty Set, Pairing, Union, Power Set, Infinity, the full Separation scheme, and Foundation, in classical first-order logic. It does not contain Replacement or Choice.

We use ordinary set encodings of ordered pairs and functions. The usual finite constructions—Cartesian products, restrictions, domains and ranges of *set relations*, and finite tagged unions—are available in this theory. For example,  $X \times Y$  is obtained by Separation from a sufficiently iterated power set of  $X \cup Y$ . No image of an arbitrary definable class function is presupposed.

All appearances of a class in the weak-base half of source 10 abbreviate a first-order formula with finitely many set parameters. An assertion quantified over formulas is an external scheme: there is one first-order assertion for each formula. There is no set of all formulas together with an unrestricted truth predicate for the universe, and no quantification over arbitrary subclasses of a proper class.

**Definition 25.2** (Definable set-indexed family). A formula  $\varphi(i, s, p)$  presents a surreal family on a set  $I$  if

$$\forall i \in I \exists! s [s \in \mathbf{No} \wedge \varphi(i, s, p)].$$

We write  $s_i$  for its unique value. This notation does not assert that  $\{(i, s_i) : i \in I\}$  is a set.

The distinction is essential. If a family is given *as a set function*, its range is a set in  $Z_f$ . If it is only given by a formula satisfying the displayed condition, forming its range is a Replacement-type step.

### 25.2 A separately named hierarchy hypothesis

**Definition 25.3** (The hierarchy axiom Hier). The following two assertions constitute Hier. First, for every ordinal  $\alpha$  there is a set function  $h$  with domain  $\alpha + 1$  satisfying

$$h(0) = \emptyset, \quad h(\zeta + 1) = \mathcal{P}(h(\zeta)), \quad h(\lambda) = \bigcup_{\zeta < \lambda} h(\zeta)$$

for the relevant successor and nonzero limit indices. Second, every set  $x$  belongs to  $h(\alpha)$  for some such  $\alpha, h$ .

The unions in this definition are unions of ranges of restrictions of a *set function*. Uniqueness and coherence of the hierarchy segments follow by induction along an ordinal. Consequently  $V_\alpha$  is unambiguously definable under Hier.

ZF proves Hier by its usual recursion and rank constructions. We do not assume that Hier follows from  $Z_f$ . Our characterization over plain  $Z_f$  is a characterization of ordinal bounding; only after adjoining Hier do we identify it with full Replacement. The model in [section 30](#) satisfies  $Z_f + \text{Hier}$  but not Replacement, so Hier itself does not trivialize the result.

### 25.3 Two different ambient settings

The first half of source 10 concerns pure set theory and the sign-sequence order. The later support-model construction takes place externally in a full well-founded universe with atoms and Choice. We write  $\text{ZFU}_R$  for urelement set theory formulated with Replacement, rather than building Collection into its name. This prevents an ambiguity in the literature: some authors use “ZFU” for a theory already containing Collection.

In the atom setting, Extensionality applies to sets, and atoms have no members. A tagged interpretation in ordinary ZFC can be used to implement the background urelement universe. No new large-cardinal assumption is needed for the constructions here. When a class model is mentioned, satisfaction is used formula by formula through relativization; a universal class truth predicate is not introduced.

## 26 Sign sequences and bounded containers

**Definition 26.1.** A surreal sign sequence is a function  $s : \alpha \rightarrow \{-, +\}$  for an ordinal  $\alpha$ . Its birthday is  $\text{b}(s) = \alpha$ . The empty sequence is 0, and the one-sign sequence  $(-)$  is denoted by  $-1$ .

We distinguish the two signs from the numerical values of these named surreals. For comparison, use the order

$$- < \text{termination} < +.$$

Two unequal sequences are compared at the first position where their signs differ or where precisely one terminates. Such a position exists and has a least ordinal index. This defines a linear order. Write  $p \preceq s$  if  $p$  is an initial segment of  $s$ , possibly equal to  $s$ .

Only this ordered tree is needed. We do not invoke the recursive definitions of surreal addition, multiplication, exponentiation, or real closedness in the weak base.

**Lemma 26.2** (Bounded sign container). *For every ordinal  $\sigma$ , the class*

$$T_\sigma = \{s \in \mathbf{No} : \text{b}(s) \leq \sigma\}$$

*is a set in  $\mathbf{Z}_f$ .*

*Proof.* Every such function is a subset of  $\sigma \times \{-, +\}$ . Separate from  $\mathcal{P}(\sigma \times \{-, +\})$  the graphs which are functions whose domains are ordinals. These domains are subsets of  $\sigma$ , hence are at most  $\sigma$ . Conversely every sign sequence of length at most  $\sigma$  occurs. This uses Power Set and Separation, not Replacement.  $\square$

**Lemma 26.3** (Set families have bounded birthdays). *If  $S$  is a set of sign sequences, there is an ordinal  $\gamma$  with  $\text{b}(s) \leq \gamma$  for every  $s \in S$ . Moreover,*

$$\lambda = \sup\{\text{b}(s) + 1 : s \in S\}$$

*is an ordinal available in  $\mathbf{Z}_f$ , and  $\text{b}(s) < \lambda$  for all  $s \in S$ .*

*Proof.* The union  $\bigcup S$  is a set relation, so its domain exists. This domain is the union of the ordinal domains of the members of  $S$ , and is therefore an ordinal  $\gamma$ . Every  $\text{b}(s)$  is at most  $\gamma$ . To avoid assuming that the displayed successor image is a set, define directly

$$\lambda = \{\xi \in \gamma + 1 : \exists s \in S (\xi \leq \text{b}(s))\}.$$

This is a downward-closed set of ordinals, equals the indicated supremum, and has the desired strict bound. When  $S$  is empty, it is 0.  $\square$

**Lemma 26.4** (All-plus ordinal codes). *For an ordinal  $\alpha$ , let  $o_\alpha$  be the all-plus sequence of length  $\alpha$ . Then  $\alpha < \zeta$  if and only if  $o_\alpha < o_\zeta$ . For every sign sequence  $s$ ,*

$$s \leq o_{b(s)} < o_{b(s)+1}.$$

*In particular,  $o_\alpha < s$  implies  $\alpha < b(s)$ .*

*Proof.* These comparisons follow at the first negative sign, or at the first termination position when no earlier negative sign occurs. Each code is constructed directly as a constant function on its given ordinal. There is no appeal to a set containing all ordinal codes.  $\square$

**Lemma 26.5** (One bounded interval contains an ordinal detector). *For each ordinal  $\alpha$ , define  $\nu_\alpha$  on  $\alpha + 2$  by*

$$\nu_\alpha(\xi) = \begin{cases} -, & \xi = 0, \\ +, & 0 < \xi < \alpha + 2. \end{cases}$$

*Then*

$$-1 < \nu_\alpha < 0, \quad \alpha < \zeta \iff \nu_\alpha < \nu_\zeta.$$

*For every negative sign sequence  $c$  of birthday  $\sigma$ ,*

$$c < \nu_\sigma.$$

*Consequently  $\nu_\alpha < c < 0$  implies  $\alpha < b(c)$ .*

*Proof.* The first sign makes the sequence negative, and the plus at position 1 makes it greater than  $-1$ . Adding more trailing plus signs increases its value. Right addition by the finite ordinal 2 is strictly increasing, giving the equivalence involving  $\alpha$  and  $\zeta$ .

A negative  $c$  has a minus at position 0. If it has another minus, then  $\nu_\sigma$  is larger at the first such position. If it has only plus signs after position 0, then it is a proper initial segment of  $\nu_\sigma$  and the latter has a plus at its termination position. Thus  $c < \nu_\sigma$ . If  $\alpha \geq \sigma$ , monotonicity would give  $\nu_\alpha \geq \nu_\sigma > c$ , proving the last assertion.  $\square$

## 27 Actual set cuts: a Replacement-free simplicity proof

**Theorem 27.1** (Set-cut interpolation with a birthday bound). *In  $Z_f$ , let  $L, R$  be sets of sign sequences with  $L < R$ , meaning  $l < r$  for every  $l \in L$  and  $r \in R$ . Empty sides are allowed. There is a unique simplest separator  $c$  satisfying*

$$L < c < R.$$

*Every separator extends  $c$  as a sign sequence. In particular,  $c$  is the unique separator of least birthday, and*

$$b(c) \leq \lambda, \quad \lambda = \sup\{b(x) + 1 : x \in L \cup R\}.$$

*Proof.* By lemma 26.3,  $\lambda$  is a set ordinal, and every endpoint has birthday below  $\lambda$ . The proof takes place inside the set  $T_\lambda$  of lemma 26.2.

For a prefix  $p$ , say it has a left obstruction if some  $l \in L$  satisfies  $p \leq l$ , and a right obstruction if some  $r \in R$  satisfies  $r \leq p$ . The two obstructions cannot occur together: they would yield  $r \leq p \leq l < r$ . If neither occurs,  $p$  is a separator. Otherwise prescribe the next sign to be plus for a left obstruction and minus for a right obstruction.

Call  $p \in T_\lambda$  a *good path* if, for every  $\zeta < b(p)$ , the restriction  $p \upharpoonright \zeta$  is not yet a separator and the sign  $p(\zeta)$  is the prescribed sign at that restriction. Good paths form a set  $G$  by Separation. The empty path belongs to  $G$ .

*Compatibility.* Any two good paths are compatible. At a first disagreement they would have the same earlier prefix but different prescribed signs, contrary to the rule's uniqueness. Hence one extends the other.

The union  $p_* = \bigcup G$  is therefore a sign sequence of length at most  $\lambda$ . It is itself good: each of its sign positions already belongs to some good path, whose earlier restrictions coincide with its own. Thus  $p_* \in G$ .

We verify the invariant

$$\begin{aligned} l \in L, p \not\leq l &\implies l < p, \\ r \in R, p \not\leq r &\implies p < r \end{aligned} \tag{27.1}$$

for every good path  $p$ . It is vacuous at the empty path. Suppose the prescribed extension is  $p^+$ , obtained by appending plus. Any left endpoint not extending  $p^+$  either was already separated from  $p$ , equals  $p$ , or takes minus at the next position; in every case it is below  $p^+$ . A left obstruction ensures some  $l \geq p$ , and all right endpoints exceed this  $l$ . Therefore a right endpoint that still extends  $p$  cannot equal  $p$  or take minus next. Right endpoints not extending  $p$  remain separated at their earlier comparison position. This proves the invariant for  $p^+$ . The minus case is symmetric.

At a limit path, an endpoint not extending that path already fails to extend some shorter prefix: use a position past its first disagreement or its termination. The invariant at that prefix persists in all later extensions. Induction along the set ordinal  $\lambda + 1$  proves (27.1).

If  $b(p_*) < \lambda$  and  $p_*$  were not a separator, appending its prescribed sign would yield a good path longer than the union of all good paths, a contradiction. If  $b(p_*) = \lambda$ , no endpoint can extend  $p_*$ , since every endpoint has smaller birthday. The invariant then says that  $p_*$  is a separator. Set  $c = p_*$ .

Finally let  $q$  be any separator. Induction along a good path shows that  $q$  extends that path. At a plus step, an extension of the current prefix that terminates or takes minus is at most the prefix and hence at most the obstructing left endpoint; it cannot be  $q$ . The minus step is symmetric. At limits, extension follows by union. Thus  $q$  extends  $c$ . Since  $c$  itself separates, it is the unique separator of least birthday.  $\square$

*Remark 27.2* (Why this is not an invocation of class recursion). The proof does not first build a class-length recursion and then take its image. It separates *all good approximations from one set container* and takes their union. Every induction is along a given set ordinal. The set-existence steps are precisely those available in  $Z_f$ .

**Proposition 27.3** (Sharpness). *The birthday bound in theorem 27.1 cannot be uniformly lowered. For a nonzero successor  $\lambda = \zeta + 1$ , the cut  $(\{o_\zeta\}, \emptyset)$  has simplest separator  $o_\lambda$ . For a nonzero limit  $\lambda$ , the cut*

$$L = \{o_\zeta : \zeta < \lambda\}, \quad R = \emptyset$$

has simplest separator  $o_\lambda$ .

*Proof.* In the successor case, a separator of birthday at most  $\zeta$  is excluded by lemma 26.4, and  $o_{\zeta+1}$  works. In the limit case,  $L$  is a set by Separation in  $T_\lambda$ ; it is not being formed by an unbounded image operation. Any separator  $c$  must have birthday at least  $\lambda$ , because  $o_\zeta < c$  gives  $\zeta < b(c)$  for every  $\zeta < \lambda$ . Again  $o_\lambda$  works. The empty cut has simplest separator 0 and bound 0.  $\square$

## 28 The equivalence theorem for definable surreal families

### 28.1 Six schemes

All formulas below may have arbitrary set parameters in the language under consideration.

**Definition 28.1.** The schemes are as follows.

- (1) OB: every definable function  $i \mapsto \alpha_i$  from a set into the ordinals has an ordinal  $\gamma$  such that  $\alpha_i < \gamma$  for all indices.
- (2) Gr: every definable set-indexed surreal family has a set graph.
- (3) UB: every definable set-indexed surreal family has a strict surreal upper bound.
- (4) SC: if  $I$  is a set and  $\forall i \in I \exists s \in \mathbf{No} \varphi(i, s, p)$ , there is a set  $S \subseteq \mathbf{No}$  with  $\forall i \in I \exists s \in S \varphi(i, s, p)$ .
- (5) DCI: if two definable families  $(l_i)_{i \in I}$  and  $(r_j)_{j \in J}$  have set index domains and satisfy  $l_i < r_j$  for all  $i, j$ , there is a surreal  $c$  with  $l_i < c < r_j$  for all  $i, j$ .
- (6) NIB: every definable set-indexed family of elements of  $(-1, 0)$  has a strict upper bound that also lies in  $(-1, 0)$ .

The right bound 0 is not an acceptable witness to NIB. This is why the last scheme is stronger than the tautological assertion that a family in  $(-1, 0)$  is bounded in  $\mathbf{No}$ .

**Theorem 28.2** (Exact weak-base equivalence). *Over  $Z_f$ ,*

$$\boxed{\text{OB} \iff \text{Gr} \iff \text{UB} \iff \text{SC} \iff \text{DCI} \iff \text{NIB}.}$$

*Proof.* OB  $\Rightarrow$  Gr. Given a definable family  $s_i$ , its birthdays are uniquely defined ordinals. Choose  $\gamma$  bounding them. Every value lies in the set  $T_\gamma$ . The graph is now a subset of  $I \times T_\gamma$ , defined by the original formula, and exists by Separation.

Gr  $\Rightarrow$  UB. The range of the set graph is a set  $S$ . By lemma 26.3, all birthdays are at most an ordinal  $\gamma$ . The all-plus sequence  $o_{\gamma+1}$  strictly bounds every member of  $S$ .

UB  $\Rightarrow$  OB. Apply UB to the definable family  $o_{\alpha_i}$ . If  $c$  strictly bounds it, lemma 26.4 gives  $\alpha_i < b(c)$  for every  $i$ .

OB  $\Rightarrow$  SC. For each  $i$ , let  $\alpha_i$  be the least birthday of a surreal witness to  $\varphi(i, s, p)$ . The least ordinal exists without a choice of witnesses for the whole family: for each fixed  $i$ , one witness supplies an ordinal interval in which Separation and well-ordering locate a minimum. The definition of  $\alpha_i$  is functional, so OB gives  $\alpha_i < \gamma$  uniformly. The set  $T_\gamma$  is then a collecting set of possible witnesses. If desired, separate from it the witnesses relevant to at least one index. No choice function selecting a particular witness is needed.

SC  $\Rightarrow$  Gr. Apply SC to a functional surreal family. Its unique values all lie in a set  $S$ . Separate the graph inside  $I \times S$ .

Gr  $\Rightarrow$  DCI. Obtain set graphs and set ranges for the two families. Apply theorem 27.1 to these actual sets.

DCI  $\Rightarrow$  NIB. For a nonempty family in  $(-1, 0)$ , use it as the left family and use the constant right family  $\{0\}$ . Any separator is automatically greater than  $-1$ . For an empty family,  $\nu_0$  works. Equivalently one can adjoin  $-1$  to the left family using a finite tagged union of index sets.

NIB  $\Rightarrow$  OB. Given ordinal values  $\alpha_i$ , apply NIB to  $\nu_{\alpha_i}$ . Its negative upper bound  $c$  yields  $\alpha_i < b(c)$  by [lemma 26.5](#).  $\square$

**Corollary 28.3** (The obstruction for an individual family). *For a given definable set-indexed surreal family, the following are equivalent in  $Z_f$ : its graph is a set; its range is contained in a set of sign sequences; and its birthdays admit an ordinal bound. Mere numerical boundedness of its values is not equivalent to these conditions.*

*Proof.* A set graph has a set range and hence bounded birthdays. A birthday bound provides the container needed to separate the graph. The counterexample to the last possible equivalence is given in [section 30](#).  $\square$

*Remark 28.4* (Restricted domains and restricted formulas). The proof preserves the index domain in most implications. The passage from a negative interval to a cut uses only a singleton right side. Thus the ordinal-detector implications already hold for a fixed domain such as  $\omega$ . One must not conclude that the  $\omega$ -restricted scheme implies Replacement on *all* domains. Similarly, the transformations of formulas are explicit, but this article does not assert optimal preservation of their position in the Lévy hierarchy.

## 29 When the test becomes full Replacement

**Theorem 29.1** (Ranked-base characterization). *Over  $Z_f + \text{Hier}$ , each of the six schemes in [theorem 28.2](#) is equivalent to full Replacement and to full Collection. No Choice is required.*

*Proof.* Replacement forms the set of ordinal outputs of a definable function and therefore gives OB.

Conversely assume OB and consider a total relation

$$\forall i \in I \exists y \varphi(i, y, p).$$

By Hier, for each  $i$  there is some ordinal  $\alpha$  such that a suitable witness lies in  $V_\alpha$ . Define

$$\alpha_i = \min\{\alpha : \exists y \in V_\alpha \varphi(i, y, p)\}.$$

This is a uniquely defined ordinal, even when the witness is not unique. As before, a least element of the nonempty ordinal class is obtained by bounding the minimization with one witness and using Separation. By OB, some  $\gamma$  bounds all  $\alpha_i$ . Monotonicity of the cumulative hierarchy shows that  $V_\gamma$  collects witnesses for every  $i$ . This proves Collection.

Finally, for a functional relation, Collection puts the unique outputs inside a set, and Separation removes extraneous members and forms the graph. Thus Collection implies Replacement. Combining with [theorem 28.2](#) proves the assertion.  $\square$

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**The precise conclusion.** Actual set-cut interpolation is a theorem of  $Z_f$ . Definable set-indexed cut interpolation is equivalent to ordinal bounding over  $Z_f$ . It is equivalent to full Replacement over the *explicitly stronger base*  $Z_f + \text{Hier}$ .

---

This is an exact diagnosis of a common informal shortcut. The expression  $\{s_i : i \in I\}$  cannot be inserted into the ordinary cut construction until its existence as a set has been justified. In a rank-complete pure universe, justifying that step for all formulas has the full strength of Replacement.

### 30 An explicit omitted cut in a rank-initial universe

We now work externally in ZFC. Let

$$\theta = \omega + \omega, \quad \text{and consider the structure } (V_\theta, \in).$$

All assertions about  $V_\theta$  below refer to its internal membership structure. In particular, “a set” in  $V_\theta$  means an element of  $V_\theta$ .

**Proposition 30.1.**  *$V_\theta$  satisfies  $Z_f + \text{Hier}$  and *Choice*, but fails *OB*, *Replacement*, and *NIB*.*

*Proof.* The ordinal  $\theta$  is a limit and is closed below itself under adding any fixed finite ordinal. Pairing, Union, and Power Set increase the ranks of the finitely many input objects by only a finite amount, so their required outputs remain below  $\theta$ . Every subset of an element of  $V_\theta$  is again in  $V_\theta$ , giving Separation for formulas interpreted in  $V_\theta$ . Infinity and Foundation hold, and Extensionality is inherited from transitivity.

For every  $\alpha < \theta$ , the hierarchy segment  $\langle V_\zeta : \zeta \leq \alpha \rangle$  has rank at most  $\alpha$  plus a fixed finite coding overhead, and therefore belongs to  $V_\theta$ . Every member of  $V_\theta$  belongs to a sufficiently high stage below  $\theta$ . This gives *Hier*. An ambient well-ordering of any  $x \in V_\theta$  is a relation on  $x$  of rank bounded by the rank of  $x$  plus a finite amount, so it also belongs to  $V_\theta$ . Hence  $V_\theta$  satisfies *Choice*.

The formula  $n \mapsto \omega + n$ , for  $n \in \omega$ , defines a total ordinal-valued family in  $V_\theta$ . Every value exists there, but these values are cofinal in  $\theta = \text{Ord}^{V_\theta}$ . They have no internal ordinal bound, so *OB* fails. Replacement fails as well.

Finally define  $s_n = \nu_{\omega+n}$ . Each individual sign sequence has birthday  $\omega + n + 2$  and, with ordinary pair coding, rank below  $\theta$ . Thus it belongs to  $V_\theta$ , and  $V_\theta$  recognizes

$$\forall n \in \omega \quad -1 < s_n < 0.$$

Given any negative  $c \in \mathbf{No}^{V_\theta}$ , its birthday  $\sigma$  is below  $\theta$ . Choose  $n$  with  $\omega + n \geq \sigma$ . By [lemma 26.5](#),

$$c < \nu_\sigma \leq \nu_{\omega+n} = s_n.$$

There is therefore no negative strict upper bound for this definable family. This refutes *NIB* and presents an omitted definable cut with right side  $\{0\}$ .  $\square$

*Remark 30.2* (What is and is not missing). The family is externally countable. Internally, however, it is only a formula assigning one existing object to each natural number; its graph is not an element of  $V_\theta$ . Every *actual set cut* in  $V_\theta$  still has an interpolant by [theorem 27.1](#). Also, 0 is an upper bound for the displayed family. What fails is an upper bound *strictly below* 0, not ordinary numerical boundedness.

This countermodel shows that the distinction survives *Choice*, the existence of all internal hierarchy stages, and the existence of each individual member of the family. The omitted resource is the ability to collect the entire definable family into one set.

[Added 4 October 2026, batch 89: the class form of this failure at every limit height, recorded in [remark 24.7](#). Part XV of [foundations-and-computation/surreal-well-orders](#) (its source 31,

batch-89 manuscript 01) uses the same witness  $n \mapsto \omega + n$  at  $\lambda = \omega + \omega$  (its Section XV.11, `swo:gcz:sec:examples`). In the two-sorted full-class rank model  $(V_\lambda, \mathcal{P}(V_\lambda))$ , for every limit  $\lambda > \omega$ , class Replacement and class Collection on an internal set  $a$  hold exactly when  $|a| < \text{cf } \lambda$  (its Lemma XV.9.2, `swo:gcz:lem:domain`), and full class Replacement holds exactly when  $\lambda$  is strongly inaccessible (its Theorem XV.9.3, `swo:gcz:thm:replacement`). The failure of Replacement above is the definable case at a height of cofinality  $\omega$ ; the failure of NIB is source 10's own. Neither manuscript cites the other.]

## 31 Urelements: rank bounds versus atom-base bounds

For an object  $x$  in a full well-founded urelement universe, its kernel  $\ker(x)$  is the set of atoms occurring in the transitive closure of  $\{x\}$ . An object is pure if its kernel is empty. For a set  $B$  of atoms define the hierarchy

$$V_0(B) = B, \quad V_{\alpha+1}(B) = \mathcal{P}(V_\alpha(B)) \cup B,$$

with unions at nonzero limit stages, and put  $V(B) = \bigcup_{\alpha \in \text{Ord}} V_\alpha(B)$  as a class abbreviation. An object lies in  $V(B)$  exactly when its kernel is contained in  $B$ .

There are now two independent resources: the height  $\alpha$  and the atom base  $B$ . In the pure case the base is always empty. With atoms, an ordinal bound on heights need not control the supports of witnesses.

**Proposition 31.1** (A two-resource witness criterion). *Suppose the required set hierarchies over atom bases and kernels exist. For a total relation on a set  $I$ , the following are equivalent:*

- (i) *there is a set collecting witnesses;*
- (ii) *there are a set  $B$  of atoms and an ordinal  $\gamma$  such that every index has a witness in  $V_\gamma(B)$ .*

*If ordinal bounding is available, condition (ii) is equivalent to the existence of a single atom base  $B$  such that every index has some witness in  $V(B)$ .*

*Proof.* A collecting set  $W$  has a kernel  $B = \ker(W)$  and belongs to some  $V_\eta(B)$ . Its members then all lie in a sufficiently high common stage over  $B$ , giving (ii). Conversely  $V_\gamma(B)$  itself is a collecting set.

For the last assertion, fix a common base  $B$  and assign to each index the least ordinal stage over  $B$  containing a witness. This assignment is functional. Ordinal bounding bounds all its values, producing a common  $\gamma$ . The reverse implication is immediate.  $\square$

In  $\text{ZFU}_R$ , Replacement supplies ordinal bounding. The resulting common-base characterization is precisely the mechanism highlighted by Glazer–Yao, Proposition 1 [21]. Our use of it is to separate what a pure surreal detector can see from what it cannot: birthdays measure ordinal height, whereas pure sign sequences carry no atom kernel at all.

## 32 Cardinal-support models and their basic axioms

Fix a full urelement universe  $\mathcal{U}$  with Choice, containing a set  $A$  of all its atoms. Let  $\kappa$  be an infinite cardinal with  $|A| \geq \kappa$ , with cardinalities computed in  $\mathcal{U}$ . Define the class

$$M_\kappa^{\ker} = \{x \in \mathcal{U} : |\ker(x)| < \kappa\}.$$

It contains every atom individually and every pure set, but does not contain the set of all atoms. Thus its atoms form a proper class *internally*, even though  $A$  is a set in the ambient universe.

The restriction is on kernel cardinality, not on the cardinality or rank of the whole object. For a permitted base  $B$ , every stage  $V_\alpha(B)$ , at every ordinal height, belongs to  $M_\kappa^{\text{ker}}$ .

**Lemma 32.1** (Kernel confinement of a unique output). *Suppose  $w, p \in M_\kappa^{\text{ker}}$ ,  $x \in w$ , and a formula has a unique output  $y \in M_\kappa^{\text{ker}}$  for  $x, p$ , when interpreted in  $M_\kappa^{\text{ker}}$ . Then*

$$\ker(y) \subseteq K, \quad K = \ker(w) \cup \ker(p).$$

*Proof.* Every permutation of  $A$  extends by rank recursion to an automorphism of  $\mathcal{U}$  and preserves  $M_\kappa^{\text{ker}}$ , because it preserves kernel cardinalities. Suppose  $a \in \ker(y) \setminus K$ . As  $\kappa$  is infinite, the union  $K \cup \ker(y)$  has size below  $\kappa$ . Choose an atom  $b$  outside this union. The transposition swapping  $a$  and  $b$  fixes  $K$  pointwise, hence fixes  $x$  and  $p$ . It moves  $\ker(y)$  and therefore moves  $y$ .

The image of  $y$  is another output satisfying the same formula in  $M_\kappa^{\text{ker}}$ , by invariance under this automorphism. This contradicts uniqueness.  $\square$

**Theorem 32.2** (The base support model). *For every such infinite  $\kappa$ ,  $M_\kappa^{\text{ker}}$  is a model of  $\text{ZFUR} + \text{AC}$ . It is transitive, has every ambient subset of each of its sets, and has exactly the same pure sets as  $\mathcal{U}$ .*

*Proof.* Membership can only decrease kernels, so the class is transitive. Subsets do not increase kernels; hence every ambient subset of an internal set is internal, and the ambient power set of an internal set is itself internal. Pairing and finite unions use only finite unions of kernels of size below an infinite  $\kappa$ , again of size below  $\kappa$ . Union similarly adds no atoms beyond those already in the input's kernel. Extensionality, Foundation, and Infinity follow immediately. Separation follows by relativizing a fixed formula to the definable class  $M_\kappa^{\text{ker}}$  and taking a subset of the given set.

For Replacement, suppose a formula assigns a unique internal output to every member of a set  $w$ , with parameter tuple coded by  $p$ . Ambient Replacement forms its range  $W$ ; relativization to  $M_\kappa^{\text{ker}}$  is a legitimate fixed ambient formula. By lemma 32.1, every output has kernel contained in  $K = \ker(w) \cup \ker(p)$ . Therefore  $\ker(W) \subseteq K$  and  $W \in M_\kappa^{\text{ker}}$ . This proves every internal Replacement instance.

For Choice, an ambient well-ordering of a set  $w \in M_\kappa^{\text{ker}}$  has kernel contained in  $\ker(w)$ , so that well-ordering belongs to  $M_\kappa^{\text{ker}}$ . Finally every pure set has empty kernel and is included; no impure object becomes pure by passing to this transitive model.  $\square$

**Lemma 32.3** (Agreement about cardinalities of internal sets). *For every  $w \in M_\kappa^{\text{ker}}$ , its cardinality as computed in  $M_\kappa^{\text{ker}}$  equals its ambient cardinality. Cofinalities of pure ordinals also agree.*

*Proof.* An ambient bijection between  $w$  and its initial ordinal has kernel contained in  $\ker(w)$ , so it is internal. Conversely an internal bijection is an ambient bijection. For cofinality, every function between pure ordinals is pure, so the two universes have exactly the same candidate cofinal maps.  $\square$

### 33 The exact Collection spectrum

Write  $\text{Collection}_{<\tau}$  for the scheme restricting index sets to cardinality below  $\tau$ . Saying that Collection fails at  $\tau$  means that one explicit formula has a total witness relation on a domain of cardinality  $\tau$  with no collecting set. It is not a quantification over a set of all schemes.

**Lemma 33.1** (The small-domain positive bound). *For every infinite  $\kappa$ ,  $M_\kappa^{\text{ker}}$  satisfies  $\text{Collection}_{<\text{cf } \kappa}$ .*

*Proof.* Let  $w \in M_\kappa^{\text{ker}}$  have size  $\lambda < \text{cf } \kappa$ , and suppose every  $x \in w$  has an internal witness to a fixed formula. In the ambient universe,  $\text{Collection}$  followed by  $\text{Choice}$  gives a family of internal witnesses  $y_x$ . Each has a kernel of size below  $\kappa$ . A union of fewer than  $\text{cf } \kappa$  sets, each of size below  $\kappa$ , has size below  $\kappa$ ; this is the usual cofinality calculation in the ambient universe with  $\text{Choice}$ . Consequently

$$|\ker(\{y_x : x \in w\})| \leq \left| \bigcup_{x \in w} \ker(y_x) \right| < \kappa.$$

The selected range is therefore an internal collecting set. By lemma 32.3, the domain-size hypothesis is the same internally and externally.  $\square$

**Theorem 33.2** (Limit-cardinal failure at the exact threshold). *Let  $\kappa$  be a limit cardinal, with  $\omega$  included in this terminology, and set  $\tau = \text{cf } \kappa$ . Then*

$$M_\kappa^{\text{ker}} \models \text{Collection}_{<\tau}, \quad M_\kappa^{\text{ker}} \not\models \text{Collection}_\tau.$$

*For every cardinal  $\lambda \geq \tau$ ,  $\text{Collection}$  also fails on a domain of size  $\lambda$ .*

*Proof.* The positive assertion is lemma 33.1. Choose a pure sequence of cardinals  $(\mu_i)_{i < \tau}$  below  $\kappa$ , cofinal in  $\kappa$ . For  $\kappa = \omega$ , use finite cardinals. The sequence and its domain belong to  $M_\kappa^{\text{ker}}$  because they are pure.

Consider the relation

$$\varphi(i, B) : \quad B \text{ is a set of atoms and } |B| = \mu_i.$$

Every  $i < \tau$  has a witness: select an ambient set  $B \subseteq A$  of size  $\mu_i$ . Both  $B$  and a bijection witnessing its cardinality have kernels of size below  $\kappa$ , so they are internal.

If  $W \in M_\kappa^{\text{ker}}$  collected such witnesses, put  $\rho = |\ker(W)| < \kappa$ . Some  $\mu_i$  exceeds  $\rho$ . But every atom set  $B \in W$  satisfies  $B \subseteq \ker(W)$  and cannot have size  $\mu_i$ . This is a contradiction.

For a larger domain, extend the same relation by giving a harmless fixed witness, such as the empty set, to the additional indices. A collecting set for the enlarged relation would collect witnesses for the original one.  $\square$

**Theorem 33.3** (The successor-cardinal case). *If  $\kappa = \mu^+$  for an infinite cardinal  $\mu$ , then  $M_\kappa^{\text{ker}}$  satisfies full  $\text{Collection}$ , with no restriction on the size of its index set.*

*Proof.* Let a formula with parameter  $p$  have a witness for every  $x \in w$ . Set  $K = \ker(w) \cup \ker(p)$ ; its size is at most  $\mu$ . Choose a set  $B \subseteq A \setminus K$  of size  $\mu$ . This is possible because  $|A| \geq \mu^+$ .

Given one internal witness  $y$ , its kernel outside  $K$  has size at most  $\mu$ . There is a permutation of  $A$  fixing  $K$  pointwise and sending these remaining atoms into  $B$ . To justify extension to a permutation, choose an injection onto a subset of  $B$  and biject the two complementary parts of  $A \setminus K$ ; both complements have cardinality  $|A|$ . The resulting automorphism fixes  $x, p$  and preserves  $M_\kappa^{\text{ker}}$ . Thus an appropriate witness exists in  $V(K \cup B)$  for every  $x$ .

For each  $x$  take the least stage of this fixed-base hierarchy containing a witness. Replacement in  $M_\kappa^{\text{ker}}$ , already proved in theorem 32.2, bounds these ordinal stages. A sufficiently high set  $V_\gamma(K \cup B)$  belongs to  $M_\kappa^{\text{ker}}$  and collects all the required witnesses.  $\square$

**Corollary 33.4** (Complete capacity classification). *For an infinite cardinal  $\kappa$ ,  $M_\kappa^{\text{ker}}$  satisfies full Collection if and only if  $\kappa$  is a successor cardinal. If it is not, the least domain cardinality at which Collection fails is  $\text{cf } \kappa$ .*

The distinction is not simply regular versus singular. For example,  $\omega$  is regular but the finite-support model fails countable Collection. A regular uncountable limit cardinal has failure first at that cardinal. A successor cardinal, by contrast, has a largest permitted infinite support size, and the common-reservoir argument works for domains of arbitrary cardinality.

### 33.1 Prescribing the first failure

**Corollary 33.5.** *For every infinite regular cardinal  $\tau$ , there is a cardinal-support model whose least Collection-failure domain has size  $\tau$ , while its pure universe is unchanged from the ambient one.*

*Proof.* Choose  $\kappa = \aleph_\tau$  and an ambient set of at least  $\kappa$  atoms. Since  $\tau$  is an infinite regular ordinal,  $\kappa$  is a limit cardinal with  $\text{cf } \kappa = \tau$ . The latter equality follows from continuity and strict increase of the aleph sequence: an index-cofinal sequence gives the upper bound, while a shorter cofinal sequence of ordinals below  $\aleph_\tau$  would yield a shorter cofinal sequence of indices below  $\tau$ . Apply [theorem 33.2](#).  $\square$

## 34 Naming an enumeration: exact failure of Replacement

Fix  $A_0 \subseteq A$  of size  $\kappa$ , and an ambient bijection

$$e : \kappa \longrightarrow A_0.$$

Its graph has kernel  $A_0$ , so it is not a set of  $M_\kappa^{\text{ker}}$ . Nevertheless it can be named as an external binary predicate  $E$  on the domain of that model. We analyze the expansion  $(M_\kappa^{\text{ker}}, E)$  with *Separation and other schemes understood in the expanded language*.

This is a controlled instance of language sensitivity. It is not a reproof of Glazer’s global-choice theorem, which concerns a different pure-set situation [\[20\]](#). The shared lesson is that adjoining a predicate can add functional definitions to which a scheme must apply.

**Proposition 34.1** (Elementary axioms survive the expansion). *The expansion  $(M_\kappa^{\text{ker}}, E)$  satisfies the urelement versions of  $\mathbf{Z}_f$  and Choice, including full Separation for formulas using  $E$ .*

*Proof.* The structural axioms and Choice are unchanged. For an expanded-language formula, replace  $E$  by its ambient set graph and relativize quantifiers to  $M_\kappa^{\text{ker}}$ . Ambient Separation gives a subset of the specified internal set. Every such subset is internal by [theorem 32.2](#), so expanded Separation holds.  $\square$

**Theorem 34.2** (The exact expanded-language threshold). *For every infinite  $\kappa$ , let  $\tau = \text{cf } \kappa$ . In  $(M_\kappa^{\text{ker}}, E)$ , both Collection and Replacement hold for all domains of cardinality below  $\tau$ , and both fail on a domain of cardinality exactly  $\tau$ .*

*Proof.* The positive argument in [lemma 33.1](#) does not use invariance under atom permutations. It uses only ambient Collection and Choice to gather fewer than  $\text{cf } \kappa$  witnesses, followed by a kernel-size calculation. It therefore works unchanged for formulas involving the ambient predicate  $E$ . This gives Collection, and hence Replacement, on all smaller domains.

Choose a pure strictly increasing cofinal sequence  $(\alpha_i)_{i < \tau}$  in  $\kappa$ . The sequence is an internal set and may be used as a parameter. The expanded language defines, for each  $i < \tau$ , the unique set

$$Y_i = e \text{ “ } \alpha_i \text{ ”}.$$

Each individual  $Y_i$  belongs to  $M_\kappa^{\text{ker}}$  because  $|\alpha_i| < \kappa$ . Its characterization uses  $E$  and the cofinal sequence, not an internally existing graph of  $e$ .

If the range  $\{Y_i : i < \tau\}$  were an internal set, its union would be the set  $A_0$ , since the  $\alpha_i$  are cofinal in  $\kappa$ . But  $A_0$  has kernel cardinality  $\kappa$  and is not internal. Replacement therefore fails for this functional family. A collecting set for these unique witnesses would have kernel containing their union  $A_0$ , so Collection fails for the same family.  $\square$

**Example 34.3** (Countable supports). Let  $\kappa = \aleph_1$ . The reduct  $M_{\aleph_1}^{\text{ker}}$  satisfies full Collection. After naming an enumeration of  $\aleph_1$  atoms, all countable-domain Collection and Replacement instances still hold, but both schemes fail at domain size  $\aleph_1$ . No pure set, sign sequence, or ordinary set function between pure objects has changed.

**Example 34.4** (Singular support capacity). Let  $\kappa = \aleph_\omega$ . In the reduct, Replacement holds on all domains but countable Collection fails. In the expansion, the sets  $e \text{ “ } \aleph_n \text{ ”}$  form a uniquely specified countable family with no internal range set. Thus countable Replacement now fails as well.

| Capacity                   | Replacement in $M_\kappa^{\text{ker}}$ | First Collection failure in $M_\kappa^{\text{ker}}$ | First failure of both in $(M_\kappa^{\text{ker}}, E)$ |
|----------------------------|----------------------------------------|-----------------------------------------------------|-------------------------------------------------------|
| $\omega$                   | All domains                            | $\omega$                                            | $\omega$                                              |
| $\aleph_1$                 | All domains                            | None                                                | $\aleph_1$                                            |
| $\aleph_\omega$            | All domains                            | $\omega$                                            | $\omega$                                              |
| $\aleph_{\omega_1}$        | All domains                            | $\omega_1$                                          | $\omega_1$                                            |
| General limit $\kappa$     | All domains                            | cf $\kappa$                                         | cf $\kappa$                                           |
| General successor $\kappa$ | All domains                            | None                                                | $\kappa$                                              |

Here “first failure” refers to the cardinality of the index domain. The table is an exact consequence of the preceding proofs, not a numerical experiment.

## 35 Why pure surreal structures cannot see the support obstruction

**Proposition 35.1** (Identity of pure structures). *The pure universe of  $M_\kappa^{\text{ker}}$  is the pure universe of  $\mathcal{U}$ . Therefore their sign-sequence classes, birthdays, prefix relations, ordinal codes, and all set-coded relations between pure objects are identical. Any specified operation defined solely within the pure universe has the same interpretation in both.*

*Proof.* Every pure object has empty kernel and is internal, and membership on these objects is unchanged. Every set made entirely from pure objects is itself pure. Thus the two pure membership structures are identical. Induction on a formula whose quantifiers are restricted to pure objects proves agreement on all such definitions. Apply this to sign sequences and the stated relations.  $\square$

This assertion is deliberately restricted to pure definitions. A formula with unrestricted quantification over impure objects can have different truth values in the two universes, even when its free variables happen to be pure.

**Theorem 35.2** (Pure-valued Collection survives definable expansions). *In  $M_\kappa^{\text{ker}}$ , expanded by any fixed finite list of ambient-definable predicates, Collection holds for every total relation whose witnesses are required to be pure. Index sets and parameters may be impure. In particular, this holds in  $(M_\kappa^{\text{ker}}, E)$ .*

*Proof.* Let  $w$  be an internal set, and suppose that for each  $x \in w$  some pure  $y$  satisfies a fixed formula in the expansion. Relativize that formula and the added predicates to the ambient universe. Ambient Collection produces a set of suitable witnesses. Separate away any extraneous nonpure members, obtaining a set  $S$  of pure witnesses covering every index. A set of pure objects is pure, so  $S \in M_\kappa^{\text{ker}}$ . This same  $S$  witnesses the original internal Collection instance.  $\square$

[Added 4 October 2026, batch 90: source 12 ([Part VIII](#)) proves, independently, the case  $\kappa = \omega$  of this theorem for the graph **Act** of a named action of a pure set-sized group on the atoms ([theorem 63.1](#)), by a second route: it bounds the least pure ranks of witnesses by ambient Replacement and separates them from one pure stage. With it, the identity of pure structures ([proposition 35.1](#)), the surreal schemes and the no-go corollary below extend to such actions and to the graded levels  $R_k$  of Replacement for outputs with at most  $k$  atoms ([proposition 63.2](#) and [corollaries 63.3](#) and [63.4](#); [remark 54.3](#)).]

**Corollary 35.3** (All six surreal schemes survive). *Every scheme OB, Gr, UB, SC, DCI, NIB holds in  $(M_\kappa^{\text{ker}}, E)$ , including its instances using  $E$  and arbitrary internal parameters, despite the failure of full Replacement in [theorem 34.2](#).*

*Proof.* Surreal sign sequences and ordinals are pure. Hence [theorem 35.2](#) gives SC and collects ordinal witnesses. Set graphs can then be separated using expanded Separation; their kernels are contained in the kernel of the index domain. The pure range sets have pure ordinal birthday bounds and pure interpolants, all of which are internal. The arguments of [theorem 28.2](#) apply directly to these sign sequences.  $\square$

There is no contradiction with [theorem 29.1](#). The hierarchy axiom used there is a cofinal pure hierarchy covering every object of an atom-free universe. No pure  $V_\alpha$  contains an atom. In the urelement setting, the varying atom base is an additional resource not controlled by ordinal bounding.

**Corollary 35.4** (No pure-surreal characterization of ambient Collection). *Across urelement models of the kind constructed here, no statement or scheme depending only on the pure surreal structure can characterize full ambient Collection. Even the enlarged package of all definable surreal-valued Collection and interpolation instances does not characterize it. Across the displayed predicate expansions, that package does not characterize full ambient Replacement either.*

*Proof.* Within an ambient universe with enough atoms, compare a successor-capacity model and a limit-capacity model. Their pure universes, and hence their pure surreal structures, are identical, while full Collection holds in the former and fails in the latter. Both have all the surreal schemes by Replacement or [theorem 35.2](#).

For Replacement, compare the same  $M_\kappa^{\text{ker}}$  with an empty interpretation of the extra predicate and with the enumeration interpretation  $E$ . The first expansion is definitionally equivalent to the reduct and satisfies full Replacement. The second fails it by [theorem 34.2](#), while satisfying all the surreal schemes by [corollary 35.3](#).  $\square$

### 35.1 Reflection and the limit of the conclusion

Full first-order reflection to transitive sets implies Collection: reflect simultaneously the totality assertion for a witness relation and the relation itself, with the index set and parameters included. The reflecting set then supplies genuine witnesses for all indices. This standard implication is also explicit in Glazer–Yao’s analysis [21]. Thus our limit-capacity models fail this full reflection principle.

We do not obtain Glazer–Yao’s finer separations between partial reflection, supertransitive partial reflection, and full reflection. Nor do we exclude a useful *mixed* surreal–atom theory. Naming atom kernels or permitting atom-labelled objects can expose exactly the data that the pure structure forgets. The obstruction concerns the proposed pure detector, not the possibility of any formalization of urelement set theory.

## 36 A formalization route for ProveIt

### 36.1 Repository alignment and inspection boundary

The inspected tree snapshot was identified by commit

2b7b388ba81a3355b19a7c2d2fe797e92f572355.

The relevant repository locations are:

| Location                                                                                 | Role in a future development                                                                                 |
|------------------------------------------------------------------------------------------|--------------------------------------------------------------------------------------------------------------|
| Algebra/SurrealNumbers/Surreal/Foundations /                                             | Sign-sequence and cut infrastructure; connect the order-level result to the existing surreal representation. |
| SetTheory/ZF/                                                                            | Object-language formulas, theories, and model semantics.                                                     |
| SetTheory/ClosureAxiomatization/                                                         | Example of maintaining both semantic directions and a deductive equivalence.                                 |
| Algebra/SurrealNumbers/docs/foundations-and-computation/surreal-fields-across-universes/ | Related but distinct universe-comparison report; useful for terminology and avoiding duplicate claims.       |

This was a targeted README and directory inspection, not a full source audit or a successful build of the repository. The repository’s existing kernel-audit claims do not transfer to the present manuscript. No purportedly checked declaration name is invented here.

[Added 3 October 2026, batch 89: the four locations exist unchanged at the placement commit 23adb85f9; the commit 2b7b388ba named above is the commit “Catalogue batches 81 to 86”. Source 10 did not inspect this report, whose Part II proves the cardinal-cutoff form of its spine (remark 24.5). Placement beside the Lean libraries confers no formal status: no statement of this part is formalized, and none of the module names below exists in the repository.]

### 36.2 Separate object-level strength from host-level resources

A theorem about ordinary Lean types can use host-level function types, images, ordinal recursion, and classical choice without recording whether these correspond to Replacement or Choice inside a

weak object theory. Consequently proving a cut theorem for a universe-polymorphic surreal type is not by itself a formal proof that  $Z_f$  proves that theorem.

The proposed formalization should have two layers. The first is a conventional order-theoretic library proving the sign comparisons and the finite-container construction. The second interprets the same constructions in arbitrary models of the explicitly listed weak axioms. A function type in the host must not silently substitute for an internal set graph.

In particular, “ $\varphi$  defines a functional family” and “there is an internal graph for  $\varphi$ ” should be different predicates. Their equivalence is one of the conclusions under investigation, not an implementation convenience.

### 36.3 Suggested modules and proof order

The following are proposed module names, not existing files.

1. **WeakSignSequences**: internal function graphs on ordinals, lexicographic comparison, prefix, bounded containers, and all-plus and negative ordinal codes.
2. **SetCutSimplicity**: the set of good paths, compatibility, its union, the separation invariant, and the unique simplest interpolant with its birthday bound.
3. **DefinableFamilySchemes**: syntactic transformations implementing the six implications, with explicit parameters and index domains.
4. **RankedReplacement**: the hierarchy axiom, least-stage witnesses, and the equivalence with Collection and Replacement.
5. **AtomKernelModels**: tagged urelement semantics, kernel operations, permutation action, and the proof of kernel confinement.
6. **CapacitySpectra**: the small-domain union bound, cofinal support-size obstruction, and the successor reservoir argument.
7. **PredicateExpansion**: interpretation of an external enumeration, expanded Separation, exact failure families, and pure-valued Collection.

The set-cut lemma should be completed before importing any full-ZF surreal construction into the weak-base argument. Its proof has a natural bounded interface: a fixed ordinal  $\lambda$  and the set  $T_\lambda$ .

### 36.4 Trust boundaries and quantification

A satisfactory formal artifact would explicitly audit at least four boundaries. First, it would specify the object theory rather than merely a list of host axioms. Second, it would distinguish each standard formula instance from a single internal assertion quantifying over formula codes. Third, it would not represent all subsets of a proper class by a host powerset. Fourth, it would distinguish the original language from its expansion by  $E$  when stating Separation and Replacement.

For the support models, one can formalize an external set-sized model and its interpreted submodel, or a definable class interpretation in a stronger metatheory. The theorem is relative: it constructs a model from an ambient model. It is not an assertion that a consistent first-order theory proves its own consistency.

## 37 Executable finite checks and reproducibility

The accompanying program, shipped as `code/10-cuts-replacement-verify_finite.py` (delivered as `code/verify_finite.py`), uses only the Python standard library. It represents a finite sign

sequence by a tuple of  $-1$  and  $+1$ , with termination compared as 0. It implements the obstruction rule from [theorem 27.1](#), then independently enumerates candidates to check separation, unique minimum birthday, and the prefix property.

The recorded run, shipped as `data/10-cuts-replacement-finite_checks.json` (delivered as `data/finite_checks.json`), completed with the following results:

| Check                                               | Number |
|-----------------------------------------------------|--------|
| Sign words of birthday at most 6                    | 127    |
| Endpoint intervals, including empty sides           | 8 256  |
| Assignments to left/right/neither at depth 2        | 2 187  |
| Separated assignments among these                   | 576    |
| Negative-code comparisons and birthday implications | 1 050  |
| Atom-transposition checks                           | 4 032  |

Every test passed. Endpoint intervals suffice for all finite sets of bounds from the depth-six endpoint universe, since a nonempty finite left set has a maximum and a nonempty finite right set has a minimum. The depth-two assignment check separately exercises the implementation on nonsingleton sets.

The atom test verifies the local combinatorial fact used by the confinement proof: a swap outside a fixed parameter support fixes that support while moving a witness support containing exactly one of the swapped atoms. It does not simulate a model of set theory or a singular cardinal.

In the delivered layout the run was

```
python3 code/verify_finite.py --output data/finite_checks.json
```

followed by two `pdflatex` passes of the delivered manuscript. With the shipped names, run the program on a copy and pass an explicit output path outside this directory (the report's `README.md` gives the commands). For this merge it was rerun with Python 3.14.4 on a copy: it passed and reproduced the recorded file exactly, apart from line endings. The tests cannot establish a transfinite induction, an axiom-scheme equivalence, an internal cardinal calculation, or a relative-consistency claim. Those depend on the mathematical proofs above. The delivered archive also contained a `README`, a proof-status note and a build script; the proof-status note is shipped as `10-cuts-replacement-PROOF_STATUS.md` and the build script as `code/10-cuts-replacement-build.sh` (it builds the delivered manuscript in the delivered layout, not this report), while the delivery `README`, the manuscript and its PDF are not shipped ([section 39](#)).

## 38 Further research questions and proposed projects

The questions below are a research agenda arising from the proved results. Their status is “not resolved in this manuscript”; no claim is made that every formulation is globally new or currently open in the literature.

**Question 38.1** (Remove or characterize the hierarchy hypothesis). Over plain  $Z_f$ , determine the exact relationship between OB and full Replacement without assuming Hier. Identify the weakest rank-coverage or tier-coverage principle that turns the equivalence in [theorem 28.2](#) into full Replacement, and give explicit models distinguishing the relevant choices of coverage axiom.

The point is not to omit Hier typographically. A useful answer must analyze the existence of ranks and hierarchy segments inside the weak model.

**Question 38.2** (Formula-complexity accounting). Determine the optimal correspondence between restricted  $\Sigma_n$  or  $\Pi_n$  instances of ordinal bounding, surreal graph existence, and interval bounding. In particular, quantify the complexity increase caused by “least birthday of a witness” and by the good-path definition.

A precise answer would turn the full-scheme result into a hierarchy of proof-theoretic characterizations.

**Question 38.3** (Domain spectra in weak pure models). For models of  $Z_f + \text{Hier}$ , classify the index-domain sizes on which the six equivalent principles hold. Which initial segments of cardinalities can occur, and how are they constrained by definable cofinal maps into the model’s ordinals?

The  $V_{\omega+\omega}$  example gives failure already on  $\omega$ , but the general domain spectrum deserves a separate analysis. Internal and external cardinalities must be kept distinct.

[Added 3 October 2026, batch 89: for the two-sorted *full-class* rank models  $(V_\lambda, \mathcal{P}(V_\lambda))$  the domain spectrum of class Replacement and Collection is computed in Part XV of [foundations-and-computation/surreal-well-orders](#): exactly the domains of size below  $\text{cf } \lambda$  ([remark 24.7](#)). The question above, for the *definable* schemes in arbitrary models of  $Z_f + \text{Hier}$ , is not answered there and stays open.]

**Question 38.4** (Beyond cardinality ideals). Replace  $|\ker(x)| < \kappa$  by membership of the kernel in a permutation-invariant ideal  $\mathcal{I}$  on the atoms. Give necessary and sufficient conditions on  $\mathcal{I}$  for Replacement, for Collection on a specified class of domains, and for a common-reservoir argument.

The cardinal-capacity calculation suggests that the relevant invariants combine union closure, available support sizes, and the action of permutations fixing parameter kernels. An arbitrary ideal need not have the fresh-atom property used in [lemma 32.1](#).

**Question 38.5** (Mixed surreal–atom witnesses). Design a natural two-sorted extension of the sign-sequence structure whose witnesses encode both a birthday and an atom base. Can a local interpolation or extension principle in this richer structure characterize full Collection without merely restating the two-resource criterion by definition?

A successful construction should remain mathematically recognizable and should explain how automorphisms act on the added data.

**Question 38.6** (Admissible language expansions). Classify predicates  $P$  for which  $(M_\kappa^{\ker}, P)$  retains Replacement or full Collection. Can the first failure be characterized in terms of the support growth of uniquely  $P$ -definable outputs? How much of the permutation group must remain available to force kernel confinement?

The enumeration predicate supplies a sharp negative case. Predicates invariant under every atom permutation supply an immediate positive mechanism, but intermediate stabilizer groups are more interesting.

[Added 3 October 2026, batch 89: for one further family of predicates, the graphs of canonical lifts of finitely many atom injections, the report [SetTheory/Cardinals/docs/reports/ordinal-s-and-order-types/naming-elementary-embeddings](#) ([remark 24.6](#)) decides exactly when the

expansion keeps Replacement and full Collection, by a criterion on the components of the named injections (Theorem 5.1, `nee:thm:profile`; Theorem 7.5, `nee:thm:collection`). This answers the question for that family only; the general classification stays open.]

[Added 4 October 2026, batch 90: answered for a second family at  $\kappa = \omega$ , the graphs **Act** of uniformly named actions of pure set-sized groups, by source 12 ([Part VIII](#); [remark 54.2](#)). Replacement holds iff every orbit is finite and only finitely many atoms lie in finite orbit types that occur finitely often ([theorem 58.1](#)); the support growth of uniquely definable outputs is measured by the levels  $R_k$ , which hold, for  $k \geq 1$ , iff every orbit is finite and only finitely many atoms have an orbit of size at most  $k$  under the centralizer  $\mathcal{Z}$  of the named action ([theorem 59.1](#)); so the group that must remain is  $\mathcal{Z}$ . Full Collection fails at  $\kappa = \omega$  for every such expansion. For finitely many named permutations this criterion is the case  $\kappa = \omega$  of the one cited in the preceding note. Arbitrary predicates, and group actions at uncountable  $\kappa$ , stay open.]

**Question 38.7** (Reflection spectra rather than Collection spectra). In controlled kernel models, determine which additional features separate partial reflection from full reflection while the pure surreal structure remains fixed. Seek a precise comparison with the more elaborate models of Glazer–Yao, rather than assuming the cardinal-capacity invariant alone controls reflection.

The present models are intentionally homogeneous and do not reproduce those finer separations. [Added 3 October 2026, batch 89: the same report shows, for the kernel models expanded by named lifts, that full transitive reflection is equivalent to Collection there (Theorem 7.7, `nee:thm:reflection`), and that partial transitive reflection of assertions  $\forall x \in w \exists y \delta$  with  $\Delta_0$  matrix is  $\Delta_0$  Collection, which fails at limit cutoffs, although each single true  $\exists y \delta$  reflects (Proposition 8.1, `nee:prop:partial`). It bears on this question but does not reproduce Glazer–Yao’s separations; the question stays open.]

**Question 38.8** (Weaker power-set assumptions). The good-path proof uses the set  $\mathcal{P}(\lambda \times \{-, +\})$ . Determine whether a restricted binary-sequence existence axiom suffices in place of full Power Set, and whether interpolation over such a base becomes equivalent to additional Collection or recursion principles.

This asks for a genuinely different lower-strength construction, not a proof that secretly retains the same function-space container.

**Question 38.9** (Constructive variants). Analyze which parts of the six-scheme equivalence survive intuitionistic logic. The current comparison rule uses a first disagreement and decidable alternatives between obstructions, while least-witness birthday selection also has classical content. A constructive version should specify which order and cut notions replace these choices.

**Question 38.10** (A proof-assistant benchmark with semantic safeguards). Implement the weak-base and expanded-language results in ProveIt, using the same internal formula language for the positive and negative directions. Include regression tests that intentionally try to confuse a definable family with a set graph, or to use a reduct’s Replacement instance after adjoining  $E$ .

Such a benchmark would test more than symbolic algebra: it would test whether an automated proof pipeline preserves the intended theory and language across an argument.

## 39 Conclusion of source 10, ledger and provenance

### 39.1 Conclusion

The phrase “a set-indexed family of surreal numbers” conceals two different assertions. A family already represented by a set graph supports the ordinary simplicity construction in Zermelo set theory without Replacement. A merely definable functional family can fail to have that graph. Uniformly repairing this failure is equivalent to ordinal bounding, and on an explicit pure cumulative-hierarchy base it is equivalent to full Replacement and Collection. The obstruction is visible inside the single interval  $(-1, 0)$ .

Urelements expose a second direction of failure. Pure sign sequences retain ordinal height but forget atom bases. Cardinal-support models keep the entire pure universe unchanged while varying the exact domain threshold for Collection. An external atom enumeration can additionally destroy Replacement, with an exact cofinality threshold, even though every pure-valued witness relation still satisfies Collection.

These results provide a concrete intersection between weak-set-theoretic research and surreal formalization. Their most useful lesson is operational: record the language, the index domain, the birthday bound, and the atom support separately. Each is mathematically consequential, and none should be supplied silently by the notation or by the proof assistant’s host theory.

### 39.2 A compact dependency and limitation ledger

| Result                               | Assumptions                                                                        | Important exclusion                                              |
|--------------------------------------|------------------------------------------------------------------------------------|------------------------------------------------------------------|
| Set-cut simplicity                   | $Z_f$ , actual sets of endpoints                                                   | No arbitrary definable images assumed.                           |
| Six-scheme equivalence               | $Z_f$ , all instances in one fixed language                                        | No assertion of full Replacement without a hierarchy hypothesis. |
| Full Replacement characterization    | $Z_f + \text{Hier}$                                                                | Pure, atom-free cumulative hierarchy.                            |
| $V_{\omega+\omega}$ example          | External ZFC                                                                       | The missing family is not an internal sequence.                  |
| $M_\kappa^{\text{ker}}$ construction | Full ambient urelement universe with Choice; $ A  \geq \kappa$ ; infinite $\kappa$ | Internal proper-class atoms are an ambient set.                  |
| Collection spectrum                  | Same ambient assumptions                                                           | Restricted by domain size, not witness rank.                     |
| Enumeration expansion                | Ambient set predicate $E$ , not an internal graph                                  | Reduct Replacement cannot be reused in the expansion.            |
| Pure-valued Collection               | Ambient-definable finite predicate expansion                                       | Does not collect arbitrary impure witnesses.                     |
| Finite checks                        | Ordinary Python execution                                                          | No transfinite or kernel-level verification.                     |

### 39.3 Bibliographic and repository audit notes

The research inspection used Glazer’s 2023/2024 global-choice paper, the February 2026 Glazer–Yao preprint, the Hamkins–Yao treatment of classical kernel models, Yao’s axiomatization paper, and the listed ProveIt pages. The live arXiv version of the Glazer–Yao paper inspected was version 1. No private professional information about Glazer was needed; the match of interests is established by his mathematical publications rather than inferred from a job title.

The limited search did not establish a published occurrence of the exact six-scheme formulation or of the combined spectrum-and-expansion package. That absence is not evidence sufficient to establish priority. Classical results and close precedents are credited in [section 24.3](#).

### 39.4 Non-claims of source 10

Every limitation stated by source 10, in its manuscript or in its proof-status note, is kept; most appear at their point of use. Collected: the manuscript is unrefereed; classical simplicity (Conway, Gonshor, Bertram) and the small-support arguments (Hamkins–Yao §1.5 with Lévy; Glazer–Yao, Proposition 1; Yao) are credited, and neither the six-scheme package nor the spectra is claimed to have priority; no Lean or Rocq development is included or claimed checked; the finite tests do not test infinite cardinals, class semantics or full axiom schemes; the full-Replacement conclusion in pure weak set theory assumes Hier; the first half uses only the sign-sequence order, not a field construction in the weak base; internal proper-class atoms can form a set in the ambient universe; cardinal computations in the support models use ambient Choice; Replacement for a reduct is not Replacement for an expanded language; pure-valued Collection is not Collection for impure witnesses; the models are relative interpretations, not consistency proofs; the expansion by  $E$  is not a reproof of Glazer’s global-choice theorem; Glazer–Yao’s reflection separations are background, not results here; the repository inspection was targeted at READMEs and directory organization, not a source audit or a build. No named conjecture is resolved.

### 39.5 Provenance of this part

**The manuscript.** Source 10 is batch-89 manuscript 03, delivered as the archive `Surreal_Cuts_Replacement_Atom_Support_Spectra.zip` in commit `7c0f2d9f9` and placed in commit `23adb85f9`. It names the repository commit `2b7b388ba` (“Catalogue batches 81 to 86”) as its inspected snapshot, and all the paths it names exist, unchanged, at the placement commit. Its manuscript (`article.tex`, 926 lines, 42 labels; a 23-page PDF), its README, its PDF and its checksum file `SHA256SUMS` (8 of 8 entries verified at placement) are not shipped; they survive in the arrival commit. Shipped, byte-identical: `10-cuts-replacement-PROOF_STATUS.md`, `code/10-cuts-replacement-verify_finite.py`, `code/10-cuts-replacement-build.sh`, `data/10-cuts-replacement-finite_checks.json` and `data/10-cuts-replacement-build_validation.json` (source 10’s own record of its 23-page build, not of this report). The Hilbert’s-tenth programme routed the batch-89 archives in commit `6aadaa4f2` (`Computability/HilbertTenthProblem/Papers/research-wip/native-stream-queue/review_topology_surreal_arrivals_20261003.md`): it read bounded ranges, ran nothing, made no correctness finding, and concluded that none of them gives a Diophantine compiler or an arithmetic saving.

**How it is printed.** The text from source 10’s title-page status box and abstract to its appendices is printed in full, in its order, as [section 24](#)–[section 39](#): the status box and the abstract open [section 24](#), its Section 1 became the next three subsections, and its two appendices became subsections of this section. Every delivered label carries the prefix `hset:zr::`; its ten questions gained labels `hset:zr:q:...`; its Theorems A–C are numbered ([theorems 24.1](#) to [24.3](#)). The symbols renamed are listed in [convention 24.4](#); the countermodel of [section 30](#) is written  $V_\theta$  instead of  $M$ ; delivered file names are replaced by the shipped ones; its references to Conway, Gonshor and Bertram use this report’s bibliography entries. Added for this report: [sections 24.4](#) and [24.5](#), the bracketed dated

notes, this subsection and the preceding one. No mathematical statement or proof of source 10 was changed.

**Where the merge had to choose.** (i) Placement: source 10 cites no report of the collection; it is placed here because its spine is the weak-base form of [theorem 10.4](#) ([remark 24.5](#)). (ii) Its Collection spectrum for the kernel models is printed in full here, with its own proofs, although the research-report collection prints the same theorem from an independent manuscript ([remark 24.6](#)). (iii) Its hierarchy axiom and set-graph scheme were renamed rather than left to collide with  $H_\kappa$  and SignGraph.

## Part VII

# Surreal universality without choice

## 40 Source 11: question, results, notation and relation to the report

This part was also added in batch 89. It prints the manuscript *Surreal Universality Without Choice: Exact coding strength, set-sized universal fields, and finite-transcendence collapse*, “A research manuscript prepared for Vladimir Reshetnikov”, October 3, 2026. It is batch-89 manuscript 04 and *source 11* in this report’s local numbering; its shipped files carry the prefix `11-choiceless-universality-`. It names the repository commit `2b7b388ba` (“Catalogue batches 81 to 86”; source 11 calls it a “tree identifier”, but it is a commit), arrived in commit `7c0f2d9f9` and was placed in commit `23adb85f9`. Its provenance and the mechanical changes made in printing it are in [section 53](#). Inside this part, “this manuscript”, “here”, “our” and “we” refer to source 11.

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**Research intersection (source 11).** Choiceless set theory and permutation models, as in the work of Elliot Glazer and Bokai Yao, meet the sign-sequence, Hahn-series, and omnific-integer foundations developed in the ProveIt repository.

**Status.** The manuscript supplies complete mathematical arguments for the stated deductions from explicitly identified classical background. It is an AI-generated, unrefereed research draft, not a Lean- or Rocq-checked development. Historical novelty is not certified. It does not claim to solve the reflection problems studied by Glazer and Yao, nor to transfer an atom-model counterexample automatically to pure set theory.

**Main message.** Freely naming independent surreal coordinates costs exactly one powerset-of-an-ordinal code. Preserving a given order is a separate requirement. In symmetric atom models, finite support places a sharp finite bound on how many independent coordinates any one representation can retain.

---

**Source 11’s summary.** We calibrate several universality assertions for the surreal numbers over set theory without the axiom of choice. For every set  $X$ , an injection  $X \rightarrow \mathbf{No}$  exists if and only if  $X$  injects into the powerset of an ordinal. An explicit double-monomial construction upgrades any such injection to algebraically independent positive infinite omnific integers. Consequently, the existence of embeddings of all free integer polynomial rings into  $\mathbf{Oz}$ , or all free real rational-function fields into  $\mathbf{No}$ , is exactly the first Kinna–Wagner principle.

For every infinite initial ordinal  $\kappa$ , we construct a set-sized real-closed subfield  $U_\kappa \subseteq \mathbf{No}$ , equinumerous with  $\mathcal{P}(\kappa)$  without choice, which realizes precisely the free real rational-function fields whose variable sets inject into  $\mathcal{P}(\kappa)$ . The cardinal upper bound uses canonical fraction and root codes rather than an unjustified choice of quotient representatives.

An orbit-factorization theorem then yields sharp bounds in permutation models. A polynomial representation into the pure surreals supported by  $s$  atoms has image-field transcendence degree at most  $s + 1$  in the basic Fraenkel model, and at most  $2s + 1$  in the ordered Mostowski model. Both bounds are attained. The latter model contains an ordered rational-function field with no embedding into its surreals, although every finitely generated-over- $\mathbb{R}$  coordinate subfield embeds in order. Compatible local embeddings would produce a global one, so the missing ingredient is coherence, not a defect of the finite algebra.

We separate coding universality from prescribed-order universality, give an ordinal convex-resolution criterion for order embeddings, identify the precise foundations required by the proofs, and propose a formalization and research program.

#### 40.1 The research question

The slogan that the surreal numbers are a universal ordered field suppresses several different quantifiers. Does one wish to assign distinct surreal coordinates to the elements of a set? To assign algebraically independent coordinates? To embed a field as a field, or as a field with its already specified order? Must the embedding belong to the universe in which the field was constructed? These questions coincide much less often when choice is unavailable.

This distinction provides a concrete meeting point between two research programs. Glazer and Yao's *Reflection Principles in ZFU* studies how assertions that agree under choice can separate when urelements and permutation symmetries are present [21]. ProveIt's surreal-number project develops sign-sequence foundations, normal forms, the real-closed surreal field, and the omnific-integer subring; it also explicitly distinguishes research claims from checked declarations [26]. Our subject is the choice cost of *representing labelled algebra inside those fields*.

The relevant weak-choice principle is classical. With the indexing convention of Karagila and Schilhan [24],

$$\text{KWP}_1 : \quad \forall X \exists \alpha \in \text{Ord} \quad X \hookrightarrow \mathcal{P}(\alpha). \quad (40.1)$$

Its coding formulation is not a new principle introduced here. Our contribution is an explicit algebraic calibration, a set-sized optimal target at each code level, and a quantitative analysis of how symmetry defeats the same constructions internally.

#### 40.2 What is proved, and what is not claimed

The principal questions answered in this manuscript are as follows.

**How much choice is required to realize arbitrary independent variables in  $\mathbf{No}$  or  $\mathbf{Oz}$ ?**

Exactly  $\text{KWP}_1$ , for the free-algebra assertions specified in [theorem 43.5](#). The result is an equivalence over ZF, not merely an implication from choice.

**Can one use an optimal set-sized real-closed target at a fixed ordinal code level?**

Yes. The field  $U_\kappa$  of [theorem 44.3](#) has the same cardinality, in the sense of an actual bijection, as  $\mathcal{P}(\kappa)$ .

### What does finite support force a polynomial representation to forget?

It identifies variables in the same stabilizer orbit. In two standard atom models this gives the sharp bounds  $s + 1$  and  $2s + 1$  of [theorems 48.1](#) and [48.3](#).

### Does finite-stage embeddability ensure a global surreal embedding?

No, in the ordered permutation model. It does so precisely when the finite-stage maps can be assembled coherently, as formalized in [theorem 49.3](#).

These are natural questions formulated and answered here; they are not presented as named, previously published open problems. The elementary coding observation, finite normal-form identities, polynomial universal property, and standard permutation-model construction are established machinery. The possible novelty lies in the exact combined statements and the quantitative refinements. A targeted literature review did not establish historical priority for those formulations.

## 40.3 Reading paths

For the main positive results, read [sections 41](#) to [44](#). For the negative results, read [sections 47](#) to [49](#). The prescribed-order issue is isolated in [sections 45](#) and [46](#). The final sections discuss proof boundaries, formalization, and research questions. All displayed embeddings into a proper class have set-sized domains and set graphs.

## 40.4 Notation of this part

**Convention 40.1** (Symbols of source 11). In this part:

| Notion in source 11                                                              | Here                                                              | Do not confuse with                                                                                                                                                             |
|----------------------------------------------------------------------------------|-------------------------------------------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Birthday of a sign sequence                                                      | $b(s)$ , an ordinal (source: $b$ )                                | the same function; Parts I–V regard its value as a surreal ordinal                                                                                                              |
| Sign sequences of length at most $\alpha$                                        | $\mathbf{No}_{\leq \alpha}$ ( $= \mathbf{No}_{< \alpha+1}$ )      | the cutoff carrier $\mathbf{No}_{< \lambda}$ of Parts I–V: at an infinite initial $\kappa$ , $\mathbf{No}_{\leq \kappa}$ also contains the sequences of length exactly $\kappa$ |
| Sign sequences of length exactly $\kappa$                                        | $\mathbb{S}_\kappa = \{-, +\}^\kappa$ (source: $S_\kappa$ )       | $S_\kappa$ , sources 05 and 06's name for the carrier $\mathbf{No}_{< \kappa}$ ( <a href="#">convention 1.5</a> )                                                               |
| Free real field on $\mathbb{S}_\kappa$                                           | $\mathcal{R}_\kappa$ (source: $K_\kappa$ )                        | the collection's $K$ , a complex field, and source 09's $K_\lambda$ (renamed $\mathbf{No}_{< \lambda}$ )                                                                        |
| Sign-to-subset code                                                              | $\text{enc}_\alpha(s)$ (source: $C_\alpha$ )                      | the relation codes $C = (\delta, r, t)$ of <a href="#">section 5</a>                                                                                                            |
| Least code level, least birthday level                                           | $\text{cl}(X)$ , $\text{bl}(X)$ (source: $\rho(X)$ , $\beta(X)$ ) | $\beta$ , the first new birthday, and $\varrho$ , the eternity bound                                                                                                            |
| A group acting on variables                                                      | $G_0$ (source: $H$ )                                              | $H_\kappa$ , the hereditary-size universe                                                                                                                                       |
| Generic ordinal variable                                                         | $\zeta$ (source: $\beta$ )                                        | $\beta$ (reserved)                                                                                                                                                              |
| An injection $X \rightarrow \mathcal{P}(\alpha)$ in <a href="#">theorem 42.1</a> | $c$                                                               | the conjugation $c$ of <a href="#">section 19</a>                                                                                                                               |
| Omnific integers; first Kinna–Wagner principle                                   | $\mathbf{Oz}$ ; $\text{KWP}_1$                                    | —                                                                                                                                                                               |
| Length of a convex resolution; limit stages                                      | $\theta$ ; $\lambda$                                              | the pairing width $\theta$ of <a href="#">section 4</a> ; the epsilon-number cutoffs $\lambda$                                                                                  |
| Permutation model; atom support size                                             | $W$ ; $s$                                                         | $s$ also denotes a sign sequence, as in the source                                                                                                                              |

Source 11's exponent sums  $E_m$  and the equivalence relations  $E_\xi$  of a convex resolution are its own two uses of  $E$ ; neither is the enumeration predicate  $E$  of [Part VI](#). Typography only: theory names (ZF, ZFA, AC, KWP) are set upright, as in the rest of this report.

## 40.5 Relation to the rest of the report and to the collection

*Remark 40.2* (The open item of [Question 21.7](#), re-scoped). [Question 21.7](#) lists, as source 06's open item, “a universal embedding theorem without Global Choice”. Source 11 answers it in part, over ZF: coordinate universality (every set of variables realized by algebraically independent surreals, or every free real rational-function field embedded in **No**) is exactly  $\text{KWP}_1$  ([theorem 43.5](#)); at each code level the set-sized real-closed field  $U_\kappa$  is an optimal target ([theorem 44.3](#)); every linear order whose underlying set is well-orderable order-embeds in **No** ([proposition 45.3](#)), and order embeddability is characterized by convex resolutions ([theorem 46.2](#)); and in a ZFA permutation model an ordered field all of whose finitely generated coordinate subfields embed has no embedding ([theorem 49.1](#)). It leaves open universality for prescribed orders and for general ordered fields without choice (its [questions 51.1](#), [51.2](#) and [51.5](#)), and any ZF transfer of the ZFA counterexample. Source 11 did not know this report.

*Remark 40.3* (Comparison with the choice-free [Section 13](#)). [Theorem 13.2](#) (source 06) identifies, over ZF, the range of the pointed ordinal-graph *interpretation* of sets: exactly HWO, the sets whose transitive closure is well-orderable. [Theorem 42.1](#) concerns *injections* of a set into **No**:  $X \hookrightarrow \mathbf{No}$  iff  $X \hookrightarrow \mathcal{P}(\alpha)$  for some  $\alpha$ . The second class is the wider one. If  $x \in \text{HWO}$  then  $x \subseteq \text{TC}(\{x\})$  is well-orderable and so injects into an ordinal; but  $\mathcal{P}(\omega)$  always injects into **No** (take  $\alpha = \omega$ ), while in a model of ZF where  $\mathcal{P}(\omega)$  is not well-orderable it is not in HWO ([proposition 13.3](#) uses the same hypothesis). Under Choice both classes are everything. A set coded by a surreal in the sense of [theorem 13.2](#) is recovered as a set; an injection into **No** only names its elements. This comparison is an observation of this merge.

*Remark 40.4* (Other reports). (i) [Theorem 43.1](#) is the case  $K = \mathbb{R}$ ,  $G = \{0\}$  of [Lemma 8.1](#) (dsn:lem:cosets, “Disjoint support cosets”, that report's source 04) of [foundations-and-computation/definable-surreals-and-omnific-integers](#): a nonzero real has normal-form support  $\{0\}$ , and distinct Conway monomials  $e_x$  have no nonzero integer relation, so the family  $\omega^{e_x}$  is algebraically independent over  $\mathbb{R}$ . Source 11's proof is printed as its own; new beside that lemma are the omnific-integer conclusion, the free-algebra calibration and the ZF framing. (ii) The report [foundations-and-computation/surreal-well-orders](#) proves over GB without set choice that Global Choice holds iff **No** has a class well-order (its [Theorem 13.1](#), swo:cf:choice; see the dated note after [remark 13.4](#)); source 11 instead works with sets and no global principle, and calibrates coordinate universality by  $\text{KWP}_1$ . That report also records the ZFC count  $|\mathbf{No}_{<\kappa}| = 2^{<\kappa}$ ; source 11's [proposition 42.3](#) is a ZF bijection at the closed bound  $\mathbf{No}_{\leq\kappa}$ . Its [Part XV](#) (batch-89 manuscript 01) studies countable global choice over Zermelo set theory: a related theme, different theorems. (iii) No other report of the collection treats Kinna–Wagner principles, permutation models or choiceless universality.

[Added 4 October 2026, batch 90: two later reports bear on this part; both are AI-assisted, unrefereed and not formalized, and neither treats the subjects of item (iii). [Part XII](#) of [foundations-and-computation/polish-models-of-omnific-arithmetic](#) (its source 18, batch-90 manuscript 03) works in a class theory with Foundation omitted, not negated, and without Global Choice. It proves that a connected real class manifold admitting a class injection into, or a class surjection from,  $\mathbf{No}^m$  is a set, of cardinality  $2^{\aleph_0}$  if nonempty (its [Theorem 163.2](#), pma:gcm:thm:surreal), using the birthday stages  $\mathbf{No}_{<\beta}$  as an ordinal exhaustion; its [Corollary 163.3](#) (pma:gcm:cor:birthdaybound) then

bounds the birthdays of the points of such a carrier by Replacement, as the proof of [theorem 42.1](#) does for a set injected into **No**, and a note there points here. No birthday cutoff uniform over all manifolds is claimed, and no theorem is shared. Lemma 3.2 (`csf:lem:independence`) of `foundations-and-computation/cantor-families-of-surreal-subfields` (batch-90 manuscript 04) proves the analogue of [theorem 43.1](#) with real algebraic exponent coefficients: the monomials  $\omega^{b\omega^q}$ , for  $q \in \mathbb{Q}$  and  $b \in \{1\} \cup \{\sqrt{p} : p \text{ prime}\}$ , are algebraically independent over the field of real algebraic numbers, by the same leading-term argument. It is likewise a case of `dsn:lem:cosets`, and a note after it credits [theorem 43.1](#). That report works in ZFC and uses the family as a transcendence basis of a countable real closed field whose relative real closures form a Cantor family of subfields, not for free-algebra universality.]

## 41 Foundations, notation, and the surreal interface

### 41.1 Set theory and classes

Unless otherwise specified, the positive results are theorems over ZF. We use classes only as definable predicates and definable operations on sets. Equivalently, one may work in a conservative class-theoretic presentation without adding set choice or global choice. The permutation-model sections work in ZFA: set theory with a *set* of atoms. This is a special setting within urelement theories, not the unrestricted proper-class-of-atoms setting of all results in [21].

A surreal number is represented by a function

$$s : \zeta \longrightarrow \{-, +\}, \quad \zeta \in \text{Ord}.$$

Its birthday is  $b(s) = \zeta$ . The totality **No** of such functions is a proper class. We write

$$\mathbf{No}_{\leq \alpha} = \bigcup_{\zeta \leq \alpha} \{-, +\}^\zeta.$$

This is a set. An expression  $f : X \rightarrow \mathbf{No}$  means that  $f$  is a set of ordered pairs, its domain is the set  $X$ , and every value satisfies the defining predicate for **No**. Replacement therefore applies to the values and to their birthdays. We never take a powerset of **No**.

In ZFA, the signs, ordinals, and sign sequences are pure sets: their transitive closures contain no atoms. Atom permutations fix each surreal number pointwise. A function whose arguments involve atoms need not be pure, even if all its values are surreals. This difference drives the counterexamples.

### 41.2 The standard surreal facts used

We use the following classical interface, with no choice principle added to the ambient set theory. Conway's foundational construction is the underlying reference [1]; Costin and Ehrlich explicitly develop their positive surreal theory in a class framework without set or class choice [25].

- (S1) **No** is an ordered real-closed field containing the canonically embedded ordered field  $\mathbb{R}$ .
- (S2) The Conway monomial map  $x \mapsto \omega^x$  is strictly increasing, has positive values, and satisfies

$$\omega^{x+y} = \omega^x \omega^y, \quad \omega^0 = 1.$$

Here  $\omega^x$  denotes Conway's monomial, not the separate Gonshor exponential function.

(S3) If  $x_1 > \cdots > x_n$  and  $r_i \in \mathbb{R} \setminus \{0\}$ , the finite normal form

$$\sum_{i=1}^n r_i \omega^{x_i}$$

is nonzero and has the sign of  $r_1$ . In particular, distinct monomials are linearly independent over  $\mathbb{R}$ .

(S4) A surreal is omnific when its normal-form exponents are nonnegative and its constant coefficient is an ordinary integer. These numbers form the subring **Oz**. In particular, a monomial  $\omega^e$  with  $e > 0$  is a positive infinite omnific integer.

(S5) For set-sized  $A, B \subseteq \mathbf{No}$  with  $A < B$ , there is a unique simplest surreal strictly between them. This is the usual set-cut construction.

Only finite normal forms are needed for the independence and symmetry theorems. The set-sized real-closed-target theorem additionally uses real closedness. No theorem about arbitrary rearrangements of infinite Hahn sums, surreal integration, or sharp product birthdays is used.

*Remark 41.1* (Three meanings of support). A polynomial has a finite *variable support*; a Hahn or Conway series has a *normal-form support*; an object in a permutation model has an *atom support*. They are different notions. In particular, a polynomial homomorphism with finite atom support can have a domain with infinitely many variables.

### 41.3 Algebra on arbitrary sets of variables

For any set  $X$ , let  $\mathbb{N}^{(X)}$  be the set of finitely supported functions  $X \rightarrow \mathbb{N}$ . The ring  $k[T_X]$  consists of finitely supported coefficient functions on  $\mathbb{N}^{(X)}$ . Addition is pointwise and multiplication is finite convolution. This definition requires no ordering or enumeration of  $X$ . If  $k$  is a field, the resulting ring is a domain, and its fraction field is denoted  $k(T_X)$ .

An embedding of  $\mathbb{R}$ -algebras fixes the specified copy of  $\mathbb{R}$ . A unital field homomorphism is automatically injective. A field embedding, unless explicitly called ordered, need not preserve a prescribed order on the source. Finite transcendence degrees below are ordinary nonnegative integers; no existence of an arbitrary infinite transcendence basis is assumed.

## 42 The exact coding strength of a surreal coordinate set

**Theorem 42.1** (Ordinal-code equivalence). *For every set  $X$ , the following are equivalent over ZF:*

$$X \hookrightarrow \mathbf{No} \quad \Longleftrightarrow \quad \exists \alpha \in \text{Ord} \quad X \hookrightarrow \mathcal{P}(\alpha).$$

*The same equivalence holds in ZFA with pure sign-sequence surreals.*

*Proof.* Suppose first that  $j : X \rightarrow \mathbf{No}$  is injective. By Replacement, the set of birthdays of its values exists. Let

$$\alpha = \sup\{b(j(x)) : x \in X\},$$

with  $\alpha = 0$  if  $X = \emptyset$ . Every value belongs to  $\mathbf{No}_{\leq \alpha}$ .

For a sign sequence  $s$  of length  $\zeta \leq \alpha$ , put

$$\text{enc}_\alpha(s) = \{2 \cdot \xi + 1 : \xi < \zeta\} \cup \{2 \cdot \xi : \xi < \zeta \text{ and } s(\xi) = +\}. \quad (42.1)$$

This is a subset of the ordinal  $2 \cdot \alpha$ . Multiplication here is ordinal multiplication on the indicated side:  $2 \cdot \alpha$  is the sum of  $\alpha$  consecutive two-point blocks. The odd position of each block records whether the sign is defined; the even position records whether it is positive. Thus the code uniquely determines both the domain and every sign, so  $\text{enc}_\alpha$  is injective. Composing with  $j$  proves the forward implication.

Conversely, given an injection  $c : X \rightarrow \mathcal{P}(\alpha)$ , let  $s_x$  be the sign sequence of exact length  $\alpha$  defined by

$$s_x(\xi) = + \iff \xi \in c(x).$$

Different codes give different surreals. This definition uses only Separation and Replacement and works unchanged when the members of  $X$  are atoms or sets involving atoms.  $\square$

*Remark 42.2* (Why definedness must be encoded). Coding a sign sequence solely by its positive positions loses information: the empty sequence and the one-symbol sequence – would both have empty code. The first half of (42.1) prevents that error. Padding by a third symbol, as in section 46, is an alternative.

**Proposition 42.3** (Exact birthday capacity). *For every infinite initial ordinal  $\kappa$ , there is a bijection*

$$\mathbf{No}_{\leq \kappa} \simeq \mathcal{P}(\kappa)$$

*in ZF. For a finite ordinal  $n$ ,*

$$|\mathbf{No}_{\leq n}| = 2^{n+1} - 1.$$

*Proof.* Subsets of  $\kappa$  inject into the sign sequences of exact length  $\kappa$ . In the other direction, (42.1) maps  $\mathbf{No}_{\leq \kappa}$  into  $\mathcal{P}(2 \cdot \kappa)$ . The ordinal  $2 \cdot \kappa$  is equipotent with  $\kappa$ : finite products of an infinite well-orderable cardinal do not require choice. Transporting subsets along such a bijection and applying Cantor–Bernstein gives the result. At finite  $n$ , sum the  $2^i$  sign sequences of each length  $0 \leq i \leq n$ .  $\square$

**Corollary 42.4** (Coincidence of least code and birthday budgets). *Let  $X$  be an infinite set admitting the embeddings in theorem 42.1. Define*

$$\text{cl}(X) = \min\{\alpha : X \hookrightarrow \mathcal{P}(\alpha)\}, \quad \text{bl}(X) = \min\{\alpha : X \hookrightarrow \mathbf{No}_{\leq \alpha}\}.$$

*Then  $\text{cl}(X) = \text{bl}(X)$ , and their common value is an infinite initial ordinal.*

*Proof.* Each minimum exists by taking the least ordinal below an existing witness. Neither is finite, since a set injecting into a finite set is finite. If an infinite witness ordinal  $\alpha$  is not initial, replace it by its initial ordinal  $\kappa < \alpha$ . Subsets of  $\alpha$  and of  $\kappa$  are equipotent. Moreover, the proof of proposition 42.3 shows  $\mathbf{No}_{\leq \alpha} \hookrightarrow \mathcal{P}(\kappa) \hookrightarrow \mathbf{No}_{\leq \kappa}$ . Thus a least witness is initial. At infinite initial ordinals the two target sets have equal capacity by proposition 42.3.  $\square$

The finite restriction matters. A three-element set fits into  $\mathbf{No}_{\leq 1}$ , whereas it does not fit into  $\mathcal{P}(1)$ .

### 43 Double monomials and free-algebra universality

**Theorem 43.1** (Uniform algebraic independence). *Let  $j : X \rightarrow \mathbf{No}$  be an injection. Define*

$$e_x = \omega^{j(x)}, \quad t_x = \omega^{e_x}. \quad (43.1)$$

*Then  $(t_x)_{x \in X}$  is algebraically independent over  $\mathbb{R}$ . Every  $t_x$  is a positive infinite omnific integer. Evaluation gives embeddings*

$$\mathbb{Z}[T_X] \hookrightarrow \mathbf{Oz}, \quad \mathbb{R}[T_X] \hookrightarrow \mathbf{No}, \quad \mathbb{R}(T_X) \hookrightarrow \mathbf{No}.$$

*All these maps are definable from  $j$  and have set graphs.*

*Proof.* The numbers  $e_x$  are distinct Conway monomials. By finite normal-form uniqueness they are linearly independent over  $\mathbb{R}$ , and in particular over  $\mathbb{Z}$ .

For  $m \in \mathbb{N}^{(X)}$ , define the finite sum

$$E_m = \sum_{x \in X} m(x)e_x.$$

If  $m \neq n$ , the finite linear combination  $\sum_x (m(x) - n(x))e_x$  is nonzero. Hence  $E_m \neq E_n$ . The multiplication law for Conway monomials gives

$$\prod_x t_x^{m(x)} = \omega^{E_m}.$$

Now take a nonzero polynomial, written after collecting like monomials as

$$p = \sum_{m \in F} a_m T^m,$$

where  $F$  is finite and every  $a_m \neq 0$ . Evaluation produces

$$p(t) = \sum_{m \in F} a_m \omega^{E_m}. \quad (43.2)$$

The exponents are distinct. They can be sorted because they form a finite subset of a linear order. The resulting finite normal form is nonzero, proving algebraic independence.

Each  $e_x > 0$ , regardless of the sign or magnitude of  $j(x)$ . Thus  $t_x = \omega^{e_x}$  has a single positive exponent and is a positive infinite element of  $\mathbf{Oz}$ . In (43.2), every nonconstant monomial has exponent  $E_m > 0$ . If the coefficients are integers, the only possible constant coefficient is an integer, so the value is omnific. Evaluation is therefore injective on both polynomial rings in the stated targets. Since nonzero denominators do not vanish, the real polynomial embedding extends uniquely to its fraction field. Replacement on the set of polynomials and fractions gives the graphs of the maps.  $\square$

**Example 43.2** (Why the second monomial layer is needed). An arbitrary injective assignment  $j$  does not make  $\omega^{j(x)}$  algebraically independent. For labels  $j(x_0) = 0$ ,  $j(x_1) = 1$ , and  $j(x_2) = 2$ , the assigned values are  $1, \omega, \omega^2$ , and two immediate polynomial relations hold. The second layer in (43.1) makes the *exponents* linearly independent before using them to build algebraically independent monomials.

**Example 43.3** (A concrete leading-term calculation). With two labels  $j(a) = 0$  and  $j(b) = 1$ , the double-monomial values are  $t_a = \omega$  and  $t_b = \omega^\omega$ . For instance,

$$T_b^2 - 3T_aT_b + T_a^5 \quad \mapsto \quad \omega^{2\omega} - 3\omega^{\omega+1} + \omega^5 > 0.$$

The three displayed exponents are distinct and decreasing. This illustrates the proof mechanism: a formal monomial becomes a Conway monomial, and the greatest surviving exponent certifies nonvanishing. The products and sums in these exponents are surreal field operations, not the ordinal multiplication used in the coding formula.

**Corollary 43.4** (Independent infinitesimal coordinates). *Under the hypotheses of [theorem 43.1](#), the numbers*

$$u_x = t_x^{-1} = \omega^{-e_x}$$

*are algebraically independent positive infinitesimals. They are not omnific integers.*

*Proof.* They are infinitesimal because  $-e_x < 0$ . A polynomial relation among finitely many inverses, multiplied by a sufficiently large product of the corresponding  $t_x$ , would give a nonzero polynomial relation among the  $t_x$ . Their negative normal-form exponents exclude them from **Oz**.  $\square$

**Theorem 43.5** (Equivalent forms of coordinate universality). *For a set  $X$ , the following conditions are equivalent over ZF:*

- (i)  $X \hookrightarrow \mathcal{P}(\alpha)$  for some ordinal  $\alpha$ ;
- (ii)  $X \hookrightarrow \mathbf{No}$ ;
- (iii)  $X \hookrightarrow \mathbf{Oz}$ ;
- (iv) there is a unital ring embedding  $\mathbb{Z}[T_X] \hookrightarrow \mathbf{Oz}$ ;
- (v) there is an  $\mathbb{R}$ -algebra embedding  $\mathbb{R}[T_X] \hookrightarrow \mathbf{No}$ ;
- (vi) there is an  $\mathbb{R}$ -field embedding  $\mathbb{R}(T_X) \hookrightarrow \mathbf{No}$ ;
- (vii) there is an  $X$ -indexed algebraically independent family over  $\mathbb{R}$  consisting of positive infinite omnific integers.

*Consequently, the universal closure over all sets  $X$  of any of these assertions is exactly  $\text{KWP}_1$ .*

*Proof.* The equivalence of (i) and (ii) is [theorem 42.1](#). From (ii), [theorem 43.1](#) gives (iv)–(vii) and also (iii). Conversely, (iii) implies (ii) by inclusion. An embedding in (iv), (v), or (vi) assigns distinct values to the distinct indeterminates: otherwise it kills the nonzero polynomial  $T_x - T_y$ . Thus each yields (ii). Algebraic independence likewise gives distinct values, so (vii) yields (ii). Universal closure and [\(40.1\)](#) prove the last statement.  $\square$

**Remark 43.6** (Not an equivalence with full choice). In pure ZF,  $\text{KWP}_1$  is weaker than AC; Cohen’s first model is a standard example satisfying the former and not the latter, as reviewed in [\[24\]](#). Thus these *free-algebra* universality assertions do not recover the well-ordering theorem. They also do not yet assert anything about preserving an independently prescribed order on  $X$  or on a field.

## 44 An optimal set-sized real-closed target

The preceding theorem places all realizations in the proper class **No**. It is useful, both mathematically and for formalization, to obtain a target that is an ordinary set and has exactly the required coding capacity.

Fix an infinite initial ordinal  $\kappa$ . Let

$$\mathbb{S}_\kappa = \{-, +\}^\kappa, \quad t_s = \omega^{\omega^s} \quad (s \in \mathbb{S}_\kappa),$$

where each  $s$  is identified with its surreal value. Set

$$\mathcal{R}_\kappa = \mathbb{R}(t_s : s \in \mathbb{S}_\kappa), \quad U_\kappa = \{u \in \mathbf{No} : u \text{ is algebraic over } \mathcal{R}_\kappa\}. \quad (44.1)$$

The notation on the right initially defines a class; the next proof shows that it is a set.

#### 44.1 A choice-free coding lemma

**Lemma 44.1** (Finite syntax at a powerset level). *Let  $\kappa$  be an infinite initial ordinal. Suppose that  $D$  and  $B$  inject into  $\mathcal{P}(\kappa)$ , that  $D$  is linearly ordered, and that  $B$  is a field with a fixed linear order on its underlying set. Then each of the following admits an injection into  $\mathcal{P}(\kappa)$ :*

$$B[T_D], \quad B(T_D), \quad \text{any ordered algebraic extension of } B(T_D).$$

*For the last assertion the extension is a set, or is a definable class in which each polynomial has only finitely many roots and ordered root indexing applies.*

*Proof.* Finite lists of subsets of  $\kappa$  can be encoded by a subset of  $\kappa$ : first tag the list length and each entry in  $\omega \times \kappa$ , then use a fixed bijection  $\omega \times \kappa \simeq \kappa$ . These are well-orderable sets, so this bijection needs no choice. Naturals and finite auxiliary flags can be accommodated by additional tags.

Every finite subset of  $D$  can be uniquely listed in increasing order. A monomial is therefore coded by an increasing finite variable list together with its positive integer exponents. Monomials are linearly ordered by comparing exponent vectors at the greatest variable where they differ. Every finite polynomial has a unique list of its nonzero terms in this order. Encoding these lists and their coefficients proves the assertion for polynomials.

A fraction requires more care. It would be invalid, without further argument, to infer that an arbitrary quotient of a  $\mathcal{P}(\kappa)$ -codable set is again  $\mathcal{P}(\kappa)$ -codable. Instead use the unique reduced representative

$$\frac{p}{q}, \quad \gcd(p, q) = 1, \quad \text{lc}(q) = 1, \quad (44.2)$$

with 0 represented as 0/1.

Here leading coefficients are taken with respect to the preceding largest-variable monomial order. They exist because a polynomial has finitely many terms. The polynomial ring in an arbitrary set of variables over a field is a unique factorization domain in the finite sense needed here: put  $p$  and  $q$  in the ring generated by their finite union of variables and use finite-variable factorization. A factor cannot introduce a new variable, since degree in that variable is additive in a polynomial domain. Gcds computed in the finite subring therefore remain gcds in the full ring. Dividing out a gcd and scaling by the leading coefficient of the denominator produces (44.2). Uniqueness follows from coprimality and cancellation in a UFD. These are existence-and-uniqueness arguments, not selections from an unstructured family of equivalence classes. Thus a fraction is injected into a pair of polynomial codes.

Finally, an algebraic element  $a$  of an ordered extension has a unique monic minimal polynomial over  $B(T_D)$ . Attach to it the position of  $a$  among the finitely many roots of that polynomial *in the extension*, in increasing order. Polynomial plus finite index uniquely determines  $a$ . This is an injection into finite lists of field elements with a natural-number tag, and hence into  $\mathcal{P}(\kappa)$ .  $\square$

*Remark 44.2.* The order on  $D$  is not required to be a well-order, and the monomial order is not claimed to well-order all monomials. Only the maximum of each finite support and compatibility with addition of exponent vectors are used. Finite-variable gcd theory can use a separate finite enumeration when proving existence; the normalization in (44.2) makes the answer independent of that temporary enumeration.

**Theorem 44.3** (Optimal code-level universal field). *Over ZF, for every infinite initial ordinal  $\kappa$ , the class  $U_\kappa$  in (44.1) is a set-sized real-closed subfield of  $\mathbf{No}$  and*

$$U_\kappa \simeq \mathcal{P}(\kappa). \quad (44.3)$$

For every set  $X$ ,

$$X \hookrightarrow \mathcal{P}(\kappa) \iff \mathbb{R}(T_X) \hookrightarrow_{\mathbb{R}} U_\kappa, \quad (44.4)$$

$$X \hookrightarrow \mathcal{P}(\kappa) \iff \mathbb{Z}[T_X] \hookrightarrow U_\kappa \cap \mathbf{Oz}. \quad (44.5)$$

The embeddings in (44.4) do not have to preserve a separately prescribed order on the source.

*Proof.* By theorem 43.1, the  $t_s$  are algebraically independent over  $\mathbb{R}$ . The set of finite rational expressions in them is a set, and their evaluation is definable. Replacement gives the set  $\mathcal{R}_\kappa$ , together with an isomorphism from  $\mathbb{R}(T_{\mathbb{S}_\kappa})$ .

For each nonzero polynomial  $p \in \mathcal{R}_\kappa[Y]$ , there are at most  $\deg p$  distinct roots in the field  $\mathbf{No}$ . Its definable class of roots is therefore a finite set. More explicitly, the assertion that a definable class has at most  $n$  members allows its members to be collected using finite induction; this is not a choice principle. The roots form a linearly ordered finite set. The function that assigns to  $(p, i)$  its  $i$ -th root, when present, and 0 otherwise is a definable function on a set. Replacement shows that the union of all these root sets is a set. It is exactly  $U_\kappa$ .

Algebraic elements are closed under field operations, so  $U_\kappa$  is an ordered field algebraic over  $\mathcal{R}_\kappa$ . If  $a > 0$  belongs to it, its positive square root in  $\mathbf{No}$  is algebraic over  $\mathcal{R}_\kappa$  and thus lies in  $U_\kappa$ . If an odd-degree polynomial has coefficients in  $U_\kappa$ , take one of its roots in the real-closed field  $\mathbf{No}$ . Its finitely many coefficients are algebraic over  $\mathcal{R}_\kappa$ , and algebraicity is transitive, so the root belongs to  $U_\kappa$ . The usual square-root/odd-degree criterion proves that  $U_\kappa$  is real closed. This is a real closure of  $\mathcal{R}_\kappa$  inside the chosen ordered field  $\mathbf{No}$ .

For the cardinal upper bound,  $\mathbb{S}_\kappa \simeq \mathcal{P}(\kappa)$  by its definition and is linearly ordered by surreal order. The reals inject into  $\mathcal{P}(\omega)$  by their rational Dedekind cuts, and thus into  $\mathcal{P}(\kappa)$ . Apply lemma 44.1, using canonical reduced fractions and minimal-polynomial/root-index codes. It gives an injection  $U_\kappa \hookrightarrow \mathcal{P}(\kappa)$ . Conversely,  $s \mapsto t_s$  injects  $\mathbb{S}_\kappa$  into  $U_\kappa$ . Cantor–Bernstein proves (44.3).

Suppose now  $c : X \hookrightarrow \mathcal{P}(\kappa)$ . Convert each code to its length- $\kappa$  sign sequence  $s_x$ , and send  $T_x$  to  $t_{s_x}$ . Algebraic independence gives the embedding into  $\mathcal{R}_\kappa \subseteq U_\kappa$ , and integer polynomial values are omnific. This proves the forward implications in (44.4) and (44.5). Either embedding sends distinct variables to distinct elements of  $U_\kappa$ ; compose with its established code to obtain  $X \hookrightarrow \mathcal{P}(\kappa)$ .  $\square$

## 44.2 What optimality means here

The theorem is sharp in a precise coding sense. Any set-sized target that contains an injective realization of  $\mathbb{S}_\kappa$  many independent variables must admit an injection from  $\mathcal{P}(\kappa)$ . Our field attains that lower bound and is real closed. We are *not* claiming that every ordered field whose underlying set injects into  $\mathcal{P}(\kappa)$  order-embeds in  $U_\kappa$ .

Nor do we claim a sharp explicit birthday bound for  $U_\kappa$ . Such a bound exists by Replacement because  $U_\kappa$  is a set. This qualitative fact must not be confused with an optimal formula for birthdays under multiplication, inversion, or taking algebraic roots.

## 45 Prescribed order is a different universality problem

Let  $L$  be a linearly ordered set. The free abelian group  $\mathbb{Z}^{(L)}$  has a largest-variable order: for distinct  $m, n$ , let  $\ell$  be the greatest member of the finite set on which they differ, and declare

$$m < n \iff m(\ell) < n(\ell). \quad (45.1)$$

This is a translation-invariant linear order. Define an order on  $\mathbb{R}[T_L]$  by saying that a nonzero polynomial is positive when the coefficient of its greatest monomial is positive. Extend it to the fraction field, denoted  $F(L)$ .

**Lemma 45.1** (Largest-term ordering). *The preceding construction makes  $F(L) = \mathbb{R}(T_L)$  an ordered field over  $\mathbb{R}$ , without requiring a well-order of  $L$ . Its variables are positive and satisfy*

$$\ell < \ell' \implies T_\ell < T_{\ell'}.$$

*Every order automorphism of  $L$  induces an order automorphism of  $F(L)$ .*

*Proof.* Translation invariance in (45.1) is immediate because adding the same vector leaves the greatest differing coordinate and the sign of the difference unchanged. A polynomial has a finite nonempty set of monomials, hence a greatest one. If  $m_0, n_0$  are the greatest monomials of nonzero  $p, q$ , then  $m_0 + n_0$  is the unique greatest exponent vector arising in their product: any strict decrease in either summand strictly decreases the sum. Consequently,

$$\text{lm}(pq) = \text{lm}(p) + \text{lm}(q), \quad \text{lc}(pq) = \text{lc}(p) \text{lc}(q).$$

This proves closure of positive elements under multiplication. For addition, unequal leading monomials retain the greater positive term, while equal leading monomials have positive coefficient sum. Exactly one of  $p, -p$  is positive for nonzero  $p$ . These observations give an ordered domain, whose order extends uniquely to its fraction field. The largest variable in  $T_{\ell'} - T_\ell$  is  $\ell'$ , with coefficient 1. All definitions are preserved by an order automorphism of  $L$ .  $\square$

**Theorem 45.2** (Exact reduction of order universality). *For every linearly ordered set  $L$ , over  $\mathbf{ZF}$ ,*

$$L \hookrightarrow_{\text{ord}} \mathbf{No} \iff F(L) \hookrightarrow_{\text{ord}} \mathbf{No}.$$

*In the forward direction the field embedding fixes  $\mathbb{R}$ .*

*Proof.* Suppose  $j : L \hookrightarrow_{\text{ord}} \mathbf{No}$ . Use  $t_\ell = \omega^{\omega^{j(\ell)}}$  as in [theorem 43.1](#). For distinct exponent vectors  $m, n$ , the sign of

$$E_m - E_n = \sum_{\ell} (m(\ell) - n(\ell)) \omega^{j(\ell)}$$

is determined by the greatest  $\ell$  where the coefficient is nonzero, because  $j$  preserves order and finite normal-form signs are determined by the greatest exponent. Thus  $m < n$  in (45.1) exactly when  $E_m < E_n$  in  $\mathbf{No}$ . The sign of an evaluated polynomial is its largest-term coefficient, so the resulting polynomial and fraction-field embeddings preserve order.

Conversely, restricting any ordered embedding  $F(L) \rightarrow \mathbf{No}$  to the strictly increasing family of variables gives an ordered injection of  $L$ .  $\square$

**Proposition 45.3** (Well-orderable domains). *Every linear order whose underlying set is well-orderable order-embeds into  $\mathbf{No}$  in ZF.*

*Proof.* Fix one well-ordering and its enumeration  $(x_\xi)_{\xi < \theta}$ . Recursively assign to  $x_\xi$  the simplest surreal between the images of earlier elements below it and the images of earlier elements above it. Those two collections are sets, are separated by the induction hypothesis, and the simplest choice is unique. Transfinite recursion and Replacement construct a set graph of length  $\theta$ . No new choice is made at any stage. The result preserves all comparisons.  $\square$

*Remark 45.4* (A gap that must not be silently bridged). An injection of the underlying set of  $L$  into  $\mathcal{P}(\alpha)$  gives it *some* surreal-induced order. It does not say that an already specified order on  $L$  is respected. Thus [theorem 43.5](#) is not, by itself, a proof that  $\text{KWP}_1$  implies ordered universality for all linear orders or all ordered fields. We leave that implication as a separately stated research question rather than conflating it with the proved equivalence.

## 46 A convex-resolution criterion for ordered embeddings

There is a choice-free combinatorial certificate for the missing prescribed-order requirement.

**Definition 46.1.** A *ternary convex resolution of length  $\theta$*  of a linear order  $L$  is a sequence  $(E_\xi)_{\xi \leq \theta}$  of equivalence relations on  $L$  such that:

- (a)  $E_0 = L \times L$  and  $E_\theta$  is equality;
- (b) the relations refine with increasing  $\xi$ , and every equivalence class is convex in  $L$ ;
- (c) at a limit  $\lambda \leq \theta$ ,  $E_\lambda = \bigcap_{\xi < \lambda} E_\xi$ ;
- (d) every  $E_\xi$ -class is partitioned into at most three  $E_{\xi+1}$ -classes.

A binary resolution is defined by replacing three with two.

**Theorem 46.2** (Order-coding certificate). *For a set-sized linear order  $L$ , the following are equivalent over ZF:*

- (i)  $L$  order-embeds into  $\mathbf{No}$ ;
- (ii)  $L$  has a ternary convex resolution of some ordinal length;
- (iii)  $L$  has a binary convex resolution of some ordinal length.

*A ternary resolution of length  $\theta$  produces an order embedding into sign sequences of exact length  $2 \cdot \theta$ .*

*Proof.* Suppose  $j : L \hookrightarrow_{\text{ord}} \mathbf{No}$ . Bound the birthdays of its range by an ordinal  $\theta$ . Pad each sign sequence to length  $\theta$  with the symbol 0 after it ends, using  $- < 0 < +$ . Surreal comparison is precisely lexicographic comparison of these padded strings. Let  $x E_\xi y$  mean that their first  $\xi$  symbols agree. Prefix classes in a lexicographic order are convex; each splits into at most three next-symbol classes, limit relations are intersections, and equality of the whole padded strings gives  $x = y$ . Thus (i) implies (ii).

Now suppose a ternary resolution is given. Inside each  $E_\xi$ -class, its finitely many  $E_{\xi+1}$ -classes inherit a linear order. Label them by 0, 1, 2 according to their number of predecessors; unused labels do not matter. This requires no selection of class representatives. For  $x \in L$ , let  $d_x(\xi)$  be the label of the child containing  $x$ .

If  $x \neq y$ , there is a least stage at which they cease to be equivalent. It is a successor stage: at a limit, the intersection condition would give an earlier separating stage. Thus their label strings first

differ at the corresponding coordinate. Convexity says that  $x < y$  precisely when the first differing label of  $x$  is smaller. Therefore  $x \mapsto d_x$  order-embeds  $L$  into  $\{0, 1, 2\}^\theta$  in lexicographic order.

Replace each digit by a two-sign block

$$0 \mapsto --, \quad 1 \mapsto -+, \quad 2 \mapsto ++.$$

Concatenating  $\theta$  blocks gives a sign sequence of length  $2 \cdot \theta$ . The replacement preserves the first differing coordinate comparison, hence gives (i). It also gives a binary convex resolution by equality of prefixes of these fixed-length sign sequences. Thus (ii) implies (iii), and (iii) implies (ii) trivially.  $\square$

The theorem does not identify an optimal resolution length. Its value is to isolate a checkable *order-sensitive* certificate, distinct from the unconstrained powerset coding of the underlying set.

## 47 Orbit factorization of polynomial representations

The negative theory begins with a general algebraic lemma that does not mention surreals, atoms, or choice principles.

Let a group  $G_0$  act on a set  $X$ . It acts on  $k[T_X]$  by permuting variables and fixing the coefficient field  $k$ . Let  $X/G_0$  denote the set of its orbits. Put

$$I_{G_0} = (T_x - T_y : x, y \text{ belong to the same } G_0\text{-orbit}). \quad (47.1)$$

**Theorem 47.1** (Orbit quotient and factorization). *There is a canonical  $k$ -algebra isomorphism*

$$k[T_X]/I_{G_0} \cong k[T_{X/G_0}].$$

*If  $D$  is a  $k$ -algebra on which  $G_0$  acts trivially and a  $k$ -algebra homomorphism  $\phi : k[T_X] \rightarrow D$  satisfies*

$$\phi(h \cdot p) = \phi(p) \quad (h \in G_0),$$

*then it factors uniquely through this quotient. If  $X/G_0$  has  $q < \omega$  members and  $D$  is a field, then*

$$\text{trdeg}_k \text{Frac}(\text{im } \phi) \leq q. \quad (47.2)$$

*Equality holds precisely when the  $q$  orbit values are algebraically independent over  $k$ ; in that case  $\ker \phi = I_{G_0}$ .*

*Proof.* The orbit map  $x \mapsto [x]$  induces a surjective homomorphism

$$\pi : k[T_X] \longrightarrow k[T_{X/G_0}].$$

It kills  $I_{G_0}$ . Conversely, in  $k[T_X]/I_{G_0}$  all variables from one orbit have the same residue. Assign to an orbit this uniquely determined residue; existence follows from the nonemptiness of the orbit, and uniqueness follows from the defining relations. This is a function defined by a unique-value condition, not a choice of representatives. The polynomial universal property gives an inverse to the induced map from the quotient. The isomorphism follows.

Invariance gives  $\phi(T_x) = \phi(T_{hx})$ , so  $I_{G_0} \subseteq \ker \phi$  and the unique factorization follows. With  $q$  orbits the image field is generated over  $k$  by their  $q$  values. Its transcendence degree is at most  $q$ . By finite field theory it equals  $q$  exactly when those values are algebraically independent. Equivalently, the factored map from  $k[T_{X/G_0}]$  is injective, which is exactly the assertion that the original kernel is  $I_{G_0}$ .  $\square$

**Corollary 47.2** (Obstruction for fraction fields). *If a  $G_0$ -orbit in  $X$  has two members, there is no invariant unital field homomorphism  $k(T_X) \rightarrow D$  to a field with trivial  $G_0$ -action.*

*Proof.* Such a map would kill the nonzero element  $T_x - T_y$  for two variables in that orbit. A unital homomorphism from a field has zero kernel.  $\square$

*Remark 47.3* (Coinvariants, not invariant polynomials). The quotient in [theorem 47.1](#) identifies variables. It is not the invariant subring  $k[T_X]^{G_0}$ , whose structure can be quite different. Confusing these two constructions would invalidate the transcendence bound.

## 48 Sharp support bounds in two permutation models

### 48.1 The model framework

Start with a ground universe of  $\text{ZFA} + \text{AC}$  with a set  $A$  of atoms. A permutation group  $G$  on  $A$  acts recursively on sets, fixing pure sets. Let the normal filter of subgroups be generated by the pointwise stabilizers

$$\text{Fix}_G(S) = \{g \in G : g(a) = a \text{ for every } a \in S\}$$

of finite subsets  $S \subseteq A$ . The hereditarily symmetric sets form the usual full Fraenkel–Mostowski model  $W$ . The permutation-model construction and its set-theoretic setting are standard; [\[21, Section 3.1\]](#) provides a recent primary-source presentation of this machinery.

Every set of  $W$  has some finite atom support  $S$ : it is fixed by every member of  $\text{Fix}_G(S)$ . Every pure ground set belongs to  $W$ , and the pure part is unchanged. In particular, the reals, ordinals, and surreals of  $W$  are pure. Each atom permutation fixes each of them individually.

A map  $\phi$  in  $W$  with pure values and support  $S$  therefore satisfies

$$\phi(g \cdot x) = g \cdot \phi(x) = \phi(x) \quad (g \in \text{Fix}_G(S)). \quad (48.1)$$

For polynomial domains the induced action permutes indeterminates. This turns the algebraic factorization theorem into a precise internal obstruction.

### 48.2 The basic Fraenkel model

Take  $A$  infinite and  $G = \text{Sym}(A)$ , with finite supports.

**Theorem 48.1** (The  $s + 1$  bound). *In the basic Fraenkel model, let*

$$\phi : \mathbb{R}[T_A] \longrightarrow \mathbf{No}$$

*be an internal  $\mathbb{R}$ -algebra homomorphism with atom support  $S$ , where  $|S| = s < \omega$ . Then*

$$\text{trdeg}_{\mathbb{R}} \text{Frac}(\text{im } \phi) \leq s + 1. \quad (48.2)$$

*All variables indexed by  $A \setminus S$  have the same image. For every finite  $S$  the bound is attained by a homomorphism supported by  $S$ . No such homomorphism is injective, and there is no internal unital field homomorphism  $\mathbb{R}(T_A) \rightarrow \mathbf{No}$ .*

*Proof.* The stabilizer  $G_0 = \text{Fix}_G(S)$  has exactly  $s + 1$  orbits on  $A$ : the  $s$  singletons in  $S$  and the infinite complement  $A \setminus S$ . Any two atoms in that complement can be swapped while all of  $S$  is fixed. Equation (48.1) and [theorem 47.1](#) prove the collapse and the bound. Two distinct atoms outside  $S$  give a nonzero difference in the kernel. The field assertion follows from [corollary 47.2](#).

For sharpness, choose a listing  $S = \{a_1, \dots, a_s\}$  of this one finite set. Assign distinct members  $z_0, \dots, z_s$  of a pure algebraically independent family, for example

$$z_i = \omega^{\omega^i}, \quad 0 \leq i \leq s,$$

by sending  $T_{a_i}$  to  $z_i$  and every variable outside  $S$  to  $z_0$ . This assignment and its polynomial evaluation graph are supported by  $S$  and hereditarily symmetric: each individual input polynomial involves only finitely many atoms, while every output is pure. Thus the map belongs to  $W$ . Its orbit values are algebraically independent, giving equality in (48.2). This construction uses only the finite choice of a listing of one given finite set, which is available without AC.  $\square$

### 48.3 The ordered Mostowski model

Now take an atom set  $A$  with a dense linear order without endpoints externally isomorphic to  $(\mathbb{Q}, <)$ , and let

$$G = \text{Aut}(A, <)$$

with the same finite-support filter. The ordered set  $(A, <)$  itself belongs to the symmetric model: its order relation is invariant under the whole group.

**Lemma 48.2** (Stabilizer orbit count). *If  $S = \{a_1 < \dots < a_s\} \subseteq A$ , the pointwise stabilizer of  $S$  has exactly  $2s + 1$  orbits:*

$$\{a_1\}, \dots, \{a_s\}, \quad (-\infty, a_1), (a_1, a_2), \dots, (a_s, \infty).$$

For  $s = 0$ , there is just the orbit  $A$ .

*Proof.* Order preservation and fixation of  $S$  prevent movement between the displayed pieces. Within any one interval, any chosen atom can be sent to any other by an order automorphism fixing all the endpoints in  $S$ . To justify existence, work in the ground universe with its fixed external countable presentation of the dense order. Extend the finite increasing partial map consisting of the fixed points of  $S$  and the proposed movement by the standard back-and-forth construction. At each alternating stage put the next unused domain or range point into the corresponding nonempty interval. Density and absence of endpoints ensure success. The resulting bijection is order preserving and fixes  $S$ . Thus each interval is a single orbit. This external argument computes the chosen permutation group; it does not put a countable enumeration of  $A$  into the symmetric model.  $\square$

**Theorem 48.3** (The  $2s + 1$  bound). *In the ordered Mostowski model, every internal  $\mathbb{R}$ -algebra homomorphism*

$$\phi : \mathbb{R}[T_A] \longrightarrow \mathbf{No}$$

*with a support  $S$  of size  $s$  satisfies*

$$\text{trdeg}_{\mathbb{R}} \text{Frac}(\text{im } \phi) \leq 2s + 1. \quad (48.3)$$

*For every finite  $S$ , equality is attained by a map supported by  $S$ . No such map is injective, and there is no internal unital field homomorphism  $\mathbb{R}(T_A) \rightarrow \mathbf{No}$ .*

*Proof.* Apply [theorem 47.1](#) to the  $2s + 1$  orbits of [lemma 48.2](#), using (48.1). Every interval contains distinct atoms, so the kernel contains a nonzero variable difference; the field assertion follows as before.

The orbits themselves have a canonical finite order, alternating intervals and singletons. Assign to these ordered orbits the algebraically independent pure values  $z_0, \dots, z_{2s}$  from the proof of [theorem 48.1](#). The resulting evaluation map is supported by  $S$ , belongs to the symmetric model, and has algebraically independent orbit values. This attains the bound.  $\square$

**Corollary 48.4** (Exact kernels at maximum dimension). *For each of the sharpness examples above, the entire kernel is generated by differences between variables in the same stabilizer orbit. There are no further polynomial relations among the orbit values.*

This follows from the last clause of [theorem 47.1](#). The upper bounds apply to any chosen support, not necessarily a smallest support. As the support size varies, maps of arbitrarily large finite image-field transcendence degree exist, but no one internal map has infinite image-field transcendence degree.

#### 48.4 Why pure coordinates recover well-orderability in these models

**Proposition 48.5.** *In either full permutation model just described, for any internal set  $X$  the following are equivalent:  $X$  injects into a pure set;  $X$  is well-orderable internally;  $X$  injects into the internal  $\mathbf{No}$ .*

*Proof.* A pure set belongs to the unchanged pure ground universe, which satisfies choice. It has a pure well-order, and that well-order also belongs to the symmetric model. An internal injection into the pure set therefore pulls back an internal well-order of  $X$ . Conversely, a well-order identifies  $X$  with an ordinal, which is pure and embeds into  $\mathbf{No}$ .

An injection into  $\mathbf{No}$  has a set-sized range; its range is a set of pure objects and is itself pure. Thus it gives an injection into a pure set. Alternatively, apply [theorem 42.1](#) and use the purity of  $\mathcal{P}(\alpha)$  in the full permutation model.  $\square$

There is no conflict with  $\text{KWP}_1$  being strictly weaker than choice in pure ZF. The preceding proposition makes an additional assumption: the unchanged pure part of these particular atom models already satisfies choice. It is not a theorem about arbitrary models of ZFA.

## 49 An ordered field with all finite stages embeddable but no global embedding

**Theorem 49.1** (Internal ordered-field counterexample). *In the ordered Mostowski model  $W$ , the set-sized ordered field*

$$F(A) = \mathbb{R}(T_a : a \in A)$$

*with the largest-variable order of [lemma 45.1](#) has no unital field homomorphism into  $\mathbf{No}^W$ , even when order preservation is not required. Nevertheless, for every finite  $J \subseteq A$ , the subfield  $F(J)$  has an internal ordered  $\mathbb{R}$ -embedding into  $\mathbf{No}^W$ , and*

$$F(A) = \bigcup_{J \in [A]^{<\omega}} F(J).$$

*Proof.* The polynomial ring, fraction-field equivalence relation, and field operations are invariant under every order automorphism of  $A$ . Each expression is hereditarily symmetric, since it uses only finitely many atoms and pure real coefficients. The largest-variable order is invariant by [lemma 45.1](#). Hence the ordered field is an object of  $W$ .

An internal field homomorphism into  $\mathbf{No}^W$  would have a finite atom support. Restricting it to  $\mathbb{R}[T_A]$  gives a map whose values are fixed by the stabilizer, whether or not the homomorphism fixes every real. In particular, two variables from one nonsingleton interval orbit have the same image. Their nonzero difference cannot be killed by a unital field homomorphism. This is a contradiction.

For a finite  $J = \{a_1 < \dots < a_n\}$ , the order-preserving map  $a_i \mapsto i$  is a set map supported by  $J$ . The construction of [theorem 45.2](#) gives an ordered embedding  $F(J) \rightarrow \mathbf{No}^W$ . Its graph is supported by  $J$  and hereditarily symmetric, so it belongs to  $W$ . Every rational expression uses only finitely many variables, proving the union identity.  $\square$

*Remark 49.2* (Scope of the counterexample). This proves a failure of ordered-field universality in a model of ZFA. It is *not* asserted to be a counterexample in pure ZF. Removing atoms while preserving a statement about embeddings into the entire internal surreal class requires a separate transfer argument; none is supplied here. The external countability of  $A$  also causes no contradiction: an external enumeration can produce an external embedding, but such an enumeration and embedding do not belong to  $W$ .

## 49.1 The exact coherence obstruction

**Theorem 49.3** (Gluing compatible coordinate embeddings). *Let  $L$  be a set-sized linear order in ZF or ZFA. A global ordered  $\mathbb{R}$ -embedding  $F(L) \rightarrow \mathbf{No}$  exists if and only if there is a set-indexed family*

$$(\phi_J : F(J) \hookrightarrow_{\text{ord}} \mathbf{No})_{J \in [L]^{<\omega}}$$

*of ordered  $\mathbb{R}$ -embeddings satisfying*

$$J \subseteq K \implies \phi_K|_{F(J)} = \phi_J. \quad (49.1)$$

*Proof.* Restrict a global embedding to obtain the family; Replacement gives its set code. Conversely, take the union of the graphs of the  $\phi_J$ . This union is a set. If an element lies in both  $F(J)$  and  $F(K)$ , compare both maps with  $\phi_{J \cup K}$  using (49.1); their values agree. Thus the union is a function on  $F(L)$ . Any finitely many arguments and their algebraic relations lie in one  $F(J)$ , so the union is an order-preserving field homomorphism. The same observation shows injectivity.  $\square$

For the field in [theorem 49.1](#), each finite stage has an ordered embedding but no compatible family exists internally. This is not a claim that first-order compactness fails, and it does not concern the existence of a model of a finitely satisfiable first-order theory. The target is the particular internal class  $\mathbf{No}$ , and the obstruction is the demanded compatibility and internal existence of its coordinate map.

## 49.2 Finite polynomial tests can all succeed

**Proposition 49.4** (Finite separation without a faithful representation). *In either of the two permutation models, every finite family of nonzero polynomials in  $\mathbb{R}[T_A]$  is simultaneously nonzero under some internal  $\mathbb{R}$ -algebra homomorphism to  $\mathbf{No}$ . Nevertheless, no one internal homomorphism is injective.*

*Proof.* Let  $J$  be the finite union of the variable supports of the given polynomials. Assign algebraically independent pure surreal values to the variables in  $J$ , and assign a fixed pure value, say 0, to every other variable. The evaluation map is supported by  $J$  and belongs to the model. Its restriction to  $\mathbb{R}[T_J]$  is injective by algebraic independence, so it preserves the nonzeroness of every polynomial in the family. Nonexistence of an injective map follows from [theorems 48.1](#) and [48.3](#).  $\square$

Equivalently, for each nonzero polynomial there exists a map not killing it, but no single map avoids all nonzero kernels. One may describe the intersection of all these kernels as zero, provided that phrase is read as a first-order quantification over maps, not as the formation of a set of all maps into a proper class.

## 50 Proof audit and relation to the existing repository

### 50.1 Dependency boundary

The main logical dependencies are short:

$$\begin{aligned} \text{sign sequences} + \text{Replacement} &\implies \text{ordinal-code equivalence,} \\ \text{finite normal forms} + \text{monomial multiplication} &\implies \text{free-algebra equivalence,} \\ \text{canonical fraction/root codes} + \text{real closedness} &\implies U_\kappa, \\ \text{polynomial orbit quotient} + \text{finite atom support} &\implies \text{sharp collapse bounds.} \end{aligned}$$

The negative construction does not assume any speculative research claim from ProveIt's other subjects. The positive algebraic construction uses only the standard surreal interface in [section 41](#). Thus an unrelated error in an unrefereed repository report would not affect the mathematical dependency chain here.

### 50.2 A ledger of theorem status

| Result                   | Mathematical status in this manuscript                                                              | Novelty / checking status                                     |
|--------------------------|-----------------------------------------------------------------------------------------------------|---------------------------------------------------------------|
| Ordinal-code equivalence | Direct proof in ZF and ZFA; exact birthday capacity included                                        | Elementary consequence of sign coding; no priority claim      |
| Double-monomial family   | Explicit finite-normal-form proof; free rings and fields calibrated                                 | Classical ingredients; combined formulation developed here    |
| Optimal field $U_\kappa$ | Set existence, real closedness, and two cardinal injections proved without quotient-choice shortcut | Candidate synthesis/refinement; independent review needed     |
| Convex resolution        | Necessary and sufficient order-coding certificate proved                                            | Related to ordered coding; historical novelty not established |
| Orbit factorization      | Full algebraic proof, with finite-dimensional equality criterion                                    | Standard universal-property mechanism                         |

| Result                      | Mathematical status in this manuscript                         | Novelty / checking status                                    |
|-----------------------------|----------------------------------------------------------------|--------------------------------------------------------------|
| $s + 1$ and $2s + 1$ bounds | Exact stabilizer counts, upper bounds, and sharp constructions | Quantitative application developed here; priority unverified |
| Ordered-field failure       | Internal counterexample in the stated ZFA permutation model    | No automatic pure-ZF transfer asserted                       |
| Coherent gluing             | Set-graph union proof; finite separation distinguished         | Standard gluing mechanism, applied to fixed target           |

No theorem in this manuscript is presented as already kernel-checked. The supplied finite tests detect mistakes in finite encodings and algebraic identities but do not prove a theorem about all ordinals, arbitrary models, or proper classes.

### 50.3 The choice issue in a Lean audit

ProveIt’s surreal README reports a standard-axiom audit allowing `propext`, `Quot.sound`, and `Classical.choice`; it maps checked statements separately from research claims [26]. That is valuable evidence about the declared Lean foundation. It is not a certificate that a theorem is provable over *object-theoretic* ZF without choice.

There are two defensible routes for the present results. One can formalize enough constructive set and ordinal coding to avoid the choice axiom in the relevant dependency chains. Alternatively, one can work in Lean’s ordinary classical metatheory while explicitly formalizing a model of ZF or ZFA and proving that the relevant internal statements follow from the designated object-theoretic axioms. Metatheoretic choice then does not silently become internal choice; the semantic or syntactic theorem must enforce that distinction.

A theorem about every type in a universe is also not literally a theorem about the proper class of all sets. The birthday-bounded targets  $\mathbf{No}_{\leq \alpha}$  and the fields  $U_\kappa$  offer useful set-sized interfaces, but a formalization must still record which universe contains each type and which internal sets it represents.

[Added 3 October 2026, batch 89: the description of `Algebra/SurrealNumbers/README.md` is accurate at the pinned commit `2b7b388ba` and at the placement commit `23adb85f9`; the README blob source `11` inspected, `18478d586`, is unchanged. No module named below exists in the repository, and no statement of this part is formalized; placement in this report confers no formal status.]

### 50.4 A proposed module decomposition

The following are suggested module responsibilities, not names of files already implemented in the repository.

#### **OrdinalSignCoding**

Define variable-length sign functions, the definedness code (42.1), its decoder, and fixed-length subset encoding. Prove the birthday-capacity theorem with explicit ordinal pairings.

#### **DoubleMonomialIndependence**

Abstract the finite-normal-form leading-term property. Prove that distinct monomials are linearly independent and that the double-monomial substitution is an injective multivariate-polynomial homomorphism. Derive the omnific range condition separately from the real-coefficient assertion.

### CanonicalFieldCodes

Normalize multivariate fractions by gcd and leading coefficient, code algebraic roots by minimal polynomial and root index, and construct the set-sized relative real closure.

### OrbitPolynomialQuotient

Formalize the group action on variables, the orbit-indexed polynomial algebra, the quotient isomorphism, and finite transcendence bounds. This module can be independent of surreal theory.

### InternalSymmetryBridge

Formalize finite-support evaluation into a pure target inside a specified permutation model. The semantic statement that an internal map has a finite support is indispensable; a theorem only about an externally chosen group action is not yet the model-theoretic conclusion.

### OrderedFreeFieldAndCoherence

Construct the largest-variable order using finite supports, prove the ordered embedding reduction, and formalize gluing of compatible finite-stage maps by a set union.

## 51 Further research questions and concrete next targets

The following are proposed research directions. Except where a cited paper explicitly supplies a status, they should not be read as claims that the literature has left each formulation open. A literature check and, where appropriate, contact with specialists should precede any priority claim.

**Question 51.1** (Pure-set prescribed-order universality). What is the exact strength over ZF of the assertion that every set-sized linear order embeds in **No**? In particular, can one construct a pure-ZF analogue of the ordered-field counterexample in [theorem 49.1](#)?

The atom proof isolates the desired obstruction but does not transfer it. A useful first target is a pure symmetric extension containing a linearly ordered set with no ordinal-length convex resolution. One must analyze its *internal* sign sequences and maps rather than merely preserve a low-rank atom surrogate.

**Question 51.2** (Does  $KWP_1$  supply an order-sensitive code?). Does  $KWP_1$  imply that every linear order admits a binary convex resolution of some ordinal length?

By [theorem 46.2](#), this is exactly the missing implication from coordinate universality to prescribed-order universality. A possible approach is to study a given order on a subset of  $\mathcal{P}(\alpha)$  and ask whether membership coordinates can be converted into a continuous sequence of convex refinements. Arbitrary coordinate cuts need not be convex, which is the concrete obstruction to the naive argument.

**Question 51.3** (Optimal convex-resolution height). For an order that embeds in **No**, how does its least binary resolution length compare with the least birthday bound of an ordered embedding?

The proof gives conversions with a factor of two at the ternary-to-binary step, but no optimality statement. Fixed-length and variable-length sign representations should be distinguished. Finite chains, ordinal orders, reversed ordinal orders, and lexicographic products provide initial test cases.

**Question 51.4** (Explicit birthday bounds for  $U_\kappa$ ). Find useful explicit ordinal bounds for the birthdays of all members of the field  $U_\kappa$  defined in [\(44.1\)](#).

Existence is immediate from its set-sizedness, but an effective bound requires controlling the two monomial operations and algebraic root formation. A conservative closure ordinal for a specified

finite-expression grammar is a reasonable first goal; a sharp product-birthday formula should not be assumed.

**Question 51.5** (More general fields at a fixed code level). Which additional hypotheses on a prescribed ordered field  $K$  with  $K \hookrightarrow \mathcal{P}(\kappa)$  guarantee an ordered embedding into  $U_\kappa$  or into a canonically enlarged target of the same coding capacity?

Free rational-function fields with induced orders are handled here. Algebraic ordered extensions, valuations with order-embeddable value groups, and fields given by a coherent tower of simple ordered extensions are natural successive cases. The existence of a chosen transcendence basis must not be silently assumed.

**Question 51.6** (Higher Kinna–Wagner levels). Is there a natural algebraic representation problem, not merely a relabelling of the definition, whose exact strength is  $\text{KWP}_n$  for  $n > 1$ ?

Ordinary surreal elements have one-level ordinal subset codes. Higher levels might arise from unordered families of coordinates, quotient presentations, or structured moduli. The challenge is to exhibit algebraically meaningful invariants whose coding genuinely requires the extra powerset level.

**Question 51.7** (Orbit profiles beyond the two examples). For a permutation group  $G$  on atoms, how precisely does the function

$$S \mapsto |A/\text{Fix}_G(S)|$$

control the transcendence spectra of pure-target polynomial representations?

The finite-orbit case is settled by [theorem 47.1](#). Infinite orbit sets introduce a new coding issue: a sharp realization requires enough independent pure coordinates indexed by the orbit set. Applying [theorem 43.5](#) to that quotient suggests an exact two-part criterion, combining orbit count and ordinal-code availability.

**Question 51.8** (Minimal supports and essential dimension). Can one recover a lower bound on the least atom-support size of a map from the transcendence degree of its image, and when is the bound exact for every map?

Our inequalities immediately give  $s \geq d - 1$  in the basic model and  $s \geq \lceil (d - 1)/2 \rceil$  in the ordered model. They are sharp as existential bounds. A finer classification would distinguish maps with redundant support from those whose support is essential and relate this to the structure of their kernels.

**Question 51.9** (Real closures under symmetry). What is the strongest ordered-algebraic enlargement of the source in [theorem 49.1](#) that can be constructed internally while retaining an explicit pure-target obstruction?

An externally selected real closure is not automatically an internal object with the desired action. A canonical construction through real-algebraic root codes, if it can be made equivariant over the ordered field, is a promising route. Any such extension would inherit nonembeddability by restriction to the original rational-function field.

**Question 51.10** (Choice strength of compatible local embeddings). For which families of subfields does the existence of an individual embedding for every member imply a compatible family of embeddings into a fixed real-closed target?

The union theorem proves what compatibility would accomplish, not that it can be obtained. Separate chains with specified extension maps, directed systems indexed by finite subsets, and unrestricted systems. A useful target is an exact weak-choice principle for one sharply specified family rather than an undifferentiated appeal to compactness.

**Question 51.11** (Residual separation and inverse limits of representations). Can the finite-separation phenomenon of [proposition 49.4](#) be organized as a failure of a natural inverse-limit existence principle, with a precise choice calibration?

For each finite variable set, consider its space of faithful coordinate assignments and restriction maps. The naive inverse system need not have surjective transition maps with a fixed birthday bound. Understanding that issue is essential before invoking any inverse-limit or compactness theorem.

**Question 51.12** (Object-theoretic formal verification). Can the ZF equivalence and the ZFA counterexample be checked in one repository development with an explicit audit of internal versus metatheoretic choice?

The shortest first milestone is the orbit polynomial quotient plus the finite-normal-form independence lemma. The hard foundational milestone is a bridge from hereditarily symmetric internal maps to stabilizer invariance, together with set-coded surreal targets. Such a development would test the semantics of the statement, not just the algebra of the displayed proof.

## 52 Reproducibility and limits of the computational evidence

The accompanying Python program, shipped as `code/11-choiceless-universality-verify_finite.py` (delivered as `verify_finite.py`), uses only the standard library and a fixed random seed. It checks the finite definedness encoding, fixed-length ternary-to-binary order encoding, distinct exponent-vector substitution, translation invariance of largest-variable order, leading terms of polynomial products, and finite permutation-orbit quotient identities. It also checks simultaneous evaluation as a ring homomorphism by exact integer arithmetic. The supplied deterministic run passed 20,595 assertions; this count is evidence only about the finite checks just described.

These checks are deliberately finite. A finite linearly ordered set has only the identity order automorphism, so its automorphism group is *not* a valid finite simulation of the ordered Mostowski model. The program does not pretend otherwise. Infinite interval transitivity is proved by the ground-model back-and-forth argument in [lemma 48.2](#); ordinal recursion, class size, and model existence are mathematical arguments, not outputs of the script.

In the delivered layout the local checks were reproduced by

```
python3 verify_finite.py
```

followed by two `pdflatex` passes of the delivered manuscript. Without `-output` the program writes `verification_results.json` beside itself; with the shipped names, run it on a copy or pass an explicit output path outside this directory (the report's `README.md` gives the commands). The recorded run is shipped as `data/11-choiceless-universality-verification_results.json`. For this merge the program was rerun with Python 3.14.4 on a copy: it passed and reproduced the recorded file exactly, apart from line endings. The delivered archive contained the source, compiled PDF, verification script and its result file, a `README`, a source-and-status note and a checksum file; of these only the script, the result file and the source-and-status note (`11-choiceless-universality-SOURCES_AND_STATUS.md`) are shipped ([section 53](#)). No font files are distributed.

## 53 Conclusion of source 11, checklist and provenance

### 53.1 Conclusion

Surreal universality is not one assertion. For unconstrained independent variables, the exact obstruction is ordinal subset coding, and double Conway monomials convert that code into omnific algebra. This conversion can be carried out inside an optimally sized real-closed field at each infinite ordinal code level. When an existing order must be preserved, a convex resolution, rather than a mere injection of underlying sets, supplies the relevant certificate.

In permutation models the obstruction becomes quantitative. A pure-valued map supported by finitely many atoms cannot distinguish variables that its stabilizer moves into one another. The number of stabilizer orbits is therefore a sharp upper bound on the independent transcendence retained by the image. In the ordered model this produces an actual ordered field with no internal surreal embedding, while every finite coordinate stage embeds and every finite polynomial test can succeed.

The resulting program is concrete: distinguish coding from order, distinguish external from internal maps, and distinguish finite satisfiability from coherent realization. These distinctions are both mathematically substantive and well suited to a proof-assistant audit of choiceless surreal algebra.

### 53.2 A compact checklist for reviewing the main proofs

A referee or formalizer can focus first on the following points.

1. In [theorem 42.1](#), check both definedness and signs in the code, the side of ordinal multiplication, and the Replacement bound for the image birthdays.
2. In [theorem 43.1](#), check that the inner monomials are linearly independent, not merely distinct; this is what separates evaluated exponent vectors. Check the positive-exponent condition separately for the omnific conclusion.
3. In [lemma 44.1](#), check uniqueness of normalized fractions in the full arbitrary-variable polynomial ring. A surjection from a coded syntax set is not enough for the cardinal upper bound without a canonical code of each value.
4. In [theorem 44.3](#), verify collection of the algebraic roots as a set, and use the root index in the actual ordered extension. No choice of one root from each polynomial is made.
5. In [theorem 45.2](#), compare the two leading-term orders rather than assuming an arbitrary algebraic embedding preserves order.
6. In [theorem 46.2](#), use continuity at limits to justify that a first separation occurs at a successor stage.
7. In [theorem 47.1](#), distinguish the orbit quotient from the invariant subring, and avoid selecting one representative from every orbit.
8. In [theorems 48.1](#) and [48.3](#), apply atom support to the *graph of the map*; pure values alone do not constrain an external map.
9. In [theorem 49.1](#), verify that the source ordered field is an internal object and that the forbidden map is internal. Do not transfer the conclusion from ZFA to ZF without proof.
10. In [theorem 49.3](#), require a single set-coded compatible family; separate existence of each finite-stage map is weaker.

### 53.3 Non-claims of source 11

Every limitation stated by source 11, in its manuscript or in its source-and-status note, is kept; most appear at their point of use. Collected: the manuscript is AI-generated and unrefereed and is not a Lean- or Rocq-checked development; the novelty of its exact formulations is not established, and its twelve questions are proposed targets, not catalogued open problems; it claims no solution of Glazer and Yao’s reflection questions or of another named problem; the atom-model counterexample is a result in ZFA, and no ZF transfer is supplied or asserted; universality for prescribed orders does not follow from  $KWP_1$  in what is proved; no sharp birthday bound for  $U_\kappa$  is claimed;  $KWP_1$  and its coding form are classical (Karagila–Schilhan), as are sign coding, normal forms, the polynomial and UFD facts, real-closed-field facts and the permutation-model construction; the finite script checks encodings and algebraic identities only, does not construct symmetric models, prove independence or check real closedness, and does not use finite chains as a model of  $\text{Aut}(\mathbb{Q}, <)$ ; the proposed modules are a roadmap; the repository was not modified, built or verified.

### 53.4 Provenance of this part

**The manuscript.** Source 11 is batch-89 manuscript 04, delivered as the archive `Glazer_Surreal_Choice_Research.zip` in commit `7c0f2d9f9` and placed in commit `23adb85f9`. It records the README blob `18478d586` of `Algebra/SurrealNumbers/README.md` (unchanged at the placement commit) and the identifier `2b7b388ba`, which it calls a “main-tree” identifier, “not asserted to be a commit identifier”; it is in fact the commit “Catalogue batches 81 to 86”. Its bibliography linked the surreal README through the moving branch `main`; the entry [26] here gives the pinned commit. Its manuscript (`article.tex`, 911 lines, 52 labels; a 22-page PDF), README, PDF and checksum file `SHA256SUMS` (6 of 6 entries verified at placement) are not shipped; they survive in the arrival commit. Shipped, byte-identical: `11-choiceless-universality-SOURCES_AND_STATUS.md`, `code/11-choiceless-universality-verify_finite.py` and `data/11-choiceless-universality-verification_results.json`. The source-and-status note keeps the delivered wording about the identifier `2b7b388ba` and mentions a profile page given to the authoring session as an identifying pointer, without a URL. The Hilbert’s-tenth programme’s routing review of commit `6aadaa4f2` covers this archive too (section 39.5).

**How it is printed.** The text from source 11’s title-page boxes and abstract to its appendix is printed in full, in its order: the title-page boxes and the abstract open section 40, its Section 1 follows them, and its appendix became a subsection of this section. Every delivered label carries the prefix `hset:kw:`, and its twelve questions gained labels `hset:kw:q:...`. The symbols renamed are listed in convention 40.1; delivered file names are replaced by the shipped ones; its references to Glazer–Yao and to Conway use this report’s bibliography entries. Added for this report: sections 40.4 and 40.5, the bracketed dated note, this subsection and the preceding one. No mathematical statement or proof of source 11 was changed.

**Where the merge had to choose.** (i) Placement: source 11 cites no report of the collection; it is placed here because it answers in part the open item of question 21.7 and is the injection counterpart of theorem 13.2 (remarks 40.2 and 40.3). (ii) Theorem 43.1 is printed with its own proof although it is a case of a lemma of another report (remark 40.4). (iii) The two new parts follow Part V, so that no existing number changes; both are printed after this report’s conclusion and before its appendices.

## Part VIII

# Named symmetries and Replacement

## 54 Source 12: question, results, notation and relation to the report

This part was added in batch 90. It prints one further manuscript, written independently of the seven above: *Named Symmetries and Replacement: An exact orbit criterion, a strict kernel hierarchy, and invariance of the pure surreal universe*, “Prepared for Vladimir Reshetnikov with ChatGPT”, 3 October 2026. It is batch-90 manuscript 05 and *source 12* in this report’s local numbering; its shipped files carry the prefix `12-named-symmetries-`. It was pinned to repository commit `8dc2592e9`, arrived in commit `a162e4386` and was placed in commit `12076b2e8`. Its provenance, the mechanical changes made in printing it, and its non-claims are in [section 66](#). Inside this part, “this manuscript”, “this article”, “this draft”, “here”, “our” and “we” refer to source 12.

Source 12 continues [Part VI](#): the “earlier companion working note” it names is source 10, which it read outside the repository and from which it assumes no theorem. Its pin predates both [Part VI](#) and the research-report collection’s report `SetTheory/Cardinals/docs/reports/ordinals-and-order-types/naming-elementary-embeddings`, whose principal theorem already contains source 12’s full criterion ([theorem 58.1](#)) for finitely many named permutations. Source 12’s introduction, written without that report, must not be read as claiming that case new; [remark 54.4](#) states what each proves.

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**Research status (source 12).** The finite-kernel construction and its classical symmetry argument are established mathematics. The exact classifications and hierarchy are proved in this manuscript as proposed refinements; their historical priority has not been established. This is an unrefereed research draft, not a claim to have resolved a named published conjecture or to have obtained independent Lean/Rocq verification. Elliot Glazer is neither a coauthor nor an endorser.

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**Source 12’s summary.** Let  $M_\omega^{\text{ker}}$  be the universe of sets and urelements with finite actual atom kernel, inside an ambient universe with Choice and a set  $A$  of infinitely many atoms. We classify which uniformly named actions of a pure set-sized group preserve Replacement. The answer has two independent parts: every atom orbit must be finite, and only finitely many atoms may belong to finite transitive orbit types occurring with finite positive multiplicity.

A sharper classification measures the atom kernel of an output. Write  $R_k$  for Replacement restricted to outputs involving at most  $k$  atoms. For  $k \geq 1$ , this holds exactly when all group orbits are finite and only finitely many atoms have centralizer orbit of size at most  $k$ . We construct, for every  $d \geq 1$ , two involutions whose separate expansions satisfy full Replacement, whereas their joint expansion satisfies precisely the finite levels  $R_k$  with  $k < d$ . All joint atom orbits are finite. A single permutation can satisfy every fixed finite level while failing full Replacement.

Every example has the same pure sets and hence the same pure sign-sequence surreal field. More strongly, Collection for pure-valued relations survives even formulas using the added symmetries and impure parameters. We give complete proofs, explicit failure witnesses, a distinction between separate symbols and a uniform evaluator, 203 finite regression checks, and a formalization roadmap.

## 54.1 Why this is an appropriate intersection

Elliot Glazer and Bokai Yao's *Reflection Principles in ZFU* studies the interaction of Collection, reflection, and urelements, including small-kernel permutation constructions [21]. Glazer's work on global versus local choice in Zermelo set theory also emphasizes sensitivity to foundational hypotheses [20]. These publications, rather than an inference from a professional profile, supply the research-interest side of the intersection.

The ProveIt repository supplies the other side: it develops surreal numbers from sign sequences and explicitly distinguishes formalized results from research drafts. Its surreal README and formalization ledger provide the relevant entry points [27]. The question pursued here is not a new algebraic identity inside **No**. It is a foundational limitation on what an unchanged, richly structured surreal universe can reveal about the surrounding universe of sets.

The classical finite-kernel model already separates Replacement from Collection. Hamkins and Yao explain this construction and attribute its Replacement argument to Lévy [22]. Our earlier companion working note (source 10, printed as [Part VI](#)) examined small-kernel models and the effect of naming an atom enumeration. Naming an enumeration is a direct way to make many atoms individually accessible. Naming an involution is much subtler: it reveals only a reversible pairing. The central question is therefore:

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**Classification problem.** Which named atom symmetries preserve Replacement in the finite-kernel universe, and how much of Replacement can fail while all pure surreal constructions remain available?

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We resolve this problem for every uniformly named action of a pure set-sized group. We also determine a strict hierarchy that records how many atoms a definable output is permitted to involve. These are questions formulated and answered here; they are not presented as previously advertised open conjectures of Glazer or Yao.

## 54.2 The main conclusions

Write  $\mathcal{Z} = \text{Aut}_\Gamma(A)$  for the group of atom permutations commuting with the named action. The two kinds of orbit must not be confused:  $\Gamma a$  is the orbit generated by the *named* symmetries, whereas  $\mathcal{Z}a$  is the orbit under symmetries that *remain available* after the action has been named.

**Exact full criterion (theorem 58.1).**

Full Replacement holds precisely when every  $\Gamma$ -orbit is finite and the union of all finitely repeated finite orbit types is finite.

**Exact bounded-kernel criterion (theorem 59.1).**

For  $k \geq 1$ , Replacement for outputs with at most  $k$  atoms holds precisely when every  $\Gamma$ -orbit is finite and  $\{a : |\mathcal{Z}a| \leq k\}$  is finite.

**Sharp hierarchy using two involutions (theorem 61.3).**

For each  $d \geq 1$ , two individually harmless involutions jointly give exactly the levels  $R_k$  for  $k < d$ . No infinite joint orbit is needed.

**The limit gap (theorem 60.1).**

One permutation with one cycle of each finite length satisfies every fixed  $R_k$  but not full Replacement.

**Surreal invariance (theorem 63.1 and corollary 63.3).**

All examples preserve the entire pure universe, and even expanded-language Collection for pure-valued relations. Consequently pure surreal-valued Replacement and set-indexed cut interpolation survive.

The logical content is stronger than an isolated bad example. The classification identifies all obstructions within the specified model family, and the hierarchy locates the first possible failure exactly.

### 54.3 Scope, attribution, and limits

The background symmetry method is classical; we do not rename it as a discovery. Our contribution within this draft is the orbit-type obstruction, the centralizer-size refinement with arbitrary parameters, and their sharp realizations. The proofs below are self-contained after the ambient set-theoretic assumptions and the standard surreal simplicity theorem have been specified. The finite computation checks only finite group-action calculations, not the infinitary metatheorems.

Glazer–Yao’s 2026 separation results concern reflection and Collection in a broader range of urelement models [21]. We do not strengthen those published separation theorems here. Our models retain Choice, and their full Collection already fails in the reduct. What changes is *Replacement in the expanded language*. This distinction prevents an apparent but false conflict with equivalences that assume Replacement as part of the base theory.

### 54.4 Notation of this part

**Convention 54.1** (Symbols of source 12). Source 12 was written without knowledge of this report, and several of its symbols already mean something else here. In this part:

| Notion in source 12                                                                                            | Here                                                                                                           | Do not confuse with                                                                                                                                                                                                                           |
|----------------------------------------------------------------------------------------------------------------|----------------------------------------------------------------------------------------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Ambient universe with atoms $A$ , Choice, Replacement and Collection                                           | $\mathcal{U} = V(A)$ (source: $U$ )                                                                            | —; the same setting as <a href="#">section 32</a>                                                                                                                                                                                             |
| Finite-kernel universe $\{x :  \ker(x)  < \omega\}$                                                            | $M_\omega^{\ker}$ (source: $M = M_{\text{fin}}(A)$ ); over the atom set $A^{(d)}$ , $M_\omega^{\ker}(A^{(d)})$ | the ground model $M$ of <a href="#">Part III</a> . It <i>is</i> the kernel model $M_\kappa^{\ker}$ of <a href="#">Part VI</a> at $\kappa = \omega$ , and the model $M_\omega(A)$ of <b>naming-elementary-embeddings</b>                       |
| Atom kernel $A \cap \text{TC}(\{x\})$                                                                          | $\ker(x)$                                                                                                      | the same operator as in <a href="#">Part VI</a> ; not a kernel of a homomorphism                                                                                                                                                              |
| Named action graph, a ternary predicate                                                                        | $\text{Act}(g, a, b)$ (source: $E$ )                                                                           | the enumeration predicate $E$ of <a href="#">section 34</a> , which names a bijection from $\kappa$ onto $\kappa$ atoms, and source 11’s $E_m$ , $E_\xi$ . <b>Act</b> is the graph of a group action, $E$ an injection of ordinals into atoms |
| Expanded structure                                                                                             | $\mathcal{M}_{\text{Act}} = (M_\omega^{\ker}, \in, \text{At}, \text{Act})$ (source: $\mathcal{M}_E$ )          | Part VI’s $(M_\kappa^{\ker}, E)$                                                                                                                                                                                                              |
| Exceptional set                                                                                                | $X_{\text{Act}}$ (source: $X_E$ )                                                                              | —                                                                                                                                                                                                                                             |
| Centralizer $\text{Aut}_\Gamma(A)$ of the named action; pointwise stabilizer of a finite $F \subseteq A$ in it | $\mathcal{Z}$ , $\mathcal{Z}_{(F)}$ (source: $H$ , $H_{(F)}$ )                                                 | $H_\kappa$ , the hereditary-size universe; source 11’s acting group $H$ (renamed $G_0$ in <a href="#">convention 40.1</a> ); $\mathbb{Z}$ , the integers; $\mathbb{Z}_f$                                                                      |
| Replacement in the expanded language; its restriction to outputs with at most $k$ atoms                        | $\text{Rep}$ ; $\text{R}_k$                                                                                    | $\text{Rep}_\sqsupset$ , one Replacement instance at $\sqsupset_\omega$ ; $\mathbb{R}$ , the reals                                                                                                                                            |

| Notion in source 12                                                                                          | Here                                              | Do not confuse with                                                                                                                                                |
|--------------------------------------------------------------------------------------------------------------|---------------------------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| The named group; codes of transitive actions; pointed codes                                                  | $\Gamma; \mathcal{T}_\Gamma, \mathcal{Q}_\Gamma$  | the collection's value groups $\Gamma$ (as in $F_\Gamma, K_\Gamma$ ); $\mathbb{Q}$ ; the component-type set $\mathcal{T}_r$ of <b>naming-elementary-embeddings</b> |
| Type strata; centralizer orbits of a pointed code                                                            | $A_t, T_q; P_{q,i}$                               | $T_\sigma$ , the sign container of <a href="#">Part VI</a> ; $T(i, a, b)$ , the evaluator relation of <a href="#">theorem 62.3</a>                                 |
| Definable selectors                                                                                          | $B(q), B_k(q, i)$                                 | the birthday structures $\mathcal{B}_\lambda$ ; the type blocks $B_t$ of <b>naming-elementary-embeddings</b>                                                       |
| Atoms with centralizer orbit of size at most $k$                                                             | $D_k$                                             | the small-type part $D_\kappa(\mathbf{f})$ of <b>naming-elementary-embeddings</b> , its counterpart for full Replacement                                           |
| Finite reservoirs; the atoms of exact period $n$ ( <a href="#">example 60.2</a> )                            | $B_F, C_F; C_n$                                   | the relation codes $C = (\delta, r, t)$ and the surcomplex structures $\mathcal{C}_\lambda$                                                                        |
| Orbit-type counts; number of cycles of length $n$ ; a rotation offset (proof of <a href="#">lemma 61.1</a> ) | $n_t, d_t, m_t; c_n; c$                           | $c$ , the conjugation                                                                                                                                              |
| The two named involutions and their rotation                                                                 | $\sigma, \tau; r = \tau\sigma$                    | the generic ordinal $\sigma$ and the cofinality $\tau = \text{cf } \kappa$ of <a href="#">Part VI</a> ; the relation word $r$ of a code                            |
| The structures with two named involutions                                                                    | $\mathcal{N}_d$                                   | the expansions $\mathcal{N}_{\kappa, \mathbf{f}}$ of <b>naming-elementary-embeddings</b> (objects of the same kind)                                                |
| Pure rank bound in the proof of <a href="#">theorem 63.1</a>                                                 | $\zeta, V_\zeta^{\text{pure}}$ (source: $\beta$ ) | $\beta$ , reserved for the first new birthday ( <a href="#">convention 1.5</a> )                                                                                   |
| Index set of the odd $n \geq 3$ (proof of <a href="#">theorem 61.3</a> )                                     | $I$                                               | $\mathcal{I}$ , the interpreted universe                                                                                                                           |
| The group $\bigoplus_{i < \omega} \mathbb{Z}/2\mathbb{Z}$ (proof of <a href="#">theorem 62.3</a> )           | $V$                                               | the universe $V$ and its stages $V_\alpha$                                                                                                                         |
| Kernel bound in <a href="#">question 65.2</a>                                                                | $\lambda \leq \kappa$ , cardinals                 | the epsilon-number cutoffs $\lambda$ of Parts I–V                                                                                                                  |

Equations are numbered within sections here, not consecutively as in the source. Typography only: source 12's sans-serif theory names (ZFC, ZFU) and its roman superscript in  $V_\zeta^{\text{pure}}$  are set as in the rest of this report. No other symbol was renamed.

## 54.5 Relation to Part VI and to the collection

*Remark 54.2* (Question [38.6](#) for named group actions). Source 10's [question 38.6](#) asks to classify the predicates  $P$  for which  $(M_\kappa^{\text{ker}}, P)$  retains Replacement or full Collection, whether the first failure is measured by the support growth of uniquely  $P$ -definable outputs, and how much of the permutation group must remain available to force kernel confinement. Source 12 answers all three parts for  $\kappa = \omega$  and for  $P$  the graph  $\text{Act}$  of a uniformly named action of a pure set-sized group.

- Replacement holds iff every  $\Gamma$ -orbit is finite and the exceptional set  $X_{\text{Act}}$  is finite ([theorem 58.1](#)).
- Support growth is measured by the levels  $R_k$ : for  $k \geq 1$ ,  $R_k$  holds iff every orbit is finite and only finitely many atoms have centralizer orbit of size at most  $k$  ([theorem 59.1](#)); the first failing level is the least  $d$  with  $D_d$  infinite, or there is none and only full Replacement fails ([corollary 59.3](#)).

- The group that remains is the centralizer  $\mathcal{Z}$ : a unique output's kernel is a union of  $\mathcal{Z}_{(F)}$ -orbits ([lemma 56.1](#)). For the trivial action  $\mathcal{Z} = \text{Sym}(A)$ , whose stabilizer  $\mathcal{Z}_{(F)}$  has the singletons of  $F$  and the infinite set  $A \setminus F$  as its orbits, so the lemma gives source 10's kernel confinement, [lemma 32.1](#), at  $\kappa = \omega$ .
- Full Collection fails at  $\kappa = \omega$  for every such expansion, already in the reduct ([proposition 55.3](#)), so the Collection half of the question has no content at this cutoff.

Source 10's enumeration predicate  $E$  is not the graph of a group action. But the uniform evaluator of [theorem 62.3](#), with the parameter  $a_0$ , defines the injection  $i \mapsto a_{e_i}$  of  $\omega$  onto a set of  $\omega$  atoms, that is, an instance of  $E$  at  $\kappa = \omega$ , and Replacement fails at domain size  $\omega$ , as [theorem 34.2](#) predicts; this comparison is an observation of this merge. The question stays open for arbitrary predicates, for group actions at uncountable  $\kappa$  ([question 65.2](#)) and for named relational structures ([question 65.3](#)). A dated note after [question 38.6](#) records this answer beside the earlier one, for named lifts of finitely many atom injections at every  $\kappa$ , in [naming-elementary-embeddings](#) ([remark 54.4](#)).

*Remark 54.3* (The companion: source 10). Source 12's setting is source 10's at  $\kappa = \omega$ , and several of its statements are cases or refinements of source 10's; both proofs are printed.

- [Proposition 55.2](#) gives, for every named action, the axioms of [theorem 32.2](#) other than Replacement at  $\kappa = \omega$ , with expanded Separation proved as in [proposition 34.1](#); Replacement of the reduct is the trivial action in [theorem 58.1](#). [Proposition 55.3](#) is the case  $\kappa = \omega$  of [theorem 33.2](#), with the same witnesses, sets of  $n$  atoms. Both sources credit the model to Lévy, through Hamkins–Yao.
- [Lemma 56.1](#) refines [lemma 32.1](#): there every atom permutation is available, and invariance forces  $\ker(y) \subseteq \ker(w) \cup \ker(p)$ ; here only the centralizer of the named action acts. [Lemma 56.2](#) isolates the last step of the proof of [theorem 32.2](#).
- [Theorem 63.1](#) is the case  $\kappa = \omega$ ,  $P = \text{Act}$ , of [theorem 35.2](#), which allows any fixed finite list of ambient-definable predicates. Source 10 collects witnesses by ambient Collection and separates the pure ones; source 12 bounds their least pure ranks by ambient Replacement and separates them from a pure stage  $V_\zeta^{\text{pure}}$ . [Proposition 63.2](#) is [proposition 35.1](#) at  $\kappa = \omega$ . The four items of [corollary 63.3](#) are, for named actions, the set-graph scheme Gr, a birthday bound, surreal Collection SC and definable-cut interpolation DCI, by the mechanism of [corollary 35.3](#).
- [Corollary 63.4](#) extends [corollary 35.4](#) in two ways: from the enumeration predicate to named group actions, and from full Replacement to the graded levels. One pure surreal universe carries the whole hierarchy of [theorem 61.3](#).
- The finite-kernel model is not a permutation model ([remark 55.1](#)); in particular it is not the basic Fraenkel model of [section 48](#), where source 11 bounds transcendence degrees by atom supports.

Dated notes after [theorem 35.2](#) and after [question 38.6](#) point here.

*Remark 54.4* (The report on named elementary embeddings). Batch-89 manuscript 05, placed in commit 23adb85f9 as the research-report collection's report [SetTheory/Cardinals/docs/reports/ordinals-and-order-types/naming-elementary-embeddings](#) (*Naming Elementary Embeddings: Atom Components, Replacement, and Reflection*; AI-assisted, unrefereed, not formalized), names the canonical lifts  $j_{f_1}, \dots, j_{f_r}$  of finitely many atom injections as function symbols in the kernel models  $M_\kappa(A) = M_\kappa^{\text{ker}}$ . Source 12 did not know it: its pin 8dc2592e9 is an ancestor of that placement.

- *The full criterion.* For permutations, naming the atom maps and naming their lifts give definitionally equivalent languages: the lift is definable from the atom map ([proposition 62.4](#)), and the lift restricted to atoms is the atom map; the uniform action of the free group on the

named maps is definable ([proposition 62.1](#)). Its orbits are that report’s components and its finite transitive types are that report’s finite component types, so that report’s small-type part  $D_\omega(\mathbf{f})$ , the union of the finite type blocks, is  $X_{\text{Act}}$  once every component is finite. That report’s Theorem 5.1(2) (`nee:thm:profile`, “Component-profile criterion”) is therefore [theorem 58.1](#) for finitely many named permutations; that report printed it in ProveIt before source 12 arrived, with an independent proof. That report also covers non-surjective injections and every uncountable  $\kappa$  (its Theorem 5.1(1)), which source 12 does not. New in source 12 are arbitrary pure set-sized group actions, the graded levels  $R_k$  ([theorem 59.1](#) and [corollary 59.3](#)), the uniform-evaluator separation ([theorem 62.3](#)) and the explicit hierarchy below.

- *One cycle of each length.* That report’s Example 5.5 (`nee:ex:finite`) is [example 60.2](#); there only the failure of Replacement is proved, while [theorem 60.1](#) adds that every  $R_k$  holds.
- *Two safe names, jointly unsafe.* That report’s Corollary 7.9 (`nee:cor:jointunsafe`) gives two permutations at the cutoff  $\aleph_1$  whose one-symbol expansions have Replacement, Collection and reflection and whose joint expansion fails Replacement. [Theorem 61.3](#) is a finite-cutoff, graded sharpening: at  $\kappa = \omega$ , two involutions with finite joint orbits fail exactly the levels  $R_k$  with  $k \geq d$ .
- *Questions.* That report’s Theorems 5.1(1) and 6.1 (`nee:thm:robust`) answer [question 65.2](#) in part, and its Theorems 7.5 and 7.7 (`nee:thm:collection`, `nee:thm:reflection`) answer [question 65.11](#) in part, both for finitely many named injections (dated notes after the two questions). Its Questions 10.7 (general atom ideals), 10.5 (other bounded-degree signatures), 10.9 (minimal complexity of failed instances) and 10.12 (machine-checked certificates) are the counterparts of [questions 65.1](#), [65.3](#), [65.6](#) and [65.12](#), and its Question 10.4 (commuting named automorphisms) meets [question 65.4](#).
- *Kunen.* As the dated note following [theorem 18.3](#) records for that report’s lifts, every named symmetry here fixes every pure set, so none bears on [theorem 18.3](#).

The two manuscripts were written independently; neither is claimed first, and both credit the finite-kernel model to Lévy through Hamkins–Yao (that report also to Yao).

[Added 4 October 2026, batch 93: since then `naming-elementary-embeddings` has a Part II (batch-93 manuscript 01, *Commuting Actions and Replacement*, labels `nee:ca:`), pinned to `a9ab9a698`, twelve minutes before this Part was written, so it does not cite this Part; this Part is prior in the repository and the proofs are independent. At  $\kappa = \omega$  several of its results are results of this Part: its Corollary 15.4 is [corollary 58.2](#); the finite-cutoff clause of its Theorem 18.2 follows from [theorem 58.1](#) with [proposition 62.1](#); its Proposition 19.1 is [proposition 62.2](#) with [corollary 58.2](#); its Proposition 19.2 at  $\kappa = \omega$  is [theorem 58.1](#) for countable groups; its Proposition 19.4 is [proposition 62.1](#); and its Theorem 19.3 makes the separation of [theorem 62.3](#) for the same group (note after that theorem). New there: every uncountable cutoff (for a finitely generated group named by generators every action keeps Replacement iff  $\text{cf } \kappa > \tau(G)$ , the number of conjugacy classes of subgroups; for a countable group with a uniform evaluator the profile criterion holds with full orbits), commuting permutations and injections (threshold  $\text{cf } \kappa > \omega$  for at most one injection,  $\text{cf } \kappa > 2^{\aleph_0}$  for two or more), the exact definable kernel for commuting permutations, and forcing stability of a fixed action. AI-assisted, unrefereed, not formalized.]

## 55 The finite-kernel universe and the expanded language

### 55.1 Ambient assumptions and set/class boundaries

Work in an ambient well-founded universe  $\mathcal{U} = V(A)$  of sets with urelements, satisfying Choice, Replacement, and Collection, with  $A$  a set of infinitely many atoms. Extensionality applies to sets, atoms have no members, and the ordinary cumulative hierarchy over  $A$  is available. The pure sets form a universe satisfying ZFC. This is an external metatheoretic setting. It can be interpreted by tagged well-founded constructions in ordinary set theory; no assertion that ZFC proves the existence of a transitive set model of itself is intended.

For every object  $x$ , including an atom, let

$$\ker(x) = A \cap \text{TC}(\{x\}), \quad M_\omega^{\ker} = \{x \in \mathcal{U} : |\ker(x)| < \omega\}.$$

The universe  $M_\omega^{\ker}$  is a definable class, not a set in  $\mathcal{U}$ . It contains every atom separately, every finite set of atoms, and every pure set. It does not contain  $A$ , or any infinite set of atoms. For a set  $w$ ,

$$\ker(w) = \bigcup_{x \in w} \ker(x). \quad (55.1)$$

For an atom  $a$ ,  $\ker(a) = \{a\}$ . The empty set and all ordinals are pure.

*Remark 55.1* (Kernel is not permutation support). The condition is that only finitely many atoms actually occur in the transitive closure. It is not the condition that an object is fixed by the pointwise stabilizer of some finite set of atoms. An infinite atom set can have empty permutation support but cannot belong to  $M_\omega^{\ker}$ . Thus  $M_\omega^{\ker}$  is not, by definition, the usual full hereditarily symmetric Fraenkel universe.

Fix a pure set-coded group  $\Gamma$ . Let an external action of  $\Gamma$  on  $A$  be given. We name its graph by a ternary relation

$$\text{Act}(g, a, b) \iff g \in \Gamma, a, b \in A, \text{ and } g \cdot a = b.$$

It is false on invalid arguments. The group code is an allowed pure parameter. The resulting structure is

$$\mathcal{M}_{\text{Act}} = (M_\omega^{\ker}, \in, \text{At}, \text{Act}).$$

The relation **Act** is an ambient set, but it need not be an element of  $M_\omega^{\ker}$ . Naming a predicate does not assert that its graph is an internal set. Later, a named atom permutation can likewise be read as its binary graph, false off the atoms. Equivalently it may be made a total function by fixing all non-atoms; the hereditary lift considered in [proposition 62.4](#) is a different, explicitly defined total function.

For every fixed formula  $\varphi$ , its interpretation in  $M_\omega^{\ker}$  is given externally by restricting quantifiers to finite-kernel objects and replacing **Act** by the given action graph. This is ordinary formula-by-formula relativization. We never use an internal truth predicate for all formulas.

### 55.2 Axioms that cannot be damaged in this construction

**Proposition 55.2.** *For every action **Act**,  $\mathcal{M}_{\text{Act}}$  satisfies the urelement versions of Extensionality, Foundation, Infinity, Pairing, Union, Power Set, Choice, and full Separation in the expanded language. Its transitive closure operation and the predicate “ $|\ker(x)| \leq k$ ” are absolute to  $\mathcal{U}$  for  $x \in M_\omega^{\ker}$  and  $k < \omega$ .*

*Proof.* The class  $M_\omega^{\ker}$  is transitive. Pairing and Union preserve finite kernels. Every ambient subset of a member  $w \in M_\omega^{\ker}$  has kernel contained in  $\ker(w)$ ; hence every such subset belongs to  $M_\omega^{\ker}$ , and the ambient power set of  $w$  itself has finite kernel. This proves Power Set and also proves expanded Separation: for a fixed formula, the externally separated subset of  $w$  still has kernel contained in  $\ker(w)$ .

An ambient well-order of  $w$  has a graph whose kernel is contained in  $\ker(w)$ , so that graph is in  $M_\omega^{\ker}$ . This proves the well-ordering form of Choice. Alternatively an ambient choice function on an internal family has no new atoms. Foundation and set Extensionality are inherited;  $\omega$  gives Infinity.

The ambient transitive closure of  $\{x\}$  has the same atom kernel as  $x$ , so it belongs to  $M_\omega^{\ker}$ . It is also the internal transitive closure, by transitivity and minimality. The finite bijections witnessing kernel sizes belong to  $M_\omega^{\ker}$ , and an internal bijection is an ambient one. This proves the final assertion.  $\square$

**Proposition 55.3.** *Full Collection fails in every  $\mathcal{M}_{\text{Act}}$ , already for a formula not using Act.*

*Proof.* For every  $n \in \omega$  there is a set of exactly  $n$  atoms in  $M_\omega^{\ker}$ . If an internal set  $v$  collected such witnesses for all  $n$ , then every one of them would be contained in the finite set  $\ker(v)$ . Taking  $n > |\ker(v)|$  is impossible.  $\square$

This is the classical obstruction [22]. Our interest is the finer behavior of unique witnesses, where automorphisms can force all outputs into one common finite atom reservoir.

### 55.3 Replacement and its kernel-bounded fragments

**Definition 55.4.** For a fixed metatheoretic  $k \in \omega$ , the scheme  $R_k$  asserts the following for every formula  $\varphi(x, y, p)$  in the expanded language and all parameters  $w, p \in M_\omega^{\ker}$ , with  $w$  a set:

$$\begin{aligned} \mathcal{M}_{\text{Act}} &\models \forall x \in w \exists! y \varphi(x, y, p), \\ \mathcal{M}_{\text{Act}} &\models \forall x \in w \forall y (\varphi(x, y, p) \rightarrow |\ker(y)| \leq k) \end{aligned}$$

imply that the range  $\{y : \exists x \in w \varphi(x, y, p)\}$  is a set in  $M_\omega^{\ker}$ . The scheme **Rep** is the same assertion without the bound on output kernels.

Finite tuples of parameters may be packed into one parameter. Range and graph versions are equivalent here: after forming the range, Separation on the product with  $w$  gives the graph, and conversely projections of an internal graph remain internal.

For fixed  $k$ ,  $R_{k+1}$  implies  $R_k$ . Every output in  $M_\omega^{\ker}$  has a finite kernel, but a definable family may have unbounded finite kernel sizes. Consequently the external collection of all schemes  $R_k$  need not imply **Rep**. A principal example below establishes that it does not.

## 56 The symmetry calculus

### 56.1 Lifting atom symmetries

An ambient permutation  $h$  of  $A$  extends uniquely by well-founded recursion to an automorphism  $\hat{h}$  of  $\mathcal{U}$ :

$$\hat{h}(a) = h(a) \quad (a \in A), \quad \hat{h}(x) = \{\hat{h}(z) : z \in x\} \quad (x \text{ a set}).$$

It preserves finite kernels and restricts to  $M_\omega^{\ker}$ . It fixes every pure object. If  $h$  fixes  $\ker(x)$  pointwise, then  $\hat{h}(x) = x$ , by induction through the transitive closure.

The atom permutations that preserve the named relation are exactly

$$\mathcal{Z} = \text{Aut}_\Gamma(A) = \{h \in \text{Sym}(A) : h(g \cdot a) = g \cdot h(a) \text{ for all } g, a\}.$$

We usually omit the hat when such a permutation acts on arbitrary objects. For a finite set  $F \subseteq A$ , let  $\mathcal{Z}_{(F)}$  be its pointwise stabilizer.

**Lemma 56.1** (Unique outputs have invariant kernels). *Suppose  $w, p \in M_\omega^{\ker}$ ,  $F = \ker(w) \cup \ker(p)$ ,  $x \in w$ , and  $y$  is the unique object satisfying  $\varphi(x, y, p)$  in  $\mathcal{M}_{\text{Act}}$ . Then every  $h \in \mathcal{Z}_{(F)}$  fixes  $y$ , and  $\ker(y)$  is a union of  $\mathcal{Z}_{(F)}$ -orbits on atoms.*

*Proof.* Every such  $h$  fixes  $x$  and  $p$ , and it is an automorphism of the expanded structure. Hence  $\varphi(x, h(y), p)$  holds. Uniqueness gives  $h(y) = y$ . Transitive closure is equivariant, so  $h[\ker(y)] = \ker(y)$ . This is exactly invariance of the kernel, which means it is a union of orbits.  $\square$

**Lemma 56.2** (Finite reservoir principle). *Fix a definable total functional relation on  $w \in M_\omega^{\ker}$ . If there is an ambient finite set  $B \subseteq A$  containing the kernel of every output, then its range belongs to  $M_\omega^{\ker}$ .*

*Proof.* Relativize the fixed defining formula to  $M_\omega^{\ker}$ . Ambient Replacement produces its range  $v$  as a set in  $\mathcal{U}$ . Equation (55.1) gives  $\ker(v) \subseteq B$ , so  $v \in M_\omega^{\ker}$ .  $\square$

The reservoir need not be definable in  $M_\omega^{\ker}$ . Its external existence is sufficient to prove the internal scheme. In contrast, to refute the scheme we must exhibit a genuinely definable internal family. This difference is why the coding arguments below are indispensable.

## 56.2 Finite transitive types and the centralizer

Assume for this subsection that all  $\Gamma$ -orbits on  $A$  are finite. Let  $t$  range over isomorphism types of finite transitive  $\Gamma$ -sets. Choose a representative  $O_t$  externally, and write

$$n_t = |O_t|, \quad d_t = |\text{Aut}_\Gamma(O_t)|, \quad m_t = \text{number of components of type } t \text{ in } A.$$

Let  $A_t$  be the union of those components. Types with  $m_t = 0$  may be ignored. The group  $\Gamma$  is fixed, so “isomorphic” always means equivariant for that fixed group, not after changing its generators by a group automorphism.

**Lemma 56.3** (Size of a centralizer orbit). *For  $a \in A_t$ , the following hold.*

1. *If  $m_t$  is infinite, then  $\mathcal{Z}a$  is infinite.*
2. *If  $m_t$  is finite, then  $|\mathcal{Z}a| = m_t d_t$ .*
3. *The automorphism group of  $O_t$  acts freely on  $O_t$ . In particular  $d_t$  divides  $n_t$ , and  $A_t$  contains  $n_t/d_t$  centralizer orbits when  $0 < m_t < \omega$ .*

*Proof.* An equivariant automorphism of a transitive  $\Gamma$ -set is determined by the image of one point: if it fixes  $a$ , it fixes every  $g \cdot a$ . Therefore the action of  $\text{Aut}_\Gamma(O_t)$  on  $O_t$  is free, so each of its orbits has  $d_t$  elements.

An element of  $\mathcal{Z}$  sends a component to an isomorphic component, and its restriction is an equivariant isomorphism. Conversely, every equivariant isomorphism from one component to another

extends to an element of  $\mathcal{Z}$ : for distinct components use the isomorphism in one direction and its inverse in the other, fixing all remaining components; for the same component extend an automorphism by the identity.

For each target component of type  $t$ , the possible images of a fixed point form an orbit of size  $d_t$ . There are  $m_t$  disjoint target components. This proves the first two assertions, and the last follows by counting.  $\square$

The factor is  $m_t d_t$ , not  $m_t n_t$ . A transitive action can have a trivial automorphism group. The odd dihedral actions below exploit precisely this possibility.

[Added 4 October 2026, batch 96: Lemma 26.3 of **naming-elementary-embeddings** (Part III, **nee:aa:lem:mult**, batch-96 manuscript 05, written later and independently, citing this Part only for **theorem 59.1**) proves the same statement for a finite tuple of named atom *injections*, whose components need not be orbits of a group, and for *infinite* components:  $|\mathcal{Z}a| = \mu_t \delta_t$  as a cardinal identity, with  $\mu_t$  the number of copies (here  $m_t$ ) and  $\delta_t = |\text{Aut}(X_t)| \leq \aleph_0$  (here  $d_t$ ), the automorphism group acting freely. It is used there, with pure-real “rooted codes” of components, to show that for every infinite cutoff  $\kappa$  the atoms occurring in objects definable from given parameters and finitely many pure reals are the component saturation of the parameters’ kernels together with the atoms whose centralizer orbit has fewer than  $\kappa$  elements (its Theorem 27.1, **nee:aa:thm:kernel**). AI-assisted, unrefereed, not formalized.]

## 57 A pure set of codes that detects finite exceptional strata

### 57.1 Labeled action codes

No canonical choice of one representative per isomorphism type is needed inside  $M_\omega^{\text{ker}}$ . Define the pure set

$$\mathcal{T}_\Gamma = \{(n, \rho) : 1 \leq n < \omega, \rho : \Gamma \rightarrow \text{Sym}(n) \text{ is a homomorphism acting transitively on } n\}. \quad (57.1)$$

It is a set because the group code, every  $n$ , and every finite symmetric group are pure, and the required function sets and their countable union exist in the pure universe. Thus  $\mathcal{T}_\Gamma \in M_\omega^{\text{ker}}$  without using any expanded-language Replacement.

For  $q = (n, \rho) \in \mathcal{T}_\Gamma$ , an *equivariant realization* is an injective map  $f : n \rightarrow A$  satisfying

$$\forall g \in \Gamma \forall i < n \quad \text{Act}(g, f(i), f(\rho(g)(i))). \quad (57.2)$$

Every such map has finite kernel and belongs to  $M_\omega^{\text{ker}}$ . Its image is a whole  $\Gamma$ -orbit: it is invariant and transitive. Conversely an isomorphism from the coded action onto a finite atom orbit is such a realization.

Define the formulas

$$\begin{aligned} \text{Type}_q(a) &\iff \text{At}(a) \wedge \exists f \exists i < n (f \text{ realizes } q \wedge f(i) = a), \\ \text{Point}_{q,i}(a) &\iff \text{At}(a) \wedge \exists f (f \text{ realizes } q \wedge f(i) = a). \end{aligned}$$

These are single first-order definitions with variables  $q, i, a$ . Quantification over  $g \in \Gamma$  is allowed because  $\Gamma$  is an internal pure set. Each realization graph is an internal finite-kernel object. Hence the definitions have exactly their intended ambient meanings.

Externally, the first formula defines the stratum  $A_t$  corresponding to  $q$ . The second defines a centralizer orbit of a point in that type, or the empty class if the type is absent. We denote these external atom subsets by  $T_q$  and  $P_{q,i}$  respectively.

## 57.2 Total finite-output selectors

**Lemma 57.1** (Definable selectors). *There are expanded-language formulas defining the following total functions on pure set domains:*

$$B(q) = \begin{cases} T_q, & T_q \text{ is finite,} \\ \emptyset, & T_q \text{ is infinite,} \end{cases} \quad q \in \mathcal{T}_\Gamma,$$

$$B_k(q, i) = \begin{cases} P_{q,i}, & |P_{q,i}| \leq k, \\ \emptyset, & |P_{q,i}| > k, \end{cases} \quad (q, i) \in \mathcal{Q}_\Gamma := \{(q, i) : q = (n, \rho) \in \mathcal{T}_\Gamma, i < n\}.$$

Every value is a finite set of atoms in  $M_\omega^{\text{ker}}$ ; the values of  $B_k$  have kernel size at most  $k$ .

*Proof.* Abbreviate the single formula

$$\Phi(q, z) \iff z \text{ is a finite set} \wedge \forall a (a \in z \leftrightarrow \text{Type}_q(a)).$$

Then define  $B(q) = y$  by

$$\Phi(q, y) \vee (y = \emptyset \wedge \neg \exists z \Phi(q, z)).$$

Set Extensionality makes the first possible value unique. It exists in  $M_\omega^{\text{ker}}$  exactly when the external stratum is finite. If it does not exist, the second disjunct gives the unique value  $\emptyset$ . When the stratum is empty the first disjunct gives  $\emptyset$  and the second is false, with no ambiguity.

For  $B_k$  replace the type predicate by the pointed predicate and require  $z$  to have at most  $k$  members. All exact finite sets involved belong to  $M_\omega^{\text{ker}}$ , so the same argument applies. The domain  $\mathcal{Q}_\Gamma$  is pure and set-sized. The functions are definable classes on internal sets; this lemma does not assert that their graphs belong to  $M_\omega^{\text{ker}}$ .  $\square$

This construction detects both finite multiplicity and the smaller pointed orbits. It uses neither an internal set of all atoms nor a choice of one atom from each component.

## 58 The exact criterion for full Replacement

Define the ambient exceptional set

$$X_{\text{Act}} = \bigcup \{A_t : 0 < m_t < \omega\}, \quad (58.1)$$

where only finite transitive orbit types occur in the union. This definition still makes sense if some other atom orbits are infinite.

**Theorem 58.1** (Full Replacement classification). *For a uniformly named action of a pure set-sized group on  $A$ ,*

$$\mathcal{M}_{\text{Act}} \models \text{Rep} \iff (\forall a \in A \ | \Gamma a| < \omega) \wedge |X_{\text{Act}}| < \omega.$$

*Proof. Necessity of finite generated orbits.* Suppose  $\Gamma a$  is infinite for some atom  $a$ . The formula  $\text{Act}(g, a, y)$ , with parameter  $a$ , defines a total atom-valued function on the pure set  $\Gamma$ . Its range is  $\Gamma a$ . No internal set can be this infinite set of atoms. Therefore Replacement fails, already with outputs of kernel size one.

*Necessity of a finite exceptional set.* Consider the definable selector  $B$  from lemma 57.1. A finite type stratum is exactly a stratum with finitely many components, since every component of that

type has the same positive finite size. Every realized finite orbit type is represented by a code in  $\mathcal{T}_\Gamma$ . Consequently

$$\bigcup_{q \in \mathcal{T}_\Gamma} B(q) = X_{\text{Act}}$$

externally. If Replacement held, the range of  $B$  would be an internal set, and its union would be the internal atom set  $X_{\text{Act}}$ . Thus  $X_{\text{Act}}$  would be finite.

*Sufficiency.* Assume every  $\Gamma$ -orbit is finite and  $X_{\text{Act}}$  is finite. Let  $w, p \in M_\omega^{\text{ker}}$  and let  $\varphi$  define unique outputs on  $w$ . Put

$$F = \ker(w) \cup \ker(p), \quad B_F = X_{\text{Act}} \cup \bigcup_{a \in F} \Gamma a.$$

This is an ambient finite set of atoms.

We claim that every atom outside  $B_F$  has infinite orbit under  $\mathcal{Z}_{(F)}$ . Such an atom  $b$  is in an infinitely repeated finite orbit type, and its component does not meet  $F$ . At most finitely many components meet  $F$ . Infinitely many isomorphic components remain disjoint from  $F$ , and equivariant swaps from the component of  $b$  to each of them belong to  $\mathcal{Z}_{(F)}$ . The images of  $b$  are therefore infinitely many distinct atoms.

By lemma 56.1, the finite kernel of each unique output is invariant under  $\mathcal{Z}_{(F)}$ . It cannot contain an atom with an infinite  $\mathcal{Z}_{(F)}$ -orbit. Thus every output kernel is contained in  $B_F$ . The finite reservoir principle now puts the range in  $M_\omega^{\text{ker}}$ , as required.  $\square$

**Corollary 58.2.** *Every action of a finite pure group preserves full Replacement. In particular, naming one atom permutation of finite order, including an arbitrary involution, preserves Replacement.*

*Proof.* Every orbit has size at most  $|\Gamma|$ . A finite group has only finitely many transitive action types, since these are represented by its coset actions on subgroups. Therefore only finitely many finite strata can contribute to  $X_{\text{Act}}$ , and each is finite.  $\square$

[Added 4 October 2026, batch 93: Corollary 15.4 of **naming-elementary-embeddings** (Part II, **nee:ca:cor:finite-group**), written later and independently, is this corollary for finitely many named permutations generating a finite group, at every infinite kernel bound  $\kappa$ , not only at  $\kappa = \omega$ .]

Taking the trivial group recovers Replacement for the classical reduct  $(M_\omega^{\text{ker}}, \in, \text{At})$ . There is one transitive type, a singleton, with infinitely many copies; hence  $X_{\text{Act}} = \emptyset$ .

*Remark 58.3* (Two distinct obstructions). Infinite generated orbits are an accessibility obstruction: one parameter already generates infinitely many atoms. Infinitely many finitely repeated finite types are a definable identification obstruction: no single orbit is large, but a pure code can identify each entire exceptional finite stratum. The latter is not excluded by local finiteness of the action.

## 59 The exact bounded-kernel criterion

For every metatheoretic  $k \in \omega$  define the ambient set

$$D_k = \{a \in A : |\mathcal{Z}a| \leq k\}.$$

When every  $\Gamma$ -orbit is finite, lemma 56.3 gives the explicit formula

$$D_k = \bigcup \{A_t : 0 < m_t < \omega, \ m_t d_t \leq k\}. \quad (59.1)$$

Also  $X_{\text{Act}} = \bigcup_{k < \omega} D_k$  in that case.

**Theorem 59.1** (Kernel-bounded Replacement classification). *For every uniformly named action as above:*

1.  $\mathcal{M}_{\text{Act}} \models R_0$  without any hypothesis on its orbits.
2. For every fixed  $k \geq 1$ ,

$$\boxed{\mathcal{M}_{\text{Act}} \models R_k \iff (\forall a \in A \ | \Gamma a| < \omega) \wedge |D_k| < \omega.}$$

*Proof.* For  $k = 0$ , every output is pure. The ambient range set is then pure and belongs to  $M_\omega^{\text{ker}}$ , so lemma 56.2 applies with the empty reservoir.

Now fix  $k \geq 1$ . An infinite  $\Gamma$ -orbit gives the atom-valued counterexample from theorem 58.1; its outputs have kernel size one. Thus  $R_k$  requires all generated orbits to be finite.

Assume this necessary condition. For each pointed code  $(q, i)$ ,  $P_{q,i}$  is either empty or one full  $\mathcal{Z}$ -orbit, by the component-isomorphism argument in lemma 56.3. Every atom occurs in at least one such pointed code. Therefore the selector  $B_k$  satisfies

$$\bigcup_{(q,i) \in \mathcal{Q}_\Gamma} B_k(q, i) = D_k.$$

If  $D_k$  were infinite,  $R_k$  would collect its finite subsets  $B_k(q, i)$  into a set, and Union would give the infinite atom set  $D_k$  in  $M_\omega^{\text{ker}}$ . This is impossible. Necessity follows.

For sufficiency suppose all generated orbits are finite and  $D_k$  is finite. Consider a definable total functional relation on  $w \in M_\omega^{\text{ker}}$ , with parameter  $p$ , all of whose outputs have kernel size at most  $k$ . Let  $F = \ker(w) \cup \ker(p)$ . Form the finite reservoir

$$C_F = D_k \cup \bigcup_{a \in F} \Gamma a \cup \bigcup \{A_t : 0 < m_t < \omega, A_t \cap F \neq \emptyset\}. \quad (59.2)$$

There are only finitely many types touched by  $F$ , and each stratum in the last union is finite. Thus  $C_F$  really is finite.

We show that every atom  $b \notin C_F$  has a  $\mathcal{Z}_{(F)}$ -orbit of size greater than  $k$ . If its type is infinitely repeated, its component misses  $F$ , and infinitely many untouched isomorphic components yield an infinite orbit under the stabilizer. If its type is finitely repeated, the whole stratum is disjoint from  $F$ , by the last term of (59.2). All equivariant automorphisms and component swaps supported on that stratum fix  $F$ , so the  $\mathcal{Z}_{(F)}$ -orbit of  $b$  equals its  $\mathcal{Z}$ -orbit. Its size exceeds  $k$  because  $b \notin D_k$ .

Every output kernel is invariant under  $\mathcal{Z}_{(F)}$  and has at most  $k$  elements. It therefore cannot contain  $b \notin C_F$ . All output kernels are contained in  $C_F$ , and the finite reservoir principle completes the proof.  $\square$

*Remark 59.2* (Why the parameter correction matters). Using only  $D_k \cup \bigcup_{a \in F} \Gamma a$  in (59.2) would be wrong. A parameter can fix one component of a finitely repeated type and thereby prevent that component from being exchanged with another. The remaining centralizer orbits can shrink. The correct argument includes the *whole finitely repeated stratum* whenever its type is touched by  $F$ . This is the essential extra step needed to prove the scheme for arbitrary impure parameters rather than merely for parameter-free definitions.

**Corollary 59.3** (First failing level). *Assume all generated orbits are finite. If a least positive integer  $d$  with infinite  $D_d$  exists, then  $\mathcal{M}_{\text{Act}}$  satisfies  $R_k$  exactly for  $k < d$ . If every  $D_k$  is finite but  $\bigcup_k D_k$  is infinite, all fixed finite levels hold and full Replacement fails.*

This yields an exact spectrum, not just lower and upper estimates.

## 60 One permutation: a complete classification and a limit gap

### 60.1 From a named permutation to a uniform action

Let  $\pi$  be a named permutation of atoms. The action of  $\mathbb{Z}$  by finite iteration is first-order definable from  $\pi$ : a finite path witnesses  $\pi^n(a) = b$  for  $n \geq 0$ , and inverse paths handle negative  $n$ . Every individual path is a finite-kernel set, regardless of whether the whole orbit is finite. Thus this definition does not use Replacement in the expanded language. Conversely  $\pi$  is the action of 1. The two presentations are definitionally equivalent for the schemes at issue.

Let  $c_n$  be the ambient number of cycles of exact length  $n$ . For a transitive  $n$ -cycle, the equivariant automorphism group is the cyclic group of order  $n$ , acting by rotations. Hence  $d_n = n$ .

**Theorem 60.1** (Single-permutation spectrum). *For a named atom permutation  $\pi$ :*

1. *Full Replacement holds if and only if all cycles are finite and*

$$\bigcup \{ \text{atoms in cycles of length } n : 0 < c_n < \omega \}$$

*is finite.*

2. *If all cycles are finite, then every fixed  $R_k$  holds.*
3. *If there is an infinite cycle,  $R_1$  fails.*

*Proof.* The first and third assertions are immediate from the classifications. For the second, the only lengths contributing to  $D_k$  satisfy  $nc_n \leq k$ . In particular  $n \leq k$ , and for each such  $n$  the whole stratum has at most  $k$  atoms. There are finitely many possible  $n$ , so  $D_k$  is finite. Apply [theorem 59.1](#).  $\square$

**Example 60.2** (Every finite level without full Replacement). Take atoms externally labeled

$$A = \{a_{n,i} : n \geq 2, i \in \mathbb{Z}/n\mathbb{Z}\}, \quad \pi(a_{n,i}) = a_{n,i+1}.$$

There is exactly one cycle of each length  $n \geq 2$ . All  $R_k$  hold, but  $X_{\text{Act}} = A$  is infinite, so full Replacement fails. An explicit failing function is

$$n \mapsto C_n = \{a : a \text{ has exact } \pi\text{-period } n\}, \quad n \geq 2.$$

Its  $n$ th value is an internal set of  $n$  atoms. Its range cannot be a set, since its union is  $A$ .

The assertion “every fixed  $R_k$  holds” is a metatheoretic statement about all the fixed-bound schemes. It must not be conflated with a schema allowing an internally varying bound  $k(x)$  across the domain. In [example 60.2](#), the bound  $k(n) = n$  is already enough to reveal the failure.

**Example 60.3** (Infinite order can still be harmless). Instead take infinitely many cycles of each positive finite length. Every finite orbit type is infinitely repeated, so  $X_{\text{Act}} = \emptyset$  and full Replacement holds. The permutation has infinite order, because its cycle lengths are unbounded. Thus finite order is sufficient but not necessary for preservation.

**Corollary 60.4** (Minimal signature for a bounded-level failure). *A single involution never destroys Replacement. A single arbitrary permutation with only finite cycles cannot destroy any fixed  $R_k$ . Two involutions suffice to destroy any prescribed positive level while preserving all lower ones, as proved next.*

## 61 Two involutions realize every finite threshold

### 61.1 The elementary infinite-orbit mechanism

First consider atoms  $a_z$  indexed externally by  $z \in \mathbb{Z}$ , with

$$\sigma(a_z) = a_{-z}, \quad \tau(a_z) = a_{1-z}.$$

Both are involutions. Each separate expansion satisfies Replacement by [corollary 58.2](#). But  $\tau\sigma(a_z) = a_{z+1}$ , so the joint expansion has an infinite generated orbit and fails  $R_1$ . This example illustrates one obstruction, but does not show that finite joint orbits are insufficient. The following construction does.

### 61.2 Odd dihedral components with repeated copies

Fix  $d \geq 1$ . Externally label the atoms by

$$A^{(d)} = \{a_{n,i,j} : n \geq 3 \text{ odd}, i \in \mathbb{Z}/n\mathbb{Z}, 0 \leq j < d\}. \quad (61.1)$$

These labels specify the ambient construction; they are *not* predicates or an enumeration inside the model. Name just two atom involutions:

$$\sigma(a_{n,i,j}) = a_{n,-i,j}, \quad \tau(a_{n,i,j}) = a_{n,1-i,j}. \quad (61.2)$$

Let  $r = \tau\sigma$ , so that  $r(a_{n,i,j}) = a_{n,i+1,j}$ . The generated group action can be taken to be that of the pure infinite dihedral group, or equivalently of the free group on the two named symbols with their involution relations. Uniform evaluation of finite words is definable by finite paths.

**Lemma 61.1** (Rigidity of an odd component). *On a single component  $\{a_{n,i,j} : i \in \mathbb{Z}/n\mathbb{Z}\}$  with  $n$  odd, an equivariant automorphism commuting with both  $\sigma$  and  $\tau$  is the identity. Between two copies with the same  $n$ , there is exactly one equivariant isomorphism: it preserves  $i$ .*

*Proof.* A bijection commuting with the rotation  $r$  has the form  $i \mapsto i + c$ . Commutation with  $\sigma$  requires

$$-i + c = -i - c \pmod{n},$$

so  $2c = 0 \pmod{n}$ . Since  $n$  is odd,  $c = 0$ . The same computation applies between two separately labeled copies.  $\square$

**Lemma 61.2** (Exact centralizer). *For the joint action (61.2),*

$$\mathcal{Z} \cong \prod_{\substack{n \geq 3 \\ n \text{ odd}}} \text{Sym}(d),$$

where the  $n$ th factor permutes the  $d$  copies of the  $n$ -component. Every  $\mathcal{Z}$ -orbit has exactly  $d$  atoms, namely

$$\{a_{n,i,j} : 0 \leq j < d\}$$

for a fixed  $n, i$ .

*Proof.* Any commuting permutation preserves the exact period under  $r$ , so it cannot exchange components with different  $n$ . It must permute the  $d$  components with a fixed  $n$ . By [lemma 61.1](#), once the target copy is chosen, the map inside the component is forced to preserve  $i$ . Conversely arbitrary copy permutations at each  $n$  commute with both named involutions. This proves the product description and the orbit formula.  $\square$

**Theorem 61.3** (A strict hierarchy from two individually harmless involutions). *For every integer  $d \geq 1$ , let*

$$\mathcal{N}_d = (M_\omega^{\ker}(A^{(d)}), \in, \text{At}, \sigma, \tau)$$

*be the structure just constructed. Then:*

1. *Each of its one-involution reducts satisfies full Replacement.*
2. *Every joint atom orbit is finite.*
3. *For every fixed  $k \in \omega$ ,*

$$\mathcal{N}_d \models R_k \iff k < d.$$

4. *A pure-domain, parameter-free definable family of  $d$ -atom sets explicitly witnesses the failure at  $d$ .*

*Proof.* The first assertion follows from [corollary 58.2](#). For each  $n, j$ , the rotation makes the  $n$ -element component transitive, and neither involution changes  $n$  or  $j$ . Hence all joint orbits are finite.

By [lemma 61.2](#),  $D_k = \emptyset$  for  $k < d$ , whereas  $D_k = A^{(d)}$  for  $k \geq d$ . The latter set is infinite. The exact spectrum follows from [theorem 59.1](#), including the always-valid level zero.

For the explicit failure, let  $I = \{n \in \omega : n \geq 3 \text{ and } n \text{ odd}\}$ . Define

$$Y_n = \{a : \text{At}(a), \sigma(a) = a, \text{ and } a \text{ has exact } r\text{-period } n\}. \quad (61.3)$$

The period test is first-order definable using a finite  $r$ -path and the condition that no smaller positive iterate returns to  $a$ . In each odd  $n$ -component,  $\sigma(i) = i$  means  $2i = 0$ , so its unique solution is  $i = 0$ . Thus

$$Y_n = \{a_{n,0,j} : 0 \leq j < d\}, \quad |\ker(Y_n)| = d.$$

For each  $n \in I$ , this finite set belongs to the model and is uniquely characterized by (61.3). The family is total and definable with no impure parameters. A set of all its values would have infinite atom kernel, since these nonempty values are pairwise disjoint for different  $n$ . Its range therefore does not exist internally.  $\square$

**Corollary 61.4** (Failure at the atom-valued level). *For  $d = 1$ , the two involutions individually preserve full Replacement but jointly fail even atom-valued Replacement, although every joint orbit is finite.*

*Proof.* In this case the unique atom fixed by  $\sigma$  and of exact  $r$ -period  $n$  is itself a definable output. These atoms form an infinite family indexed by the pure set  $I$ .  $\square$

### 61.3 What the construction explains

For  $d > 1$ , there is no parameter-free way to choose one of the  $d$  atoms in  $Y_n$  uniformly: the remaining  $\text{Sym}(d)$  symmetry can exchange them. A finite parameter set can break this symmetry in only finitely many component types. The corrected reservoir argument in [remark 59.2](#) handles

those finitely many exceptions. Therefore arbitrary parameter use does not turn the failure at  $d$  into a failure below  $d$ .

No appeal to the failure of Choice is involved. Every internal set is well-orderable. The difficulty is that the collection of all  $Y_n$  is not an internal set to which Choice can first be applied. This is exactly the distinction between individual finite existence and set-indexed collection of those objects.

| Expansion                              | Joint orbits | Fixed $R_k$     | Full Rep |
|----------------------------------------|--------------|-----------------|----------|
| One finite-order permutation           | Finite       | All             | Yes      |
| One cycle of each length $n \geq 2$    | Finite       | All             | No       |
| Infinitely many cycles of every length | Finite       | All             | Yes      |
| Two reflections of $\mathbb{Z}$        | Infinite     | Only $k = 0$    | No       |
| Two involutions on $A^{(d)}$           | Finite       | Exactly $k < d$ | No       |

## 62 Separate symbols, uniform evaluation, and named automorphisms

### 62.1 Finite tuples of permutations

**Proposition 62.1.** *For finitely many named atom permutations  $\pi_1, \dots, \pi_s$ , their uniform action by words in a pure free group is definable by finite-path formulas. Consequently [theorems 58.1](#) and [59.1](#) apply to finite-permutation languages without adding a new set-existence assumption.*

*Proof.* A group word has finite length and uses generator symbols or their inverses. For a fixed word and starting atom, each next point exists uniquely. Its finite evaluation trace is an ambient finite set of atoms and therefore an internal object. A single formula says that a trace follows the word, with inverse steps expressed by the reversed generator relation. The endpoint is unique. This defines the evaluator uniformly on the pure set of reduced words. Conversely each generator action is a special case of the evaluator. Formula translation preserves all the schemes.  $\square$

### 62.2 An infinite signature is not a uniform evaluator

**Proposition 62.2** (Finite-symbol principle). *For an arbitrary family of separately named atom permutations, full Replacement holds in the resulting first-order language if and only if it holds in every finite-symbol reduct. The same assertion holds for each fixed  $R_k$ .*

*Proof.* Every first-order formula contains only finitely many nonlogical symbols. Its scheme instance therefore occurs in one finite-symbol reduct. The converse follows by restricting the language.  $\square$

This does not authorize quantifying over the names of all those symbols. Such quantification becomes possible only after an additional uniform evaluator has been supplied.

**Theorem 62.3** (Separate involutions safe; a uniform evaluator unsafe). *There is a countable language naming involutions  $(\tau_i)_{i < \omega}$  such that the expansion satisfies full Replacement, but adding a single relation  $T(i, a, b)$  for their uniform evaluation destroys  $R_1$ .*

*Proof.* Let  $V = \bigoplus_{i < \omega} \mathbb{Z}/2\mathbb{Z}$  be the pure group of finitely supported binary sequences. Take atoms  $a_v$  labeled externally by  $v \in V$ , and define

$$\tau_i(a_v) = a_{v+e_i}.$$

Each  $\tau_i$  is an involution. Any finite list of these generators acts through a finite group. Its expansion satisfies full Replacement by [corollary 58.2](#). The finite-symbol principle gives Replacement for the entire language of separate symbols.

Now add the relation  $T(i, a, b) \leftrightarrow b = \tau_i(a)$  for  $i \in \omega$ . With parameter  $a_0$ , it defines on the pure set  $\omega$  the function

$$i \mapsto a_{e_i}.$$

These are distinct atoms. Its range is not an internal set, so  $R_1$  fails. The external labels  $v$  were never part of the original language; the added evaluator is exactly the new uniform information used in this argument.  $\square$

[Added 4 October 2026, batch 93: Theorem 19.3 of [naming-elementary-embeddings](#) (Part II, [nee:ca:uni:two\\_point](#)), written later and independently, makes the same separation for the same group  $V = \bigoplus_{i < \omega} \mathbb{Z}/2\mathbb{Z}$  with a different witness: each block of the action is the two-point action of one nonzero character, so every orbit has at most two atoms and the failure comes from a uniform formula distinguishing stabilizers, not from an infinite orbit. It holds in the kernel models  $M_\kappa^{\text{ker}}$  for every  $\kappa$  with  $\text{cf } \kappa \leq 2^{\aleph_0}$ , the evaluator is amenable, and for  $\text{cf } \kappa > 2^{\aleph_0}$  every uniformly named action of  $V$  keeps Replacement. At  $\kappa = \omega$  that witness keeps  $R_1$  and fails  $R_2$  by [theorem 59.1](#) (an observation recorded there at the write), whereas the regular action above fails  $R_1$ .]

### 62.3 The same results for automorphisms of the membership universe

The named atom permutations can also be viewed as named automorphisms of the whole universe  $M_\omega^{\text{ker}}$ . This is not a different source of pathology.

**Proposition 62.4** (Definability of a hereditary lift). *For a named atom permutation  $\pi$ , the graph of its canonical hereditary lift  $\hat{\pi}$  is first-order definable over  $M_\omega^{\text{ker}}$  without assuming expanded Replacement. The analogous assertion holds uniformly for a pure set-coded group action.*

*Proof.* For a given object  $x$ , use the internal set  $D = \text{TC}(\{x\})$ . In the ambient universe, the graph of  $\hat{\pi} \upharpoonright D$  has kernel contained in

$$\ker(x) \cup \pi[\ker(x)],$$

which is finite. Therefore this graph belongs to  $M_\omega^{\text{ker}}$ .

A formula characterizes it as a function  $f$  on the transitive set  $D$  such that  $f(a) = \pi(a)$  for atoms in  $D$  and, for every set  $z \in D$ ,

$$\forall u (u \in f(z) \leftrightarrow \exists v \in z u = f(v)),$$

with  $f(z)$  required to be a set. Well-founded induction proves uniqueness. Asking for such a function with  $f(x) = y$  therefore defines  $y = \hat{\pi}(x)$ . This formula uses an existing graph on one transitive closure, not a class-sized graph. For a fixed  $g \in \Gamma$ , replace  $\pi(a)$  by the relation  $\text{Act}(g, a, b)$ ; the argument is uniform in  $g$ .  $\square$

Thus the hierarchy may equivalently be stated as a hierarchy produced by naming two involutive automorphisms of the membership universe. Their individual well-founded recursive behavior is not enough to guarantee that Replacement survives their joint naming.

## 63 Why pure surreal mathematics cannot see these failures

### 63.1 A stronger pure-output theorem

**Theorem 63.1** (Expanded Collection for pure-valued relations). *Let  $\text{Act}$  be any named action, including one violating the criteria above. Let  $w, p \in M_\omega^{\text{ker}}$ , with  $w$  a set, and let  $\varphi$  be an expanded-language formula. If*

$$\mathcal{M}_{\text{Act}} \models \forall x \in w \exists y (y \text{ is pure} \wedge \varphi(x, y, p)),$$

*then there is a pure set  $b \in M_\omega^{\text{ker}}$  such that*

$$\mathcal{M}_{\text{Act}} \models \forall x \in w \exists y \in b \varphi(x, y, p).$$

*In particular, full Replacement for pure-valued functions holds, even with impure parameters and occurrences of  $\text{Act}$ .*

*Proof.* Relativize the fixed formula to  $M_\omega^{\text{ker}}$  in the ambient universe. For each  $x \in w$ , let  $\alpha_x$  be the least pure rank at which a pure witness appears. This least ordinal exists because a witness exists. Ambient Replacement bounds the ordinals  $\alpha_x$  by an ordinal  $\zeta$ . Now define in  $\mathcal{U}$

$$b = \{y \in V_\zeta^{\text{pure}} : \exists x \in w \varphi^{M_\omega^{\text{ker}}}(x, y, p)\},$$

choosing  $\zeta$  sufficiently large that every least-rank witness belongs to  $V_\zeta^{\text{pure}}$ . This is an ambient set of pure objects, hence is itself pure and belongs to  $M_\omega^{\text{ker}}$ . It collects witnesses for the internal relation. No global choice of witnesses, and no internal expanded Replacement, has been used.

When the relation is functional, its exact ambient range consists entirely of pure objects and is a pure set. Alternatively use Separation after the preceding Collection assertion to obtain that range internally.  $\square$

The proof is valid for any fixed additional predicates that are externally definable from ambient set parameters, not merely actions. Its strength comes from retaining all pure ranks, not from any special property of the centralizer.

### 63.2 The sign-sequence surreal class is unchanged

Use the pure sign-sequence presentation

$$\mathbf{No} = \bigcup_{\alpha \in \text{Ord}} \{-, +\}^\alpha,$$

where  $-$  and  $+$  are two fixed distinct pure objects. A surreal number is a set-coded function of ordinal length, and its birthday is that length. This is the familiar sign-expansion presentation, also central to the repository's construction [2, 27].

**Proposition 63.2** (Pure invariance). *The structures  $\mathcal{U}$  and  $M_\omega^{\text{ker}}$  have identical pure sets, pure ordinals, and pure sign-sequence surreal numbers. Any relation or operation defined by a fixed formula whose quantifiers are restricted to the pure universe has the same interpretation in both.*

*Proof.* An ambient object is pure exactly when its kernel is empty, so every ambient pure object lies in  $M_\omega^{\text{ker}}$ ; conversely every object pure in  $M_\omega^{\text{ker}}$  is ambiently pure by [proposition 55.2](#). Membership on these objects is unchanged. Induction on formulas restricted to the pure domain proves the last assertion. In particular the ordinal and sign-sequence definitions are identical.  $\square$

This applies to the ordinary pure constructions of the surreal ordered field, its complexification, and the omnific integers. It does not validate every unrefereed claim in a repository report, and it does not identify an atom-enriched presentation with the pure presentation without an explicit coding theorem.

**Corollary 63.3** (Surreal-valued Replacement and cuts). *For every named action  $\text{Act}$ , all of the following hold in  $\mathcal{M}_{\text{Act}}$ , for expanded-language formulas with arbitrary internal parameters.*

1. *Every definable set-domain surreal-valued function has an internal set range and graph.*
2. *The birthdays of its values are bounded by an ordinal.*
3. *Every definable relation on an internal set with surreal witnesses has an internal set collecting surreal witnesses.*
4. *Given definable total functions  $\ell : I \rightarrow \mathbf{No}$  and  $r : J \rightarrow \mathbf{No}$  on internal sets, with  $\ell(i) < r(j)$  for all  $i, j$ , there is a unique simplest surreal strictly between all the lower and upper values.*

*Proof.* Surreal values are pure, so the first and third assertions follow from [theorem 63.1](#). The range of birthday lengths is a pure set of ordinals; its union, or a larger successor ordinal, bounds every birthday. For the last assertion, collect both ranges into actual pure sets  $L, R$ . They satisfy  $L < R$  in the unchanged surreal order. The standard simplicity theorem supplies the unique shortest sign sequence between these set bounds [2]. Its pure code belongs to  $M_{\omega}^{\text{ker}}$ .  $\square$

### 63.3 A non-recoverability statement

**Corollary 63.4.** *Pure surreal-valued Replacement, pure surreal-valued Collection, and definable set-indexed surreal cut interpolation do not together imply  $R_1$  in the urelement setting of this article. More generally, the entire hierarchy of [theorem 61.3](#) is compatible with one and the same pure surreal universe.*

*Proof.* Apply [corollary 63.3](#) to  $\mathcal{N}_1$ , which fails  $R_1$ . For the second assertion, build the countable atom sets  $A^{(d)}$  over a fixed pure universe. All resulting structures have that same pure universe and its canonical surreal class; their kernel-bounded spectra are different by [theorem 61.3](#).  $\square$

There is no contradiction with a theorem that recovers full Replacement from ordinal or surreal bounding over a *pure* base theory with additional hierarchy axioms. The present ambient theory has objects with nonempty atom kernels. The missing sets would contain infinitely many atoms, whereas every pure surreal code has empty kernel. The type of the output matters.

## 64 Finite verification and a formalization roadmap

### 64.1 The finite algorithm

The accompanying program, shipped as `code/12-named-symmetries-verify_orbit_spectra.py`, uses only the Python standard library. A finite action is supplied as a tuple of permutations on  $\{0, \dots, N-1\}$ . Its generated components are found by graph traversal. To find every equivariant isomorphism between two transitive components of size  $n$ , the algorithm tries each possible image of one base point and propagates it along generator and inverse-generator edges. Inconsistent assignments are rejected.

**Proposition 64.1** (Correctness of the base-image algorithm). *For two finite transitive actions with the same generator labels, the base-image propagation algorithm returns exactly their equivariant bijections. With  $s$  generators and  $n$  points in each action it uses  $O(sn^2)$  elementary transition checks, with  $O(n)$  working memory per candidate beyond stored input and output.*

*Proof.* Any equivariant map is forced by the image of one point, because every point is reached by a generator word. Therefore a genuine isomorphism follows the algorithm’s propagation and is never rejected. Conversely a successful propagation gives a total map commuting with every generator and inverse. Its image is a nonempty invariant subset of the transitive target, hence is the whole target. Equal finite cardinalities then make it bijective. There are  $n$  candidate base images; each propagation visits at most  $n$  points and checks  $2s$  edges per point. This gives the stated bound.  $\square$

To compute centralizer orbits on a disjoint union, the program considers all equivariant isomorphisms between pairs of generated components. Each such isomorphism extends to a global commuting permutation, as in [lemma 56.3](#); their point images therefore give the exact centralizer orbit relation. This avoids enumerating the potentially factorial-size centralizer itself.

## 64.2 Executed checks and what they establish

The delivered script was run successfully and produced the record shipped as `data/12-named-symmetries-verification_results.json`. It reports **203 checks**, including **35 independent brute-force comparisons** with enumeration of every permutation on domains of size at most seven. The cases include regular cycles, odd dihedral components with repeated copies, finite disjoint unions of distinct odd lengths, and malformed input rejection. Even dihedral lengths provide negative controls: the local centralizer has two elements rather than one, so the oddness hypothesis is tested rather than silently ignored.

| Odd cycle length $n$ | Copies $d$ | Centralizer orbit size | Fixed atoms per $n$ |
|----------------------|------------|------------------------|---------------------|
| 3                    | 1          | 1                      | 1                   |
| 3                    | 5          | 5                      | 5                   |
| 9                    | 2          | 2                      | 2                   |
| 9                    | 9          | 9                      | 9                   |
| 19                   | 5          | 5                      | 5                   |
| 19                   | 9          | 9                      | 9                   |

A finite truncation itself has only finitely many atoms and does *not* exhibit a failure of Replacement. The computations check the finite component facts used in the infinite proofs. The passage from infinitely many components to nonexistence of an internal range is proved mathematically in [theorem 61.3](#), not numerically tested.

## 64.3 A staged Lean/Rocq development

The repository distinguishes mathematical statements, executable checks, and audited formal proofs [\[27\]](#). A compatible implementation should preserve that separation.

**Stage 1: finite actions.** Formalize the facts that equivariant maps from a transitive action are determined by a base image, that the automorphism group acts freely, and that its orbit size multiplies by the number of repeated components. Then formalize the two odd-dihedral permutations and their centralizer orbit calculation. This stage needs no surreal theory and no urelement model theory.

**Stage 2: the kernel and automorphism interface.** Use a well-founded set-with-atoms representation, with an explicit atom sort disjoint from sets. Prove transitive-closure absoluteness, the kernel union identity, and hereditary extension of atom permutations. The finite reservoir lemma should have separate hypotheses for ambient range existence and for a finite atom bound. In particular, it must not assume the very expanded Replacement scheme that it is intended to prove.

**Stage 3: formulas and relativization.** Use explicit first-order syntax for  $\in$ ,  $\text{At}$ , and the action relation. Prove the automorphism-invariance theorem by induction on formulas. Code the pure set  $\mathcal{T}_\Gamma$  of finite actions and the selectors from lemma 57.1. This is the necessity half of the classification and is not supplied merely by verifying group-theoretic facts.

**Stage 4: the classification.** Derive theorems 58.1 and 59.1, retaining the parameter-sensitive reservoir (59.2) as a named lemma. A theorem statement should quantify over each formula and its finite parameter tuple, rather than over an unproved class of arbitrary internal functions.

**Stage 5: pure output and surreal specialization.** Prove theorem 63.1 at the universe level and only then specialize to the pure sign-sequence representation. A universe-bounded Lean type of surreals is a set-sized approximation in a larger metatheory; it should not be silently treated as an internal set of all class surreals. The interface must state which universe bounds are being used.

**Trust boundary.** No Lean or Rocq proof files are claimed in this package. A program checking finitely many permutations does not certify the set-theoretic metatheorems. Conversely, a formal theorem with a hypothesis explicitly assuming the finite reservoir property would certify a conditional implication, not the actual action classification, until that property is discharged.

## 65 Further research questions

The following are proposed continuation problems. They are open within the scope of this manuscript; no claim is made that each is absent from the wider literature.

**Question 65.1** (General ideals of allowed atom kernels). Replace finite kernels by an ideal  $\mathcal{I}$  on  $A$ . Which conditions on an action and its parameter stabilizers are necessary and sufficient for expanded Replacement in  $M_{\mathcal{I}}^{\text{ker}} = \{x : \text{ker}(x) \in \mathcal{I}\}$ ? A sufficient condition should bound all permissible invariant output kernels in one member of  $\mathcal{I}$ . The main difficulty is an equally sharp, definable necessity argument.

[Added 4 October 2026, batch 90: this is source 10’s question 38.4, specialized to named group actions, and the counterpart of Question 10.7 (“General atom ideals”) of naming-elementary-embeddings, asked there for named lifts. None of the three is answered. Added 4 October 2026, batch 93: Question 23.9 of Part II of that report (nee:ca:q:ideals) asks it again for its group actions; it is

not answered either. Added 4 October 2026, batch 96: Question 37.6 of Part III (`nee:aa:q:ideals`) asks it for finite tuples of named injections, in rooted-code and reservoir form; not answered.]

**Question 65.2** (Infinite kernel bounds and singular thresholds). For  $|\ker(x)| < \kappa$ , classify Replacement restricted to outputs of kernel size below  $\lambda \leq \kappa$ . Determine where cofinality enters and whether an orbit-size formula analogous to (59.1) remains exact. Infinite equivariant realization graphs are no longer automatically allowed, so the coding lemma itself must be reanalyzed.

[Added 4 October 2026, batch 90: answered in part, for finitely many named atom injections instead of group actions and for full Replacement only, by `naming-elementary-embeddings` ([remark 54.4](#)). For every uncountable  $\kappa$ , its Theorem 5.1(1) (`nee:thm:profile`) shows that Replacement survives the naming exactly when the union  $D_\kappa(\mathbf{f})$  of the component-type blocks of cardinality below  $\kappa$  has cardinality below  $\kappa$ : a criterion of the kind asked for, at the level  $\lambda = \kappa$ . Its Theorem 6.1 (`nee:thm:robust`) locates the cofinality: for every infinite  $\kappa$  and  $|A| \geq \kappa$ , every expansion by one name keeps Replacement iff  $\text{cf } \kappa > \omega$ , and every expansion by  $r \geq 2$  names iff  $\text{cf } \kappa > 2^{\aleph_0}$ . Still open: the levels  $\lambda < \kappa$ , actions of arbitrary pure groups at uncountable  $\kappa$ , and whether a centralizer-orbit formula like (59.1) stays exact. Added 4 October 2026, batch 93: Part II of the same report answers “actions of arbitrary pure groups at uncountable  $\kappa$ ” for countable groups and full Replacement: with a uniform evaluator the criterion is the component-profile criterion with full orbits as components and conjugacy classes of stabilizers as types (its Proposition 19.2, `nee:ca:uni:profile`), and with generator names of a finitely generated group every action keeps Replacement iff  $\text{cf } \kappa > \tau(G)$  (its Theorem 18.2, `nee:ca:grp:robust`). Still open: uncountable groups, the levels  $\lambda < \kappa$ , and a centralizer-orbit formula. Added 4 October 2026, batch 96: for finitely many named injections, Part III of the same report gives a centralizer-orbit formula at the level  $\lambda = \kappa$ : for  $\kappa > \omega$  the union  $D_\kappa(\mathbf{f})$  above, whose cardinality decides Replacement, is exactly the set of atoms whose orbit under the centralizer of the named maps has fewer than  $\kappa$  elements (its Corollary 27.2 with Lemma 26.3, `nee:aa:cor:finitenkernel`, `nee:aa:lem:mult`). Still open: the levels  $\lambda < \kappa$ , uncountable groups, and such a formula for actions not given by finitely many named maps.]

**Question 65.3** (Arbitrary relational atom structures). Let a relational structure  $\mathcal{A}$  be named on the atoms, rather than a group action. Is a useful exact criterion expressible in terms of finite orbits of  $\text{Aut}(\mathcal{A})_{(F)}$  together with definability of their families? The group-action proof has a special advantage: finite transitive actions admit a pure set of uniform codes. General structures may lack a comparable finite-orbit coding theorem.

**Question 65.4** (Restricted generator classes). Two involutions already realize every finite threshold. Under what additional restrictions—commuting generators, bounded exponent of the generated group, or a prescribed finitely generated group—does such a hierarchy remain possible? Commuting involutions generate a finite group, so [corollary 58.2](#) gives a genuine obstruction in that case.

[Added 4 October 2026, batch 90: Question 10.4 (“Commuting named automorphisms”) of `naming-elementary-embeddings` asks for the sharp threshold for a fixed number of commuting named permutations, at uncountable cutoffs. The remark on commuting involutions above is its case  $\kappa = \omega$  for involutions. Neither question is answered. Added 4 October 2026, batch 93: that report’s Question 10.4 is now answered (its Theorem 15.3, `nee:ca:thm:commuting`: every fixed number of commuting named permutations keeps Replacement for every action iff  $\text{cf } \kappa > \omega$ ), and for a prescribed finitely generated group named by generators its Theorem 18.2 (`nee:ca:grp:robust`) gives full Replacement for every action iff  $G$  is finite at  $\kappa = \omega$  and iff  $\text{cf } \kappa > \tau(G)$  above. The graded hierarchy asked about here, under these restrictions, remains open.]

**Question 65.5** (Effective presentations). For computable atom actions of a fixed finitely generated group, determine the descriptive or computability-theoretic complexity of the full and bounded-kernel criteria. Local finiteness, multiplicity of finite types, and eventual finiteness of exceptional strata introduce different quantifier patterns. The current finite checker is not a decision procedure for infinite presentations.

**Question 65.6** (Complexity of the failure formulas). Find optimal syntactic bounds for the selectors  $B$  and  $B_k$ , and for the explicit period witnesses  $Y_n$ . Does the hierarchy remain strict for specific restricted formula classes, such as bounded formulas in a carefully chosen language? The coding of exact periods and complete type strata must be included in any complexity accounting.

[Added 4 October 2026, batch 93: Question 23.10 of Part II of `naming-elementary-embeddings` (`nee:ca:q:complexity`, “Complexity of a failed instance”) asks the same for its negative examples; neither is answered. Added 4 October 2026, batch 96: Question 37.8 of Part III (`nee:aa:q:complexity`) asks it for the type selector and the failed instances of its commutative-monoid presentations; not answered.]

**Question 65.7** (Joining safe expansions). Characterize pairs of locally finite actions whose separate expansions preserve Replacement and whose joint expansion also does. Our examples show that separate preservation and finite joint orbits are not sufficient. An effective criterion should describe how joint orbit types split the infinitely repeated types of the reducts.

[Added 4 October 2026, batch 96: Section 28.2 of `naming-elementary-embeddings` (Part III, its source 03) gives, at  $\kappa = \omega$ , the involutions  $s(n) = 1 - n$  and  $t(n) = -n$  on atoms indexed by  $\mathbb{Z}$ : each keeps Replacement alone, and jointly they fail it, since  $t$  defines the atom 0 and  $s \circ t$  is the shift. Their joint orbit is infinite, so the pair is not of the kind asked about here, whose joint orbits are finite ([theorem 61.3](#)); the question stays open.]

**Question 65.8** (Uniform evaluators in larger signatures). Determine exactly which internal coding or amenability conditions make a family of separately named operations equivalent to one uniformly indexed operation. [Theorem 62.3](#) shows that the passage can change even atom-valued Replacement. A language translation theorem must account for this distinction explicitly.

[Added 4 October 2026, batch 93: answered in part by Part II of `naming-elementary-embeddings`: for a finitely generated group the uniform evaluator is definable from the generator names and a pure word selector, so it never changes Replacement (its Proposition 19.4, `nee:ca:uni:fg`, at every kernel bound); for  $\bigoplus_{\omega} \mathbb{Z}/2\mathbb{Z}$  it can, even with orbits of at most two atoms and an amenable evaluator (its Theorem 19.3); and for actions with finite orbits universal preservation under the evaluator is governed by the number of conjugacy classes of finite-index subgroups, at  $\kappa = \omega$  by finiteness of the profinite completion (its Corollary 19.5, `nee:ca:uni:finite_orbit`). The exact coding or amenability conditions remain open.]

**Question 65.9** (Quantitative reservoirs). For a given action satisfying  $R_k$ , bound the size of the reservoir  $C_F$  in terms of  $k$ , the number of parameter atoms, and the finite types they meet. Can such bounds support a compositional proof rule for definable functions, with a mechanically tracked atom budget?

**Question 65.10** (Atom-enriched surreal encodings). Which enrichments of the pure surreal presentation can detect the missing atom collections? Pure-valued schemes cannot do so by [corollary 63.4](#). Any proposed enrichment should state whether it supplies a uniform encoding of atoms by pure objects, because that addition can itself destroy the preservation results it is meant to test.

**Question 65.11** (Reflection and Collection in expanded models). Investigate analogues of the orbit criterion for models with larger kernels in which Collection holds before expansion. How do naming actions and uniform evaluators interact with the Tail and Small Violations of Choice hypotheses studied by Glazer and Yao? Our finite-kernel examples cannot answer this, because they already fail Collection in the reduct.

[Added 4 October 2026, batch 90: answered in part, for finitely many named atom injections, by `naming-elementary-embeddings` ([remark 54.4](#)). At an uncountable successor cardinal  $\kappa$ , where the kernel model  $M_\kappa^{\text{ker}}$  satisfies Collection before the expansion ([theorem 33.3](#)), full Collection survives the naming exactly when fewer than  $\kappa$  component types are realized (its Theorem 7.5, `nee:thm:collection`), and full transitive reflection is equivalent to Collection in these expansions (its Theorem 7.7, `nee:thm:reflection`); its Corollary 7.9 (`nee:cor:jointunsafe`) and Example 7.10 show that two names safe one at a time need not be safe together. Still open: actions of arbitrary pure groups, uniform evaluators, and the interaction with Glazer–Yao’s Tail and Small Violations of Choice hypotheses.]

**Question 65.12** (A fully audited formalization). Carry out the five stages of [section 64](#), with a theorem ledger separating finite computation, semantic metatheorems, and the surreal specialization. The most informative first milestone is the graded reservoir lemma with parameters: it is both reusable and the point at which a plausible abbreviated proof most easily becomes incorrect.

[Added 4 October 2026, batch 93: Question 23.12 of Part II of `naming-elementary-embeddings` (`nee:ca:q:formal`, “Formal schema independence certificates”) asks for a formal certificate beginning with the distinct-period example and the two-point uniformity separation; neither is answered. Added 4 October 2026, batch 96: Question 37.12 of Part III (`nee:aa:q:formal`) asks for a formal chain from a presentation certificate to the exact Replacement verdict; not answered.]

## 66 Conclusion of source 12, audit and provenance

### 66.1 Conclusion

The same finite-kernel universe can behave very differently depending on which symmetries are named. The exact obstruction to full Replacement is not simply an infinite generated orbit. A locally finite action can also expose infinitely many finite exceptional strata through a pure set of action codes.

At bounded kernel size, the remaining symmetry is decisive. Centralizer orbits of size at most  $k$  are precisely the atoms accessible to invariant outputs with that atom budget, up to finitely many types affected by parameters. This gives an exact classification and a strict hierarchy realized by two involutions. A different phenomenon occurs at the limit: all fixed finite budgets can be safe while unbounded finite budgets are not.

Throughout these separations the pure universe, including the canonical surreal field, remains unchanged. Even formulas mentioning the named symmetries retain pure-valued Collection. Consequently an extensive theory of pure surreal numbers is not, by itself, an adequate test of ambient Replacement in the presence of urelements. The output sort, the parameter kernels, and the uniformity of the language are essential parts of the theorem.

## 66.2 Proof dependencies and a compact audit

| Result                                     | Mathematical dependencies                                      | Principal audit issue                                            |
|--------------------------------------------|----------------------------------------------------------------|------------------------------------------------------------------|
| Full criterion                             | Invariant kernels; finite reservoirs; type selector            | Finite orbit types alone do not suffice; multiplicity matters.   |
| Bounded criterion                          | Centralizer orbit sizes; pointed selector; corrected reservoir | Parameters can shrink finite centralizer orbits.                 |
| Two-involution hierarchy                   | Odd-dihedral rigidity; $d$ repeated copies; bounded criterion  | External labels must not be added to the language.               |
| All finite levels without full Replacement | Single-cycle centralizer; unbounded finite cycle lengths       | Fixed metatheoretic bounds differ from a varying internal bound. |
| Uniform-evaluator separation               | Finite-symbol principle; finite generated subgroups            | Separate operation names cannot be quantified over uniformly.    |
| Pure-output theorem                        | Ambient formula relativization; pure rank bounds               | No internal truth predicate or expanded Replacement is assumed.  |
| Surreal specialization                     | Identical pure universes; pure-output theorem; simplicity      | Class surreals are not treated as an internal set.               |

The central mathematical claims do not depend on any unrefereed theorem from ProveIt. The repository supplies motivation, a relevant pure presentation, and a potential formalization destination. The external foundational assumptions are fixed in [section 55](#); all scheme failures are witnessed by explicit formulas or the uniform selectors in [section 57](#).

## 66.3 Source and reproducibility record

The repository snapshot used for the source comparison is commit

8dc2592e951b3402951060a15b61d65162aad0f.

The inspected surreal README has blob hash

18478d58670c384953fb85dcd1a6d931c644aa32.

The article does not claim a full repository audit or a build of that snapshot. The mathematical source comparison used Glazer–Yao v1 (25 February 2026), Hamkins–Yao v2 (5 December 2022, subsequently published in 2024), and Glazer’s 2023 preprint. The earlier companion note is a project-local working manuscript, not a peer-reviewed source.

The delivered package contained the manuscript’s L<sup>A</sup>T<sub>E</sub>X source, the compiled PDF, the finite verification script and its actual JSON output, a source-audit note, and a Makefile, and was run from the package directory as

```
python3 verify_orbit_spectra.py
latexmk -pdf -interaction=nonstopmode -halt-on-error article.tex
```

The verification script requires Python 3.10 or newer and no third-party modules. Compilation of the delivered manuscript required the standard packages named in its source, including `newpxtext`, `newpxmath`, and `tcolorbox`. Font files are not included.

[Added 4 October 2026, batch 90: in this report the script, its record, the audit note and the Makefile are shipped byte-identical as `code/12-named-symmetries-verify_orbit_spectra.py`,

`data/12-named-symmetries-verification_results.json`, `12-named-symmetries-source_audit.md` and `code/12-named-symmetries-Makefile`; the manuscript's source and PDF are not shipped. Without `--output` the script writes `verification_results.json` into the current directory, so run it with an explicit output path outside this directory; the report's README gives the command. The README blob above is also the blob of the surreal README at the placement commit.]

## 66.4 Non-claims of source 12

Every limitation stated by source 12, in its manuscript, its README or its source audit, is kept; most appear at their point of use. Collected: the manuscript is an unrefereed research draft prepared with ChatGPT; the finite-kernel model and its symmetry method are classical (Lévy, through Hamkins–Yao) and are not claimed as discoveries; the classifications and the hierarchy are proposed refinements whose historical priority has not been established; no named published conjecture is claimed solved, and Glazer–Yao's reflection and Collection separations are neither strengthened nor presented as open; Elliot Glazer is neither a coauthor nor an endorser; no Lean or Rocq verification is claimed, and no proof files are included; the 203 finite checks validate finite centralizer and witness calculations only, not the infinite model constructions or the Replacement schemes, and a finite truncation of an example does not refute Replacement; the ambient metatheory has Choice, there is no claim that ZFC proves the existence of a transitive set model of itself, and formulas are relativized one at a time, with no internal truth predicate; the models retain Choice and already fail full Collection in the reduct, so what changes is Replacement in the expanded language; pure invariance does not validate any unrefereed claim of a repository report and does not identify an atom-enriched presentation with the pure one; the central claims do not depend on any unrefereed theorem of ProveIt; the research questions are open within the manuscript's scope, with no claim that each is absent from the literature; the repository was inspected read-only at the pin, not audited or built, and no repository file was modified. This merge adds one: for finitely many named permutations, [theorem 58.1](#) is not new in ProveIt ([remark 54.4](#)).

## 66.5 Provenance of this part

**The manuscript.** Source 12 is batch-90 manuscript 05, delivered as the archive `Named_Symmetries_and_Replacement.zip` (335,558 bytes; inner directory `named_symmetries_replacement/`) in commit `a162e4386` and placed in commit `12076b2e8`. It names the repository commit `8dc2592e9` as its snapshot, a merge commit of 3 October 2026 that is an ancestor of the batch-89 placement `23adb85f9`; at the pin neither [Part VI](#) nor `namings-elementary-embeddings` existed, which is why source 12 calls source 10 a project-local working note (it read it outside the repository) and does not cite that report. The surreal README it inspected has the same blob `18478d586` at the pin and at the placement, and its statements about it (the sign-sequence construction and the distinction between formalized results and AI-assisted research reports) are true. Its manuscript (`article.tex`, 915 lines, 51 labels; a 23-page PDF), README, PDF and checksum file `MANIFEST.sha256` (7 of 7 entries verified at placement) are not shipped; they survive in the arrival commit. Shipped, byte-identical: `12-named-symmetries-source_audit.md`, `code/12-named-symmetries-verify_orbit_spectra.py`, `code/12-named-symmetries-Makefile` and `data/12-named-symmetries-verification_results.json`. The Hilbert's-tenth programme routed the six batch-90 archives in commit `bcc1a4438` (`Computability/HilbertTenthProblem/Papers/research-wip/native-stream-queue/review_polish_partx_c9bc70d8f.md`): for this one it verified the archive hash and read only the delivery README (lines 1–71), ran nothing, made

no correctness finding, and recorded a finite-atom-kernel Replacement result with finite centralizer diagnostics and no arithmetic compiler.

**How it is printed.** The text from source 12’s title-page abstract and status box to its appendices is printed in full, in its order: the status box and the abstract open [section 54](#), its Section 1 follows them as three subsections, and its two appendices became subsections of this section. Every delivered label carries the prefix `hset:ns:`; its twelve research questions, numbered 1–12 there, are Questions 65.1–65.12 here, with labels `hset:ns:q:...`, and its section numbers 2, ..., 12 are 55, ..., 65, so that its Theorem 5.1 is [theorem 58.1](#). The symbols renamed are listed in [convention 54.1](#); equations are numbered within sections; delivered file names are replaced by the shipped ones, and the sentence of the reproducibility record describing the delivered package is put in the past tense; its citations of the companion note became references to [Part VI](#), and its other references use this report’s bibliography entries. Added for this report: [sections 54.4](#) and [54.5](#), the label of the remark “Kernel is not permutation support”, the bracketed dated notes, this subsection and the preceding one. No mathematical statement or proof of source 12 was changed.

**Where the merge had to choose.** (i) Placement: source 12 answers [question 38.6](#) for named group actions at  $\kappa = \omega$  and continues source 10’s Section 35 ([remarks 54.2](#) and [54.3](#)), and it has surreal content; it was therefore placed here rather than as a second part of `naming-elementary-embeddings`, whose principal theorem it generalizes at  $\kappa = \omega$  ([remark 54.4](#)). (ii) Its full criterion is printed with its own proof, although for finitely many permutations it is that report’s Theorem 5.1(2), and its pure-valued Collection theorem with its own proof, although it is a case of [theorem 35.2](#). (iii) Its predicate  $E$  was renamed `Act`, rather than left beside source 10’s enumeration predicate  $E$ , and its centralizer  $H$  was renamed  $\mathcal{Z}$ . (iv) The part follows [Part VII](#), so that no existing number changes, and is printed before this report’s appendices.

## Part IX

# Surreal-only foundations: theory-level bi-interpretations, the native omnific language and the class boundary

## 67 Sources 13 and 14: question, results, notation and relation to the report

This part was added in batch 95. It prints, merged into one text, two further manuscripts written independently of each other and of the twelve above, on the same assignment and on the same day, 4 October 2026:

- *source 13, Surreal and Omnific Foundations: Bi-Interpretations with Set Theory and the Boundary at Classes*, “Research study prepared for Vladimir Reshetnikov”, batch-95 manuscript 01, pinned to repository commit 722337445; it is the base of the merge, and its local file prefix `13-omnific-foundations-` is used for provenance only, since it ships no code or data;

- *source 14, Surreal-Only Foundations and the Set–Class Boundary: Explicit axiomatizations, omnific quotient birthdays, bi-interpretations, and the obstruction at NBG*, “Prepared for Vladimir Reshetnikov”, “Research study developed with ChatGPT”, batch-95 manuscript 02, pinned to commit `bd1de458b`; its shipped files carry the prefix `14-surreal-only-nbg-`.

Both archives arrived in commit `ec91f8c7c` and were placed in commit `350b9a954`. Their provenance, the mechanical changes made in printing them, the places where the merge had to choose, and their non-claims are in [section 80](#). The running text is source 13’s, in its order; source 14’s results are inserted where they belong, each marked “(source 14)”. A result proved by both is printed once, with both sources named, and a genuinely different proof of the other is kept as a marked second route. Inside this part “this article”, “here”, “our” and “we” refer to the source of the passage in which they occur; statements added in this merge are marked as such.

Both sources re-derive this report’s [theorem 18.1](#), the announced global theorem of Chen, Hamkins and Yang, and both credit it and this report. What they add is a *theory-level* form of it (an explicit recursively axiomatized theory with numerical individuals, bi-interpretable with ZFC), the same reconstruction in the native language of the omnific integers, and an analysis of the boundary at class theory. [Section 67.5](#) states what each adds to this report and to the collection.

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**Principal conclusion (source 13).** There are explicit, recursively axiomatized first-order theories in which the only individuals are surreal numbers, or alternatively omnific integers, and which are bi-interpretable with ZFC. Arithmetic, order, and birthday precedence suffice for the constructions given here. Full NBG/GBC cannot be recovered uniformly when the forward interpretation uses only set-coded individuals of the same universe. A faithful class-theoretic version retains classes of numbers as an additional sort.

**Research status (source 13).** The surreal-field result was already announced by Chen, Hamkins, and Yang, and is discussed in the repository. This article supplies an explicit omnific-integer construction, axioms specifying the two inverse interpretations, and precise class-theoretic obstructions. Proofs are mathematical proofs; no new theorem in this article is claimed to have been checked by a proof assistant. Literature priority is not asserted for the derived results. Repository baseline: <https://github.com/VladimirReshetnikov/ProveIt>, inspected commit `722337445aa691a90e005ff8f619fd8ce5397103`.

---

**Main conclusion (source 14).** A recursively axiomatized first-order theory whose individuals are surreal numbers can recover *all sets*, with definable inverse interpretations. An explicit omnific-integer version is obtained by adjoining the quotient-birthday operation. Recovering *all classes of an NBG model* is a different requirement: it cannot be achieved by a theory uniformly interpretable in ZFC, assuming  $\text{Con}(\text{ZFC})$ . A second class sort provides an exact positive replacement.

**Attribution and proof status (source 14).** The fundamental birthday-enriched surreal/ZFC bi-interpretation is due to the announced work of Junhong Chen, Joel David Hamkins, and Ruizhi Yang, not to this report. The report supplies detailed reconstructions and further deductions; it makes no priority claim for those deductions. Proofs are mathematical, not Lean-verified. Finite executable checks are supplementary only. Repository audit pinned to `bd1de458b7145f535cfcefbef47d851c2e82463`. No repository files have been modified.

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**Source 13’s abstract.** Can set theory be formulated with numbers as its only objects, in the same strong sense in which a suitable theory of hereditarily finite sets is bi-interpretable with Peano

arithmetic? For surreal numbers, the answer depends on the language and on whether set theory means its set universe or also its proper classes. We separate these issues and prove explicit positive and negative results.

Our constructive focus is the ordered ring of omnific integers equipped with the relation of strictly earlier birthday. We recover ordinals and signs in this language and show that an omnific integer can encode an arbitrary bounded set of ordinals: each bit occupies a block of ordinal length  $\omega$ , with any intervening gaps filled uniformly. Pointed well-founded extensional graphs then interpret the whole set universe. The sign sequence of an original integer supplies a definable inverse to the reconstructed integer structure. This yields a first-order bi-interpretation at the structure level and, by an explicit recursive axiom construction, a theory  $T_{\mathbf{Oz}}$  bi-interpretable with ZFC. The same framework gives a surreal-only theory  $T_{\mathbf{No}}$ , without assuming a final intrinsic axiom list from the announced Chen–Hamkins–Yang programme.

The class boundary is stronger than a mere cardinality intuition about proper classes. Established model constructions give a model of GBC with countably many sets and uncountably many classes. Hence no uniform bi-interpretation of full GBC can send every model to a structure whose individuals are finite tuples of its original sets, even with arbitrary class-dependent relations and quotient coding. We also explain the reflection obstruction to interpreting GBC in ZFC, the precise limits of the diagonal argument in predicative class theory, and a positive two-sorted theory of numbers and their classes. The final sections distinguish proved constructions from inherited results, describe formalization tasks, and propose further research.

**Source 14’s abstract.** The analogy between Ackermann coding of hereditarily finite sets and arithmetic suggests replacing finite binary strings by ordinal-length surreal sign sequences. We examine exactly what this accomplishes. An explicit first-order code for rooted well-founded extensional graphs recovers the set universe from the ordered surreal field with birthday. A round-trip axiom construction turns this structural reconstruction into a recursively axiomatized theory bi-interpretable with ZFC, without assuming an unchecked list of intrinsic surreal axiom schemes. We transfer this construction to the omnific integers using the binary operation  $h(a, b) = b(a/b)$ , with a convention at zero denominator. We then prove a reflection obstruction: no theory uniformly interpretable in ZFC can interpret the usual finitely axiomatized NBG theory, assuming  $\text{Con}(\text{ZFC})$ . An independent class-diagonal argument rules out uniform numerical names for every available class with an elementary evaluation relation. Finally, we construct a reversible correspondence between class realizations over a set model and appropriate class realizations over its surreal structure. This yields numerical versions of GBC and KM with a necessary class sort. The outcome is a precise positive answer at the set level, a qualified negative answer to the native one-sorted NBG proposal, and a research program concerning smaller omnific languages, intrinsic axioms, choice, class recursion, and formal verification.

## 67.1 The question and the answer

The motivating analogy is sound once the finite set theory is specified carefully. An hereditarily finite set has a natural-number code, and suitable finite set theories and PA can recover one another together with definable isomorphisms for both compositions. The transfinite analogue asks for substantially more than an injection between informal collections of mathematical objects.

There are three independent requirements:

1. the individuals of the proposed theory must really correspond to the surreal numbers, or the omnific integers, of the universe under study;
2. the language must allow the representation to be read and the relevant operations to be defined;
3. both interpretations must recover their inputs by definable isomorphisms, including every class object when the target is class theory.

The first two requirements can be met for ZFC in a natural numerical language. The third prevents a direct extension to the full class sort of NBG.

**The same analogy, as source 14 puts it.** A natural number can encode a finite set by its binary digits; a surreal number can encode a subset of *any ordinal* by its signs. The second statement supplies codes for arbitrary well-orderable membership graphs, not merely for finite graphs. It is therefore capable of reaching the entire universe of sets under Choice. What it does not supply is a set-sized sign sequence for an arbitrary proper class. Two qualifications to the motivating premise are worth making at once. “Finitary” should refer to finite objects or finite descriptions, not to arbitrary objects merely studied by finitary proofs. Moreover, the PA/finite-set correspondence requires an appropriately axiomatized theory of hereditarily finite sets. Removing Infinity from ZF and adding its negation is not, without further attention to induction and transitive containment, an interchangeable description of every such theory. The careful finite-set formulations and their bi-interpretations are discussed in [30] and [31, §6.3].

Source 14 distinguishes four requirements throughout.

- (i) **Set reconstruction:** can an enriched numerical structure interpret  $(V, \in)$ , with a definable round trip?
- (ii) **Axiomatic equivalence:** is there an effective first-order numerical theory bi-interpretable with ZFC?
- (iii) **Class reconstruction:** can ordinary numerical individuals also recover the class objects of an arbitrary NBG model?
- (iv) **Literal categoricity:** do the axioms have only the externally intended proper-class numerical universe as a model?

The first two have positive answers. The third encounters genuine obstructions, made precise below. The fourth is not a property of an ordinary consistent first-order theory with infinite models. Its set-sized models can be nonstandard, just as models of PA can be nonstandard. Bi-interpretability concerns these models as well as the intended class-sized structure; it is not a categoricity claim.

**A precise summary (source 13).** Throughout this part  $\prec_b$  denotes *strict birthday precedence*, the relation of lemma 2.1:

$$x \prec_b y \iff b(x) < b(y).$$

It does not denote proper-prefix simplicity, and it is not the elementary substructure relation  $\prec$  of Parts I–V. Source 13 uses the finite language

$$\mathcal{L}_b = \{0, 1, +, -, \cdot, <, \prec_b\}.$$

The constants and unary minus could be replaced by definitions.

| Question                             | Answer and scope                                                                                                                              | Status                                                    |
|--------------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------|-----------------------------------------------------------|
| Pure ordered surreal field           | Does not interpret even Robinson arithmetic; its elementary theory is RCF.                                                                    | Established obstruction                                   |
| Surreal field with birthdays         | Interprets the whole set universe, with definable inverse compositions.                                                                       | CHY announcement; prior repository proof                  |
| Omnific ring with birthdays          | The same set-universe reconstruction works using ordinal blocks of signs.                                                                     | Explicit proof in this article                            |
| Recursive numerical axioms           | $T_{\mathbf{No}} \simeq_{\text{bi}} \text{ZFC}$ and $T_{\mathbf{Oz}} \simeq_{\text{bi}} \text{ZFC}$ , with both theories in $\mathcal{L}_b$ . | Explicit construction here                                |
| Full NBG/GBC on the original numbers | No uniform bi-interpretation with a forward domain coded by the original sets.                                                                | Corollary of established model constructions, proved here |
| Numbers and classes of numbers       | A faithful class lift is bi-interpretable with GBC; analogous lifts handle stronger class theories.                                           | Construction and proof here                               |

The positive axiom systems below are not asserted to be complete, categorical, or finitely axiomatized. They have the same interpretive strength as ZFC, including its possible nonstandard models and independence phenomena. Their quantifiers range only over numbers; sets are represented by equivalence classes of numerical codes.

**Theorem 67.1** (Summary of the constructions and obstructions; source 14). *In the precise interpretation conventions of section 68:*

- (1) *There is a recursively axiomatized theory  $\text{SA}_{\text{rt}}$  in the finite language  $\mathcal{L}_s = \{0, 1, +, -, \cdot, <, b\}$ , with its canonical interpretation on surreal numbers, that is parameter-free bi-interpretable with ZFC.*
- (2) *There is such a theory  $\text{OA}_{\text{qb}}$  on omnific integers in the language  $\mathcal{L}_z = \{0, 1, +, -, \cdot, <, h\}$ , where canonically*

$$h(a, b) = \begin{cases} b(a/b) & b \neq 0, \\ 0 & b = 0. \end{cases}$$

*It is parameter-free bi-interpretable with  $\text{SA}_{\text{rt}}$  and with ZFC.*

- (3) *If  $\text{Con}(\text{ZFC})$ , no theory uniformly interpretable in ZFC interprets the usual NBG theory. In particular, neither  $\text{SA}_{\text{rt}}$  nor  $\text{OA}_{\text{qb}}$  interprets NBG.*
- (4) *With an additional sort of classes of numerical codes, there are bi-interpretable numerical presentations of GBC, and similarly of KM. These presentations retain, rather than eliminate, class objects.*

Part (1) reconstructs an already announced result, using a deliberately explicit axiomatic presentation. Parts (2)–(4) are proved in this report as extensions and applications. None is advertised as an independently verified new theorem in the literature. The important outcome for the research question is the separation between (1) and (3), not a claim that all forms of set theory have secretly become the same theory. (Source 14.) In this merge, (1) is [theorem 73.4](#) for  $\mathbf{No}$  (source 14’s presentation is [theorem 73.6](#)), (2) is [theorem 74.7](#), (3) is [theorem 75.2](#) and [corollary 75.3](#), and (4) is [theorem 76.5](#) and [corollary 76.7](#). Source 13 proves the omnific case of (1)–(2) in the weaker language  $\mathcal{L}_b$ , without  $h$  ([theorem 73.4](#)); by [proposition 74.5](#) the two omnific languages are definitionally equivalent.

**What the negative statement does not say (source 14).** The hypothesis “uniformly interpretable in ZFC” is essential. It includes a canonically defined numerical domain and operations

whose proposed axioms are all provable in ZFC. It excludes adding an arbitrary relation of independent class-theoretic strength and merely calling its arguments numbers. It also excludes coding the classes of a small universe by numbers belonging to a strictly larger universe while silently changing the reference universe. We do not prove that every conceivable structure externally relabelled by surreal numbers is incapable of carrying NBG. External relabelling can hide essentially arbitrary information and is not an intrinsic numerical foundation. For natural, same-universe codes with elementary evaluation, [theorem 75.7](#) gives a separate obstruction even in class theory. These qualifications are part of the answer, rather than loopholes to be suppressed.

## 67.2 What was already known

Chen, Hamkins, and Yang announced the bi-interpretability of  $\langle \mathbf{No}, +, \cdot, <, <_b \rangle$  with  $\langle V, \in \rangle$ , and an associated first-order theory SA with ZFC, in presentations during 2025–2026. The available slides mark the axiom programme as work whose details were still being checked. The relevant announcements and slides are primary sources [7, 8]. We do not present that surreal result as new, and we do not assume that an unpublished final axiomatization has been verified.

There are also earlier approaches connecting surreals and set-theoretic foundations. Rangel and Mariano develop an algebraic set-theoretic programme of surreal algebras [16]; Bertram discusses set-based representations of Conway numbers and games [17]. Their purposes and chosen structures differ from the first-order bi-interpretations considered here.

The current ProveIt corpus already discusses the CHY programme, develops birthday-based reconstructions of hereditary set universes, and studies ambiently definable surreal and omnific codes. In particular, the full surreal reconstruction and the distinction between local Choice and Global Choice are prior material. The specific additional construction developed here carries the reconstruction out within the *native omnific ring with its own birthday relation*. The class obstruction also answers a different question from reconstructing  $\langle V, \in \rangle$ . The exact overlaps are recorded in [section 80.5](#).

**The same background, as source 14 records it.** Chen, Hamkins, and Yang announced a first-order surreal arithmetic bi-interpretable with ZFC in 2025–2026. The key additional structure is birthday comparison; the underlying field alone is insufficient. Hamkins’s March 2026 slides describe graph coding and candidate axioms. They label the project as work in progress and explicitly warn that details were still being checked [7, 8]. We therefore distinguish the structural coding theorem from any particular proposed minimal list of intrinsic axiom schemes. The requested repository already contains substantial work on this exact background. At the audited revision, the directory `Algebra/SurrealNumbers/docs/foundations-and-computation/birthday-cutoffs-and-hereditary-sets/` contains a merged report whose README credits the global construction to Chen–Hamkins–Yang and describes an explicit full-universe proof. Its cutoff results, graph encodings, and further weak-foundation discussions are not discoveries of the present investigation. The nearby `definable-surreals-and-omnific-integers` report also explicitly credits that precedent [41]. The README reports no Lean implementation mapping for its `hset:` labels. We have not built or independently audited that entire report, and do not import its unverified cutoff claims as axioms. The present report instead uses the classical sign-sequence and normal-form constructions [1, 2], gives the coding argument needed here, and isolates three follow-on issues: a concrete omnific language, reflection at the NBG boundary, and reversible transport of the class collection. Ordinary interpretation terminology follows [31]. For class-theory conventions and realizations we use [34].

There is also an older question about intrinsic axioms for omnific integers. Jeřábek's answer distinguishes a useful recursively axiomatized fragment from the *complete true theory*, which cannot be recursively axiomatized because it contains arithmetic [37]. Our  $\text{OA}_{\text{qb}}$  is an incomplete foundational theory in an enriched language, not the complete theory of the intended omnific ring. Thus it does not evade that obstruction. (Source 14.)

### 67.3 What counts as a result here

Theorems below labeled as proved here are accompanied by proofs, not by claims of bibliographic priority. The most substantial constructions are the omnific block interpretation, the corresponding recursive theory in the native numerical language, and the application of models with different sizes of their set and class sorts. These resolve precise versions of the question posed in this article. They are not advertised as solutions to previously published open problems without a verified priority history. (Source 13.)

### 67.4 Notation of this part

**Convention 67.2** (Symbols of sources 13 and 14). Sources 13 and 14 used different letters for the same notions, and several of their letters already mean something else in this report. In this part:

| Notion                                            | Here (source 13; source 14)                                                                                                                                                           | Do not confuse with                                                                                                                                                                             |
|---------------------------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Strict birthday precedence<br>$b(x) < b(y)$       | $x \prec_b y$ (13: $x \prec y$ ); 14 uses the birthday function $b$ (14: $b$ )                                                                                                        | the elementary substructure relation $\prec$ of Parts I–V ( $\mathcal{B}_\lambda \prec \mathcal{B}_\eta$ , $H_\lambda \prec H_\eta$ ); proper-prefix simplicity ( <b>found:rem:simplicity</b> ) |
| Equal birthday                                    | $\text{SameB}(x, y)$ (13: $\text{SameB}$ ); $E_b$ in lemma 2.1                                                                                                                        | —                                                                                                                                                                                               |
| Ordinal predicate                                 | $\text{Ord}(a)$ (13: $\text{Ord}$ via $\text{SameB}$ ; 14: $b(x) = x$ )                                                                                                               | the same predicate as in Parts I–V; equivalent by lemma 2.1                                                                                                                                     |
| Languages                                         | $\mathcal{L}_b = \{0, 1, +, -, \cdot, <, \prec_b\}$ (13: $\mathcal{L}_b$ );<br>$\mathcal{L}_s = \{0, 1, +, -, \cdot, <, b\}$ ,<br>$\mathcal{L}_z = \{0, 1, +, -, \cdot, <, h\}$ (14)  | $\mathcal{L}_b$ and $\mathcal{L}_s$ are definitionally equivalent on <b>No</b> and on <b>Oz</b> (lemma 69.1)                                                                                    |
| Full birthday structure on <b>No</b>              | $\mathcal{B} = (\mathbf{No}; 0, 1, +, -, \cdot, <, b)$ (14: $\mathcal{S}$ )                                                                                                           | the cutoff structures $\mathcal{B}_\lambda$                                                                                                                                                     |
| Numerical domain, <b>No</b> or <b>Oz</b>          | $\mathbf{D}$ (13: $D$ )                                                                                                                                                               | 14's packed-code class $D$ (here $\text{UC}$ ); the diagonal classes $D$ (here $\Delta$ )                                                                                                       |
| Interpretations: numbers to sets; sets to numbers | $\mathcal{I}_\mathbf{D}, \mathcal{J}_\mathbf{D}$ (13: $I_D, K_D$ ); $\mathcal{I}, \mathcal{J}$ (14)                                                                                   | the same roles as $\mathcal{I}, \mathcal{J}$ in theorem 18.1                                                                                                                                    |
| Canonical-number predicate                        | $\text{Is}_\mathbf{D}(s)$ (13: $D_D(s)$ )                                                                                                                                             | —                                                                                                                                                                                               |
| Prefix and sign predicates                        | $\text{Pref}, \text{Plus}, \text{Read}$ (13); $\text{Pref}, \text{Bit}$ (14: $\text{Pref}, \text{Bit}$ )                                                                              | this report's $\text{Pre}$ (another correct prefix formula, remark 4.2) and $\text{Bit}$ , which $\text{Plus}$ and 14's $\text{Bit}$ agree with                                                 |
| Block map, block, code length                     | $f(\beta) = \omega\beta, W_\beta, L_\alpha$ (13: $f, I_\beta, L_\alpha$ )                                                                                                             | $\mathcal{I}$ , the interpreted universe                                                                                                                                                        |
| Pairings and lengths                              | $\pi_\kappa(i, j) = \kappa i + j$ (13);<br>$\varpi(\alpha, \beta) = (\alpha + \beta)^2 + \alpha$ ,<br>$\Lambda_{14}(\delta) = (\delta + \delta)^2 + \delta + 1$ (14: $\pi, \Lambda$ ) | $p_\theta$ and $\Lambda(\theta) = \theta^2 + \theta$ of lemma 4.3; $\varpi$ is source 06's polynomial of lemma 4.5                                                                              |

| Notion                                                          | Here (source 13; source 14)                                                                                                                                                                    | Do not confuse with                                                                                 |
|-----------------------------------------------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-----------------------------------------------------------------------------------------------------|
| Codes                                                           | $c = (\nu_c, r_c, z_c)$ , root $r_c$ (13);<br>$c = (\delta, r, e)$ , root $r$ (14)                                                                                                             | $C = (\delta, r, t)$ of <a href="#">section 5</a> , where $r$ is the relation word and $t$ the root |
| Validity, equality and membership of codes                      | Valid, Eq, Mem (13: Valid, Eq, Mem; 14: Code, $\sim_*$ , $\varepsilon_*$ )                                                                                                                     | Code, $\sim$ , $\in_{\mathcal{I}}$ of Parts I–V (same roles for those codes)                        |
| Map codes; their base                                           | $Y_h$ ; $\nu_{cd}$ (13: $H_h$ ; $K$ )                                                                                                                                                          | the hereditary universes $H_\kappa$ ; a complex field $K$                                           |
| Root collapse of a code                                         | $a_c$ , $\text{col}(c)$ (13: $a_c$ , $\text{col}$ ; 14: $v(c)$ , $\text{Coll}(c, a)$ )                                                                                                         | the collector $\text{Coll}(C)$ of Part I                                                            |
| Code of the ordinal $\alpha$ (14)                               | $c_\alpha^{\text{ord}}$ (14: $O(\alpha)$ )                                                                                                                                                     | the ordinal code $\text{OC}(a)$ of Parts I–V; source 13’s relation $\text{OG}(\alpha, e)$           |
| Numerical comparisons                                           | $\text{Cmp}_{\mathbf{D}}(x, c)$ (13: $C_D$ ); $\rho(x, c)$ (14)                                                                                                                                | $C$ , a relation code; $\mathcal{C}$ , the surcomplex structure; $\varrho$ , the eternity bound     |
| Theories                                                        | $\mathbf{T}_{\mathbf{No}}$ , $\mathbf{T}_{\mathbf{Oz}}$ , $\mathbf{T}_{\mathbf{D}}^{\text{cl}}$ (13); $\mathbf{SA}_{\text{rt}}$ , $\mathbf{OA}_{\text{qb}}$ , $\mathbf{SAC}_{\text{GBC}}$ (14) | $\mathbf{SA}$ , the announced Chen–Hamkins–Yang theory, whose axioms neither source verifies        |
| Four-place fraction-birthday predicate                          | $\mathbf{FB}(a, b, c, d)$ (13: $B$ )                                                                                                                                                           | $\mathcal{B}$ , $\mathcal{B}_\lambda$ ; dsn’s pairing $B(a, b)$ ; Part VIII’s selector $B(q)$       |
| Unary code class of the class lift; its equivalence             | $\mathbf{UC}$ ; Eq (13: $C$ , $E$ ; 14: $D$ , $\sim_*$ )                                                                                                                                       | —                                                                                                   |
| Diagonal classes                                                | $Q$ , $\Delta$ (13: $C$ , $D$ ; 14: $C$ , $D$ )                                                                                                                                                | —                                                                                                   |
| Shear map and coefficient ( <a href="#">proposition 74.12</a> ) | $\vartheta$ , $\text{co}_\delta(a)$ , $o_\delta = \omega^{\omega^\delta}$ (13: $\theta$ , $c_\delta(a)$ , $\Omega$ )                                                                           | the conjugation $c$ ; dsn’s omega map $\Omega$                                                      |
| Models of class theory                                          | $(M, \mathcal{X})$ , $N = \mathbf{D}^M$ , $\mathcal{Y}$                                                                                                                                        | the outer models $M \subseteq N$ of Part III                                                        |
| Class theories                                                  | GB (no choice), GBc (+ set Choice), GBC (+ Global Choice); KM; NBG (one-sorted GBC)                                                                                                            | 13’s “GB” is GB; 14’s “GB” printed GBc and is GBc ( <a href="#">convention 68.1</a> )               |

Natural (Hessenberg) sums and products of ordinals are the native ring operations and are written  $+$ ,  $\cdot$  as in the sources; this report’s  $\oplus$ ,  $\otimes$  denote the same operations. Source 14’s ordinary ordinal operations are written  $+_{\text{ord}}$ ,  $\cdot_{\text{ord}}$ . Equations are numbered within sections. Typography only: the sans-serif theory names of both sources are set in roman. No other symbol was renamed.

## 67.5 Relation to the report and to the collection

*Remark 67.3* (The global theorem of Part IV, and Parts I–V). For  $\mathbf{D} = \mathbf{No}$ , [theorems 71.1](#) and [72.2](#) (source 13) and [theorem 71.3](#) and [proposition 72.4](#) (source 14) are two further written proofs of [theorem 18.1](#), after source 07’s printed there and source 05’s bisimulation version; both sources credit Chen–Hamkins–Yang and this report, and neither is claimed first. Their codes are those of [section 5](#) with a different pairing: source 13’s  $\pi_\kappa$  is the map  $p_\theta$  of [lemma 4.3](#), source 14’s  $\varpi$  is source 06’s polynomial ([lemma 4.5](#)). Source 13’s ordinal and prefix lemmas ([lemmas 69.1](#) and [69.2](#)) re-prove [lemmas 2.1](#) and [4.1](#), now also in  $\mathbf{Oz}$ ; its [proposition 71.5](#) is [theorem 13.2](#) with [proposition 13.3](#) (source 13 credits this report). The status note after [theorem 18.1](#) says that the proof there “is not an audit . . . of their proposed axiomatization”. That stays true: [theorem 73.4](#) supplies *an* explicit recursive axiomatization of the round trip, not the intrinsic axioms announced by Chen–Hamkins–Yang, which neither source verifies ([question 79.2](#)). A dated note after [theorem 18.1](#) points here.

*Remark 67.4* (Axiomatics and the minimal signature). [Question 21.7](#) asks for definitional equivalence, or its failure, between the schemes Et, Pack and published axiom formulations of surreal arithmetic

“once finalized”. Nothing here is finalized: the theories of [section 73](#) are explicit and recursive, but they are built from the interpretation ([theorem 73.1](#)), and an intrinsic list is [question 79.2](#). The question’s other items (ill-founded models, quantifier elimination, choiceless universality, formalization) meet [questions 79.8, 79.17 and 79.19](#) and are not answered. The “Minimal signature” question of [section 21](#) asks which reducts of  $\{0, 1, +, -, \cdot, <, b\}$  still *interpret* the hereditary sets; source 13’s [question 79.3](#) asks which reducts support a *canonical bi-interpretation*. [Proposition 74.2](#), an observation of this merge, shows that the second question is a definability question: a reduct, or any structure on **D** with set-definable primitives, is canonically bi-interpretable with  $V$  exactly when it defines  $<_b$  together with the ring operations. The one-way question is untouched. Dated notes after the two questions point here.

*Remark 67.5* (Definable operations in the collection’s other reports). Part III of **definable-sur-reals-and-omnific-integers** (labels `dsn:op:`, its source 06, pinned to 1404038df and written independently) interprets  $(V, \in)$  parameter-free in  $\mathfrak{S}_\Omega = (\mathbf{No}; +, \cdot, <, \mathbf{Oz}, \Omega)$ , with  $\Omega$  the omega map, by the same three validity clauses (well-foundedness, extensionality and root accessibility tested by subset quantification) as [section 71](#), but with vertex sets read from the supports of unit-coefficient series and edges read through a pairing of supports (its Theorem 37.1, `dsn:op:thm:V-interpretation`). It has *no* comparison map back to the original numbers; its Question 42.1 (`dsn:op:q:biinterpretation`, “The missing comparison map”) names “a sufficiently strong intrinsic birthday relation” as a promising route and the birthday-enriched announcement as a comparison target. By [corollary 74.1](#) the classes **Oz** and the graph of  $\Omega$  are  $\mathcal{L}_b$ -definable in **No**, so  $\mathfrak{S}_\Omega$  is a reduct of a definitional expansion of  $(\mathbf{No}, \mathcal{L}_b)$ , and the theorems of this part are that comparison target. They do not answer the question, which concerns the reduct; but by [proposition 74.2](#) its specific form, a definable relation sending each original surreal to the class of codes of its own sign sequence, is equivalent to the definability of  $<_b$  in  $\mathfrak{S}_\Omega$ , with the same parameters. Same coding scheme, different theorem; source 13’s native **Oz** codes (sign blocks) and dsn’s (Hahn supports) are different mechanisms. A dated note after [theorem 16.5](#) records dsn’s transfer theorem.

Source 13 also credits the ambient omnific codes and hulls of **definable-surreals-and-omnific-integers** (`dsn:thm:bounded-code`, `dsn:thm:hull`); its requirement here is that both directions be first-order definable in the native birthday language. Its [proposition 74.12](#) is a weaker consequence of `opa:thm:parameters` and `opa:sc:thm:undef` of **surreal/omnific-preserving-automorphisms**, which it credits: there a strong real-linear automorphism preserving **Oz** and fixing any given set and  $\mathbb{R}$  sends the ordinal  $\omega^{\omega^\beta}$  to a nonmonomial, and  $\text{Mon} \cap \mathbf{Oz}$  is not definable over any set of parameters in the pure ordered ring. The fraction-field fact [proposition 74.3](#) is `odg:thm:fractions` of **surreal/omnific-diophantine-geometry**, proved in Lean ([section 78.1](#)); the pure ring **Oz** parameter-free interprets the field **No** with  $\mathbb{R}$  and its order (`odg:def:thm:realrecovery`), is bi-interpretable with  $(\mathbf{No}, \mathbf{Oz})$  (`dsn:op:prop:native-omnific`, Proposition 35.1 of dsn), and defines  $\mathbb{Z}$  and  $\mathbb{N}$  (`odg:def:cor:arithmetic`), which confirms source 14’s remark that the pure ring “already has arithmetic strength” ([remark 74.11](#)). The failure of the unrestricted cut principle in [corollary 77.2](#) is `found:prop:allcuts` of the foundations report, and [section 75.1](#) agrees with `found:rem:conservativity`.

*Remark 67.6* (Answered across the two sources). Source 14’s main open question, [question 79.6](#) (“Is  $h$  definable in  $(\mathbf{Oz}; 0, 1, +, -, \cdot, <, b \restriction \mathbf{Oz})$ ?”), is answered positively by source 13’s [corollary 74.4](#); the derivation is [proposition 74.5](#), added in this merge. Hence source 14’s theory  $\text{OA}_{\text{qb}}$  and source 13’s  $\text{TOz}$  live in definitionally equivalent languages, and the item “elimination of binary quotient birthday in favor of unary birthday” of **14-surreal-only-nbg-SOURCES\_AND\_STATUS.md** is resolved, by a long interpretation-based definition; source 13’s [question 79.5](#) (a short definition) stays open. The set-recovery part of source 14’s question “Removing Choice” ([question 79.9](#)) is [theorem 13.2](#): the

recovered sets are exactly HWO (also [proposition 71.5](#)). Source 14’s [question 79.4](#) is answered in part by [proposition 74.14](#).

*Remark 67.7* (The Hilbert’s-tenth programme’s intake). The Hilbert’s-tenth programme authenticated both archives at their arrival and read bounded spans of them in commit 196e11d19 (Computability/HilbertTenthProblem/Papers/research-wip/native-stream-queue/review\_new\_foundations\_f300069cf.md; its “archive 1” is source 14 and its “archive 2” source 13). It found no correction in the spans read: source 13’s prefix and block arguments “check out” conditional on the cited omnific sign and initiality facts, which it did not check; source 14’s reflection argument keeps its premises. It observed the cross-source answer of [remark 67.6](#) as a lead, without certifying it, and that the model-cardinality and class-lift claims remained source claims for it. It also records that even a finite bit set receives an ordinal-length code in source 13’s storage (for a finite length  $n \geq 1$  the birthday is  $\omega(n + 1)$ ), and that the formulas quantify over all earlier-birthday numbers and all coded subsets, so no bounded integer encoding, operation-count improvement or Diophantine compiler follows. This merge checks the proofs it left unread ([section 80](#)); the unconditional parts stay conditional on the imported inputs listed in [section 79.4](#).

## 68 Logical and foundational conventions

### 68.1 Interpretation is stronger than encoding

An interpretation of a theory  $S$  in a theory  $T$  specifies a definable domain  $D(\bar{x})$ , an equivalence relation  $E(\bar{x}, \bar{y})$ , and definitions of the relations and function graphs of  $S$  on  $D/E$ . The tuples have some fixed finite length. The theory  $T$  must prove the conditions making these data an interpretation and must prove the translations of every axiom of  $S$ . Quotient equality is often essential [32].

If  $\mathcal{I}$  constructs an  $S$ -model from a  $T$ -model and  $\mathcal{J}$  constructs a  $T$ -model from an  $S$ -model, bi-interpretability additionally requires uniform definable isomorphisms

$$A \cong \mathcal{J}(\mathcal{I}(A)), \quad M \cong \mathcal{I}(\mathcal{J}(M)).$$

Merely obtaining some set-theoretic universe from the numbers and some number system from the sets is insufficient. An arbitrary surplus predicate on the numbers, for example, may be lost on the first translation and prevent the second isomorphism.

All positive interpretations here are parameter-free. When discussing negative results we explicitly allow finite parameters if the argument allows them. No proof chooses a global representative from every code equivalence class.

**The same conventions, as source 14 states them.** An interpretation of a theory  $T$  in a theory  $U$  consists of finitely many formulas defining a domain  $D \subseteq M^n$ , an equivalence relation  $\sim$  on  $D$ , and the relations and operations of  $T$  on the quotient. The theory  $U$  proves nonemptiness, congruence, and every translated axiom of  $T$ . We allow a fixed finite number of sorts and pieces, since these can be handled by the same syntax. The main positive constructions are parameter-free. For clarity,  $\mathcal{I}(N)$  will denote the set structure interpreted in a numerical structure  $N$ , and  $\mathcal{J}(M)$  the canonical numerical structure interpreted in a set structure  $M$ . A bi-interpretation requires *definable isomorphisms*

$$M \cong \mathcal{I}(\mathcal{J}(M)), \quad N \cong \mathcal{J}(\mathcal{I}(N)),$$

uniformly in the models, and proved in the respective theories. Merely showing that models can be encoded in both directions does not suffice. Likewise, equality of consistency strength and conservativity do not suffice. Finite axiomatizability is one invariant of bi-interpretability [31].

Quotients are important. An arbitrary set can have many enumerations of its transitive closure. We will not select one enumeration globally. Instead, different codes represent the same interpreted object when a first-order formula recognizes the appropriate isomorphism. The interpretation never quantifies over a proper-class quotient as a new set. In an internal class construction it uses representatives; in a set-sized model its quotient is a set in the metatheory. (Source 14.)

## 68.2 Proper-class structures are handled formula by formula

In a universe satisfying ZFC, **No** is a definable proper class of sets, represented here by sign functions

$$s : \alpha \longrightarrow \{-, +\}, \quad \alpha \in \text{Ord}.$$

Every individual sign function is a set. Its birthday is  $\alpha$ . Writing a first-order formula about **No** means relativizing its quantifiers to this definable class and replacing its numerical symbols by their set-theoretic definitions. This is a syntactic translation. It does not require a set containing all surreals or a truth predicate for the whole universe.

For a model  $M \models \text{ZFC}$ , let  $\mathbf{No}^M$  and  $\mathbf{Oz}^M$  mean these constructions *as computed inside*  $M$ . If  $M$  is countable externally, so are these interpreted domains. Internally,  $M$  regards them as proper classes. If  $M$  is nonstandard, its internally well-founded graphs need not be externally well-founded. The interpretation and collapse theorems are applied internally in  $M$ ; external well-foundedness is never silently assumed.

Source 14 adds: a surreal number is represented by a *set-sized* function  $s : \alpha \rightarrow \{-, +\}$ , where  $\alpha$  is an ordinal. Its birthday is  $\alpha$ , identified with the all-plus surreal of length  $\alpha$ . Thus every individual surreal is a set, while the total collection **No** is a proper class. This is precisely the reason that a domain of all surreals is a legitimate definable domain of interpretation in ZFC but is not a set asserted to exist by ZFC. All proofs of set constructions below are uniform theorems of ZFC. Consequently they apply internally to nonstandard models. An externally ill-founded relation which a nonstandard model regards as well-founded is not a counterexample: the interpreted collapse is the model's internal collapse. We never infer external well-foundedness merely from an internal first-order formula.

## 68.3 Sets versus classes

**Convention 68.1** (Class-theory names). Following `found:sub:gbconvention` of the foundations report, in this part GB is the two-sorted Gödel–Bernays theory with set and class variables, class extensionality, elementary class comprehension (class parameters allowed, quantified variables restricted to sets) and class replacement, with no choice assumption; GBc is GB with the Axiom of Choice for sets; GBC is GB with Global Choice (and hence set Choice). NBG is the usual one-sorted presentation of GBC. KM adds comprehension for formulas with class quantifiers. Henkin semantics is used for the class sort.

Source 13 used “GB” for GB, stating set Choice “when it matters”; source 14’s macro for “GB” printed “GBc” and meant GBc, “Choice for sets, but without Global Choice”. Each statement below names the theory its proof uses: the diagonal arguments need only GB; the class lifts need GBc, because the ordinal-graph codec uses set Choice ([proposition 71.5](#)); GB without set Choice

is [question 79.13](#). Where a source stated a result under a stronger convention than its proof uses, both are said at the point of use.

We use GBC for the usual two-sorted Gödel–Bernays theory with Global Choice. A model has the form

$$\mathcal{M} = (M, \mathcal{X}), \quad \mathcal{X} \subseteq \mathcal{P}(M)$$

externally, with sets in the first sort and classes in the second. Class extensionality identifies a class by its members. Elementary class comprehension permits set quantifiers and class parameters but no bound class quantifiers. Class replacement and Global Choice are included. The standard one-sorted formulation of NBG with the corresponding choice convention is equivalent to this presentation. KM allows class quantifiers in comprehension. These conventions remove a frequent ambiguity in the name NBG [22, 35]. (Source 13.)

Two distinctions are fundamental. First,  $\mathcal{X}$  is part of the model, not something determined by  $M$  alone. Second, a proper class is not an individual surreal number of  $M$ : the latter is a set-sized sign function. The fact that all surreals together form a proper class does not make that class a container for all classes as individual members.

Source 14 states the background thus: the usual NBG presentation with Global Choice is treated as a one-sorted presentation of GBC. None of the principal obstruction arguments depends on the terminological preference between these presentations. We use Henkin semantics for the class sort, not the requirement that all external subclasses be present. A model is written  $(M, \mathcal{X})$ , where  $M$  is the set model and  $\mathcal{X}$  its specified class collection. An elementary class formula may have set and class parameters but no bound class variables. Class comprehension for such formulas, together with class replacement, is sufficient for our elementary transports and diagonal argument. Full impredicative comprehension belongs to the KM discussion.

## 68.4 The finite analogy, stated accurately

For hereditarily finite sets the Ackermann map is

$$a(x) = \sum_{y \in x} 2^{a(y)}.$$

Its inverse reads binary digits recursively. This gives the familiar membership test on  $\mathbb{N}$ . The axiomatic equivalence requires an adequate finite set theory, including the appropriate induction or transitive-containment principle; replacing Infinity by its negation without checking the remaining formulation is not enough to justify every version of the claim [30]. In ProveIt the deductive PA/HF bi-interpretation is formalized in `Logic/Interpretability/PAHF` (its Ackermann membership core is `AckermannHFCore.lean`); see `Logic/README.md`.

Our transfinite codes will use well-founded graphs rather than a transfinite binary expansion of a single ordinal. Different labelings of the same transitive closure are identified. This is why quotient interpretations replace the unique natural-number code available in the finite Ackermann picture.

## 69 Surreal structure visible inside the omnific integers

### 69.1 The numerical facts used

We use the sign-sequence construction of **No**, with the lexicographic comparison in which a missing sign lies between  $-$  and  $+$ . Conway addition and multiplication are the usual recursively defined

operations. The resulting ordered field is real closed. Every surreal has a unique Conway normal form

$$x = \sum_{i < \lambda} r_i \omega^{a_i}, \quad r_i \in \mathbb{R} \setminus \{0\}, \quad a_i > a_j \ (i < j),$$

with set-sized, reverse well-ordered support. The  $\omega$ -power here is Conway's monomial map, not ordinary real exponentiation.

The omnific integers form the subring

$$\mathbf{Oz} = \left\{ n + \sum_{i < \lambda} r_i \omega^{a_i} : n \in \mathbb{Z}, \ a_i > 0 \right\}.$$

Equivalently,  $x \in \mathbf{Oz}$  precisely when  $x$  is the simplest surreal strictly between  $x - 1$  and  $x + 1$ . These are classical descriptions [1, 2, 33].

Two sign-theoretic facts will be crucial:

- (i)  $\mathbf{Oz}$  is initial in the surreal sign tree: every prefix of an omnific sign sequence is omnific.
- (ii) A surreal is omnific if and only if its sign sequence has no two adjacent positions carrying opposite signs. Thus signs may change at limit positions, but not between  $\gamma$  and  $\gamma + 1$ .

These facts are recorded in Ehrlich–Kaplan, arXiv version 1, Section 2.2, Lemma 2.1 and the proof of Lemma 2.3; the latter refers to Gonshor, Theorem 8.1, p. 111 [19, 2]. The version matters because later versions reorganize that section. They are imported, not re-proved, in source 13 and in this merge (section 79.4); every statement below about the native omnific structure depends on them.

Every ordinal is an omnific integer. The restrictions of surreal addition and multiplication to ordinals are natural, or Hessenberg, addition and multiplication. They are commutative and obey ordered-ring identities. They must not be confused with ordinary ordinal addition and multiplication.

## 69.2 Birthdays and ordinals are first-order definable

For the next lemmas,  $\mathbf{D}$  is either  $\mathbf{No}$  or  $\mathbf{Oz}$ , with all quantifiers ranging over  $\mathbf{D}$ . Define

$$\text{SameB}(x, y) \iff \neg(x \prec_b y) \wedge \neg(y \prec_b x), \quad (69.1)$$

$$\text{Ord}(a) \iff \forall y (\text{SameB}(y, a) \longrightarrow y \leq a). \quad (69.2)$$

**Lemma 69.1** (Ordinal recognition; sources 13, 14). *In either canonical structure,  $\text{Ord}(a)$  holds exactly for the embedded ordinals. The birthday  $b(x)$  is the unique  $a$  such that  $\text{Ord}(a) \wedge \text{SameB}(x, a)$ . Hence on  $\mathbf{No}$  and on  $\mathbf{Oz}$  the languages  $\mathcal{L}_b$  and  $\mathcal{L}_s$  are definitionally equivalent.*

*Proof.* Among all sign sequences of length  $\alpha$ , the sequence consisting of  $\alpha$  plus signs is greatest. It is the ordinal  $\alpha$ , and belongs to both domains. Every other length- $\alpha$  sequence is smaller at its first minus sign. Thus the greatest member of the birthday fiber is exactly  $\alpha$ . The last sentence is the observation that  $x \prec_b y$  is  $b(x) < b(y)$ .  $\square$

This is lemma 2.1 for  $\mathbf{D} = \mathbf{No}$ ; the case  $\mathbf{Oz}$  uses only that the all-plus sequence of each length is an ordinal, hence omnific. Source 14 proves it as a remark (“Birthday function versus birthday comparison”): the primitive  $b$  is definitionally interchangeable, on the canonical structure, with the relation  $x \prec_b y \iff b(x) < b(y)$  in the CHY announcement; in the reverse direction, with  $x \equiv_b y$  for  $\neg(x \prec_b y) \wedge \neg(y \prec_b x)$ ,  $b(x)$  is the unique greatest element, in numerical order, of the fibre  $\{y : y \equiv_b x\}$ , since among all signs of a fixed ordinal length the all-plus sequence is greatest;

“thus the change from comparison to a birthday function is not an additional coding assumption.” Source 14 also notes that  $b(x) = x$  defines the nonnegative ordinal surreals.

In subsequent formulas  $b$  is an abbreviation for this definable function, not an additional primitive. In particular, ordinal variables can be specified by  $\text{Ord}$ , and ordinary ordinal comparison is just the restriction of  $<$ . The ordinal  $\omega$  is definable as the least positive limit ordinal. For example, a nonzero ordinal  $a$  is limit when

$$\forall u(\text{Ord}(u) \wedge u < a \longrightarrow u + 1 < a).$$

Here  $u + 1$  is the ordinary ordinal successor, which agrees with its natural sum with 1.

### 69.3 Recovering the entire sign sequence

Define

$$\text{Pref}(u, z) \iff \neg(z \prec_b u) \wedge \forall v(v \prec_b u \longrightarrow (v < u \iff v < z)). \quad (69.3)$$

**Lemma 69.2** (Prefix test; sources 13, 14). *For  $u, z \in \mathbf{D}$ ,  $\text{Pref}(u, z)$  holds if and only if the sign sequence of  $u$  is a prefix, possibly the whole sequence, of that of  $z$ .*

*Proof.* The first conjunct says  $b(u) \leq b(z)$ . If  $u$  is a prefix, comparison with any shorter sign sequence  $v$  is decided before the end of  $u$ , so  $v < u$  and  $v < z$  agree.

Conversely, suppose the lengths are ordered as above but the sequences are not in the prefix relation. Let  $\gamma < b(u)$  be their first difference, and let  $v$  be their common prefix of length  $\gamma$ . The next sign of one is  $+$ , and that of the other is  $-$ . Consequently  $v$  lies strictly between  $u$  and  $z$ , contradicting the second conjunct. For  $\mathbf{D} = \mathbf{Oz}$ , initiality guarantees that this witness  $v$  is still in the quantified domain.  $\square$

(69.3) is source 02’s prefix formula of [remark 4.2](#), and the lemma is [lemma 4.1](#) with that formula; the case  $\mathbf{D} = \mathbf{Oz}$  is new and uses initiality, (i) above. Source 14 states the same lemma in  $\mathcal{B}$  with  $b$  (its formula is (69.3) with  $b(p) \leq b(x)$  and  $b(z) < b(p)$ ), and its proof adds: if there is no disagreement, the birthday inequality says precisely that  $p$  is a prefix, which also handles the empty prefix.

For ordinal  $a$ , put

$$\text{Plus}(z, a) \iff \text{Ord}(a) \wedge a \prec_b z \wedge \exists u(\text{SameB}(u, a) \wedge \text{Pref}(u, z) \wedge u < z). \quad (69.4)$$

**Lemma 69.3** (Sign test; sources 13, 14). *If  $a < b(z)$  is ordinal, then  $\text{Plus}(z, a)$  holds exactly when the  $a$ -th sign of  $z$  is  $+$ . It is false for ordinals outside the domain of that sign sequence.*

*Proof.* Initiality supplies the unique prefix  $u = z \restriction a$ . Its comparison with  $z$  is determined by the next sign:  $u < z$  exactly when that sign is  $+$ .  $\square$

Source 14’s bit predicate, for ordinal  $\alpha < b(x)$ ,

$$\text{Bit}(x, \alpha) \iff \exists p[\text{Pref}(p, x) \wedge b(p) = \alpha \wedge p < x], \quad (69.5)$$

declared false for nonordinal  $\alpha$  or  $\alpha \not< b(x)$ , is the same predicate: the unique prefix of length  $\alpha$  lies below  $x$  exactly when its next sign is plus. It is also this report’s Bit of (4.2).

Notice the quantifier scope. The test reads signs of every original omnific integer. It does not depend on the integer having any specially chosen code format. This fact is what will recover the whole numerical carrier in the second interpretation.

**Corollary 69.4** (Full bounded subset coding; source 14). *For every ordinal  $\theta$ , the surreal numbers of birthday exactly  $\theta$  are in definable bijection with  $\mathcal{P}(\theta)$ , via*

$$x \mapsto \{\alpha < \theta : \text{Bit}(x, \alpha)\}.$$

*In particular, universal quantification over subsets of a fixed ordinal can be replaced by first-order quantification over numbers of that birthday.*

*Proof.* The characteristic signs of any subset of  $\theta$  form a set-sized function of domain  $\theta$ , hence a surreal. Equality of all signs is equality of the corresponding surreal. No choice principle is needed for this lemma.  $\square$

This is (4.3) for the full class. In **Oz** the numbers of one birthday do not code every subset (an omnific sign sequence has no adjacent opposite signs); [theorem 70.2](#) replaces the corollary there.

## 70 Ordinal bit storage in a single omnific integer

An arbitrary sign sequence need not be omnific:  $+-$  represents  $1/2$ . Nevertheless, arbitrary bits can be stored in omnific integers by separating their transitions with limit positions.

### 70.1 The block construction

All products and sums in the next display are *native surreal operations on ordinal elements*. Define

$$f(\beta) = \omega \cdot \beta, \quad L_\alpha = f(\alpha) + \omega, \quad W_\beta = [f(\beta), f(\beta) + \omega).$$

The intervals refer to the usual order of ordinals.

**Lemma 70.1** (Properties of the blocks). *The function  $f$  is strictly increasing. Each  $W_\beta$  has ordinary ordinal order type  $\omega$ , and*

$$f(\beta) + \omega = f(\beta + 1).$$

*The intervals  $W_\beta$  are pairwise disjoint. For  $\beta > 0$ ,  $f(\beta)$  is a nonzero limit ordinal.*

*Proof.* Strict increase follows from the ordered-field law for multiplication by the positive number  $\omega$ . Distributivity gives the displayed identity, since native  $\beta + 1$  is the ordinal successor.

In Cantor normal form, natural multiplication by  $\omega$  shifts each exponent  $\xi$  to  $1 \oplus \xi$ . These exponents are positive. Thus  $f(\beta)$  has no finite term, and, when nonzero, is a limit ordinal. Because there is no term with exponent below 1, natural addition of  $\omega$  agrees in this case with ordinary right-addition of  $\omega$ . Hence the interval has order type  $\omega$ .

If  $\beta < \gamma$ , then  $\beta + 1 \leq \gamma$ , so the right endpoint of  $W_\beta$  is at most the left endpoint of  $W_\gamma$ .  $\square$

**Theorem 70.2** (Omnific bit storage). *For every ordinal  $\alpha$  and every set  $A \subseteq \alpha$ , there is an omnific integer  $z_{\alpha,A}$  such that*

$$\text{b}(z_{\alpha,A}) = L_\alpha, \quad \beta \in A \iff \text{Plus}(z_{\alpha,A}, \omega \cdot \beta) \quad (\beta < \alpha).$$

*The relation  $z = z_{\alpha,A}$ , when  $A$  is given by its bits, has a uniform first-order specification in  $\mathcal{L}_b$ .*

*Proof.* Define a sign sequence with domain  $L_\alpha$  by

$$z_{\alpha,A}(\gamma) = + \iff \exists \beta \in A (f(\beta) \leq \gamma < f(\beta) + \omega). \quad (70.1)$$

All other positions receive  $-$ . This is a set-sized function formed by Separation. No choice is involved.

It has no adjacent opposite signs. A change into a positive block can occur only at its left endpoint, a limit ordinal unless the endpoint is 0. A change out of such a block can occur only at its right endpoint, again a limit ordinal. No successor position can therefore be the first position after a sign change. The omnific sign criterion implies  $z_{\alpha,A} \in \mathbf{Oz}$ . The bit read at  $f(\beta)$  is precisely the membership bit of  $\beta$  in  $A$ , by disjointness of the blocks.

For first-order specification, require the indicated birthday and, for every ordinal  $\gamma < L_\alpha$ , require

$$\text{Plus}(z, \gamma) \iff \exists \beta < \alpha (\text{Plus}(z, f(\beta)) \wedge f(\beta) \leq \gamma < f(\beta) + \omega).$$

All ordinal quantifiers are guarded by Ord. This states that each selected block is positive and everything else is negative. Prescribing the selected bits then determines the word uniquely.  $\square$

*Remark 70.3* (Limit gaps are part of the construction). Natural multiplication is not generally continuous in an ordinal argument. For example,

$$\sup_{\beta < \omega^\omega} (\omega \cdot \beta) = \omega^\omega, \quad \omega \cdot \omega^\omega = \omega^{\omega+1},$$

where the products in parentheses and on the right are natural. The gap between these two ordinals is substantial. The word (70.1) fills such gaps with minus signs. Treating the blocks as automatically contiguous at all limits would be a mistake. The proof requires only disjointness and limit endpoints, which have been established.

**Example 70.4.** For  $A = \{0, 2\} \subseteq 3$ , the code has length equal to the native product  $4\omega$ , or ordinary ordinal  $\omega \cdot_{\text{ord}} 4$ , and four successive blocks of length  $\omega$ , with signs

$$\underbrace{+ \cdots +}_\omega \quad \underbrace{- \cdots -}_\omega \quad \underbrace{+ \cdots +}_\omega \quad \underbrace{- \cdots -}_\omega.$$

The last block is the fixed negative terminal block. This notation describes a sign function, not an assertion that the numerical value is a simple alternating sum of four copies of  $\omega$ .

We henceforth use the native formula

$$\text{Read}(z, \beta) \iff \text{Ord}(\beta) \wedge \text{Plus}(z, \omega \cdot \beta). \quad (70.2)$$

For every bounded ordinal set, a suitable  $z$  gives its characteristic predicate. Arbitrary  $z$  are also permitted: restricting the read predicate to any fixed ordinal always gives an ordinary set. One need not require graph codes to use the normalized block format.

For  $\mathbf{D} = \mathbf{No}$ , all these statements remain true because the constructed words belong to the omnific subclass. Thus the same interpretation formulas can be used in both numerical domains. (In  $\mathbf{No}$  alone [corollary 69.4](#) would allow the plain bit predicate; the block format is what makes one set of formulas serve both domains.)

## 70.2 A small pairing lemma avoids extra ordinal machinery

**Lemma 70.5** (Native ordinal pairing). *For a positive ordinal  $\kappa$ , the function*

$$\pi_\kappa(i, j) = \kappa \cdot i + j$$

*is injective on  $\kappa \times \kappa$ , with range below the native product  $\kappa^2$ .*

*Proof.* If  $i < i'$ , ordinal discreteness gives  $i + 1 \leq i'$ . Hence

$$\kappa i + j < \kappa(i + 1) \leq \kappa i' \leq \kappa i' + j' \quad (j, j' < \kappa).$$

For equal first coordinates, addition cancels. Finally,  $\kappa i + j < \kappa(i + 1) \leq \kappa^2$ . The argument uses only ordered-ring laws and the fact that native addition by 1 is ordinal successor.  $\square$

This is lemma 4.3 ( $\pi_\kappa = p_\kappa$ ; there the bound is  $\Lambda(\kappa) = \kappa^2 + \kappa$ ).

Consequently, graphs of relations and functions between bounded ordinal domains can themselves be represented by a single numerical bit code. The large base is a coding device; it is not the graph's vertex set. Padding a well-founded extensional graph with extra isolated vertices would generally destroy extensionality.

**Lemma 70.6** (Ordinal pairing; source 14). *The polynomial*

$$\varpi(\alpha, \beta) = (\alpha + \beta)^2 + \alpha \tag{70.3}$$

*is injective on  $\text{Ord}^2$ . For  $\alpha, \beta < \delta$ , its values are below the ordinal*

$$\Lambda_{14}(\delta) = (\delta + \delta)^2 + \delta + 1.$$

*Proof.* Natural sums and products of ordinals are ordinals. Write  $s = \alpha + \beta$ . Then  $0 \leq \alpha \leq s$ , so

$$s^2 \leq \varpi(\alpha, \beta) \leq s^2 + s.$$

If the sum for another pair is  $t > s$ , ordinal discreteness gives  $t \geq s + 1$  in surreal order. Therefore

$$t^2 \geq (s + 1)^2 = s^2 + 2s + 1 > s^2 + s.$$

Thus the intervals associated with distinct sums are disjoint. Equal pairing values force equal sums, then equal first coordinates by cancellation, and finally equal second coordinates. The asserted bound follows by monotonicity in the ordered field. The proof uses neither ordinary ordinal division nor a global well-order of all sets.  $\square$

The polynomial is source 06's  $\varpi$ , and the injectivity argument is that of lemma 4.5; source 14's contribution is the explicit bound  $\Lambda_{14}$ . Source 14 uses it in  $\mathcal{B}$ , without blocks: a relation  $E \subseteq \delta^2$  is encoded by a single number  $e$  of birthday  $\Lambda_{14}(\delta)$ , with

$$iEj \iff \text{Bit}(e, \varpi(i, j)).$$

At unused positions put minus signs. The length is fixed even for an empty relation. A rectangular relation  $R \subseteq \delta \times \eta$  can similarly be put in a sign sequence of birthday  $\Lambda_{14}(\delta + \eta)$ . This supplies function graphs and isomorphism witnesses without introducing set variables. (Source 14.)

### 70.3 Canonical finite-tuple encodings

**Lemma 70.7** (Native tuple codec). *For each fixed positive integer  $m$ , there is a parameter-free definable injection  $\mathbf{D}^m \rightarrow \mathbf{D}$  with a definable inverse on its image.*

*Proof.* For  $(x_0, \dots, x_{m-1})$ , let  $\alpha_i = \mathbf{b}(x_i)$  and choose the uniquely specified ordinal

$$\kappa = \max(2m + 1, \alpha_0 + 1, \dots, \alpha_{m-1} + 1).$$

In a  $\kappa$ -by- $\kappa$  array, use row  $i < m$  for a single marker in column  $\alpha_i$ , and row  $m + i$  for the plus-sign positions of  $x_i$ . Set every other entry to zero. Flatten the array with  $\pi_\kappa$ , set every index below  $\kappa^2$  outside the image of  $\pi_\kappa$  to zero, then apply the normalized block construction with  $\alpha = \kappa^2$ .

All these requirements have fixed first-order descriptions for fixed  $m$ , by the preceding lemmas. The output word is unique. Its birthday determines  $\kappa$ , because  $\kappa \mapsto f(\kappa^2) + \omega$  is strictly increasing on positive ordinals. The marker rows recover the lengths, and the other rows recover every sign. The inputs are therefore unique.  $\square$

This lemma will let the class-theoretic construction use classes of single numbers even when the set interpretation is conveniently written using triples. Source 14's packing of tuples (lemma 76.1) reaches the same end by a set-theoretic sign construction translated through the interpretation.

## 71 Recovering the set universe from numerical graphs

We now work in either of the canonical  $\mathcal{L}_b$ -structures on  $\mathbf{D} = \mathbf{No}$  or  $\mathbf{D} = \mathbf{Oz}$ . Every quantifier in the definitions below ranges over numbers. Bounds such as  $i < \nu$  mean that  $i$  is also required to satisfy  $\text{Ord}(i)$ .

### 71.1 Valid codes

A candidate code is a triple

$$c = (\nu_c, r_c, z_c),$$

where  $\nu_c$  is a nonzero ordinal and  $r_c < \nu_c$  is an ordinal. Put  $\kappa_c = \nu_c + 1$ , and define

$$R_c(i, j) \iff i, j < \nu_c \wedge \text{Read}(z_c, \kappa_c i + j). \quad (71.1)$$

The orientation is *member to containing set*:  $i R_c j$  will mean that the collapse of  $i$  belongs to the collapse of  $j$ .

The formula  $\text{Valid}(c)$  adds the following conditions.

**Well-foundedness.**

$$\forall w \left[ (\exists i < \nu_c \text{Read}(w, i)) \longrightarrow \exists i < \nu_c (\text{Read}(w, i) \wedge \forall j < \nu_c (\text{Read}(w, j) \rightarrow \neg R_c(j, i))) \right]. \quad (71.2)$$

Every nonempty subset of  $\nu_c$  is tested, because every such subset has a numerical code by theorem 70.2.

**Extensionality.**

$$\forall i, j < \nu_c \left[ (\forall k < \nu_c (R_c(k, i) \leftrightarrow R_c(k, j))) \longrightarrow i = j \right]. \quad (71.3)$$

### Accessibility from the root.

$$\begin{aligned} \forall w \Big[ & (\text{Read}(w, r_c) \wedge \forall i, j < \nu_c ((\text{Read}(w, j) \wedge R_c(i, j)) \rightarrow \text{Read}(w, i))) \\ & \longrightarrow \forall i < \nu_c \text{Read}(w, i) \Big]. \end{aligned} \quad (71.4)$$

This says that the only predecessor-closed subset containing the root is the whole graph. It is equivalent to every vertex being reachable from the root by a finite sequence of downward membership edges. The displayed version avoids introducing a separate finite-path codec.

These are finite first-order formulas after the previously defined abbreviations are expanded. Their subset quantification is numerical quantification over  $w$ , not a second-order quantifier.

Source 14's code is a triple  $c = (\delta, r, e)$  with  $\delta > 0$  an ordinal, root  $r < \delta$ , and  $e$  coding  $E_c \subseteq \delta^2$  by  $\varpi$ ; its three conditions are the same: every nonempty subset  $A \subseteq \delta$ , quantified by its birthday- $\delta$  sign code  $t$ , has an  $E_c$ -minimal element,

$$\forall t [\text{b}(t) = \delta \wedge \exists i < \delta \text{Bit}(t, i) \longrightarrow \exists i < \delta (\text{Bit}(t, i) \wedge \forall j < \delta [\text{Bit}(t, j) \rightarrow \neg(j E_c i)])];$$

distinct vertices have distinct predecessor sets; and every downward-closed subset of  $\delta$  which contains  $r$  is all of  $\delta$  ("root generation"), downward closure being  $\forall i, j < \delta [(j \in A \wedge i E_c j) \rightarrow i \in A]$ . Its  $\text{Code}(c)$  is the conjunction of these, including minus padding at unused positions; root generation is preferred to paths because it avoids prematurely introducing a code for natural-number sequences. These are the conditions of [section 5](#).

## 71.2 Equality and membership of represented sets

Let  $c, d$  be valid codes. To code a candidate map between their vertices, choose the definable ordinal

$$\nu_{cd} = \max(\nu_c, \nu_d) + 1$$

and read the relation

$$Y_h(i, j) \iff i < \nu_c \wedge j < \nu_d \wedge \text{Read}(h, \nu_{cd}i + j).$$

A finite formula says that  $Y_h$  is the graph of a total injective map from  $\nu_c$  to  $\nu_d$ . Additional formulas express surjectivity, root preservation, and preservation and reflection of edges. Define  $\text{Eq}(c, d)$  by the existence of such a bijection satisfying

$$Y_h(r_c, r_d)$$

and, for all pairs of matched vertices,

$$Y_h(i, j) \wedge Y_h(i', j') \longrightarrow (R_c(i, i') \iff R_d(j, j')).$$

Thus  $\text{Eq}$  is rooted graph isomorphism.

For membership, it is useful to permit a predecessor-closed embedding instead of an isomorphism onto the whole target. Require the total injective map to preserve and reflect edges:

$$Y_h(i, j) \wedge Y_h(i', j') \longrightarrow (R_c(i, i') \iff R_d(j, j')).$$

In addition, require it to satisfy

$$Y_h(i, j) \wedge R_d(k, j) \longrightarrow \exists i' < \nu_c Y_h(i', k).$$

Its image is then closed under all predecessors in the target graph. Define

$$\text{Mem}(c, d) \iff \exists h \exists j < \nu_d \left[ \begin{array}{l} Y_h \text{ is such a predecessor-closed embedding,} \\ Y_h(r_c, j) \wedge R_d(j, r_d) \end{array} \right]. \quad (71.5)$$

The verbal phrase stands for the finite conjunction just specified. It can be expanded without any additional quantifier sort.

**Source 14's membership.** For a vertex  $j$  of  $d$ , its downward hull is

$$\text{Hull}_d(j, i) \iff \text{every downward-closed subset containing } j \text{ contains } i,$$

subsets again being numerically quantified;  $c \in_* d$  means that for some  $j \in_d r_d$  there is a root-preserving isomorphism from  $c$  onto the induced graph on the hull of  $j$  in  $d$ . Surjectivity here means onto that hull, not onto the entire domain of  $d$ ; a rectangular code expresses this condition without relabelling the hull by a new ordinal. Source 14's  $c \sim_* d$  is  $\text{Eq}(c, d)$ . (Observation of this merge:) An edge-preserving and reflecting embedding with predecessor-closed image containing  $j$  and sending the root to  $j$  has image exactly the hull of  $j$ , by accessibility of the source graph; so the two membership formulas are equivalent on valid codes.

### 71.3 Collapse and its range

**Theorem 71.1** (The graph interpretation; sources 13, 14). *In ZFC, the formulas Valid, Eq, Mem define an interpretation  $\mathcal{I}_D$  satisfying*

$$\mathcal{I}_D(\mathbf{D}) = (\{c : \text{Valid}(c)\} / \text{Eq, Mem}) \cong (V, \in).$$

*The isomorphism is the root value under Mostowski collapse and is parameter-free definable in set theory.*

*Proof.* For a valid  $c$ , the decoded relation is a set relation on an ordinal. The subset coverage in [theorem 70.2](#) makes [\(71.2\)](#) and [\(71.3\)](#) exactly well-foundedness and extensionality, so Mostowski collapse gives a unique isomorphism

$$F_c : (\nu_c, R_c) \longrightarrow (T_c, \in)$$

onto a transitive set  $T_c$ . Let  $a_c = F_c(r_c)$ . Accessibility gives

$$T_c = \text{TC}(\{a_c\}).$$

Indeed, closing the root under finitely many predecessor steps corresponds to its transitive closure, and [\(71.4\)](#) makes this the whole domain. This closure itself need not be finite.

If  $c, d$  are rooted-isomorphic, uniqueness of collapse gives  $a_c = a_d$ . Conversely, equal root values give  $T_c = T_d = \text{TC}(\{a_c\})$ , so  $F_d^{-1}F_c$  is a rooted isomorphism. Its graph is a subset of a bounded ordinal rectangle and therefore has a numerical code. Consequently

$$\text{Eq}(c, d) \iff a_c = a_d.$$

A predecessor-closed embedding in [\(71.5\)](#) identifies the source collapse with the transitive closure of its image root. One can see this directly by well-founded induction: closure of the image under predecessors and preservation and reflection of edges give

$$F_c(i) = F_d(Y_h(i))$$

for every source vertex. If its root maps to  $jR_d r_d$ , then  $a_c = F_d(j) \in a_d$ . Conversely, if  $a_c \in a_d$ , let  $j$  be the unique vertex of  $d$  whose value is  $a_c$ . The map obtained from the two collapses embeds  $c$  onto the predecessor closure of  $j$ , and again has a numerical graph code. Thus

$$\text{Mem}(c, d) \iff a_c \in a_d.$$

Finally, given any set  $a$ , apply Choice to the set  $\text{TC}(\{a\})$ , choose a bijection from an ordinal  $\nu$  to this transitive closure, and transport membership and the distinguished element  $a$ . The resulting graph is well-founded, extensional, and accessible. Pairing and bit storage produce its numerical code. This establishes surjectivity.

The relation defining  $F_c$  is an ordinary set-theoretic formula asserting that a set function is the collapse of the specified graph. Its root value therefore defines the asserted isomorphism.  $\square$

For  $\mathbf{D} = \mathbf{No}$  this is the forward half of [theorem 18.1](#); for  $\mathbf{D} = \mathbf{Oz}$  it is new in the collection.

**Lemma 71.2** (Collapse and range; source 14). *Every valid code has a unique Mostowski collapse. Its root collapses to a set, denoted  $a_c$ . Every set occurs as  $a_c$  for some valid code. Moreover, the full collapsed vertex set is  $\text{TC}(\{a_c\})$ .*

*Proof (source 14).* The relation is a well-founded extensional relation on the set  $\delta$ . Well-founded recursion defines

$$F(i) = \{F(j) : jE_c i\}.$$

Extensionality, by well-founded induction, makes  $F$  injective. The range is transitive, and root generation says that it is exactly the transitive closure of the singleton containing  $F(r)$ . Uniqueness follows from the same recursion.

Conversely, for a set  $a$ , Choice supplies an ordinal  $\delta$  and a bijection between  $\delta$  and the set  $\text{TC}(\{a\})$ . Pull back membership, choose the vertex representing  $a$  as root, and use [lemma 70.6](#) to encode the relation. This code is well-founded, extensional, and root-generated. The enumeration is chosen for this one set only. A global definable choice of enumerations is not used.  $\square$

**Theorem 71.3** (Interpretation of the set universe; source 14). *The formulas above give a fixed finite-dimensional interpretation  $\mathcal{I}$  satisfying*

$$\mathcal{I}(\mathcal{B}) = (\text{Code}/\text{Eq}, \in_*) \cong (V, \in).$$

*More precisely, in the background set theory the definable relation  $\text{col}(c) = a$ , asserting that the root collapse of  $c$  is  $a$ , is the graph of an isomorphism on equivalence classes.*

*Second proof (source 14).* By uniqueness of collapse and root generation,  $c \text{ Eq } d \iff a_c = a_d$ . If an isomorphism exists, the collapses agree. In the reverse direction, equal root collapses give the same transitive closure of a singleton, and composing the inverse collapse bijections gives the required root-preserving isomorphism.

The collapse of the hull of  $j$  is  $\text{TC}(\{F_d(j)\})$ . Consequently the membership definition says exactly  $c \in_* d \iff a_c \in a_d$ . Existence of a code for every set is [lemma 71.2](#). The two displayed equivalences prove the equivalence-relation and congruence conditions, as well as the asserted isomorphism. Every condition is a fixed first-order formula, not a scheme quantifying over formulas.  $\square$

**Example 71.4** (The set  $2 = \{\emptyset, \{\emptyset\}\}$ ; sources 13, 14). Use vertices 0, 1, 2, root 2, and edges

$$R = \{(0, 1), (0, 2), (1, 2)\}.$$

With base  $\kappa = 4$ , the positive relation positions are  $\{1, 2, 6\} \subseteq 16$ . Any omnific code  $z$  satisfying  $\text{Read}(z, j) \leftrightarrow j \in \{1, 2, 6\}$  on  $j < 16$  gives a valid triple  $(3, 2, z)$ . Relabeling the three vertices changes the raw triple but not its equivalence class. The empty set has a one-vertex graph with no edges. These two examples already explain why native numerical equality and interpreted set equality have different jobs. (Source 13.)

In source 14's coding the same graph has pairing positions  $\varpi(0, 1) = 1$ ,  $\varpi(0, 2) = 4$ ,  $\varpi(1, 2) = 10$ . Put plus signs there and minus signs at the other positions in a sequence of length  $\Lambda_{14}(3) = 40$ . The long padding is intentionally not an efficient finite code; its benefit is a uniform polynomial bound at every ordinal. (Source 14.)

**Why this is not merely a global relabelling (source 14).** The domain, code equivalence, and membership formulas are read directly from arithmetic, order, and birthday. Neither a global map from  $V$  to ordinals nor a primitive membership relation on numbers was added. The code quotient avoids the hidden global choice that a unique canonical number for every set would require. Conversely, the construction does not claim an intrinsic interpretation from the pure real-closed field: birthday carries indispensable information.

## 71.4 The exact use of Choice

**Proposition 71.5** (Range over ZF). *Over ZF, the same ordinal-graph collapse has range exactly*

$$\text{HWO} = \{a : \text{TC}(\{a\}) \text{ is well-orderable}\}.$$

*Its surjectivity onto  $V$  is equivalent to the Axiom of Choice.*

*Proof.* A collapse with ordinal domain makes its transitive range well-orderable, so its root value belongs to HWO. Conversely, a well-order of  $\text{TC}(\{a\})$  gives an ordinal labeling and hence a code. If the collapse is onto  $V$ , every set belongs to HWO; in particular every set is a subset of a well-orderable transitive closure and is itself well-orderable. This is equivalent to Choice. The converse was proved in [theorem 71.1](#).  $\square$

This is the choice-sensitive boundary of this particular interpretation, already identified in this report's [theorem 13.2](#) and [proposition 13.3](#) (source 06's), which source 13 credits. It is not an impossibility theorem for every conceivable numerical interpretation of ZF. Nor must HWO be a model of ZF: if  $\mathcal{P}(\omega)$  is not well-orderable, every individual real belongs to HWO but the full collection  $\mathcal{P}(\omega)$  does not, and the internal power-set axiom at  $\omega$  fails. (For  $\mathbf{D} = \mathbf{Oz}$  the block codes are built without Choice, [theorem 70.2](#), so the omnific range is HWO too.)

Global Choice is absent from the positive proof. Each set merely has some graph code. A definable global choice of *one* such code for each set would be additional structure: a selected graph would let one choose the member with least vertex label from any nonempty represented set. The quotient interpretation does not make this selection.

## 72 Recovering every original number

The previous theorem establishes mutual interpretation only after we also recall that set theory defines  $\mathbf{D}$  and its arithmetic. We now verify the extra condition needed for bi-interpretability.

### 72.1 The canonical set-theoretic interpretation

Fix the standard parameter-free set-theoretic definitions:

- $\text{Is}_{\mathbf{No}}(s)$ :  $s$  is a sign function with ordinal domain;
- $\text{Is}_{\mathbf{Oz}}(s)$ :  $\text{Is}_{\mathbf{No}}(s)$ , with the omnific sign criterion;
- the Conway arithmetic relations on these sign functions;
- lexicographic numerical order and comparison of domain ordinals.

Write  $\mathcal{J}_{\mathbf{D}}$  for this interpretation of the numerical language in set theory. Thus  $\mathcal{J}_{\mathbf{D}}(M)$  is the canonical  $\mathcal{L}_{\mathbf{b}}$ -structure on  $\mathbf{D}^M$ .

Inside the numerical interpretation of sets we may apply these same fixed formulas, translating them through  $\mathcal{I}_{\mathbf{D}}$ . For example,  $\text{Is}_{\mathbf{D}}^{\mathcal{I}_{\mathbf{D}}}(c)$  says that the set represented by  $c$  is internally one of the relevant canonical sign functions. There is no quantification over all definitions or all truths. Only fixed first-order formulas are being translated.

**The set-to-number interpretation, as source 14 describes it.** Let  $\mathcal{J}$  be the usual definable construction of surreal numbers inside set theory. Its domain is the set-coded functions from an ordinal into  $\{0, 1\}$ , with zero representing minus and one representing plus. Comparison is lexicographic with an absent sign lying between minus and plus. Birthday is the domain ordinal, represented again as an all-plus sign sequence. The ring operations are the definable Conway operations. For completeness, their definability here does not require a global satisfaction predicate or an elementary class recursion principle. Canonical options of a number are its shorter prefixes on the appropriate side. Addition is determined recursively by

$$x + y = \{x^L + y, x + y^L \mid x^R + y, x + y^R\}.$$

Multiplication uses the usual lower expressions  $x^L y + x y^L - x^L y^L$ ,  $x^R y + x y^R - x^R y^R$ , and upper expressions  $x^L y + x y^R - x^L y^R$ ,  $x^R y + x y^L - x^R y^L$ . For any fixed inputs the required recursion has set-sized input data, so ordinary set recursion gives definable operation graphs. The least-birthday separator of a set cut is itself set-definable. These are the standard arithmetic constructions [1, 2].

### 72.2 Linking native ordinals to represented ordinals

Define  $\text{OG}(\alpha, e)$  as follows:  $\alpha$  is a native ordinal,  $e$  is a valid set code, and  $e$  is equivalent to a graph with domain  $\alpha + 1$ , root  $\alpha$ , and edges

$$iRj \iff i < j < \alpha + 1.$$

Existence of a code for this graph follows from bit storage. The quantifier over that code avoids selecting a representative.

**Lemma 72.1.** *For a native ordinal  $\alpha$  and a valid code  $e$ , under the collapse isomorphism,*

$$\text{OG}(\alpha, e) \iff a_e = \alpha.$$

*In particular, every code representing the ordinal  $\alpha$  satisfies  $\text{OG}(\alpha, e)$ .*

*Proof.* The displayed graph is the membership graph of the transitive set  $\alpha + 1$ , with root  $\alpha$ . Its collapse is the identity. Equivalence of valid graph codes is equality of root values.  $\square$

### 72.3 An explicit formula for the second isomorphism

In the language of set theory, let  $\text{Dom}(s, \lambda)$  assert that the function  $s$  has domain  $\lambda$ , and let  $\text{Val}_+(s, \beta)$  assert that its value at  $\beta$  is the chosen set representative of  $+$ . For definiteness, use 0 for  $-$  and 1 for  $+$ .

Define  $\text{Cmp}_{\mathbf{D}}(x, c)$  to say:

- (i)  $c$  is valid and  $\text{Is}_{\mathbf{D}}^{\mathcal{I}_{\mathbf{D}}}(c)$ ;
- (ii) with  $\lambda = \text{b}(x)$ , there is a code  $e$  such that  $\text{OG}(\lambda, e)$  and  $\text{Dom}^{\mathcal{I}_{\mathbf{D}}}(c, e)$ ;
- (iii) for every native ordinal  $\beta < \lambda$  and every code  $e$  with  $\text{OG}(\beta, e)$ ,

$$\text{Val}_+^{\mathcal{I}_{\mathbf{D}}}(c, e) \iff \text{Plus}(x, \beta).$$

These clauses define a single first-order numerical formula. Code variables stand for fixed triples and can be expanded into three ordinary variables each. The quantifier binding  $\lambda$  has scope over both the domain and sign clauses: formally their conjunction is placed inside  $\exists \lambda (\text{Ord}(\lambda) \wedge \text{SameB}(x, \lambda) \wedge \dots)$ .

**Theorem 72.2** (Full numerical round trip; sources 13, 14). *For  $x \in \mathbf{D}^M$  and valid  $c$ , in every  $M \models \text{ZFC}$ , evaluated internally,*

$$\text{Cmp}_{\mathbf{D}}(x, c) \iff a_c = x$$

*as equality of canonical sign-function sets. Consequently  $\text{Cmp}_{\mathbf{D}}$  defines an isomorphism*

$$\mathcal{J}_{\mathbf{D}}(M) \cong \mathcal{J}_{\mathbf{D}}(\mathcal{I}_{\mathbf{D}}(\mathcal{J}_{\mathbf{D}}(M)))$$

*on the entire original numerical domain.*

*Proof.* By lemma 72.1, native ordinal  $\beta$  and every code  $e$  used for it have the same value  $\beta$  after collapse. The domain clause therefore says that  $a_c$  has domain  $\text{b}(x)$ , and the last clause says that its signs agree with  $x$  at every position of that domain. Sign-function extensionality gives  $a_c = x$ . The converse follows from the same observations.

Existence of an image follows from theorem 71.1 applied to the set  $x$ . The image is unique modulo Eq; distinct original numbers have distinct images. Every reconstructed canonical number has an original preimage because collapse is onto and preserves the fixed definition of  $\text{Is}_{\mathbf{D}}$ .

Finally, the canonical set-theoretic definitions of  $+$ ,  $-$ ,  $\cdot$ ,  $<$ ,  $\prec_b$ ,  $0$ , and  $1$  are invariant under the collapse isomorphism. Thus the comparison just proved preserves every numerical symbol. Equivalently,  $\text{Cmp}_{\mathbf{D}}$  is the inverse of the map induced by  $\mathcal{J}_{\mathbf{D}}$  on the collapse isomorphism.  $\square$

**Corollary 72.3** (Structure-level bi-interpretation). *The canonical structures*

$$(\mathbf{No}, 0, 1, +, -, \cdot, <, \prec_b) \quad \text{and} \quad (\mathbf{Oz}, 0, 1, +, -, \cdot, <, \prec_b)$$

*are each parameter-free bi-interpretable, formula by formula in ZFC, with  $(V, \in)$ .*

The surreal instance gives a proof in the setting of the previously announced CHY result (theorem 18.1). The omnific instance follows from the additional block-storage lemma and initiality. No use of a monomial-map predicate, fractional birthday predicate, membership primitive, or auxiliary set sort is made in its native language. (Source 13.)

**Source 14's comparison.** For an ordinal surreal  $\alpha$ , let  $c_\alpha^{\text{ord}}$  denote the code with vertex domain  $\alpha + 1$ , root  $\alpha$ , and relation  $iEj \iff i < j$ . Its root collapse is the set ordinal  $\alpha$ . This is a definable code: its relation is padded using  $\Lambda_{14}(\alpha + 1)$ . Writing  $c_\alpha^{\text{ord}}$  is an abbreviation for the formula defining that triple, not an additional primitive function. Define  $\rho(x, c)$  to mean that  $c$  is a set code which, in the interpreted set universe  $\mathcal{I}(\mathcal{B})$ , is a function with domain  $c_{b(x)}^{\text{ord}}$ , values in the interpreted set  $\{0, 1\}$ , and satisfies

$$c(c_\alpha^{\text{ord}}) = 1 \iff \text{Bit}(x, \alpha) \quad (\alpha < b(x)). \quad (72.1)$$

For the values zero and one in this formula use the interpreted von Neumann integers, not the original ring elements. The distinction avoids a type error. Evaluation, functionhood, and domain are ordinary fixed set-theoretic formulas translated by  $\mathcal{I}$ . Consequently (72.1), together with these conditions, is a single first-order  $\mathcal{L}_s$ -formula in  $x$  and the triple  $c$ . It names a specific code for each ordinal where  $\text{Cmp}_\mathbf{D}$  quantifies over all codes satisfying OG; the two formulas define the same relation on valid codes.

**Proposition 72.4** (Definable numerical recovery; source 14). *ZFC proves that  $\rho$  induces an isomorphism*

$$\mathcal{B} \cong \mathcal{J}(\mathcal{I}(\mathcal{B})).$$

*Together with col, it witnesses a structural bi-interpretation between the canonical birthday-enriched surreal structure and the set universe.*

*Second proof of the round trip for No (source 14).* Under the collapse isomorphism from [theorem 71.3](#),  $c_\alpha^{\text{ord}}$  becomes the actual ordinal  $\alpha$ . Thus a code satisfying  $\rho(x, c)$  collapses to exactly the sign function defining  $x$ . This proves existence and uniqueness modulo Eq, and also injectivity. Every sign function in the recovered set universe is the sign function of a surreal in the original canonical domain, giving surjectivity.

Order and birthday are determined by the same signs on both sides. The set-theoretic definitions of the Conway operations are the same formulas in the two set universes related by collapse. An isomorphism preserves every fixed definable construction; hence it preserves these operation graphs. Equivalently, one can induct on the relevant set-sized recursion tables. The resulting bijection therefore preserves  $0, 1, +, -, \cdot, <, b$ , not just the underlying order.  $\square$

## 73 Explicit formal axiomatic theories

The structure-level result is not yet a displayed axiom system. This section gives one, using fixed first-order formulas and an effective translation of the ZFC axiom schemes. Its correctness does not depend on the final form of any unpublished intrinsic surreal-arithmetic axiomatization.

### 73.1 A general principle

**Theorem 73.1** (Axioms for inverse interpretations; sources 13, 14). *Let  $S$  be a recursively axiomatized theory in a finite language. Let  $K$  define, in  $S$ , a structure in a finite language  $L$ . Suppose that:*

- (i) *fixed  $L$ -formulas  $I$  interpret the language of  $S$ ;*
- (ii)  *$S$  proves admissibility of  $I$  in its  $K$ -structure and proves a fixed definable isomorphism  $I(K(M)) \cong M$ ;*

(iii) a fixed  $L$ -formula  $C$ , evaluated in the  $K$ -structure, is proved in  $S$  to give an isomorphism

$$K(M) \cong K(I(K(M))).$$

There is a recursively axiomatized  $L$ -theory  $T$  bi-interpretable with  $S$ . It consists of

$$T = \text{Adm}(I) \cup \{\sigma^I : \sigma \text{ is an axiom of } S\} \cup \{\text{Iso}(C)\}. \quad (73.1)$$

The first and last expressions denote finite lists of first-order sentences.

*Proof.* By its first two groups of axioms,  $T$  admits  $I$  as an interpretation of  $S$ . Conversely,  $S$  proves the  $K$ -translations of the admissibility sentences by hypothesis. It proves  $(\sigma^I)^K$  because  $I(K(M))$  is definably isomorphic to its own universe. It proves the translated isomorphism assertions by the last hypothesis. Thus  $K$  interprets  $T$  in  $S$ . The source-side comparison and the explicitly asserted comparison  $C$  are the two definable isomorphisms required.

All translations are finite syntactic substitutions and relativizations. The languages are finite, so admissibility and preservation of their symbols require only finitely many sentences. Enumerating the axioms of  $S$  and translating them enumerates the axioms of  $T$ .  $\square$

In this proof  $K$  is a construction, not the letter  $K$  of a field; the instances below take  $K = \mathcal{J}_D$ .

**Lemma 73.2** (Round-trip axiom construction; source 14). *Let  $U$  be a recursively axiomatized theory. Suppose  $\mathcal{J}$  is a fixed definable construction of structures in a finite language  $L$  inside  $U$ . Suppose fixed formulas  $\mathcal{I}$ ,  $\eta$ , and  $\rho$  satisfy, provably in  $U$ :*

$$\mathcal{I}(\mathcal{J}(M)) \models U, \quad \eta : M \cong \mathcal{I}(\mathcal{J}(M)), \quad \rho : \mathcal{J}(M) \cong \mathcal{J}(\mathcal{I}(\mathcal{J}(M))).$$

*Then there is a recursively axiomatized  $L$ -theory  $T$  bi-interpretable with  $U$ , whose canonical models are the  $\mathcal{J}(M)$ .*

*Second proof (source 14).* Let  $\text{Val}(\mathcal{I})$  be the finite set of interpretation-validity sentences for the domain, quotient equivalence, and congruence formulas. Let  $\text{Iso}(\rho)$  be the finite collection saying that  $\rho$  is an isomorphism from the original  $L$ -structure onto the structure obtained by the composite  $\mathcal{J} \circ \mathcal{I}$ . After unwinding the fixed interpretations, these are  $L$ -sentences. They say that  $\rho$  is total, single-valued modulo interpreted equality, injective, surjective, and preserves each primitive symbol of  $L$ . Take

$$T = \text{Val}(\mathcal{I}) \cup \{\sigma^{\mathcal{I}} : \sigma \in \text{Ax}(U)\} \cup \text{Iso}(\rho). \quad (73.2)$$

This is a recursively enumerable axiom set in a finite language. It proves that  $\mathcal{I}$  interprets  $U$ . Conversely the hypotheses prove that  $\mathcal{J}$  interprets  $T$  in  $U$ . The two definable isomorphisms are respectively  $\eta$ , already proved in  $U$ , and  $\rho$ , asserted by the coherence axioms in  $T$ . Thus the two theories are bi-interpretable.

There is no assertion of a global truth predicate in (73.2). The middle term is an external, effectively specified axiom scheme; only individual axioms are translated.  $\square$

**Theorem 73.1** and **lemma 73.2** are the same recipe, proved independently:  $\text{Adm}(I) = \text{Val}(\mathcal{I})$  and  $C = \rho$ .

The isomorphism assertions are *axioms*, not consequences asserted without proof from a small ordered-ring fragment. The source-side proof that they hold in the intended number structure is the substantive requirement. Here it was discharged by [theorem 72.2](#). (Source 13.) Source 14 adds that it is easy to confuse a statement about the intended structure with a first-order theory having the required equivalence in *all* models; the recipe closes that gap.

### 73.2 The two numerical theories

**Definition 73.3.** For  $\mathbf{D} = \mathbf{No}, \mathbf{Oz}$ , define  $T_{\mathbf{D}}$  by (73.1), using  $S = \text{ZFC}$ , the numerical language  $\mathcal{L}_b$ , the graph interpretation  $\mathcal{I}_{\mathbf{D}}$ , the canonical number interpretation  $\mathcal{J}_{\mathbf{D}}$ , and the signwise formula  $\text{Cmp}_{\mathbf{D}}$ . Write  $T_{\mathbf{No}}$  and  $T_{\mathbf{Oz}}$  for these theories.

Here are the three axiom groups in explicit form.

**Group A: quotient admissibility.** Assert:

1.  $\exists c \text{ Valid}(c)$ ;
2.  $\text{Eq}$  is reflexive, symmetric, and transitive on valid codes;
3. if  $\text{Eq}(c, c')$  and  $\text{Eq}(d, d')$ , then  $\text{Mem}(c, d) \leftrightarrow \text{Mem}(c', d')$ .

Code variables are fixed triples. These are finite first-order sentences in the numerical language.

**Group B: the translated set axioms.** Take the usual axioms of ZFC, including every Separation and Replacement instance. Replace equality by  $\text{Eq}$ , membership by  $\text{Mem}$ , and each set quantifier by a quantifier over valid triples. For example,

$$\exists x \varphi(x) \mapsto \exists \nu_x \exists r_x \exists z_x (\text{Valid}(\nu_x, r_x, z_x) \wedge \varphi^{\mathcal{I}}).$$

Universal quantifiers are relativized correspondingly.

**Group C: the numerical comparison is an isomorphism.** Let  $\text{Is}_{\mathbf{D}}^{\mathcal{I}}(c)$  assert that  $c$  represents a canonical  $\mathbf{D}$ -number in the interpreted universe. Assert the universal closures of

$$\begin{aligned} \text{Cmp}_{\mathbf{D}}(x, c) &\longrightarrow \text{Valid}(c) \wedge \text{Is}_{\mathbf{D}}^{\mathcal{I}}(c), \\ \forall x \exists c \text{ Cmp}_{\mathbf{D}}(x, c), \\ (\text{Cmp}_{\mathbf{D}}(x, c) \wedge \text{Eq}(c, d)) &\longrightarrow \text{Cmp}_{\mathbf{D}}(x, d), \\ (\text{Cmp}_{\mathbf{D}}(x, c) \wedge \text{Cmp}_{\mathbf{D}}(x, d)) &\longrightarrow \text{Eq}(c, d), \\ (\text{Cmp}_{\mathbf{D}}(x, c) \wedge \text{Cmp}_{\mathbf{D}}(y, d) \wedge \text{Eq}(c, d)) &\longrightarrow x = y, \\ \text{Valid}(c) \wedge \text{Is}_{\mathbf{D}}^{\mathcal{I}}(c) &\longrightarrow \exists x \text{ Cmp}_{\mathbf{D}}(x, c). \end{aligned}$$

Valid-code guards are understood for every occurrence of  $\text{Eq}$ . The equality  $x = y$  in the injectivity axiom is *raw numerical equality*, not code equivalence.

For preservation of addition, assert

$$\text{Cmp}_{\mathbf{D}}(x, c) \wedge \text{Cmp}_{\mathbf{D}}(y, d) \wedge \text{Cmp}_{\mathbf{D}}(z, e) \longrightarrow (x + y = z \longleftrightarrow \text{Add}_{\mathbf{D}}^{\mathcal{I}}(c, d, e)).$$

Include the corresponding sentences for multiplication and negation, and for constants 0, 1. Finally, assert

$$\begin{aligned} \text{Cmp}_{\mathbf{D}}(x, c) \wedge \text{Cmp}_{\mathbf{D}}(y, d) &\longrightarrow (x < y \longleftrightarrow \text{Less}_{\mathbf{D}}^{\mathcal{I}}(c, d)), \\ \text{Cmp}_{\mathbf{D}}(x, c) \wedge \text{Cmp}_{\mathbf{D}}(y, d) &\longrightarrow (x \prec_b y \longleftrightarrow \text{Earlier}_{\mathbf{D}}^{\mathcal{I}}(c, d)). \end{aligned}$$

All predicates on the right are translations of fixed set-theoretic definitions of the canonical numerical structure.

This describes an actual recursive axiom list. It does not say to include all truths of the intended numbers, and it does not introduce the displayed abbreviations as new primitive symbols.

**Theorem 73.4** (Numerical foundations for ZFC; sources 13, 14). *The theories  $T_{\mathbf{No}}$  and  $T_{\mathbf{Oz}}$  are parameter-free bi-interpretable with ZFC:*

$$T_{\mathbf{No}} \simeq_{\text{bi}} \text{ZFC} \simeq_{\text{bi}} T_{\mathbf{Oz}}.$$

*Every  $M \models \text{ZFC}$  yields the canonical model  $\mathcal{J}_{\mathbf{D}}(M)$  of  $T_{\mathbf{D}}$ . Every  $A \models T_{\mathbf{D}}$  is definably isomorphic to the full canonical  $\mathbf{D}$ -structure of the set universe  $\mathcal{I}_{\mathbf{D}}(A)$  which it interprets.*

*Proof.* Apply theorem 73.1 using theorems 71.1 and 72.2. The last sentence is exactly the Group C comparison.  $\square$

**Definition 73.5** (An explicit surreal foundational theory; source 14).  $\text{SA}_{\text{rt}}$  is the theory (73.2) obtained by taking  $U = \text{ZFC}$ , the code formulas of section 71 in source 14's form, the construction  $\mathcal{J}$  above, and the formula  $\rho$  in (72.1).

**Theorem 73.6** (Surreal-only set foundations; source 14). *The theory  $\text{SA}_{\text{rt}}$  is parameter-free bi-interpretable with ZFC. Every model  $N \models \text{SA}_{\text{rt}}$  is definably isomorphic to  $\mathcal{J}(\mathcal{I}(N))$ , where  $\mathcal{I}(N) \models \text{ZFC}$ . In particular, it contains no unintended extra numerical component invisible to the interpreted set universe.*

*Proof (source 14).* Theorem 71.3 and proposition 72.4 verify the hypotheses of lemma 73.2. The last statement is precisely the surjectivity and injectivity clause for  $\rho$ .  $\square$

$\text{SA}_{\text{rt}}$  and  $T_{\mathbf{No}}$  are the same recipe applied to two presentations of the same interpretation, in the definitionally equivalent languages  $\mathcal{L}_{\text{s}}$  and  $\mathcal{L}_{\text{b}}$  (lemma 69.1); both are bi-interpretable with ZFC, hence with each other. This definition is intentionally not a claim that the candidate intrinsic SA axiom list in the CHY slides [8] has been fully verified here. It is a different, mechanically specifiable presentation. It sacrifices axiomatic elegance to make the theory-level round trip explicit. Its primitive vocabulary remains numerical; sets occur only as interpreted codes. (Source 14.)

### 73.3 The meaning of the axioms

There is no hidden set sort in  $T_{\mathbf{Oz}}$ . A quantified triple  $(\nu, r, z)$  consists of three omnific integers, and the assertion that it represents a set is a numerical formula. For example, the singleton operation on represented sets is specified by

$$\text{Singleton}(c, a) \iff \forall d (\text{Valid}(d) \rightarrow (\text{Mem}(d, c) \leftrightarrow \text{Eq}(d, a))),$$

with validity of  $a, c$  included. Pairing uses  $\text{Eq}(d, a) \vee \text{Eq}(d, b)$ ; union uses  $\exists e (\text{Valid}(e) \wedge \text{Mem}(e, a) \wedge \text{Mem}(d, e))$ . For valid  $p, a$ , the power-set condition is

$$\forall c [\text{Valid}(c) \rightarrow (\text{Mem}(c, p) \leftrightarrow \forall d (\text{Valid}(d) \rightarrow (\text{Mem}(d, c) \rightarrow \text{Mem}(d, a))))].$$

These operations are on code equivalence classes. Native numerical addition remains ordinary omnific addition; it is not being redefined as set union.

Source 14 displays several translated axioms the same way, variables  $a, b, c, u, v, w$  abbreviating valid triples:

$$\begin{aligned} \text{Extensionality: } & \forall a, b [\forall c (\text{Mem}(c, a) \leftrightarrow \text{Mem}(c, b)) \rightarrow \text{Eq}(a, b)], \\ \text{Pairing: } & \forall a, b \exists c \forall u [\text{Mem}(u, c) \leftrightarrow (\text{Eq}(u, a) \vee \text{Eq}(u, b))], \\ \text{Union: } & \forall a \exists b \forall u [\text{Mem}(u, b) \leftrightarrow \exists v (\text{Mem}(v, a) \wedge \text{Mem}(u, v))], \\ \text{Power Set: } & \forall a \exists b \forall u [\text{Mem}(u, b) \leftrightarrow \forall v (\text{Mem}(v, u) \rightarrow \text{Mem}(v, a))]. \end{aligned}$$

Separation for a set-language formula  $\varphi$  becomes  $\forall a, \bar{p} \exists b \forall u [\text{Mem}(u, b) \leftrightarrow (\text{Mem}(u, a) \wedge \varphi^{\mathcal{I}}(u, \bar{p}))]$ . Replacement becomes the statement that the image of an interpreted set under any translated definable functional relation is represented by a code. Infinity asserts the existence of a code for an inductive set; Foundation, Empty Set, and Choice translate in the same fixed syntactic way. All quantifiers ultimately range over surreal individuals.

Group B supplies the set existence assertions. Group C ensures that the numerical carrier and its operations agree with the number system of that universe. It rules out spurious extra copies of numbers or numerical structure that disappears upon decoding.

An intrinsic theory based on short arithmetic, generation, and birthday-bounding principles could be more illuminating and easier to use. The theory here instead settles the precise *existence* question with transparent logical guarantees. Reducing these axiom groups is a further problem, not a premise of the theorem.

### 73.4 Provability and independence transfer

**Corollary 73.7.** *For every set-theoretic sentence  $\sigma$  and numerical sentence  $\varphi$ ,*

$$\text{ZFC} \vdash \sigma \iff \text{T}_{\mathbf{D}} \vdash \sigma^{\mathcal{I}_{\mathbf{D}}}, \quad \text{T}_{\mathbf{D}} \vdash \varphi \iff \text{ZFC} \vdash \varphi^{\mathcal{J}_{\mathbf{D}}}.$$

*Proof.* The forward implications are interpretation of proofs. Translate back for the reverse implications and use the corresponding definable isomorphism to identify the double translation with the original sentence.  $\square$

Thus questions about CH, large cardinals, or other set-theoretic assertions have numerical translations with exactly the same provability status. This is representational equivalence; it does not decide any statement independent of ZFC. Consistency of any one of the three theories is equivalent to consistency of the others.

Every additional set-theoretic axiom  $\sigma$ , such as a large-cardinal assertion, can be added as  $\sigma^{\mathcal{I}}$ . The same round trips then give

$$\text{ZFC} + \sigma \text{ bi-interpretable with } \text{SA}_{\text{rt}} + \sigma^{\mathcal{I}}.$$

This is a consequence of the established interpretation, not evidence for  $\sigma$  or a new consistency proof. (Source 14; the same holds for  $\text{T}_{\mathbf{D}}$ , and [lemma 73.8](#) states the general fact.)

**Finite axiomatizability.** Assuming consistency, these numerical theories are incomplete and undecidable. They are not finitely axiomatizable either. Finite axiomatizability is preserved by a bi-interpretation between finite languages: choose a finite source fragment proving the translated finite axiom list and the finite assertion that the double interpretation is an isomorphism, then translate proofs back to show that this fragment proves the entire source theory. But reflection proves consistency of each finite fragment of ZFC; a finite axiomatization of all of ZFC would therefore contradict the second incompleteness theorem. (Source 13; source 14's [lemma 75.4](#) proves the preservation statement in full.)

### 73.5 Why this is more than an external relabeling

An external bijection between two domains permits transport of arbitrary relations, but supplies no native way to read them. Our formulas use the actual numerical arithmetic, order, and birthday

relation. The graph relation is defined by signs at specified numerical positions. The second comparison matches the full original sign function, not an externally chosen enumeration.

Nevertheless, the axiom system is deliberately built from the interpretation. It is reasonable to want a shorter arithmetic presentation. The distinction is between a proved formal existence theorem and a desirable simplification of its axioms. In particular, nothing here proves that an arbitrary ordered ring with an unrelated well-founded relation must recover set theory.

**Lemma 73.8** (Persistence under translated extensions; added in this merge). *Let  $S$  and  $T$  be bi-interpretable by  $\mathcal{I}$  (interpreting  $S$  in  $T$ ) and  $\mathcal{J}$  (interpreting  $T$  in  $S$ ), with the definable isomorphisms provable in  $S$  and  $T$  respectively. For every set  $\Sigma$  of sentences of the language of  $S$ , the theories  $S + \Sigma$  and  $T + \{\sigma^{\mathcal{I}} : \sigma \in \Sigma\}$  are bi-interpretable by the same  $\mathcal{I}$ ,  $\mathcal{J}$  and isomorphisms. If  $T$  has a computably enumerable axiom set,  $\Sigma$  is computably enumerable, and the translation  $\sigma \mapsto \sigma^{\mathcal{I}}$  is computable, then the second theory is computably axiomatizable.*

*Proof.*  $\mathcal{I}$  interprets  $S + \Sigma$  in  $T + \Sigma^{\mathcal{I}}$  by construction. For  $\mathcal{J}$  it remains to see that  $S + \Sigma$  proves  $(\sigma^{\mathcal{I}})^{\mathcal{J}}$  for  $\sigma \in \Sigma$ : this sentence says that  $\sigma$  holds in  $\mathcal{I}(\mathcal{J}(M))$ , which  $S$  proves equivalent to  $\sigma$  by the definable isomorphism  $M \cong \mathcal{I}(\mathcal{J}(M))$ . The isomorphisms are unchanged, and their properties remain provable in the larger theories. Under the effective hypotheses, dovetail an enumeration of the axioms of  $T$  with the computable translations of an enumeration of  $\Sigma$ . This enumerates an axiom set of  $T + \Sigma^{\mathcal{I}}$ .  $\square$

*Remark 73.9* (Effective hypotheses in the persistence lemma). The merge of 4 October 2026 ended the lemma with “if  $\Sigma$  is recursive, so is the second theory.” Without an effective axiomatization of the starting theory this is false: take  $S = T = \text{Th}(\mathbb{N})$ , identity interpretations, and  $\Sigma = \emptyset$ . The target remains true arithmetic, which is not computably axiomatizable: otherwise it would be a consistent, complete, computably axiomatized extension of arithmetic, contrary to incompleteness. The corrected clause also states effectivity of translation; fixed-formula interpretations between the finite languages used here satisfy that condition. The bi-interpretation assertion itself needs neither of these computability assumptions.

This is classical and is used below for the class variants ([corollary 76.7](#)); source 14 gives the same argument in the proof of [theorem 76.5](#).

## 74 Consequences for definability and the choice of language

### 74.1 Every ambient set definition has a native numerical translation

**Corollary 74.1** (Definability transfer). *Let  $\mathbf{D} = \mathbf{No}$  or  $\mathbf{Oz}$ . For each first-order set-theoretic formula  $\psi(s_1, \dots, s_n)$ , there is a parameter-free formula  $\psi_{\mathbf{D}}(x_1, \dots, x_n)$  in  $\mathcal{L}_{\mathbf{b}}$  such that ZFC proves*

$$\mathbf{D}(x_1) \wedge \dots \wedge \mathbf{D}(x_n) \longrightarrow (\psi(x_1, \dots, x_n) \leftrightarrow \psi_{\mathbf{D}}(x_1, \dots, x_n)).$$

*Here the arguments on the set-theoretic side are the canonical sign-function sets representing the numbers.*

*Proof.* Use the formula

$$\exists c_1 \dots \exists c_n \left( \bigwedge_{i=1}^n \text{Cmp}_{\mathbf{D}}(x_i, c_i) \wedge \psi^{\mathcal{I}_{\mathbf{D}}}(c_1, \dots, c_n) \right).$$

The collapse isomorphism and [theorem 72.2](#) give exactly the claimed equivalence.  $\square$

Thus the native birthday language defines much more than a convenient membership coding. It can express every set-theoretically definable relation on its numerical domain. For example, in **No** it defines the usual monomial map and surreal exponential by translating their fixed set-theoretic definitions. In **Oz** it defines their graphs where both arguments lie in **Oz**, and it defines ordinary ordinal addition and multiplication as relations on the native ordinals. Additional numerical parameters are carried through the same comparison.

This is a separate translation for each formula. It is not a single formula evaluating the truth of every formula code. Consequently it supplies no truth predicate and does not evade Tarski's theorem. Class-quantified formulas of a stronger class theory are also outside the assertion unless the class lift is used.

**Proposition 74.2** (Canonical bi-interpretations of reducts and expansions; added in this merge). *Let  $\mathcal{A} = (\mathbf{D}; R_1, \dots, R_k)$ , with  $\mathbf{D} = \mathbf{No}$  or  $\mathbf{Oz}$ , be a structure each of whose finitely many primitives  $R_i$  is given by a fixed parameter-free set-theoretic formula on the canonical sign functions, and let  $\bar{p}$  be a finite tuple from  $\mathbf{D}$ . The following are equivalent, provably in ZFC formula by formula:*

- (a) *there are an interpretation  $\mathcal{I}'$  of set theory in  $\mathcal{A}$  and a relation  $F(x, c)$ , both first-order in  $\mathcal{A}$  with parameters  $\bar{p}$ , and a set-theoretically definable isomorphism  $G : \mathcal{I}'(\mathcal{A}) \cong (V, \in)$ , such that  $F(x, c) \iff G(c) = x$  (each number goes to the codes of its own sign sequence);*
- (b) *the symbols  $0, 1, +, -, \cdot, <, \prec_b$  are definable in  $\mathcal{A}$  with parameters  $\bar{p}$ .*

*If they hold,  $\mathcal{A}$  is bi-interpretable with  $(V, \in)$ , with the canonical set-to-number interpretation, and definitionally equivalent, over  $\bar{p}$ , to  $(\mathbf{D}, \mathcal{L}_b)$ . Moreover (a) then holds for every interpretation  $\mathcal{I}'$  that is  $\mathcal{A}$ -definable over  $\bar{p}$  and has a set-theoretically definable isomorphism onto  $(V, \in)$ .*

*Proof.* (a) $\Rightarrow$ (b). Let  $\psi(\bar{s})$  be a set-theoretic formula defining one of the symbols on canonical sign functions. Then  $\psi(\bar{x})$  holds iff  $\exists \bar{c} (\bigwedge_i F(x_i, c_i) \wedge \psi^{\mathcal{I}'}(\bar{c}))$ , because  $\psi^{\mathcal{I}'}(\bar{c})$  holds iff  $\psi(G\bar{c})$  does, and  $Gc_i = x_i$ . This is an  $\mathcal{A}$ -formula over  $\bar{p}$ .

(b) $\Rightarrow$ (a). Substitute the definitions of the symbols of  $\mathcal{L}_b$  into Valid, Eq, Mem and Cmp $_{\mathbf{D}}$ ; theorems 71.1 and 72.2 give (a) with  $\mathcal{I}' = \mathcal{I}_{\mathbf{D}}$  and  $F = \text{Cmp}_{\mathbf{D}}$ . Each  $R_i$  is  $\mathcal{L}_b$ -definable by corollary 74.1, so the two structures are definitionally equivalent over  $\bar{p}$ ; the comparison Cmp $_{\mathbf{D}}$  preserves every  $R_i$ , being the identity on numbers after collapse, so the canonical set-to-number interpretation of  $\mathcal{A}$ 's language, composed with  $\mathcal{I}_{\mathbf{D}}$ , gives a bi-interpretation. For another  $\mathcal{I}'$  with  $G' : \mathcal{I}'(\mathcal{A}) \cong V$ , the relation “the  $\mathcal{I}_{\mathbf{D}}$ -code  $c$  and the  $\mathcal{I}'$ -code  $d$  represent the same set” is  $\mathcal{A}$ -definable over  $\bar{p}$ : inside the set universe  $\mathcal{I}_{\mathbf{D}}(\mathcal{A})$  the structure  $\mathcal{A}$  is reconstructed as  $\mathcal{J}_{\mathbf{D}}$  of it, transported along Cmp $_{\mathbf{D}}$ , and ZFC proves that  $V \cong \mathcal{I}'(\mathcal{A})$  by the definable  $G'$ ; composing gives  $F$  for  $\mathcal{I}'$ .  $\square$

Three cases matter here. (1) For a reduct of  $\mathcal{L}_b$  containing  $\prec_b$ , a canonical bi-interpretation exists exactly when the reduct defines the missing ring symbols; source 06's corollary 12.6 is a positive instance at uncountable cardinal cutoffs, for  $(\mathbf{No}_{<\kappa}; <, b, G_{\varpi})$ . (2) For dsn's  $\mathfrak{S}_{\Omega} = (\mathbf{No}; +, \cdot, <, \mathbf{Oz}, \Omega)$  (remark 67.5), which names the ring operations, the specific form of dsn:op:q:biinterpretation is equivalent to the definability of  $\prec_b$  in  $\mathfrak{S}_{\Omega}$ ; whether  $\prec_b$  is so definable is not decided here. (3) For the pure ordered ring of **Oz**, (b) fails even with parameters (corollary 74.13); proposition 74.14 below excludes more, namely any comparison, canonical or not. The proposition concerns comparisons sending each number to codes of itself; it says nothing about one-way interpretations, the subject of the “Minimal signature” question of section 21.

## 74.2 Recovering fractions and their birthdays

The classical fraction-field fact provides a useful comparison with the direct construction.

**Proposition 74.3** (Fractions; sources 13, 14). *Every surreal is a fraction of two omnific integers, with positive denominator. Thus the natural map  $\text{Frac}(\mathbf{Oz}) \rightarrow \mathbf{No}$  is an ordered-field isomorphism.*

*Proof.* For  $x = \sum_i r_i \omega^{a_i}$ , choose a positive ordinal  $\beta$  larger than every  $-a_i$ . The support is a set, so such an ordinal exists. Put  $b = \omega^\beta$ . Then  $b \in \mathbf{Oz}$ ,  $b > 0$ , and

$$bx = \sum_i r_i \omega^{\beta+a_i} \in \mathbf{Oz},$$

since every exponent is positive. Hence  $x = (bx)/b$ . The case  $x = 0$  is immediate.  $\square$

*Second proof (source 14).* For  $x \neq 0$ , write its normal form as  $x = \sum_{\gamma \in S} r_\gamma \omega^\gamma$ . The support  $S$  is a set. There is a positive ordinal surreal  $\delta$  larger than every  $-\gamma$  for  $\gamma \in S$ . For example, first bound the birthdays of those numbers, then choose a larger ordinal; a sufficiently long all-plus sequence exceeds every sequence in that birthday bound. Let  $q = \omega^\delta$  and  $p = qx$ . All exponents  $\delta + \gamma$  of  $p$  are positive. Hence  $p, q \in \mathbf{Oz}$ , and  $x = p/q$ . The case  $x = 0$  uses  $0/1$ . Injectivity of the natural fraction map follows from the integral-domain property of  $\mathbf{Oz}$ .  $\square$

This fact and stronger simultaneous denominator-clearing results already appear in the collection, including Lean formalizations (`odg:thm:fractions`; [section 78.1](#)). It must not be confused with recovery of the birthday relation. Taking a fraction field from the *pure* ring adds no birthday information. (Source 13.) Source 14 adds: this standard fact is also noted by Jeřábek [37]; the proof emphasizes its *whole-support* bound. Shifting merely the leading exponent would not suffice when the support extends arbitrarily far down within a set-sized range. The assertion also does not follow for an arbitrary integer part of a real-closed field: the fraction field of  $\mathbb{Z}$  is  $\mathbb{Q}$ , not  $\mathbb{R}$ . (The Hilbert’s-tenth intake adds that the shift requires an ordinal bound for every exponent of a set-sized support and is not a bounded integer operation.)

One could instead enrich  $\mathbf{Oz}$  with the four-variable predicate

$$\text{FB}(a, b, c, d) \iff b, d > 0 \wedge b(a/b) < b(c/d).$$

It is invariant under the usual fraction equivalence  $ad = bc$ , with the appropriate pairs of numerators and denominators. The fraction interpretation would then directly recover the birthday-equipped surreal field.

The native result is stronger in its choice of primitives. [Corollary 74.1](#) gives:

**Corollary 74.4.** *The relation FB is already parameter-free definable in  $(\mathbf{Oz}, 0, 1, +, -, \cdot, <, \prec_b)$ .*

*Proof.* Canonical surreal division and comparison of the domains of its sign functions have fixed set-theoretic definitions. Apply [corollary 74.1](#) to their composite on four omnific arguments.  $\square$

The resulting formula can be large. Finding a short direct arithmetic description remains a useful simplification problem ([question 79.5](#)); the existence of a definition is settled here. (Source 13.)

**Proposition 74.5** (Quotient birthday from unary birthday; added in this merge). *Source 14’s quotient birthday  $h(a, b)$ , with  $h(a, 0) = 0$ , is parameter-free definable in  $(\mathbf{Oz}; 0, 1, +, -, \cdot, <, b \restriction \mathbf{Oz})$ . Consequently the languages  $\mathcal{L}_z$  and  $\mathcal{L}_b$  are definitionally equivalent on  $\mathbf{Oz}$ , and [question 79.6](#) has a positive answer.*

*Proof.* On  $\mathbf{Oz}$ ,  $b \mid \mathbf{Oz}$  and  $\prec_b \mid \mathbf{Oz}$  are interdefinable (lemma 69.1), so corollary 74.4 applies. For an ordinal  $\gamma$ ,  $\gamma/1 = \gamma$  and  $b(\gamma) = \gamma$ . Hence for  $b > 0$ ,

$$h(a, b) = \gamma \iff \text{Ord}(\gamma) \wedge \neg \text{FB}(a, b, \gamma, 1) \wedge \neg \text{FB}(\gamma, 1, a, b),$$

since the two negated conjuncts say  $b(a/b) = \gamma$ ; for  $b < 0$  put  $h(a, b) = h(-a, -b)$ , because  $a/b = (-a)/(-b)$ ; and  $h(a, 0) = 0$ . Conversely  $x \prec_b y \iff h(x, 1) < h(y, 1)$  defines  $\prec_b$  on  $\mathbf{Oz}$  from  $h$ .  $\square$

The definition inherits the length of corollary 74.4 and its dependence on the imported omnific sign facts (through theorem 72.2 for  $\mathbf{Oz}$ ). Source 14's warning that neither the fraction construction alone nor knowledge of the two input birthdays settles the question stays correct: the definition goes through the whole interpretation.

### 74.3 A second omnific route: quotient birthday (source 14)

Source 14 reaches the omnific integers through fractions instead of native blocks. An omnific integer is a surreal with normal form

$$z = \sum_{\gamma > 0} r_\gamma \omega^\gamma + n, \quad n \in \mathbb{Z}, \quad (74.1)$$

where the nonzero support is a set and is reverse well-ordered, and the coefficients  $r_\gamma$  are real. Zero support and negative coefficients are allowed. This is the usual omnific ring  $\mathbf{Oz}$ .

Define a binary operation on  $\mathbf{Oz}$  by

$$h(a, b) = \begin{cases} b(a/b), & b \neq 0, \\ 0, & b = 0. \end{cases} \quad (74.2)$$

The output really is an omnific integer: birthdays are ordinal surreals, and ordinal surreals are omnific integers. Put

$$\mathcal{O}_{\text{qb}} = (\mathbf{Oz}; 0, 1, +, -, \cdot, <, h).$$

The subscript records the extra quotient-birthday information. We do *not* assert that  $h$  is definable in the pure ordered ring. (The letter  $\mathcal{O}$  is not the collection's valuation ring.)

Use pairs  $(a, b)$  with  $b > 0$  as fraction codes, and define

$$\begin{aligned} (a, b) \sim_F (c, d) &\iff ad = cb, \\ (a, b) <_F (c, d) &\iff ad < cb, \\ (a, b) +_F (c, d) &= (ad + cb, bd), \\ (a, b) \cdot_F (c, d) &= (ac, bd), \\ b_F([(a, b)]) &= [(h(a, b), 1)]. \end{aligned}$$

For the actual structure, equal fractions have equal quotient birthdays, so the last operation is well-defined. These fixed formulas interpret  $\mathcal{B}$  in  $\mathcal{O}_{\text{qb}}$ .

For the reverse interpretation,  $\mathbf{Oz}$  is definable in  $\mathcal{B}$ . One method is to translate the fixed set-theoretic normal-form definition (74.1) through the already established set interpretation and  $\rho$ . This avoids any assumption that an ordered-field formula alone recognizes the normal form. The operation  $h$  is then definable using field division and  $b$ . Call the resulting  $\mathcal{L}_s$ -formula  $\text{Omnific}(x)$ .

**Theorem 74.6** (Omnific quotient-birthday bi-interpretation; source 14). *The structures  $\mathcal{O}_{\text{qb}}$ ,  $\mathcal{B}$ , and  $(V, \in)$  are parameter-free bi-interpretable by the constructions above.*

*Proof.* [Proposition 74.3](#) identifies the interpreted fraction field with all of  $\mathbf{No}$ , including birthday by definition of  $h$ . Starting with  $\mathcal{B}$ , the number-to-fraction round trip is defined by the relation  $a = bx$ , with  $a, b \in \mathbf{Oz}$  and  $b > 0$ . It is single-valued modulo fraction equivalence and onto, and respects every primitive.

Starting with  $\mathcal{O}_{\text{qb}}$ , the ring-to-fractions map  $z \mapsto [(z, 1)]$  identifies exactly the original omnific integers with the omnific predicate in the recovered field. The formula for  $h$  is preserved because it uses the same quotient and birthday. This gives both definable round trips. Composition with [proposition 72.4](#) gives the set-theoretic assertion.  $\square$

With [proposition 74.5](#),  $\mathcal{O}_{\text{qb}}$  is a definitional expansion of  $(\mathbf{Oz}, \mathcal{L}_{\text{b}})$ , so this theorem and the omnific case of [corollary 72.3](#) are two routes to one structural statement: source 13's by native sign blocks, without fractions, source 14's by fractions, with  $h$ . Source 14's route does not use the omnific sign criterion; it uses the normal-form definition (74.1) and [proposition 74.3](#).

**An axiom system on the omnific sort (source 14).** Source 14 presents this as “an explicit alternative to simply applying [lemma 73.2](#) again”. Define  $\text{OA}_{\text{qb}}$  in  $\mathcal{L}_{\text{z}}$  by these axiom groups:

**O1.** The ordered-integral-domain axioms, with  $0 < 1$ .

**O2.**  $h(a, 0) = 0$ , and fraction invariance:

$$bd \neq 0 \wedge ad = cb \quad \longrightarrow \quad h(a, b) = h(c, d).$$

**O3.** Every axiom of  $\text{SA}_{\text{rt}}$ , translated to the positive-denominator fraction interpretation above.

**O4.** In that interpreted surreal structure, for every fraction  $u$ ,

$$\text{Omnific}(u) \quad \Longleftrightarrow \quad \exists z (u = [(z, 1)]).$$

Here Omnific is the fixed formula just constructed.

These are first-order axiom groups, and O3 is an effective scheme. The equivalence in O4 is crucial: merely requiring a real-closed fraction field would allow many other subrings and would not ensure recovery of the original omnific sort.

**Theorem 74.7** (An effective omnific foundational theory; source 14).  *$\text{OA}_{\text{qb}}$  is recursively axiomatized and parameter-free bi-interpretable with  $\text{SA}_{\text{rt}}$ , hence with ZFC. Its canonical interpretation has only omnific-integer individuals and has the intended  $h$  operation.*

*Proof (source 14).* O1 and O2 make the fraction construction and its birthday operation well-defined. O3 makes it a model of  $\text{SA}_{\text{rt}}$ . O4 says that the reverse omnific interpretation recovers exactly the original domain. The map  $z \mapsto [(z, 1)]$  preserves the ordered ring operations.

For  $b \neq 0$ , fraction invariance identifies any representation of  $[(a, 1)]/[(b, 1)]$  with  $(a, b)$ . Thus its interpreted birthday is  $[(h(a, b), 1)]$ ; at zero denominator O2 gives the stated convention. So the map preserves  $h$  as well. The converse direction follows inside  $\text{SA}_{\text{rt}}$  by transfer of [proposition 74.3](#) and [theorem 74.6](#) through its numerical round trip. All formulas are fixed and all axiom translations effective.  $\square$

*Remark 74.8* (O1–O4 is an instance of the recipe; added in this merge). O1–O4 is [theorem 73.1](#) applied with  $S = \text{SA}_{\text{rt}}$ , the fraction interpretation as  $I$ , and the comparison  $z \mapsto [(z, 1)]$  as  $C$ : O1

and O2 are the admissibility sentences, O3 the translated axioms, and O4 the one clause of  $\text{Iso}(C)$  (surjectivity) that is not provable from O1–O3; injectivity and preservation of  $+$ ,  $\cdot$ ,  $<$  and  $h$  follow from O1–O2 as the proof shows. So it is not an alternative to the recipe but a presentation of it in which most isomorphism clauses are derived; its transparency is that of O4. By [proposition 74.5](#) the same recipe in  $\mathcal{L}_b$  is  $\text{TO}_z$  up to a definitional translation of  $h$ .

*Remark 74.9* (An example and a warning; source 14, re-scoped). For ordinary integers,  $h(1, 2) = 2$ , since  $1/2$  has sign sequence  $+-$ . Hence quotient birthday is not the birthday of the numerator or denominator alone. Giving an axiom system with  $h$  does not settle whether the weaker language consisting of ordered-ring arithmetic and the unary restricted birthday  $z \mapsto b(z)$  already suffices. That precise reduction remains a research question here. (Source 14.) [Merge: it is settled positively by source 13’s [corollary 74.4](#), through [proposition 74.5](#); see [question 79.6](#).]

#### 74.4 Why the pure ordered surreal field is insufficient

**Proposition 74.10** (Classical decidability obstruction; sources 13, 14). *The pure ordered field  $(\mathbf{No}, 0, 1, +, -, \cdot, <)$  does not first-order interpret Robinson arithmetic  $Q$ , even with a fixed finite tuple of parameters. In particular it cannot interpret ZFC.*

*Proof.* Its elementary theory is the complete decidable theory RCF. Suppose that fixed formulas, with parameters  $p$ , interpret  $Q$ . Admissibility and the finitely many axioms of  $Q$  give a single ordered-field formula  $\theta(p)$  true in this field.

In the language with constant symbols  $c$ , the consistent theory  $\text{RCF} + \theta(c)$  is decidable: deciding whether it proves  $\varphi(c)$  amounts to deciding

$$\text{RCF} \vdash \forall c(\theta(c) \rightarrow \varphi(c)).$$

The proposed interpretation would therefore give a decidable consistent extension of  $Q$ , contradicting its essential undecidability. Since ZFC interprets  $Q$ , it is excluded as well.  $\square$

This is an instance of the general interpretation/undecidability principle discussed by Friedman [\[32\]](#). The extra birthday relation is carrying logically substantial structure. (Source 13.)

*Second proof, parameter-free (source 14).* The surreal field is real closed, and the complete theory of real-closed ordered fields is decidable by quantifier elimination. If it interpreted Robinson arithmetic, take the inverse image of its complete decidable theory under the effective interpretation translation. This would be a consistent decidable complete extension of Robinson arithmetic, contradicting essential undecidability. Since ZFC interprets that arithmetic, it cannot be interpreted either.

If birthday were ordered-field definable, substituting its definition in our set-code formulas would supply precisely the forbidden interpretation. Equivalently,  $b(x) = x$  would define the ordinals, whereas one-variable sets in a real-closed field are finite unions of points and intervals. The ordinal subclass has neither form.  $\square$

Source 14 states it as: the ordered-field theory of the surreal numbers does not interpret ZFC or Robinson arithmetic; in particular, no ordered-field formula defines a birthday expansion sufficient for the constructions above. The standard real-closed-field background is also highlighted in the CHY announcement. The argument is *not* a no-go theorem for the pure omnific ring: it is not a field, and it already has arithmetic strength. Nor does it show that every enrichment weaker than birthday must fail. Those are separate model-theoretic questions. (Source 14.) This report’s

**proposition 12.5** (sources 05, 07) proves the nondefinability of birthday, also in every o-minimal expansion of the field, at epsilon-number cutoffs and formula by formula on **No**.

*Remark 74.11* (The pure omnific ring has arithmetic strength). Source 14's last remark is confirmed in the collection: the pure ring **Oz** parameter-free defines  $\mathbb{Z}$  and  $\mathbb{N}$  with their arithmetic (`odg:def:cor:arithmetic`), so it interprets  $Q$  and its theory is undecidable; it interprets the field **No** with  $\mathbb{R}$  and the order (`odg:def:thm:realrecovery`), the order of **Oz** is definable in its pure unital ring language, and  $(\mathbf{Oz}; +, -, \cdot, 0, 1)$  is bi-interpretable with  $(\mathbf{No}, \mathbf{Oz})$  (`dsn:op:prop:native-omnific`). None of these interprets set theory.

## 74.5 The pure omnific ring loses ordinal and birthday information

No parallel decidability assertion about the pure omnific ring is assumed. A different argument proves a narrower but strong obstruction. The repository contains more extensive results about omnific-preserving surreal automorphisms; the following proof isolates a consequence relevant here [40, 36].

**Proposition 74.12** (Automorphisms fixing prescribed parameters). *For every set  $P \subseteq \mathbf{No}$ , there is a nonidentity parameter-definable ordered-field automorphism  $\Phi$  of **No** which fixes  $P$ , preserves **Oz**, and sends an ordinal to a nonordinal.*

*Proof.* Collect the exponents occurring in the Conway normal forms of members of  $P$ , and then collect the supports of the normal forms of those exponents. These are sets. Choose a positive ordinal  $\delta$  greater than every member of the second collection.

For any surreal  $a$ , let  $\text{co}_\delta(a) \in \mathbb{R}$  be the coefficient of  $\omega^\delta$  in its normal form, or zero when that monomial is absent. Define

$$\vartheta(a) = a + \frac{1}{2} \text{co}_\delta(a).$$

Coefficient extraction is additive. Since  $\delta > 0$ , adding a real does not change  $\text{co}_\delta(a)$ , so  $\vartheta^{-1}(a) = a - \frac{1}{2} \text{co}_\delta(a)$ . This is an additive automorphism.

It preserves order. When  $\text{co}_\delta(a) \neq 0$ , the leading monomial of  $a$  is at least  $\omega^\delta$ , hence is infinite; changing only a real term does not change its sign. When the coefficient is zero,  $a$  is fixed. Apply this to differences to obtain strict order preservation.

Lift  $\vartheta$  to normal forms:

$$\Phi\left(\sum_i r_i \omega^{a_i}\right) = \sum_i r_i \omega^{\vartheta(a_i)}.$$

An additive order automorphism of the exponent group preserves reverse well-ordered supports and the sums of exponents appearing in multiplication. The displayed map is therefore an ordered-field automorphism, with inverse given by the lift of  $\vartheta^{-1}$ . This standard Hahn-series lifting argument is described in [36].

Every exponent occurring in  $P$  has zero  $\omega^\delta$ -coefficient by the choice of  $\delta$ . Consequently  $\Phi$  fixes  $P$  pointwise. The map  $\vartheta$  fixes 0 and preserves positive exponents;  $\Phi$  fixes real coefficients. It therefore preserves the normal-form criterion for **Oz**.

Finally, the ordinal  $o_\delta = \omega^{\omega^\delta}$  is sent to

$$\Phi(o_\delta) = \omega^{\omega^\delta + 1/2}.$$

The exponent lies strictly between the ordinal  $\omega^\delta$  and its successor, so is not ordinal. An ordinal's Conway normal form is its Cantor normal form and has only ordinal exponents. Uniqueness of normal form therefore shows that  $\Phi(o_\delta)$  is not ordinal.  $\square$

The conclusion follows from `opa:thm:parameters` of `surreal/omnific-preserving-automorphisms`, whose automorphism also fixes  $\mathbb{R} \cup \{\omega\}$ , is strong and real-linear, and sends the ordinal  $\omega^{\omega^\beta}$  to a nonmonomial ([remark 67.5](#)); the shear is a short self-contained route, with no priority claim.

**Corollary 74.13.** *The embedded ordinals and birthday precedence are not definable in the pure ordered omnific ring, even with finitely many numerical parameters. The analogous nondefinability holds in the pure ordered surreal field.*

*Proof.* Apply the proposition with the parameter tuple included in  $P$ . Any definable predicate is preserved by every automorphism fixing its parameters. The ordinal class is not preserved. If birthday precedence were definable, (69.2) would define the ordinals, a contradiction.  $\square$

**Proposition 74.14** (No bi-interpretation of the pure omnific ring with  $V$  on the canonical numbers; source 13, made a proposition in this merge). *There are no interpretation  $\mathcal{I}'$  of set theory in the pure ordered ring  $\mathcal{R} = (\mathbf{Oz}; 0, 1, +, -, \cdot, <)$  and relation  $F$ , both first-order in  $\mathcal{R}$  with a finite tuple  $\bar{p}$  of parameters, together with a set-theoretically definable (with parameters) isomorphism  $G : (V, \in) \cong \mathcal{I}'(\mathcal{R})$ , such that  $F$  is an isomorphism of  $\mathcal{R}$  onto the canonical ordered ring  $\mathcal{J}(\mathcal{I}'(\mathcal{R}))$  of the interpreted universe. In other words, the pure omnific ring is not bi-interpretable, even with parameters, with the actual well-founded universe when the set-to-number direction is the canonical construction. The same holds for the pure ordered field  $\mathbf{No}$ .*

*Proof.* Suppose  $\mathcal{I}'$ ,  $F$ ,  $\bar{p}$  and  $G$  existed. Apply [proposition 74.12](#) with  $P$  the set of entries of  $\bar{p}$ ; it gives a nonidentity ordered-ring automorphism  $\Phi$  of  $\mathcal{R}$  fixing  $\bar{p}$ , definable in set theory with a parameter.

Because the formulas of  $\mathcal{I}'$  use only the parameters  $\bar{p}$ , the map  $\Phi_{\mathcal{I}'} : [\bar{c}] \mapsto [\Phi\bar{c}]$  is a well-defined automorphism of  $\mathcal{I}'(\mathcal{R})$ , definable in set theory with parameters. Then  $\pi = G^{-1} \circ \Phi_{\mathcal{I}'} \circ G$  is a definable automorphism of  $(V, \in)$ . By  $\in$ -induction  $\pi$  is the identity: if  $\pi(y) = y$  for every  $y \in x$ , then  $z \in \pi(x) \iff \pi^{-1}(z) \in x$ , and  $\pi^{-1}(z) \in x$  gives  $z = \pi(\pi^{-1}(z)) = \pi^{-1}(z) \in x$ , while  $z \in x$  gives  $z = \pi(z) \in \pi(x)$ . So  $\Phi_{\mathcal{I}'}$  is the identity of  $\mathcal{I}'(\mathcal{R})$ , and therefore so is the map it induces on  $\mathcal{J}(\mathcal{I}'(\mathcal{R}))$ , which is defined inside  $\mathcal{I}'(\mathcal{R})$  by fixed set-theoretic formulas.

The graph of  $F$  is defined by a ring formula over  $\bar{p}$ , so it is preserved by  $\Phi$  acting on numbers and  $\Phi_{\mathcal{I}'}$  acting on codes:  $F(\Phi x) = \Phi_{\mathcal{I}'}(F(x)) = F(x)$  for every  $x$ . Since  $F$  is injective,  $\Phi$  is the identity, a contradiction. The field  $\mathbf{No}$  is treated in the same way.  $\square$

Source 13 gave this argument as a paragraph after [corollary 74.13](#): “The same construction rules out a bi-interpretation between the pure omnific ring and the actual well-founded set universe whose set-to-number direction is the canonical construction. Indeed the displayed automorphism is set-theoretically definable with a parameter. A proposed interpretation would transport it to a definable automorphism of  $(V, \in)$ . Foundation and induction on rank force every such automorphism to be the identity. Definability of the second comparison then forces the original numerical automorphism to be the identity as well, a contradiction. Finite interpretation parameters cause no problem because they can be included in  $P$ .” The argument is complete; the proof above writes out its three steps. It uses only that  $\mathcal{R}$  has, over every finite tuple, a set-definable automorphism that is not the identity; it does not use [corollary 74.13](#). For comparisons that send each number to codes of itself, [proposition 74.2](#) with [corollary 74.13](#) gives the same conclusion by definability.

This does *not* prove that the pure omnific ring cannot interpret some model of set theory. Nor does it exclude every noncanonical theory-level interpretation one could formulate. The negative

assertion is deliberately about definability and the canonical inverse direction. (Source 13.) The remaining question is [question 79.4](#).

By contrast, the actual birthday-equipped structures are rigid: an automorphism preserves the definable ordinals, fixes this well-ordered class pointwise, and then fixes every birthday and every sign by (69.4). This last argument concerns the actual well-founded universe; it does not assert that all nonstandard models of  $T_{\mathbf{D}}$  have no automorphisms. (Source 13.)

## 75 The obstruction at the class boundary

The preceding construction recovers  $(V, \in)$ . It does not recover an arbitrary class part  $\mathcal{X}$  of a model  $(M, \mathcal{X}) \models \text{GBC}$ . This distinction cannot be removed by saying that **No** itself is a proper class. (Source 13.)

### 75.1 Conservativity is not interpretability

The obstruction below is stronger than failure of bi-interpretability. Its proof uses ordinary reflection and incompleteness, not an assumption that proper classes have an external cardinality. (Source 14.)

**Lemma 75.1** (Reflection produces models of finitely axiomatized targets; source 14). *If a finitely axiomatized first-order theory  $U$  has a uniform finite-dimensional interpretation in ZFC, then ZFC proves that  $U$  has a set-sized model. The statement also holds for finitely many sorts, definable quotient domains, and interpretations with a provably nonempty domain of finitely many admissible set parameters.*

*Proof.* Fix the interpretation formulas. Let  $\tau$  be the conjunction of the translations of a finite axiom basis of  $U$  and the finite interpretation-validity sentences. By hypothesis  $\text{ZFC} \vdash \tau$ . The Lévy reflection theorem [38], applied to this finite collection of formulas and its subformulas, proves the existence of a rank segment  $V_\alpha$  satisfying  $\tau$ .

The interpreted domain is now a definable subset of a finite power of the set  $V_\alpha$ . Separation forms it as a set, and the interpreted relations are sets as well. Its equivalence classes form a set quotient. The congruence and axiom clauses of  $\tau$  show that the resulting structure is a model of the finite theory  $U$ . Multiple sorts are handled one at a time.

With parameters, first take witnesses from the provably nonempty parameter domain and reflect above a rank containing them. Equivalently include the assertion that admissible parameters exist in the finite reflected conjunction. The same set construction follows.  $\square$

**Theorem 75.2** (No interpretation of NBG in ZFC; sources 13, 14). *Assume  $\text{Con}(\text{ZFC})$ . Then the usual NBG theory, equivalently GBC, is not uniformly first-order interpretable in ZFC, even allowing fixed finite tuple domains, quotients, and admissible set parameters. Consequently no theory bi-interpretable with ZFC, including  $T_{\mathbf{No}}$  or  $T_{\mathbf{Oz}}$ , interprets GBC.*

*Proof (source 14).* NBG has a finite first-order axiom basis. Its usual set-sort restriction interprets every axiom of ZFC; the corresponding syntactic proof translation is effective and its correctness is formalizable in ordinary arithmetic. Finite axiomatizability of GBC/NBG is the standard background fact recorded, for example, in [34, Corollary 3.11 and its proof].

If ZFC interpreted NBG, lemma 75.1 would make ZFC prove the existence of a set model of NBG and therefore prove  $\text{Con}(\text{NBG})$ , by soundness for set-sized structures. The standard proof translation from ZFC into the set sort of NBG gives, already formalizably in arithmetic,

$$\text{Con}(\text{NBG}) \longrightarrow \text{Con}(\text{ZFC}).$$

Thus ZFC would prove its own consistency, contradicting Gödel's second incompleteness theorem [39] under the assumed consistency. The last sentence of the theorem follows because interpretations compose.  $\square$

*Second proof (source 13).* Use a finite axiomatization of GBC. If an interpretation existed, a finite fragment  $\Gamma$  of ZFC would prove its admissibility and the translations of those finitely many axioms. Reflection in ZFC gives a set model of  $\Gamma$ . Applying the interpretation produces a set model of GBC, whose set part is a model of ZFC. Thus ZFC would prove  $\text{Con}(\text{ZFC})$ , contrary to the second incompleteness theorem. Interpretations compose, giving the consequence.  $\square$

Source 13 states the theorem as “If ZFC is consistent, it does not interpret GBC”, with the consequence for  $\text{T}_{\text{No}}$ ,  $\text{T}_{\text{Oz}}$ , and attributes the argument to Hamkins and Yao [22, Section 4]. That attribution was checked for this merge against arXiv version 2 (5 December 2022) of [22]: Section 4, “Class theory”, states that GBC and ZFC, if consistent, “are not mutually interpretable”, because GBC is finitely axiomatizable, an interpretation of it in ZFC would use only finitely many axioms of ZFC, ZFC proves the consistency of its finite fragments, and so the interpretation would violate the second incompleteness theorem. It is compatible with the usual conservativity and equiconsistency results. Conservativity concerns sentences in the set language; a class-theoretic interpretation would have to supply the entire class domain. (Source 13.)

One can equivalently say that a set model of NBG has a set-sized set-sort reduct satisfying all ZFC axioms. The syntactic version of the proof is useful because it makes clear that the assertion about *all* axiom instances is not an illicit appeal to a truth predicate for  $V$ . The translation of formal proofs is a finitary arithmetical operation. (Source 14.)

**Corollary 75.3** (The canonical numerical no-go theorem; source 14). *Assuming  $\text{Con}(\text{ZFC})$ , if a numerical theory  $T$  is uniformly interpretable in ZFC, then  $T$  does not interpret NBG. In particular, neither  $\text{SA}_{\text{rt}}$  nor  $\text{OA}_{\text{qb}}$  is bi-interpretable with NBG.*

*Proof.* An interpretation of NBG in  $T$  could be composed with the interpretation of  $T$  in ZFC, contradicting theorem 75.2.  $\square$

This includes any proposed numerical axiom system for which a fixed ZFC definition of the native numerical structure is proved to satisfy each axiom. If one removes that provability hypothesis, the corollary no longer applies automatically. A complete theory or an expansion by arbitrary predicates may carry additional information which ZFC cannot verify. One must analyze such a proposal separately rather than cite this corollary without its hypothesis. (Source 14.)

**A second diagnostic: finite axiomatizability (source 14).**

**Lemma 75.4** (Finite axiomatizability transfers; source 14). *For theories in finite languages, ordinary bi-interpretability preserves finite axiomatizability.*

*Proof.* Suppose  $T$  and  $U$  are bi-interpretable and  $U$  has a finite basis. Inside  $T$ , take the finitely many statements which ensure that its interpretation of  $U$  is valid and satisfies that basis, and that the specified round-trip relation is an isomorphism. A finite subset  $T_0 \subseteq T$  proves all these statements.

If  $M \models T_0$ , then the interpreted model  $\mathcal{I}(M)$  satisfies  $U$ . Interpreting back gives  $\mathcal{J}(\mathcal{I}(M)) \models T$ . The round-trip isomorphism, already secured by  $T_0$ , makes  $M \models T$ . Hence  $T_0$  axiomatizes  $T$ . The reverse implication is symmetric.  $\square$

Assuming consistency, ZFC is not finitely axiomatizable: a hypothetical finite basis would reflect to a set model, making ZFC prove its own consistency. Therefore  $\text{SA}_{\text{rt}}$  and  $\text{OA}_{\text{qb}}$  (and  $\text{T}_{\text{No}}$ ,  $\text{T}_{\text{Oz}}$ ) are not finitely axiomatizable either. NBG is finitely axiomatizable. This gives a second proof that the displayed numerical theories are not bi-interpretable with NBG. Unlike [theorem 75.2](#), this diagnostic alone does not rule out a one-way interpretation.

The reflection reasoning is classical. Its application here should not be misrepresented as a new inconsistency result or as a newly discovered separation of ZFC and NBG. It explains exactly why an otherwise successful surreal encoding cannot be promoted to the requested class theory merely by changing its name. (Source 14.)

**Why conservativity does not contradict the obstruction (source 14).** GBC/NBG is conservative over ZFC for set-language sentences. This means that adding class quantification does not produce new theorems *in that restricted language*. It does not provide a uniform interpretation of every class object by set objects, and does not require the two theories to have matching finite axiomatizability. For a fixed set model, a suitable collection of definable classes may supply a class-theoretic expansion in appropriate circumstances. That external model construction is not the same as a first-order internal interpretation using a uniform truth-evaluation formula. In particular, Global Choice and the exact allowed class collection must not be silently inserted. The reflection proof pinpoints the difference without appealing to an informal “there are more classes” slogan. The foundations report states conservativity in the same way (`found:rem:conservativity`).

## 75.2 A stronger obstruction when the individuals are original sets

The reflection argument concerns a target theory already bi-interpretable with ZFC. The next result allows arbitrary numerical relations, including ones depending on the original class part. Its only structural restriction is that the numerical individuals remain coded by finitely many original sets.

We use two established model-theoretic facts. A countable GBC-realization  $\mathcal{X}$  of a countable  $M$  has a proper countable extension with the same sets, obtained by adding a class Cohen generic. Also, the union of an increasing chain of GBC-realizations of the same  $M$  is a GBC-realization. See Williams, Theorem 1.61(5), the proof of Theorem 1.65, and the class-forcing discussion preceding them [34]; the related realization framework is developed in [35]. (Both are imported, [section 79.4](#).)

**Lemma 75.5** (Different external sizes of the two sorts). *Assuming  $\text{Con}(\text{GBC})$ , there is a model  $(M, \mathcal{X}) \models \text{GBC}$  with  $M$  countable and  $|\mathcal{X}| = \aleph_1$ , as measured externally.*

*Proof.* Start with a countable model  $(M, \mathcal{X}_0)$ . By recursion on the external ordinals  $\alpha < \omega_1$ , choose countable GBC-realizations of the same  $M$  with

$$\mathcal{X}_\alpha \subsetneq \mathcal{X}_{\alpha+1}.$$

At a successor use a proper countable class extension. At a nonzero limit put  $\mathcal{X}_\lambda = \bigcup_{\alpha < \lambda} \mathcal{X}_\alpha$ . This is countable because  $\lambda$  is countable, and is a realization by the union fact.

Now put  $\mathcal{X} = \bigcup_{\alpha < \omega_1} \mathcal{X}_\alpha$ . This union again satisfies GBC. For clarity, elementary comprehension at the union uses only finitely many class parameters; all occur in some earlier stage, where the required class already exists. A class function in the union likewise occurs in one stage, which verifies class replacement for it. A global choice class from the initial stage remains available. These observations explain why the union argument applies here.

There are at most  $\aleph_1$  classes in this union. The strict successor steps supply  $\aleph_1$  distinct classes, so equality holds.  $\square$

**Theorem 75.6** (No uniform class recovery from set-coded individuals). *Assume Con(GBC). There is no uniform first-order bi-interpretation between GBC and a one-sorted theory  $U$  whose forward interpretation, in every  $(M, \mathcal{X}) \models \text{GBC}$ , has domain a definable quotient of a subclass of  $M^n$ , for some fixed finite  $n$ .*

*In particular, no such bi-interpretation has forward domain the canonical  $\mathbf{No}^M$  or  $\mathbf{Oz}^M$ , or fixed finite tuples or quotients of these, even if its relations depend on  $\mathcal{X}$ .*

*Proof.* Apply the proposed interpretations to the model of lemma 75.5. The forward domain is countable, because it is a quotient of a subset of  $M^n$ . The interpreted  $U$ -structure is therefore countable.

Every sort of any ordinary first-order interpretation in a countable structure is again a quotient of a subset of a finite Cartesian power, hence countable. Thus the inverse interpretation produces a GBC-model with countable class sort. A bi-interpretation would require that model to be isomorphic to the original  $(M, \mathcal{X})$ . Its class sort cannot be isomorphic to the uncountable  $\mathcal{X}$ , a contradiction.  $\square$

The conclusion tolerates very liberal definitions: the forward domain and relations may use class parameters, bound class quantifiers, and arbitrary finite tuple coding. None of those devices enlarges a countable domain. The proof requires no inaccessible cardinal or transitive model. Its consistency assumption is the ordinary consistency assumption for the class theory.

The theorem is uniform over all models, as a theory bi-interpretation must be. It does not claim that two particular countable structures cannot be externally relabeled by a bijection. Nor does it prohibit a one-sorted presentation in which some individuals represent genuine class objects and therefore are not canonical set-sized surreal numbers of the original universe. (Source 13; source 14 has no counterpart of this theorem.)

### 75.3 What diagonalization does and does not show

There is a useful direct diagonal argument, but its comprehension hypothesis must be stated. The preceding argument compares theories; this one works inside a given class universe and clarifies the intended semantics of “only numerical objects” (source 14).

**Theorem 75.7** (No uniform set-code enumeration of classes; sources 13, 14). *Work in GB. Let  $Q$  be a class of sets, and let  $E(x, c)$  be an elementary formula, possibly with fixed class parameters. There cannot be a coding scheme satisfying*

$$\forall A \exists c \in Q \forall x [x \in A \leftrightarrow E(x, c)], \quad (75.1)$$

*where  $A$  ranges over all classes of the model and  $x$  ranges over its sets. Codes need not be unique.*

*Proof.* Elementary comprehension forms the diagonal class

$$\Delta = \{c \in Q : \neg E(c, c)\}.$$

By (75.1), some  $d \in Q$  codes  $\Delta$ . Substituting  $x = d$  gives

$$d \in \Delta \iff E(d, d), \quad d \in \Delta \iff \neg E(d, d),$$

a contradiction. Injectivity or unique codes were never used.  $\square$

Source 14 stated the theorem in GBc; its proof, and source 13's, use only elementary comprehension, so it is printed over GB. Source 13's form ("No class-valued universal decoder"): in GB, let  $Q$  be a class of sets and let  $E \subseteq Q \times V$  be an available class relation; its sections  $E_c = \{x : E(c, x)\}$  cannot include every class. Its proof forms  $\Delta = \{c \in Q : \neg E(c, c)\}$ ; if  $\Delta = E_d$  for some  $d \in Q$ , then  $d \in \Delta \leftrightarrow d \notin \Delta$ . The two forms are the same theorem, an available class relation being one given by an elementary formula with a class parameter.

**Corollary 75.8** (Ordinary surreal and omnific codes are insufficient; source 14). *No elementary evaluation scheme assigns a surreal number, an omnific integer, a fixed finite tuple of them, or an arbitrary set-sized package of them to every class of sets in the same GB universe.*

*Proof.* Each proposed code is a set, so apply theorem 75.7 to the class of all valid packages. Coding equality by an equivalence relation does not help, since the proof permits multiple codes.  $\square$

For subclasses of **No** rather than of  $V$ , a version with codes in **No** has the identical proof: the diagonal lies within **No**. Fixed finite numerical tuples can also be packed into a single surreal, as shown in lemma 76.1. The essential issue is the presence of a *single elementary evaluator* whose sections exhaust the available classes. The theorem does not assert that an externally countable model cannot have its classes enumerated externally; it asserts that the required internally available evaluation is impossible. (Source 14.)

This immediately excludes a universal decoder whose evaluation relation is an elementary formula with class parameters. It does not by itself exclude every arbitrary formula with bound class quantifiers: predicative comprehension need not turn that formula into an available class relation. Suppressing this distinction would overstate the argument. The model-cardinality theorem above avoids this restriction. (Source 13.)

In KM, full comprehension removes that loophole.

**Proposition 75.9** (Impredicative coding obstruction). *In KM, no formula  $\Phi(X, c)$ , even with bound class quantifiers and parameters, can assign every class  $X$  some set code  $c$ , with each code assigned to at most one class.*

*Proof.* Full comprehension forms

$$E(c, x) \iff \exists X (\Phi(X, c) \wedge x \in X).$$

It then forms  $\Delta = \{c : \neg E(c, c)\}$ . Choose a code  $d$  for  $\Delta$ . Uniqueness of the class assigned to  $d$  gives  $E(d, x) \leftrightarrow x \in \Delta$ , and hence the diagonal contradiction.  $\square$

Source 14 makes the same point: the same diagonal works in KM for evaluators whose definitions involve class quantifiers, because KM has the corresponding comprehension; in GB, the elementary hypothesis is indispensable: one must not form a class by a formula for which the base theory lacks comprehension.

**Definable classes do not evade the diagonal.** Coding formulas and their finite parameters is not a universal solution either. Such codes naturally describe definable classes, whereas general GBC-models have additional classes. A single evaluator definable in the original set language and correct for all formulas of that language would yield a truth definition forbidden by Tarski. An additional truth class can exist in stronger class theories, but it does not enumerate all classes. Particular formulas can be translated, as in [corollary 74.1](#); a uniform truth evaluator is a different object. (Source 13.) Source 14: for each fixed formula  $\varphi(x, p)$ , the class it defines can be specified by that formula and its parameter. This is a nonuniform metalinguistic observation. To turn all such descriptions into the objects of a numerical first-order theory, one needs a uniform predicate which evaluates arbitrary formula codes. For the whole set universe, that is exactly where Tarski's undefinability phenomenon enters. A global truth predicate may be added as extra class structure in a stronger theory, but then it was not obtained merely by decoding an ordinary number in the original elementary language. Restricting the complexity of formulas can give partial satisfaction relations and fragmentary coding; it does not give the forbidden unrestricted ([75.1](#)).

## 75.4 The precise conclusion for the original question

For ZFC, numbers with birthdays suffice. For full NBG/GBC, the usual numbers of the same set universe do not suffice as the entire individual domain of a uniform bi-interpretation. Allowing richer numerical operations or a predicate for Global Choice does not bypass [theorem 75.6](#).

One can retain full class theory by allowing classes of numbers, as the next section shows. Alternatively one can change what counts as a number, for example by admitting certain class-length sign objects. Such objects are not ordinary Conway surreals: a surreal has an ordinal, set-sized sign domain. Any proposed generalized-number programme must specify this changed ontology and establish its own arithmetic. (Source 13; source 14's [proposition 77.3](#) and [question 79.18](#) make the same point.)

## 76 A faithful theory of numbers and their classes

There is a positive class-theoretic analogue, provided that classes are retained as objects of a second sort. The number sort can consist entirely of surreals or entirely of omnific integers. (Source 13.) Source 14 puts it as: the correct class-level analogue of replacing sets by numerical codes is to replace classes of sets by *classes of numerical codes*; the important point is that this can be done reversibly, with both round trips and without a hidden canonical representative choice.

### 76.1 From classes of sets to classes of numbers

Fix  $\mathbf{D} = \mathbf{No}$  or  $\mathbf{Oz}$ , and let  $(M, \mathcal{X}) \models \text{GBC}$ . Put

$$N = \mathbf{D}^M, \quad \mathcal{Y} = \{A \in \mathcal{X} : A \subseteq N\}.$$

Equip  $N$  with its canonical numerical structure and use ordinary class membership between  $N$  and  $\mathcal{Y}$ . The expression defining  $\mathcal{Y}$  specifies an interpreted class sort. It does not assert that the collection of all these classes is itself a class object.

Use [lemma 70.7](#) to replace each valid triple by a single number. Let  $\text{UC} \subseteq N$  be the resulting definable class of unary set codes, with equivalence relation  $\text{Eq}$  and represented membership inherited from  $\mathcal{I}_{\mathbf{D}}$ . Write

$$\text{col} : \text{UC}/\text{Eq} \longrightarrow M$$

for the collapse isomorphism.

A numerical class  $B \in \mathcal{Y}$  represents a class of sets precisely when it is contained in  $\mathbf{UC}$  and is Eq-saturated:

$$\begin{aligned} \text{Sat}(B) \iff & \forall n (n \in_N B \rightarrow n \in \mathbf{UC}) \\ & \wedge \forall c, d \in \mathbf{UC} (\text{Eq}(c, d) \rightarrow (c \in_N B \leftrightarrow d \in_N B)). \end{aligned}$$

Both containment and saturation are part of this first-order condition. The recovered set sort is  $\mathbf{UC}/\text{Eq}$ ; the recovered class sort consists of the  $B$  satisfying  $\text{Sat}$ . Set-class membership is simply  $c \in B$ , well-defined on the quotient.

**Lemma 76.1** (Definable numerical packing; source 14). *In  $\mathcal{B}$  there is a parameter-free definable injection  $P : \mathbf{No}^2 \rightarrow \mathbf{No}$  with a definable image and definable inverse coordinates on its image. Hence the same is true for every fixed finite power of  $\mathbf{No}$ .*

*Proof.* Here is an explicit set-theoretic sign construction. Replace each sign of the first input by a two-sign block: plus becomes  $++$ , minus becomes  $+-$ . Append a delimiter block  $--$ , and then append the similarly encoded second input, with no second delimiter. Blocks are indexed in ordinal order. For inputs of lengths  $\alpha, \beta$ , the result has ordinary ordinal length

$$2 \cdot_{\text{ord}} \alpha +_{\text{ord}} 2 +_{\text{ord}} 2 \cdot_{\text{ord}} \beta.$$

Every data block starts with plus, and precisely the delimiter block starts with minus. Decomposing an ordinal into consecutive blocks of length two is unique, including at limit stages. The delimiter therefore recovers the first input length and all its signs, and the remaining blocks recover the second input. The empty-input cases are included.

This is a fixed set-definable construction, as are the validity test for its image and its two decoders. Translate those formulas through  $\mathcal{I}$  and use  $\rho$  to identify interpreted sign functions with original surreal numbers. [Proposition 72.4](#) makes the resulting functions  $\mathcal{L}_s$ -definable. Iterating the construction packs a fixed finite tuple. The ordinal operations in this proof are explicitly *ordinary* ordinal operations inside the interpreted set theory; they are not mistaken for the native ring operations.  $\square$

Source 14 fixes a packing of triples and lets its class  $D$ , here  $\mathbf{UC}$  (for  $\mathbf{D} = \mathbf{No}$ ), be the packed valid set codes, with  $\text{Eq}$  and  $\text{Mem}$  transported to them; its  $\text{Sat}(B)$  is the displayed condition:  $B \subseteq \mathbf{UC}$  and  $\forall d, e \in \mathbf{UC} [\text{Eq}(d, e) \rightarrow (d \in B \leftrightarrow e \in B)]$ . These are *saturated* classes of codes, meaning that they contain all representatives of every set they represent. The two packings differ in mechanism: source 13's native codec ([lemma 70.7](#)) is a block word with marker rows, source 14's a delimiter construction translated through the interpretation; either works, since only definability of the injection and of its inverse is used.

**Lemma 76.2** (Classes are recovered exactly; sources 13, 14). *The following maps are definable and inverse:*

$$\begin{aligned} X &\longmapsto B_X = \{c \in \mathbf{UC} : \text{col}(c) \in X\}, \\ B &\longmapsto X_B = \{x \in M : \exists c \in \mathbf{UC} (\text{col}(c) = x \wedge c \in B)\}. \end{aligned}$$

*They establish an isomorphism between the original class sort and the saturated numerical classes, compatible with the collapse map on sets.*

*Proof.* The graph of collapse is an elementary set-theoretic definition. Both displayed classes therefore exist by elementary comprehension with the given class parameter. The class  $B_X$  is saturated because equivalent codes have the same collapse.

Surjectivity of collapse gives  $X_{B_X} = X$ . For saturated  $B$ , if  $c \in B_{X_B}$ , some  $d \in B$  has the same collapse as  $c$ ; then  $\text{Eq}(c, d)$ , and saturation gives  $c \in B$ . The converse is immediate, so  $B_{X_B} = B$ . Membership preservation follows directly from the definitions.  $\square$

No representative is selected from each equivalence class. This is why the construction requires no additional global choice of graph codes. The source's Global Choice class is transported as one of its ordinary class objects. (Source 13.)

Source 14 states the lemma as its *exact class correspondence*, over GBc: for  $(M, \mathcal{X}) \models \text{GBc}$  with  $N = \mathcal{B}^M$  its internally constructed numerical structure and  $\mathcal{Y} = \{B \in \mathcal{X} : B \subseteq \mathbf{No}^M\}$  the available subclasses of the numerical domain, it puts

$$A^\# = \{d \in \text{UC}^M : \exists a (\text{col}(d) = a \wedge a \in A)\} \quad (76.1)$$

for  $A \in \mathcal{X}$ , a member of  $\mathcal{Y}$  by elementary comprehension and saturated, and, for saturated  $B \in \mathcal{Y}$ ,

$$B^\flat = \{a \in M : \exists d \in B \text{ col}(d) = a\}. \quad (76.2)$$

These definitions are relative to  $M$ ; for nonstandard models they use its internally defined collapse relation. So  $A^\# = B_A$  and  $B^\flat = X_B$ .

**Theorem 76.3** (Exact class correspondence; source 14). *The maps  $A \mapsto A^\#$  and  $B \mapsto B^\flat$  are mutually inverse between  $\mathcal{X}$  and the saturated members of  $\mathcal{Y}$ . They preserve membership in the recovered set sort. In particular,*

$$(A^\#)^\flat = A, \quad (B^\flat)^\# = B.$$

*Second proof (source 14).* Every set  $a$  has at least one valid code, and all of its valid codes are equivalent. Therefore a set belongs to  $(A^\#)^\flat$  exactly when it belongs to  $A$ .

If a valid code  $d$  belongs to  $(B^\flat)^\#$ , then some  $e \in B$  has the same root collapse. Hence  $\text{Eq}(d, e)$ , and saturation gives  $d \in B$ . The converse is immediate. The membership assertion is the equivalence

$$a \in A \iff d \in A^\# \quad \text{whenever } \text{col}(d) = a.$$

All classes used are obtained by elementary formulas with existing class parameters. No class choice or global selection of a single code has been used.  $\square$

## 76.2 An explicit two-sorted axiom system

The language has the numerical symbols  $\mathcal{L}_b$ , a class sort, and a relation  $n \in_N A$  between numbers and classes. It has no primitive set-membership relation between numbers.

**Definition 76.4.** The theory  $\text{T}_\mathbf{D}^{\text{cl}}$  consists of:

1. the axioms of  $\text{T}_\mathbf{D}$  on the number sort;
2. class extensionality:

$$\forall A \forall B [\forall n (n \in_N A \leftrightarrow n \in_N B) \rightarrow A = B];$$

3. elementary numerical comprehension, namely

$$\exists A \forall n (n \in_N A \leftrightarrow \varphi(n, \bar{p}, \bar{B}))$$

for each formula  $\varphi$  with number quantifiers, number parameters, and class parameters, but no bound class quantifiers;

4. every axiom of GBC, translated by taking UC/Eq as the set sort and the saturated numerical classes as the class sort.

In the final group, equality of represented sets is Eq; membership of represented sets is the graph-decoding relation; and membership in a represented class is  $\in_N$ . Class quantifiers are relativized to Sat. In particular, this group includes the translations of class replacement and Global Choice. Elementary comprehension alone must not be substituted for those requirements.

Every clause is a first-order sentence or a recursive first-order scheme in the two-sorted language. The unary codec, its image, and the comparison formulas are native definitions, whose required properties follow from  $T_D$ .

### 76.3 Recovering the entire numerical class sort

One must still check a second class round trip. The saturated subclasses of UC recover  $\mathcal{X}$ , but the numerical structure originally had *all* available subclasses  $B \subseteq N$ , not only saturated ones. (Source 14.)

**Theorem 76.5** (The class lift; sources 13, 14). *For  $D = \mathbf{No}$  and  $D = \mathbf{Oz}$ ,*

$$T_D^{\text{cl}} \simeq_{\text{bi}} \text{GBC}.$$

*The forward interpretation uses the canonical numbers of the set universe together with its available classes of those numbers.*

*Proof.* From a GBC-model, construct  $(N, \mathcal{Y})$  as above. The number sort satisfies  $T_D$ . Class extensionality and numerical comprehension are inherited from the original model by translating the numerical symbols to their elementary set definitions. By lemma 76.2, the saturated-code interpretation is isomorphic to the whole original GBC-model, so every translated class axiom holds. This proves one interpretation and its source-side comparison.

Conversely, from a model  $(N, \mathcal{Y})$  of  $T_D^{\text{cl}}$ , take its interpreted GBC-model. The numerical comparison  $\text{Cmp}_D$  gives the required isomorphism between  $N$  and that model's canonical number structure. It remains to recover *every* original numerical class.

For  $A \in \mathcal{Y}$ , use elementary numerical comprehension to form

$$B_A = \{c \in \text{UC} : \exists n (n \in_N A \wedge \text{Cmp}_D(n, c))\}.$$

Here  $\text{Cmp}_D$  has been rewritten using the unary codec. The class  $B_A$  is saturated and contains only codes for reconstructed numbers. It is therefore a class of numbers in the reconstructed class theory. Conversely, from any such  $B$ , form

$$A_B = \{n \in N : \exists c \in \text{UC} (\text{Cmp}_D(n, c) \wedge c \in_N B)\}.$$

The totality, surjectivity, injectivity, and equivalence-invariance of  $\text{Cmp}_D$  show that these maps are inverse. Class extensionality turns equality of memberships into equality of classes. Together with  $\text{Cmp}_D$ , they preserve numerical operations and number–class membership. This supplies the second full definable isomorphism.  $\square$

**Source 14's second round trip, and its form of the theorem.** For an arbitrary  $B \subseteq N$ , source 14 uses the numerical isomorphism  $\rho$  to form its image among the surreals of  $\mathcal{I}(N)$ ; its corresponding class of interpreted sets is represented by

$$\widehat{B} = \{d \in \text{UC} : \exists x \in B \rho(x, d)\}, \quad (76.3)$$

with  $\rho(x, d)$  read after unpacking  $d$ . This is a saturated class in  $\mathcal{Y}$ , and in the recovered set universe it is a class of surreal sign functions. Pulling it back along  $\rho$  returns exactly  $B$ :

$$x \in B \iff \exists d \in \widehat{B} \rho(x, d).$$

Existence follows from the totality of  $\rho$ ; uniqueness of the original  $x$  follows from its injectivity. Thus arbitrary numerical classes, not just sets and saturated code classes, are recovered. ( $\widehat{B}$  is source 13's  $B_A$  with  $A = B$ .)

**Theorem 76.6** (Class-level bi-interpretation; source 14). *There are fixed first-order two-sorted interpretations giving*

$$(M, \mathcal{X}) \longleftrightarrow (\mathcal{B}^M, \mathcal{Y})$$

*with definable inverse round trips for both the individual and class sorts. The construction is valid over GBc; it carries Global Choice, full KM comprehension, and additional specified class principles to their exact translated counterparts.*

*Second proof (source 14).* Interpret numerical individuals in  $(M, \mathcal{X})$  by  $\mathcal{J}$ , and numerical classes by those class objects contained in the numerical domain. In the opposite direction use UC/Eq for sets, Mem for set membership, and saturated numerical classes for classes of sets. The first round trip is col on individuals and (76.1)–(76.2) on classes.

The second is  $\rho$  on individuals and (76.3) on arbitrary numerical classes. The preceding proofs establish totality, equality, surjectivity, and preservation of the membership relations. Class extensionality turns agreement of members into equality of class objects. Every formula used has only individual quantifiers apart from its free class parameters, so GBc comprehension suffices for all required class constructions.

For an additional axiom or scheme, translate it through these fixed interpretations. The two definable isomorphisms identify its round-trip translation with the original. This proves exact preservation rather than an informal analogy.  $\square$

A first-order axiomatization of the numerical class theory is obtained by the same recipe as lemma 73.2, now with two sorts: include interpretation validity, the translated class-theory axioms, and the finite individual-and-class round-trip coherence conditions. Call the GBC version  $\text{SAC}_{\text{GBC}}$ . It is bi-interpretable with NBG under our convention, and it has a finite axiom basis because NBG does (lemma 75.4). There is no conflict with section 75.1: it is not a one-sorted theory uniformly interpreted in ZFC; it includes a class sort. (Source 14.) [Merge:  $\text{SAC}_{\text{GBC}}$  and  $\text{T}_{\text{No}}^{\text{cl}}$  are bi-interpretable, both being bi-interpretable with GBC;  $\text{T}_{\text{No}}^{\text{cl}}$  as displayed (with the full scheme of  $\text{T}_{\text{No}}$  on the number sort) is equivalent to a finite theory by lemma 75.4.]

The omnific version is obtained in the same way. Finite tuples of omnific integers are themselves definably packable into one omnific integer after the established bi-interpretation: first pack in the surreal sort, then apply a fixed definable injection  $\text{No} \rightarrow \text{Oz}$ . One example is

$$x \mapsto \omega^{1+x/\sqrt{1+x^2}},$$

whose exponent lies strictly between 0 and 2. The image consists of positive-exponent monomials, and injectivity follows from strict monotonicity. The omega-map and inverse on its image are available in the recovered set structure; no assertion of pure-ring definability is being made. Alternatively one may simply retain finite tuples in the interpretation, which avoids this optional packing step. (Source 14; source 13's native codec lemma 70.7 works in **Oz** directly.)

**The class collection is genuine additional information (source 14).** The transformation does not replace  $\mathcal{X}$  by all external subclasses. It transports the particular classes the model actually has. Distinct class realizations of the same  $M$  cannot collapse to the same  $\mathcal{Y}$ : theorem 76.3 would recover the same  $\mathcal{X}$  from it. More precisely, for two GBc realizations over one fixed  $M$ ,

$$\mathcal{X}_1 \subseteq \mathcal{X}_2 \iff \mathcal{Y}_1 \subseteq \mathcal{Y}_2. \quad (76.4)$$

The forward direction is restriction. For the reverse direction, apply  $A \mapsto A^\#$  in the first realization and then  $B \mapsto B^b$  in the second. Hence the partially ordered collection of class realizations is preserved, not erased. This gives an exact interface for transferring questions about class realizations to numerical foundations; compare the general model-theoretic study in [34]. (Merge, spelling out the reverse direction:  $A^\#$  is defined from  $A$  and  $\text{col}$  by an elementary formula, and  $\text{col}$  is the same set-theoretic relation in both realizations, which have the same sets; so  $A^\# \in \mathcal{Y}_1 \subseteq \mathcal{Y}_2$  and  $(A^\#)^b = A$  computed in the second realization puts  $A$  in  $\mathcal{X}_2$ .)

## 76.4 Variants and limitations

The same construction works for GB with set Choice but without Global Choice, that is GBc, deleting the Global Choice requirement from the translated source theory. The present ordinal-graph codec still uses set Choice; GB without it needs a different analysis. (Source 13.)

For KM with a ZFC set part, add impredicative numerical class comprehension and translate the corresponding KM axioms. The same maps prove the analogous bi-interpretation. Class quantifiers in the numerical comprehension scheme are translated to class quantifiers in the source, where full comprehension is available. Global Choice is included exactly when it is included in the chosen source version. (Source 13.)

Further class principles, such as elementary transfinite recursion, can be transported in the same manner by adding their explicit translations. Determining short native formulations of those principles is a separate task. (Source 13.)

Source 13 sketched these three variants (“the same maps prove”); source 14's theorem 76.6 states them with the argument of its last paragraph. The merge makes them a corollary with a proof.

**Corollary 76.7** (Class variants; added in this merge). *Let  $\mathbf{D} = \mathbf{No}$  or  $\mathbf{Oz}$ .*

- (i) *Over GBc: the theory  $T_{\mathbf{D}}^{\text{cl},c}$ , defined as in definition 76.4 with GBc in place of GBC in item 4, is bi-interpretable with GBc.*
- (ii) *For every set  $\Sigma$  of sentences or schemes in the language of class theory, in particular for Global Choice and for elementary transfinite recursion ETR, the extension of the numerical class theory by the translations  $\Sigma^{\mathcal{I}}$  is bi-interpretable with the corresponding extension of GBc or GBC.*
- (iii) *Let  $T_{\mathbf{D}}^{\text{KM}}$  be  $T_{\mathbf{D}}^{\text{cl}}$  with item 3 strengthened to impredicative numerical comprehension (formulas with bound class quantifiers). Then  $T_{\mathbf{D}}^{\text{KM}} \simeq_{\text{bi}} \text{KM}$  (with Global Choice), and likewise without Global Choice over the GBc variant.*

*Proof.* (i) The proof of [theorem 76.5](#) (equivalently of [theorem 76.6](#)) uses, on the source side, elementary comprehension, class extensionality and the surjectivity of collapse, which needs set Choice ([proposition 71.5](#)); Global Choice enters only as one of the translated axioms. (ii) is [lemma 73.8](#) applied to (i) or to [theorem 76.5](#).

(iii) By (ii), KM is bi-interpretable with the numerical class theory plus the translations of the KM comprehension instances; it remains to see that this extension and  $T_D^{KM}$  have the same theorems. In the forward model, a native impredicative instance  $\{n \in N : \varphi(n, \bar{p}, \bar{B})\}$ , with class quantifiers over subclasses of  $N$ , is translated to a set-theoretic formula whose class quantifiers range over classes contained in  $N$ ; KM comprehension gives the class, so the forward interpretation satisfies impredicative numerical comprehension and KM interprets  $T_D^{KM}$ . Conversely, a translated KM instance asks for a saturated  $B$  with  $\forall c \in UC (c \in_N B \leftrightarrow \psi^I(c, \dots))$ , where the class quantifiers of  $\psi^I$  range over saturated classes. Impredicative numerical comprehension gives  $B = \{n : n \in UC \wedge \psi^I(n, \dots)\}$ , and  $B$  is saturated because  $\psi^I$  respects Eq in its code variable (congruence of the interpretation: its atomic formulas respect Eq, and its class quantifiers range over saturated classes). The definable isomorphisms are those of [theorem 76.5](#), unchanged.  $\square$

What stays open: GB without set Choice ([question 79.13](#)), and short *native* formulations of class replacement, Global Choice and ETR, as opposed to their translations ([questions 79.14](#) and [79.15](#)).

As usual, a two-sorted theory can be rewritten in a one-sorted language using disjoint tags and sort predicates. That operation does not make the class individuals ordinary surreal numbers. It changes the logical notation while retaining both kinds of object. The positive theorem is therefore consistent with the obstruction in [theorem 75.6](#). (Source 13.)

## 77 Set size, cuts, and alternative readings (source 14)

**Theorem 77.1** (Birthday boundedness detects set size; source 14). *In GBc, for an available class  $A \subseteq \mathbf{No}$ , the following are equivalent:*

- (i)  $A$  is a set;
- (ii) there is an ordinal  $\alpha$  such that  $b(x) < \alpha$  for every  $x \in A$ .

*The same assertion holds for classes of omnific integers with their native birthday  $h(x, 1)$ .*

*Proof.* If  $A$  is a set, class replacement applied to the birthday function makes  $\{b(x) : x \in A\}$  a set of ordinals, which has a strict ordinal bound. Conversely,

$$\mathbf{No}_{<\alpha} = \bigcup_{\beta < \alpha} \{-, +\}^\beta$$

is a set. An available class contained in this set is a set by class separation. Restricting to the definable omnific subclass gives the second assertion.  $\square$

The proof uses no Choice, so the equivalence holds in GB as well; source 14 stated it in GBc. Consequently, if  $F$  is an available numerical class function and  $A$  is birthday-bounded, class replacement implies that  $F[A]$  is birthday-bounded. This is a direct native numerical form of a replacement consequence. We are not claiming that this sentence by itself, over an unspecified weak background, axiomatizes all of class replacement. That stronger equivalence would have to account for coding choices and the existence of the relevant rank and sign levels. (Source 14.) This report's Part VI proves the sharp weak-base form for definable families: over Zermelo set theory without

Replacement, bounding the birthdays of definable surreal families is equivalent to ordinal bounding, and with a hierarchy axiom to Replacement (theorems 28.2 and 29.1).

**Corollary 77.2** (The correct scope of the surreal cut principle; source 14). *Available classes  $L, R \subseteq \mathbf{No}$  with birthday-bounded union and  $L < R$  have a unique least-birthday separator. The assertion cannot be extended to all separated proper-class pairs.*

*Proof.* The boundedness hypothesis makes both sides sets by theorem 77.1, so the ordinary surreal set-cut construction applies. For the failure without boundedness, take  $L = \text{Ord}$  and  $R = \emptyset$ . Every surreal lies below some ordinal, so there is no number larger than all members of  $L$ .  $\square$

This illustrates why “every definable cut is filled” is an unsafe axiom unless the allowed cuts are explicitly bounded. Similarly, a proper-class family of compatible initial sign segments need not be a single surreal number. (Source 14; the failure is `found:prop:allcuts` of the foundations report, and Part VI’s theorem 27.1 is the Replacement-free set-cut theorem.)

**Proposition 77.3** (Proper-class branches are not additional surreals; source 14). *Let  $A \subseteq \text{Ord}$  be an available class. For every ordinal  $\alpha$ , its characteristic signs below  $\alpha$  determine a surreal  $s_\alpha$  of birthday  $\alpha$ , and  $s_\beta$  is a prefix of  $s_\alpha$  when  $\beta < \alpha$ . No ordinary surreal has all these  $s_\alpha$  as prefixes.*

*Proof.* Class separation gives  $A \cap \alpha$  as a set, so the characteristic sign function exists. Compatibility is immediate. A surreal containing all the prefixes would have birthday at least every ordinal, which is impossible for a set-sized sign function.  $\square$

Allowing such proper-class branches as objects would therefore change the ontology. It may be an interesting new theory of generalized sign objects, but it is not a first-order theory whose only individuals are the ordinary Conway surreal numbers. (Source 14.)

## 77.1 A larger universe can code a smaller class model

There is a genuine positive construction which should not be confused with the same-universe proposal. Suppose the ambient metatheory has a strongly inaccessible cardinal  $\kappa$ . Then

$$(V_\kappa, \mathcal{P}(V_\kappa))$$

is a set-sized model of the usual class axioms, including KM comprehension and Global Choice. All of its class objects are sets in the larger ambient universe. Since  $|V_\kappa| = \kappa$ , an external bijection  $e : \kappa \rightarrow V_\kappa$  turns each class of that model into a sign sequence of length  $\kappa$ , hence into an *ambient* surreal number.

The verification of the model assertion uses inaccessibility in the usual concrete way. Sets and their power sets stay below  $\kappa$ ; a class function on a member of  $V_\kappa$  has range of cardinality less than  $\kappa$ , whose ranks are bounded by regularity. Every external subclass needed for comprehension belongs to  $\mathcal{P}(V_\kappa)$ . This is a legitimate example, not a consistency proof from ZFC alone. (Merge: Global Choice holds because a well-order of  $V_\kappa$  of type  $\kappa$ , induced by  $e$ , belongs to  $\mathcal{P}(V_\kappa \times V_\kappa)$  and is coded as a class.)

The codes of length  $\kappa$  do *not* belong to  $\mathbf{No}^{V_\kappa}$ . Indeed the latter has cardinality  $\kappa$ , whereas the full class collection of the displayed model has cardinality  $2^\kappa$ . No fixed finite power or quotient of  $\mathbf{No}^{V_\kappa}$  can recover all those classes by an isomorphism. Using the larger numerical universe solves a different, relative coding problem. It does not produce a bi-interpretation of the ambient set universe with its full class universe. (Source 14.)

## 77.2 External relabelling and added predicates

If a model and a chosen set of numerical labels have the same external cardinality, a bijection can transport its relations to those labels. This works for essentially any carrier, not specifically for surreal numbers. It need not preserve native addition, multiplication, birthday, or internal definability. In the proper-class setting the existence of a suitable higher-level bijection is itself an additional assumption.

A proposed relation  $R$  on numbers can also be postulated to have the entire strength of set or class membership. Unless its numerical meaning and the inverse interpretation are given, this is a relabelling of the desired foundation. The constructions in this article avoid that by using a concrete birthday operation, explicit graph formulas, and proved round trips. The NBG obstruction explains why that standard cannot simply be extended to all class objects in the same way. (Source 14.)

## 77.3 Second-order numerical logic

One may phrase [theorem 76.6](#) as a theory with numerical individual variables and second-order variables for classes of numbers. Under Henkin semantics this is a notational variant of the two-sorted theory already constructed. In that sense it is a perfectly sound “surreal-based NBG.” It is not a first-order theory with only numerical objects, since its second-order variables retain the class collection.

Replacing Henkin semantics by full second-order semantics is another substantial change. It requires all external subclasses of the numerical domain and no longer represents arbitrary NBG class realizations. Accordingly full semantics must not be used to claim the first-order bi-interpretability established here. (Source 14.)

# 78 Formalization programmes and finite checks

## 78.1 What is, and is not, machine verified

The repository’s formalization ledger explicitly distinguishes written statements, mathematical review, and completed Lean declarations. That distinction is preserved here [\[40\]](#). The module `Algebra/SurrealNumbers/Surreal/Foundations/OmnificSupportBounds.lean` contains the following declarations in the namespace `Surreal.Foundations.SignSequence`:

- `omnific_monomial_clearing`;
- `surreal_eq_omnific_fraction`;
- `omnificSurrealIsFractionRing`.

The sign-sequence infrastructure also supplies birthdays, ordinal embeddings, signs, and prefixes. At the inspected commit, the birthday-cutoff report stated that none of its `hset`: labels had a Lean implementation mapping. (Source 13; both statements are true at the placement, and the three declarations are mapped to the fraction-field instance in `Algebra/SurrealNumbers/docs/FORMALIZATION.md` as proved. That covers [proposition 74.3](#) only; nothing else in this part is formalized.)

This article does not report a fresh build of those targets. More importantly, the explicit native interpretation and its inverse-comparison theorem have not been supplied as new Lean proofs in this research task. A compiled PDF is not a mathematical proof-assistant verification.

A practical mechanization should proceed in the following dependency order. (Source 13.)

1. **Prove the native ordinal and sign definitions.** Formalize `Ord`, `SameB`, `Pref`, `Plus` and their equivalence to the canonical semantic notions, over both **No** and **Oz**. The omnific initiality witness in the prefix proof is a key obligation.
2. **Formalize the block function.** Keep natural and ordinary ordinal operations as distinct typed operations. Prove disjointness and limit endpoints before constructing the sign word. Include the negative filling of gaps and all unused pairing indices.
3. **Formalize bounded relation codes.** Prove the pairing lemma, graph-of-function clauses, and the unary finite-tuple injection. The actual vertex ordinal must remain distinct from the larger matrix base.
4. **Connect graph validity to collapse.** Use the coverage theorem for all bounded subsets to identify the first-order well-foundedness test with set-theoretic well-foundedness. Prove root accessibility and the equality and membership comparison lemmas. The membership embedding must reflect as well as preserve edges.
5. **Prove both comparison maps.** Formalize the ordinal graph relation `OG`, then `CmpD`, and prove sign-function equality. Verify raw injectivity, surjectivity onto every reconstructed number, and preservation of all language symbols.
6. **Formalize syntax and axiom translation.** Represent the finite numerical language, quotient interpretations, recursive axiom schemes, and `Iso(CmpD)`. A semantic construction inside one ambient type universe should not be mistaken for an internal interpretation in every model.
7. **Lift to class theories separately.** Formalize saturated numerical classes, both class comparison maps, and translated class replacement. Keep the model-cardinality obstruction in the metatheory; it is not an internal statement that a model's proper class has an external cardinality.

The first five steps establish the mathematical construction. The sixth verifies the precise first-order bi-interpretability claim. Combining them is preferable to treating informal code representations as if they already constituted an interpretation.

## 78.2 A universe-safe implementation plan (source 14)

A proof-assistant development should separate three levels. First, define sign sequences whose ordinal domains are small in a fixed universe, and prove the prefix, bit, and polynomial-pairing lemmas. Second, define rooted graphs, the code equivalence relation, and the membership relation on the quotient. Third, define first-order syntax, interpretation composition, and the explicit round-trip maps.

For a Lean-style implementation, a type of surreals small in universe  $u$  generally lives in a larger universe; it must not be asserted to be an element of itself. A quotient of codes is a quotient type in the appropriate metatheoretic universe, not a new set in the represented object theory. Similarly, a class collection is additional structure; using the full ambient type of predicates as the class sort would silently impose stronger external comprehension than an arbitrary Henkin model of GB possesses.

The central implementation milestones are:

- F1.** Formalize (69.3), (69.5), and (70.3), including ordinal discreteness under natural arithmetic.
- F2.** Prove the well-founded extensional collapse interface and the root-hull membership theorem without choosing global representatives.
- F3.** Implement interpretation syntax and prove the generic lemma 73.2, including finite-dimensional quotient flattening.

- F4.** Formalize the number/sign-function relation  $\rho$ , including operation preservation, then derive  $\text{SA}_{\text{rt}}$ .
- F5.** Formalize fraction equivalence and quotient birthday, and prove O4 is exactly the required omnific recovery condition.
- F6.** Implement the saturated-class lift and separately formalize the reflection and diagonal obstructions in suitable metatheories.

This report does not supply Lean proofs for these milestones. Existing repository code may support individual ingredients, but a matching filename or theorem label is not evidence of a completed formalization of the interpretation theorem.

### 78.3 Executable finite checks (source 14)

The accompanying program, shipped as `code/14-surreal-only-nbg-verify_finite.py`, uses only the Python standard library. It checks the prefix formula on a finite sign tree, the pairing on finite ordinals and on a finite family of Cantor-normal-form polynomials, padding and decoding of finite graph codes, equality and membership by graph isomorphism, finite tuple packing, dyadic quotient-birthday invariance, and the finite diagonal identity. Its record, shipped as `data/14-surreal-only-nbg-verification.json`, records the actual counts.

| Finite check                                       | Cases verified  |
|----------------------------------------------------|-----------------|
| Prefix pairs                                       | 16,129          |
| Finite / polynomial ordinal pairs                  | 6,400 / 4,096   |
| Labelled valid rooted graphs                       | 207             |
| Graph equality / membership comparisons            | 42,849 / 42,849 |
| Tuple-packing round trips                          | 961             |
| Dyadic fraction-invariance checks                  | 1,225           |
| Boolean matrices checked for the diagonal identity | 66,066          |

These tests are designed to detect indexing, orientation, padding, normalization, and quotient mistakes. They do not establish transfinite well-foundedness, ZFC consistency, interpretation of an infinite axiom scheme, or the class-theoretic theorems. The displayed mathematical proofs, not a finite test run, carry those claims. (Source 14.)

[Added 4 October 2026, batch 95: the record also lists 642 sign-bit checks, 75 increasing-DAG candidates and 511 dyadic sign-birthday checks, which the table omits. For this merge the shipped program was rerun with Python 3.14.4 on a copy outside the repository: it passed in about two seconds and reproduced the shipped record exactly, apart from line endings (CRLF on Windows). Without an argument it writes `verification.json` next to itself, so it must be run on a copy; the report’s README gives the command. Source 13 ships no program.]

## 79 Further questions and research

The questions below are proposed research directions. Unless a published source is explicitly named, they are not claimed to be previously published open problems, and no assertion of exhaustive novelty is made. Several are simplification or classification problems rather than existence problems. (Source 13.) Source 14’s questions are tasks left unresolved by that report; they are not all asserted to be previously published open problems, no claim of absence from the literature is made, and each is stated so that a future result can be checked against a definite language or base theory. Questions

asked by both sources are printed once, with both credited. The last subsection lists, under the standing rule on unproved claims, the inputs both sources import without proof.

## 79.1 Intrinsic axioms and minimum structure

**Question 79.1** (Short intrinsic omnific axioms; source 13). Can one replace the translated ZFC schemes and the full comparison axioms by a transparent list of arithmetic, initiality, generation, and birthday-bound principles in the native language?

$T_{Oz}$  settles the recursive-axiomatization existence problem. It leaves open the most attractive presentation of that theory. A candidate shorter system must prove that every original number is recovered; proving only that graph codes form some model of set theory does not suffice. Assuming consistency, a finite axiomatization in a finite language is ruled out by bi-interpretability, so the realistic target is a small collection of natural schemes.

**Question 79.2** (Intrinsic axioms versus round-trip axioms; source 14). Can one prove, in a fully checked first-order development, that a natural list of order, birthday, recursion, bounded-cut, replacement, and power-set principles is equivalent to  $SA_{rt}$ ? The target is to eliminate the explicit coherence axioms of (73.2) by proving them from intrinsic numerical principles. A useful deliverable would be a literal axiom list and a formal derivation of both round trips, not only their verification in the intended class structure.

Source 14 closes its list thus: the most direct next mathematical target is [question 79.6](#), which asks whether the explicit positive omnific construction can be made much more intrinsic; the most valuable foundational verification target is [question 79.2](#); both have a concrete success criterion, and neither requires claiming a breakthrough before a proof is checked.

Both questions contain the unverified part of the Chen–Hamkins–Yang programme: its candidate intrinsic axioms for  $SA$  ([section 79.4](#)), and they meet the first item of [question 21.7](#).

**Question 79.3** (Reducts of the birthday language; source 13). Which proper reducts of  $\{0, 1, +, -, \cdot, <, \prec_b\}$  still support a canonical bi-interpretation with ZFC? In particular, can multiplication be eliminated from the native omnific construction?

The current proof uses multiplication to address blocks and matrix entries. Recovering that addressing mechanism in a reduct would be a concrete first step. Conversely, a suitable family of automorphisms or a model-theoretic tameness result could prove that a proposed reduct is insufficient. Merely replacing birthday precedence by the definable birthday function is a change of presentation, not a new open question. (Source 13.) [Merge: by [proposition 74.2](#), for comparisons sending each number to codes of itself the question asks which reducts define the remaining symbols of  $\mathcal{L}_b$ ; it overlaps the “Minimal signature” question of [section 21](#), which asks for one-way interpretations, and Question 42.3 of [definable-surreals-and-omnific-integers](#), “How much of the signature is necessary?”, for  $\mathfrak{S}_\Omega$ .]

**Question 79.4** (Pure omnific interpretation strength; sources 13, 14). What is the exact first-order interpretation strength of the pure ordered omnific ring? Can it interpret a set theory through a genuinely different, noncanonical reconstruction? (Source 13.) Does the pure ordered omnific ring interpret the set universe, with a uniform definable reverse reconstruction, or can such an interpretation be ruled out? This is not the question of recursive axiomatizability of its complete theory, and the real-closed-field decidability argument does not apply to this ring. (Source 14.)

**Corollary 74.13** excludes recovering the canonical birthday predicate by a pure-ring formula. The automorphism argument also excludes the stated canonical bi-interpretation with the actual well-founded  $V$  ([proposition 74.14](#)). Neither statement resolves the broader interpretation question. Work on this problem should keep those three claims separate. (Source 13.) [Merge: answered in part. The pure ring interprets  $(\mathbb{N}, +, \cdot)$  and the field **No** with  $\mathbb{R}$  ([remark 74.11](#)), so its strength is at least arithmetic; a reconstruction with the canonical set-to-number direction is excluded by [proposition 74.14](#); whether it interprets ZF or ZFC at all, by a noncanonical reconstruction, is open.]

**Question 79.5** (Short fractional-birthday definitions; source 13). Find a direct, reasonably short formula defining  $\text{FB}(a, b, c, d)$  in the native omnific birthday structure, and compare its quantifier complexity with the interpretation-based definition.

Existence is already proved in [corollary 74.4](#). A substantially shorter definition could make fraction-field transfers practical and reveal arithmetic constraints on birthdays that the graph construction hides.

**Question 79.6** (Unary birthday on omnific integers; source 14). Is the quotient-birthday operation  $h(a, b)$  definable in

$$(\mathbf{Oz}; 0, 1, +, -, \cdot, <, b \upharpoonright \mathbf{Oz})?$$

A positive answer would remove the binary enrichment from [theorem 74.7](#). A negative answer should produce models or symmetries agreeing on the weaker structure but disagreeing on a quotient birthday. Neither the field fraction construction alone nor knowledge of the two input birthdays settles this.

[Added 4 October 2026, batch 95: answered positively by source 13’s [corollary 74.4](#), through [proposition 74.5](#); the definition is interpretation-based and long, and [question 79.5](#) asks for a short one. Source 14 called this “the most direct next mathematical target”.]

**Question 79.7** (Quantifier complexity of the omnific predicate; source 14). What is the least first-order complexity of  $\text{Omnific}(x)$  in  $\mathcal{L}_s$ ? The definition obtained through normal forms and set codes is effective but expensive. An economical birthday-and-order characterization could substantially simplify O4 and subsequent formalization.

The omnific sign criterion (ii) of [section 69](#), if read with Plus, gives a candidate:  $x \in \mathbf{Oz}$  iff no ordinal  $\gamma$  with  $\gamma + 1 < b(x)$  has  $\text{Plus}(x, \gamma) \not\leftrightarrow \text{Plus}(x, \gamma + 1)$ . This merge records it as a candidate only: it is a short  $\mathcal{L}_s$ -formula whose correctness is the imported criterion, and its exact complexity is not computed here.

## 79.2 Weaker foundations and bounded universes

**Question 79.8** (Choice-free numerical foundations; source 13). Is there a canonical numerical bi-interpretation for ZF without Choice using a different decoder? If so, what replaces ordinal-labeled transitive closures?

The present decoder has exact range HWO. This is a proved limitation of that decoder, not an impossibility theorem for every conceivable one. Any proposed replacement must explain how it represents sets whose transitive closures are not well-orderable, while keeping each individual code a standard set-sized number.

**Question 79.9** (Removing Choice; source 14). Which choice-free universes can be recovered by ordinal-sign graph codes? The proof of [lemma 71.2](#) uses a well-ordering of each transitive closure. In ZF this may fail. Determine the exact class of sets recovered by the same interpretation, and whether a modified numerical language removes the obstruction without quietly adding a well-ordering principle.

[Added 4 October 2026, batch 95: the first part is answered, and was already answered at source 14's pin: the recovered sets are exactly HWO ([theorem 13.2](#), source 06's, and [proposition 71.5](#)). The modified-language part is [question 79.8](#), still open; this report's [remark 13.4](#) asks the same.]

**Question 79.10** (Birthday cutoffs; source 13). For which ordinals or cardinals  $\kappa$  does the native structure on  $\mathbf{Oz}_{<\kappa} = \{x \in \mathbf{Oz} : b(x) < \kappa\}$  recover a natural hereditary set universe? What closure hypotheses are exact?

The repository already studies related surreal cutoffs [40]. The omnific codec adds explicit bounds involving  $\alpha \mapsto \omega\alpha + \omega$  and  $\kappa \mapsto \kappa^2$ , all naturally computed. Limit discontinuities must be included in any sharp estimate. One must also check whether the chosen cutoff is closed under the arithmetic symbols of the proposed structure. (Source 13.) [Merge: for  $\mathbf{No}_{<\lambda}$  this is Parts I–III of this report ([Part I](#)); the omnific cutoffs are not treated there.]

**Question 79.11** (Fragments and axiom strength; source 13). Which numerical schemes correspond exactly to Separation, Replacement, Power Set, and weaker collection principles? Can natural fragments of omnific arithmetic recover admissible or other restricted set universes?

A useful outcome would be a table of equivalences with proofs in both directions, not just upper bounds obtained by translating all of set theory. The finite PA/HF correspondence suggests an approach through induction and coding closure, but transfinite well-foundedness and Choice introduce additional issues. (Source 13.) [Merge: for the surreal field this report has partial answers: eternity is Replacement at cardinal cutoffs and the translated Power Set sentence detects strong limits (Part II), and over Zermelo set theory definable birthday bounding is ordinal bounding ([theorems 28.2](#) and [29.1](#)).]

**Question 79.12** (Selecting codes and choice principles; sources 13, 14). What extra principle is equivalent to a definable selection of a graph code for every represented set? What weaker selections can be made on bounded families? (Source 13.) Under what additional definable choice principles can the quotient Code/Eq be replaced by a definable subclass of unique numerical representatives? Determine whether the resulting presentation is *definitionally equivalent* to the associated set theory, rather than merely bi-interpretable. Adding an arbitrary Global Choice class as a parameter changes the theory and must be recorded. (Source 14, “Canonical representatives and definitional equivalence”.)

The main construction needs existence of some code and deliberately avoids selecting one globally. A selected ordinal labeling of each transitive closure gives a way to choose a least-labeled member of each nonempty set. Conversely, a suitable Global Choice class can be used to choose labelings. A precise analysis should distinguish class parameters, parameter-free definability, and bounded selection schemes. (Source 13.)

**Question 79.13** (Class theory without set Choice; source 13). Is there a numerical class lift bi-interpretable with GB without set Choice? The present ordinal-graph codec uses set Choice ([proposition 71.5](#)); GB without it “needs a different analysis”.

### 79.3 Class theories, models, and proof complexity

**Question 79.14** (Native class axioms; sources 13, 14). Find compact numerical formulations of class replacement, Global Choice, and elementary transfinite recursion, and prove their exact equivalence to the translated principles in  $T_D^{\text{cl}}$ . (Source 13.) Over a carefully specified weak numerical two-sorted base, characterize which birthday-boundedness and packing principles are equivalent to class replacement, elementary comprehension, and Power Set. The implication proved in [section 77](#) is not yet such a reverse mathematics result. (Source 14, “Minimal native class axioms”).

The two-sorted existence theorem is established. The remaining task is to expose the mathematical content of the class principles in numerical terms. Strengthening comprehension changes the available classes, so GBC, its recursion extensions, and KM should be analyzed separately rather than grouped under one unspecified class theory. (Source 13.) [Merge: the translated forms are bi-interpretable with their sources by [corollary 76.7](#); the question asks for native forms.]

**Question 79.15** (Elementary transfinite recursion; source 14). Formulate a native numerical version of ETR, and prove its exact equivalence to the class-theoretic principle under [theorem 76.6](#). Distinguish set-sized recursion, already used in surreal construction, from recursion with class-sized stages or class values. The latter must not be silently assumed in a GB proof.

**Question 79.16** (Class realizations and forcing; source 14). Use the inclusion-preserving correspondence ([76.4](#)) to transfer concrete realization and class-forcing questions to surreal-based GBC. Which numerical predicates correspond to adjoining a new amenable class without adding sets? This could make precise what information remains invisible to the underlying birthday field.

**Question 79.17** (Nonstandard models in numerical terms; source 13). How can familiar invariants of set-theoretic models be recognized intrinsically in models of  $T_{\text{Oz}}$ ? What do nonstandard ordinals, ill-founded external membership, and different class realizations look like in the numerical presentation?

Bi-interpretability already transports the relevant model-theoretic information and definable comparisons. The research task is a useful intrinsic description, not another existence proof. In particular, rigidity of the actual numerical universe must not be extrapolated to all countable or nonstandard models. (Source 13; it meets the item “ill-founded models” of [question 21.7](#).)

**Question 79.18** (Generalized sign objects; source 14). Can a theory of proper-class sign branches be axiomatized so that it extends, rather than contradicts, ordinary surreal arithmetic, while admitting a meaningful partial arithmetic? Specify its ambient higher-order foundation before discussing a universe of all such branches. It is not enough to write a Conway cut with a proper-class option set and assume a number exists.

**Question 79.19** (Effective proof translations and formal proof; sources 13, 14). What are the formula-length and proof-length costs of the explicit interpretations? Can graph, ordinal, and function coding be modularized without concealing hypotheses? (Source 13.) Construct a proof-assistant implementation of the code interface, including actual first-order formula syntax, and measure the cost of translating natural set-theoretic statements. Can common set operations be compiled into substantially smaller formulas than the generic normal-form and graph-code route produces? (Source 14, “Formal proof and proof complexity”).

The translations are effective, but effectiveness does not imply low complexity. Fixed abbreviations, definitional extensions, and a reusable syntactic library could reduce formalization overhead. None of these improvements would make the resulting theory decidable. (Source 13.)

**Question 79.20** (Automorphisms of richer arithmetic; source 13). How do the native birthday results interact with the published automorphism questions for surreal exponentiation and Conway’s monomial map?

Kaplan, Krapp, and Serra pose questions about nontrivial exponential 1-automorphisms, automorphisms preserving the monomial map, and extensions of order automorphisms of generalized epsilon numbers [36, Questions 5.4, 5.6, 5.7]. The shear automorphisms used here need not preserve those extra operations, so they do not answer those questions. Comparing the exact published hypotheses with the repository’s constructions is a worthwhile next step; the present article makes no claim to have resolved them. (Source 13.)

## 79.4 Inputs imported without proof

Under the standing rule on unproved claims, the following inputs, used by sources 13 and 14 and kept by this merge, are recorded as imported. Each is a published or announced result, credited at its point of use; none was re-proved here, and each item says what is missing for a self-contained account in the collection.

- (1) *Omnific sign criterion and initiality* ((i), (ii) of [section 69](#); Ehrlich–Kaplan arXiv v1 §2.2, Lemma 2.1 and the proof of Lemma 2.3, after Gonshor, Theorem 8.1). Every native **Oz** result of this part depends on it: [lemma 69.2](#) in **Oz**, [theorem 70.2](#), [theorem 72.2](#) and [theorem 73.4](#) for **Oz**, [corollary 74.4](#) and [proposition 74.5](#). Source 14’s quotient-birthday route does not. Missing: a proof in the collection, or a formalization against the sign-sequence library; neither the Hilbert’s-tenth intake nor this merge checked the cited proofs.
- (2) *Hahn lifting of exponent automorphisms* (the step “an additive order automorphism of the exponent group ... is an ordered-field automorphism” in [proposition 74.12](#); Kaplan–Krapp–Serra [36]). Missing: a written proof that the lifted map is onto **No** and multiplicative on Conway normal forms; the collection’s `surreal/omnific-preserving-automorphisms` uses such lifts for its stronger `opa:thm:parameters`.
- (3) *Class-forcing facts* for [lemma 75.5](#): a countable GBC-realization of a countable  $M$  has a proper countable extension with the same sets (a class Cohen generic), and unions of increasing chains of GBC-realizations of one  $M$  are realizations (Williams [34], Theorem 1.61(5), the proof of Theorem 1.65). Source 13 checks the union step for comprehension, replacement and Global Choice; the extension step is imported.
- (4) *Finite axiomatizability of GBC/NBG* ([theorem 75.2](#); Williams, Corollary 3.11 and its proof), and the Lévy reflection theorem and Gödel’s second incompleteness theorem [38, 39]: classical.
- (5) *Attribution of the reflection argument to Hamkins–Yao, Section 4*: checked for this merge and correct ([theorem 75.2](#)).
- (6) *Jeřábek’s discussion* of omnific axiomatization and of the complete theory of **Oz** [37], cited by source 14 for  $\text{Frac}(\mathbf{Oz}) = \mathbf{No}$  (proved here) and for the non-recursive axiomatizability of the complete theory (not used in any proof).
- (7) *The Chen–Hamkins–Yang intrinsic axioms* for SA: announced, with the slides marking details as still being checked; neither source verifies them, and [theorems 73.4](#) and [73.6](#) do not depend on them. This is [question 79.2](#).
- (8) *That  $(V_\kappa, \mathcal{P}(V_\kappa))$  satisfies KM with Global Choice for inaccessible  $\kappa$*  ([section 77.1](#)): standard; source 14 sketches the verification.

No claim of either source was found to be wrong. The KM, GBc and ETR class lifts that both sources sketched are proved in [corollary 76.7](#); source 13’s paragraph on the pure omnific ring is [proposition 74.14](#).

## 80 Conclusions of sources 13 and 14, audit and provenance

### 80.1 Final assessment (source 13)

The finite-set analogy extends fully to the set universe once the numerical language retains constructional information. Arithmetic, order, and birthday precedence allow an omnific integer to store arbitrary bounded ordinal data. Well-founded extensional graph codes reconstruct sets; signwise comparison reconstructs the original numbers. The explicit recursive axiom construction turns these facts into

$$\mathbf{T}_{\mathbf{No}} \simeq_{\text{bi}} \mathbf{ZFC} \simeq_{\text{bi}} \mathbf{T}_{\mathbf{Oz}}.$$

The corresponding statement for full class theory needs a different ontology. A uniform interpretation whose numerical individuals are all original sets cannot recover every possible class part of a GBC-model. Retaining classes of numbers gives a faithful two-sorted theory, with explicit comparison maps and no global selection of code representatives.

The constructive omnific result and the class boundary are the most useful outcomes of this investigation. They clarify what has been proved, which language features are doing the work, and which further questions concern genuine simplification or new foundational strength. They do not settle an independent set-theoretic statement or establish a literature-priority claim.

### 80.2 Conclusions and limits of the result (source 14)

There is a rigorous and natural set-level realization of the proposed analogy. Ordinal-length signs replace finite bits; birthday makes the sign information first-order accessible; well-founded graph collapse recovers arbitrary sets; and explicit coherence supplies the second round trip. An omnific-integer implementation is available through fraction reconstruction and the quotient-birthday operation.

The resulting theories are ZFC-equivalent, not NBG-equivalent. For uniform theories interpretable in ZFC, reflection and incompleteness forbid even a one-way interpretation of the finitely axiomatized NBG theory. For native same-universe class names with elementary evaluation, the diagonal argument independently forbids assigning an ordinary set-sized numerical code to every class.

The correct NBG-level construction is reversible but retains a class sort. It replaces sets by numerical graph codes and classes by saturated classes of codes, and it recovers arbitrary numerical subclasses in the second round trip. The distinction is substantive: the class realization is preserved rather than discarded.

The report therefore answers the formulation problem with an explicit positive construction and a precisely delimited negative theorem. It does not claim to originate the Chen–Hamkins–Yang bi-interpretation, to settle the pure omnific-ring problem, to certify a minimal intrinsic SA axiom system, or to prove consistency of any of these foundations. Its proposed follow-on results have complete mathematical arguments here, but their novelty has not been independently certified and their proofs have not been checked in a proof assistant.

### 80.3 Formula expansion and type checking

**Quantifier and domain conventions (source 13).** In numerical formulas, every unqualified variable ranges over  $\mathbf{D}$ . The abbreviation  $\forall i < \nu \varphi$  means  $\forall i ((\text{Ord}(i) \wedge i < \nu) \rightarrow \varphi)$ . Existential

bounded ordinal quantifiers use conjunction instead of implication. The relation  $x \leq y$  is  $x < y \vee x = y$ . A definable function such as  $b(x)$  can be eliminated by introducing a variable and its unique defining relation.

A code variable  $c$  stands for three numerical variables  $(\nu_c, r_c, z_c)$ . Both  $\text{Eq}(c, d)$  and  $\text{Mem}(c, d)$  include  $\text{Valid}(c) \wedge \text{Valid}(d)$  as conjuncts. Thus they are false on invalid arguments. No equality of quotient objects is substituted for raw numerical equality inside the definitions of the codes.

For a candidate  $h$  for a map from  $\nu_c$  to  $\nu_d$ , the required clauses are:

$$\begin{aligned} \text{Total}(h) : & \quad \forall i < \nu_c \exists j < \nu_d Y_h(i, j), \\ \text{Functional}(h) : & \quad \forall i < \nu_c \forall j, k < \nu_d ((Y_h(i, j) \wedge Y_h(i, k)) \rightarrow j = k), \\ \text{Injective}(h) : & \quad \forall i, k < \nu_c \forall j < \nu_d ((Y_h(i, j) \wedge Y_h(k, j)) \rightarrow i = k), \\ \text{Surjective}(h) : & \quad \forall j < \nu_d \exists i < \nu_c Y_h(i, j). \end{aligned}$$

Edge preservation and reflection are the biconditional in [section 71](#). Equality of set codes uses all four displayed conditions and root preservation. Membership uses the first three, edge preservation and reflection, predecessor closure, and the root-edge condition in [\(71.5\)](#). These formulas use a fixed number of variables regardless of the ordinal sizes of the graphs. Their quantification over map codes  $h$  is still first-order quantification over numbers.

**The second comparison as one formula (source 13).** Suppressing only the fixed triple expansion, the formula  $\text{Cmp}_{\mathbf{D}}(x, c)$  can be written

$$\begin{aligned} \text{Valid}(c) \wedge \text{Is}_{\mathbf{D}}^{\mathcal{T}\mathbf{D}}(c) \wedge \exists \lambda [ & \text{Ord}(\lambda) \wedge \text{SameB}(x, \lambda) \\ & \wedge \exists e (\text{OG}(\lambda, e) \wedge \text{Dom}^{\mathcal{T}\mathbf{D}}(c, e)) \\ & \wedge \forall \beta < \lambda \forall e (\text{OG}(\beta, e) \rightarrow (\text{Val}_{+}^{\mathcal{T}\mathbf{D}}(c, e) \leftrightarrow \text{Plus}(x, \beta)))]. \end{aligned}$$

The source predicates  $\text{Dom}$ ,  $\text{Val}_{+}$ ,  $\text{Is}_{\mathbf{D}}$  refer to the fixed sign-function representation. Their translations are effective substitutions. They do not name a satisfaction relation.

**Formula-expansion and type-checking checklist (source 14).** A literal compiler for the definitions of source 14 can follow the following fixed rules. They make precise why mathematical abbreviations do not add a hidden sort of sets to  $\text{SA}_{\text{rt}}$ . Replace  $\text{Ord}(x)$  by  $b(x) = x$ . Replace  $\alpha < \delta$  in an ordinal-bound quantifier by the conjunction of this formula with the order inequality. Replace  $\text{Pref}$ ,  $\text{Bit}$ ,  $\varpi$ , and  $\Lambda_{14}$  by [\(69.3\)](#), [\(69.5\)](#), and the indicated polynomials. Replace every variable for a subset of  $\delta$  by one numerical variable constrained to have birthday  $\delta$ . Replace every rectangular relation variable by one numerical variable of birthday  $\Lambda_{14}(\delta + \eta)$ , with its membership predicate read at  $\varpi(i, j)$ . Extra unused bits in witnesses can either be ignored or fixed to minus; the two versions define the same existence claims.

A vertex map from  $\delta$  onto a definable subset  $H \subseteq \eta$  is a rectangular relation  $R$  subject to

$$\begin{aligned} \forall i < \delta \exists! j < \eta [ & H(j) \wedge R(i, j)], \\ \forall j < \eta [ & H(j) \rightarrow \exists! i < \delta R(i, j)], \\ \forall i < \delta \forall j < \eta [ & R(i, j) \rightarrow H(j)]. \end{aligned}$$

Preservation of edges is

$$R(i, j) \wedge R(i', j') \longrightarrow (i E_c i' \leftrightarrow j E_d j').$$

Preservation of the root adds  $R(r_c, r_d)$ , or  $R(r_c, j_0)$  for the hull rooted at  $j_0$ . These conditions provide the full expanded content of the isomorphism abbreviations used in [section 71](#). (Here  $H$  is a definable vertex set, not a hereditary universe.)

For a set formula, translate atomic equality by Eq, membership by Mem, Boolean connectives homomorphically, and each quantifier by a quantifier over three numerical coordinates restricted by Code. Capture-free substitution uses renamed bound variables. A composite interpretation substitutes these finite formulas again; its tuple dimension stays finite. Isomorphism preservation for a function symbol is a formula about its graph, so no interpretation of partial terms is required.

The following distinctions should be explicit in an implementation: 0, 1 as surreal ring constants; 0, 1 as interpreted set integers; an ordinal as an all-plus number; the code  $c_\alpha^{\text{ord}}$  for that ordinal as a set; the original number  $x$ ; and a set code  $c$  for its sign function in  $\rho(x, c)$ . Conflating any pair of these is a common source of false round-trip proofs.

## 80.4 Proof ledgers

**What each part of the proof supplies (source 13).**

| Required obligation                       | Where it is discharged                                                                                                  |
|-------------------------------------------|-------------------------------------------------------------------------------------------------------------------------|
| Native ordinal recognition and signs      | <a href="#">Lemmas 69.1 to 69.3</a> .                                                                                   |
| Coverage of all bounded subsets           | <a href="#">Theorem 70.2</a> , including limit gaps.                                                                    |
| Definable relation and tuple coding       | <a href="#">Lemmas 70.5 and 70.7</a> ; all unused indices are zero.                                                     |
| A well-defined set interpretation         | <a href="#">Theorem 71.1</a> , using collapse and edge reflection.                                                      |
| Surjectivity onto the set universe        | Set Choice in <a href="#">theorem 71.1</a> ; exact ZF range in <a href="#">proposition 71.5</a> .                       |
| Recovery of every original number         | <a href="#">Lemma 72.1</a> and <a href="#">theorem 72.2</a> , without a code-format restriction on the original number. |
| An effective theory, not just a structure | <a href="#">Theorems 73.1 and 73.4</a> and <a href="#">definition 73.3</a> .                                            |
| Recovery of the full class sort           | Containment and saturation, followed by both maps in <a href="#">lemma 76.2</a> and <a href="#">theorem 76.5</a> .      |
| The limit of a number-only class theory   | <a href="#">Theorem 75.6</a> , with its stated same-universe domain restriction.                                        |

The mathematical proofs have been checked for these specific obligations during preparation of the article. The ledger is not a replacement for external peer review or a machine-checked formalization. (Source 13.)

**Proof and provenance ledger (source 14).**

| Result                        | Basis                                                                   | Status and limitation                                                                         |
|-------------------------------|-------------------------------------------------------------------------|-----------------------------------------------------------------------------------------------|
| Birthday field / set universe | <a href="#">Sections 69 and 71</a> and <a href="#">proposition 72.4</a> | Reconstructed proof of the CHY precedent; no new-priority claim.                              |
| Round-trip axiomatization     | <a href="#">Lemma 73.2</a> and <a href="#">theorem 73.6</a>             | Explicit alternative presentation; not a verification of every candidate intrinsic SA scheme. |

| Result                     | Basis                                                  | Status and limitation                                                                                                                                                |
|----------------------------|--------------------------------------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Omnific quotient birthdays | <a href="#">Theorems 74.6 and 74.7</a>                 | Full proof with a named binary operation; pure-ring and unary-birthday reductions not established [merge: the unary reduction is <a href="#">proposition 74.5</a> ]. |
| NBG obstruction            | <a href="#">Theorem 75.2 and corollary 75.3</a>        | Classical reflection/incompleteness deduction; conditional on $\text{Con}(\text{ZFC})$ ; uniform interpretations only.                                               |
| Class diagonal             | <a href="#">Theorem 75.7</a>                           | Direct proof; comprehension must cover the evaluation formula.                                                                                                       |
| Class lift                 | <a href="#">Theorems 76.3 and 76.6</a>                 | Both class round trips proved; the extra class sort is retained.                                                                                                     |
| Set-size criterion         | <a href="#">Theorem 77.1</a>                           | Proved over GB; not claimed as a complete weak-base                                                                                                                  |
| Executable tests           | <code>code/14-surreal-only-nbg-verify_finite.py</code> | reverse-mathematics axiomatization. Finite sanity checks only; no transfinite or formal foundational verification.                                                   |

**Added in this merge.** [Proposition 74.2](#), [proposition 74.5](#), [lemma 73.8](#), [corollary 76.7](#) and [remark 74.8](#) are observations of this merge, each with its proof; [proposition 74.14](#) is source 13's argument made a numbered proposition; [section 67](#) (except the two status boxes and abstracts), [convention 68.1](#), the bracketed dated notes, and this section's audit and provenance subsections are the merge's text.

## 80.5 A versioned provenance record (source 13)

The repository was inspected at commit `722337445aa691a90e005ff8f619fd8ce5397103`. This is a targeted mathematical and source audit, not a certification of every claim in that large corpus. The repository was not modified.

The following distinctions are necessary for an accurate assessment of what this article adds.

| Topic                       | Prior repository material                                                                                                                                  | Treatment here                                                                                                       |
|-----------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------|----------------------------------------------------------------------------------------------------------------------|
| Full surreal reconstruction | The birthday-cutoff article attributes the full theorem to CHY and develops sign tests, ordinal graph coding, collapse, and hereditary-universe results.   | Acknowledged prior result; a common construction is given that also works natively in <b>Oz</b> .                    |
| Choice boundary             | The same article identifies the collapse range HWO, the equivalence of surjectivity with Choice, and the possible failure of Power Set in HWO.             | Reproved for the present codec; not claimed as a new Choice theorem.                                                 |
| Omnific coding              | The definability article gives ambient set-theoretic codes into omnific integers, including bounded interval encodings and definable fraction-field hulls. | The extra requirement here is that both directions be first-order definable in the native omnific birthday language. |

| Topic                      | Prior repository material                                                                                                            | Treatment here                                                                             |
|----------------------------|--------------------------------------------------------------------------------------------------------------------------------------|--------------------------------------------------------------------------------------------|
| Automorphisms              | The omnific-preserving-automorphism material gives stronger parameter-fixing and nondefinability results.                            | A short normal-form shear proof isolates the birthday obstruction; no priority claim.      |
| Classes                    | The foundations article distinguishes the set universe from its class realizations and discusses the limitations of class quotients. | An explicit uniform-domain obstruction and a two-sorted class lift are proved.             |
| Fraction fields            | The ledger maps omnific denominator clearing and the surreal fraction-field instance to Lean declarations.                           | Used as established repository infrastructure, separate from the new graph interpretation. |
| Arithmetic and finite sets | The logic portion develops Ackermann coding and finite-set/arithmetic interpretations.                                               | The motivating analogy is retained with its axiomatic qualifications.                      |

The principal inspected documents are [40]. Within the birthday-cutoff article, relevant labels include `hset:thm:chy`, `hset:lem:prefix`, `hset:lem:pairing`, `hset:lem:collapse`, `hset:thm:collapse-range`, and `hset:thm:choice`. In the definability article, `dsn:thm:bounded-code` and `dsn:thm:hull` identify related ambient constructions.

The targeted audit did not locate the native omnific birthday bi-interpretation with the explicit recursive axiom construction of [section 73](#), nor the precise model-cardinality theorem of [theorem 75.6](#). That is an audit result about the inspected material, not proof of worldwide novelty. A future submission should repeat the search against the then-current repository and literature and seek feedback from the authors of the closest work. Online sources were consulted on 4 October 2026; version-specific claims are identified explicitly where numbering or research status matters. (Source 13.)

[Added 4 October 2026, batch 95: every repository statement of this record was checked at the pin and is true; the cited labels resolve in this article as [theorems 5.3](#), [13.2](#) and [18.1](#) and [lemmas 4.1](#), [4.3](#) and [5.1](#) (the last two concern the cutoff ranges  $H_\kappa$  and the ZF range HWO respectively; Power Set in HWO is [proposition 13.3](#)). The audit result stays true at the placement: the collection had neither theorem, its closest material being the pure-ring/field bi-interpretation of `surreal/omnific-diophantine-geometry` and `dsn`’s interpretation in  $\mathfrak{S}_\Omega$ , written later ([remark 67.5](#)).]

## 80.6 Non-claims of sources 13 and 14

Every limitation stated by either source, in its manuscript, its README or its source-and-status note, is kept; most appear at their point of use. Collected.

**Source 13.** The Chen–Hamkins–Yang result is not presented as new, and no unpublished final axiomatization is assumed verified;  $\mathbf{T_D}$  is not claimed complete, categorical or finitely axiomatized; no literature priority is asserted, no previously published open problem is claimed solved, and novelty is an audit result about the inspected material only; no proof assistant was used, no new theorem is claimed machine-checked, no fresh Lean build was made, and a compiled PDF is not a verification; HWO is the limit of this decoder only, not an impossibility theorem for every interpretation of ZF, and HWO need not model ZF; no global code selection is made; definability transfer is per formula, supplies no truth predicate, and excludes class-quantified formulas; the pure-**Oz** result does not

exclude every (noncanonical) interpretation of set theory; rigidity concerns the actual universe, not all models of  $\mathbf{T_D}$ ; the class no-go does not forbid external relabelling of particular countable models or one-sorted presentations with class individuals; the GB diagonal does not cover formulas with bound class quantifiers; the class lift keeps a class sort; GB without set Choice is open; the shear automorphisms do not answer Kaplan–Krapp–Serra’s Questions 5.4, 5.6 and 5.7; nothing independent of ZFC is decided; the main constructions received “independent proof reviews during preparation”, which are distinct from external peer review or formalization.

**Source 14.** The CHY bi-interpretation is not this report’s; no priority is claimed for its deductions; nothing is Lean-verified and no Lean development or fully serialized first-order syntax compiler is included; the finite checks are supplementary and do not verify transfinite interpretations, infinite schemes, class comprehension, reflection or consistency; the negative theorem needs “uniformly interpretable in ZFC” and excludes neither external relabelling, added predicates of independent strength, nor a larger universe;  $\mathbf{SA_{rt}}$  does not verify CHY’s candidate axioms;  $\mathbf{OA_{qb}}$  is not the complete theory of  $\mathbf{Oz}$  and does not evade Jeřábek’s obstruction;  $h$  is not claimed definable in the pure ring; the pure-field argument is not a no-go theorem for the pure omnific ring; the reflection argument is classical and not a new separation of ZFC and NBG; the finite-axiomatizability diagnostic alone does not exclude a one-way interpretation; the diagonal concerns internally available evaluators; full second-order semantics is not claimed; [theorem 77.1](#) is not a reverse-mathematics axiomatization; the inaccessible example is not a same-universe solution nor a consistency proof from ZFC; the repository audit was targeted, with no nonduplication or novelty guarantee, and it did not build or audit this report or import its cutoff claims as axioms.

**This merge** adds: for  $\mathbf{D = No}$  the structural theorems are [theorem 18.1](#) again ([remark 67.3](#)); [proposition 74.5](#) rests on the imported omnific sign criterion; [proposition 74.2](#) does not decide whether  $\prec_b$  is definable in  $\mathbf{dsn}$ ’s  $\mathfrak{S}_\Omega$ .

## 80.7 Provenance of this part

**The manuscripts.** Source 13 is batch-95 manuscript 01, delivered as the archive `surreal_omnific_foundations.zip` (566,548 bytes; inner directory `surreal_omnific_foundations/`), and source 14 is batch-95 manuscript 02, delivered as `Surreal_Only_Foundations_and_NBG.zip` (381,702 bytes; inner directory `Surreal_Only_Foundations/`). Both arrived in commit `ec91f8c7c`, the batch-94A placement, whose message does not mention them (they were staged into the shared index while that commit was made), and both were placed in commit `350b9a954`. Source 13 names the repository commit `722337445` as its baseline (“Record the standing rule on unproved and wrong claims in the intake procedure”, 4 October 2026), with bibliography links pinned to it; source 14 names `bd1de458b` (“New research reports”, 4 October 2026), and its search excerpts came from the earlier indexed revision `fb2287290`. All three are ancestors of the placement, and at each this article was the one of Parts I–VIII. Both describe this report’s README and source audit accurately. Source 13’s manuscript (`surreal_omnific_foundations.tex`, 2,765 lines, 59 labels, 12 questions, 21 bibliography entries; a 34-page PDF) and its `README.txt` ship no code or data; source 14’s manuscript (`article.tex`, 1,836 lines, 54 labels, 11 questions; a 27-page PDF), README, PDF and checksum file `SHA256SUMS` (8 of 8 entries verified at placement) are not shipped. They survive in the arrival commit. Shipped, byte-identical, all from source 14: `14-surreal-only-nbg-SOURCES_AND_STATUS.md`, `code/14-surreal-only-nbg-verify_finite.py`, `code/14-surreal-only-nbg-build.sh`, `data/14-surreal-only-nbg-verification.json` and

`data/14-surreal-only-nbg-build_validation.json`. The prefix `13-omnific-foundations-` names no file. The Hilbert’s-tenth intake ([remark 67.7](#)) authenticated both archives.

*Remark 80.1* (Delivered label-count correction). The publication described source 13 as having “76 labels” and source 14 as having “70 labels.” A fresh inventory of the immutable delivered L<sup>A</sup>T<sub>E</sub>X sources gives 59 and 54 literal label declarations, respectively. The old numerals are retained here as incorrect provenance counts. This correction does not change the publication’s 151 new merged labels or identify source declarations with generated cross-reference entries.

**How they are printed.** Source 13’s text from its title-page boxes and abstract to its appendix is printed in full, in its order, as the spine: Sections 1–13 are [section 67](#) to [section 80](#), its Section 11 split between [section 78.1](#) (formalization) and [section 80.5](#) (provenance record), its appendix in [sections 80.3](#) and [80.4](#), its closing note on online sources at the end of [section 80.5](#). Source 14’s text is printed in full and inserted by subject: its Sections 1–2 in [section 67](#), 3 in [section 68](#), 4 in [sections 69](#) and [70](#), 5 in [section 71](#), 6 in [sections 72](#) and [73](#), 7 in [section 74.3](#), 8 in [section 74](#), 9–10 in [section 75](#), 11 in [section 76](#), 12–13 in [section 77](#), 14 in [section 79](#), 15 in [section 78](#), 16 and its appendices here. Shared results are printed once (the duplicates are named at each result: graph interpretation, round trip, recipe, fractions, RCF obstruction, NBG obstruction, diagonal, class correspondence and lift), with the other source’s statement or proof as a marked second route; the theories  $T_{\mathbf{No}}$  and  $SA_{rt}$  are printed side by side as two presentations. Every label of the part carries the prefix `hset:sf::`; the delivered labels are not kept one to one, since a shared result carries one label. Both sources’ questions gained labels `hset:sf:q::`; questions asked by both are merged (pure **Oz**, selection of codes, native class axioms, proof cost), and the two intrinsic-axiom and the two choice questions are printed next to each other. The symbols renamed are listed in [convention 67.2](#); equations are numbered within sections; delivered file names are replaced by the shipped ones; the two sources’ references use this report’s bibliography, with the entries [\[31, 32, 30, 34, 35, 36, 33, 37, 38, 39\]](#) and the pinned repository entries added. No mathematical statement or proof of either source was changed, except that [theorems 75.7](#) and [77.1](#) are stated over GB where source 14 wrote GBc (its proofs use no Choice; said at the point of use).

**Where the merge had to choose.** (i) Placement: both sources re-derive [theorem 18.1](#) and bear on [question 21.7](#) and the “Minimal signature” question, so they were placed here as one addition, rather than as a new report or as a part of `definable-surreals-and-omnific-integers`, whose Part III uses a different signature. (ii) Base: source 13, whose omnific theory uses the weaker language  $\mathcal{L}_b$ , where source 14 needs  $h$ ; source 14’s omnific route is kept in full as a second route ([section 74.3](#)). (iii) Source 14’s sharper reflection lemma carries the proof of the NBG obstruction, with source 13’s proof as second proof; source 14’s arbitrary-class round trip is printed inside the class lift. (iv) Class names follow the foundations report ([convention 68.1](#)); source 13’s elementary-precedence symbol  $\prec$  is  $\prec_b$  here, because this report’s  $\prec$  is elementary substructure; source 14’s  $\Lambda$  and  $\pi$  and source 13’s  $B$ ,  $D$ ,  $C$ ,  $K_D$ ,  $I_\beta$ ,  $H_h$  and  $\Omega$  are renamed ([convention 67.2](#)). (v) Standing-rule items: source 13’s prose argument on the pure omnific ring is complete and became [proposition 74.14](#); the sketched KM, GBc and ETR class lifts are proved in [corollary 76.7](#); GB without set Choice is [question 79.13](#); imported inputs are listed in [section 79.4](#); the Hamkins–Yao attribution was checked and is correct; no claim was found wrong. (vi) The part follows Part VIII, so that no existing number changes, and is printed before this report’s appendices.

## A Formula expansion and size checks

### A.1 What “first-order” means in the coding proof

Every abbreviation expands to a finite formula in  $\{0, 1, +, -, \cdot, <, b\}$ . Bounded ordinal quantifiers are shorthand for ordinary scalar quantifiers with guards  $\text{Ord}(x)$  and  $x < \delta$ . A set variable in an interpreted membership formula becomes a triple of scalar variables. Equality becomes  $\sim$ , membership becomes  $\in_{\mathcal{I}}$ , and quantifiers are relativized to  $\text{Code}$ :

$$(u = v)^{\mathcal{I}} = (C_u \sim C_v), \quad (u \in v)^{\mathcal{I}} = (C_u \in_{\mathcal{I}} C_v), \quad (\exists u \varphi)^{\mathcal{I}} = \exists \delta_u \exists r_u \exists t_u (\text{Code}(C_u) \wedge \varphi^{\mathcal{I}}),$$

commuting with the connectives. Induction on formulas proves  $H_{\kappa(\lambda)} \models \varphi(x_{C_1}, \dots) \iff \mathcal{B}_\lambda \models \varphi^{\mathcal{I}}(C_1, \dots)$  for valid codes, independently of the representatives.

For example, a graph code  $f$  on  $\delta \times \epsilon$ , embedded in the square of width  $\theta = \delta + \epsilon + 1$ , uses

$$F(i, j) \iff i < \delta \wedge j < \epsilon \wedge \text{Bit}(f, p_\theta(i, j)).$$

It is a total function exactly when

$$\forall i < \delta \exists j < \epsilon (F(i, j) \wedge \forall k < \epsilon (F(i, k) \longrightarrow k = j)).$$

Injectivity is  $\forall i, i' < \delta \forall j < \epsilon (F(i, j) \wedge F(i', j) \rightarrow i = i')$ , and surjectivity is  $\forall j < \epsilon \exists i < \delta F(i, j)$ . Occurrences of  $f(i)$  in the main text can therefore be eliminated by these unique-value clauses. Root preservation and edge correspondence add only finitely many quantifiers. The graph code’s birthday is  $\Lambda(\theta)$ ; all plus positions outside the relevant rectangle can be required absent, which makes canonical relation coding a definable operation when the intended relation is specified by a formula and its bit string exists.

The dependency order of the abbreviations is

| Predicate                     | Earlier definitions needed               | Quantification simulated                             |
|-------------------------------|------------------------------------------|------------------------------------------------------|
| $\text{Ord}(a)$               | $b(a) = a$                               | Embedded ordinals                                    |
| $\text{Pre}(x, y)$            | $<, b$                                   | Literal prefix test                                  |
| $\text{Bit}(s, a)$            | $\text{Ord}, \text{Pre}, <, b$           | One sign position                                    |
| $E_C$                         | $\text{Bit}, p_\delta$                   | Edges of a code                                      |
| $\text{Code}(C)$              | shape, clean bits, (5.1)–(5.3)           | Pointed membership graphs; all subsets of one domain |
| $\sim, \in_{\mathcal{I}}$     | Code, $E$ , coded maps                   | Interpreted equality and membership                  |
| $\text{OC}(a), \text{VisOrd}$ | $\text{Ord}, \text{Bit}, p_\theta, \sim$ | Literal ordinal as a set code                        |
| Coll                          | Code, $\in_{\mathcal{I}}, \text{VisOrd}$ | Collection of literal ordinals                       |

The table follows source 02’s ledger, with this report’s names. A tuple variable is eliminated by replacing it with ordinary variables; “exactly one” has its usual two-quantifier expansion; variables are renamed before substitution to prevent capture. The specification is an ordinary finite formula, albeit a long one.

### A.2 Every potentially large construction has a stated bound

| Object                                             | Bound used in the proof                                                                                            |
|----------------------------------------------------|--------------------------------------------------------------------------------------------------------------------|
| Subsets tested in well-foundedness                 | Individual sign strings of birthday $\delta$ ; no collection of that whole fiber is required.                      |
| A relation on a code domain $\delta$               | Sign birthday $\Lambda(\delta) = \delta^2 + \delta < \lambda$ .                                                    |
| An isomorphism or member embedding                 | Square width $\delta + \epsilon + 1 < \lambda$ , with the corresponding $\Lambda$ bound.                           |
| Collapse value of a rooted code                    | Transitive closure has exactly $ \delta $ members.                                                                 |
| An arithmetic computation                          | At most $\mu$ word vertices or table cells, hereditary size $\mu$ , where $\mu$ bounds the inputs and is infinite. |
| Rank labels in that computation                    | Below $\mu^+$ , hence each has cardinality at most $\mu$ .                                                         |
| Replacement image after eternity                   | One bound $\max(\aleph_0,  \delta ,  \varrho ) < \kappa$ , not an arbitrary singular-cardinal union.               |
| A packed birthday- $\alpha$ fiber                  | Exists exactly if $2^{ \alpha } < \kappa(\lambda)$ .                                                               |
| The old-power-set parameter                        | Hereditary size $(2^{ \alpha })^M$ for infinite $\alpha$ .                                                         |
| Actual power set of a set of hereditary size $\mu$ | At most $2^\mu$ ; this need not be below $\kappa$ , and no earlier row needs it to be.                             |

## B An essential negative regression

In the membership definition, edge preservation alone is insufficient. Take the code for  $x = \emptyset$  and the three-node chain coding  $y = \{\{\emptyset\}\}$ . Map the sole node of the first code to the middle node of the chain, which codes  $\{\emptyset\}$ . The image root is a child of the target root, and the one-node induced subgraph has no edges, just like the source. But  $\emptyset \notin \{\{\emptyset\}\}$ . The predecessor-full clause (5.6) rejects this false witness, because the middle node has a predecessor omitted from the image. The finite suites of sources 06, 07 and 09 confirm this counterexample.

## C A small concrete membership code

Let  $x = \{\emptyset, \{\emptyset\}\}$ , the von Neumann ordinal 2 (source 02's example). Its transitive closure with the root included has the three nodes  $0 \leftrightarrow \emptyset$ ,  $1 \leftrightarrow \{\emptyset\}$ ,  $2 \leftrightarrow x$ . Take  $\delta = 3$ , root  $t = 2$ , and a relation word of length  $\Lambda(3) = 12$ . The edges are  $(0, 1)$ ,  $(0, 2)$  and  $(1, 2)$ ; under  $p_3(u, v) = 3u + v$  the plus positions are 1, 2, 5, and all other positions of the length-12 word are minus. Its collapse is the identity on the ordinal 3, with root value 2.

A different labeling can put  $x$  at label 0, the empty set at label 2, and  $\{\emptyset\}$  at label 1. The relation word then changes, and the root is no longer the largest label. Nevertheless the isomorphism code identifies the two roots, so interpreted equality cannot be literal equality of the triples. The example also shows why label order must not be confused with membership order in the countable-ordinal overflow of corollary 5.6.

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